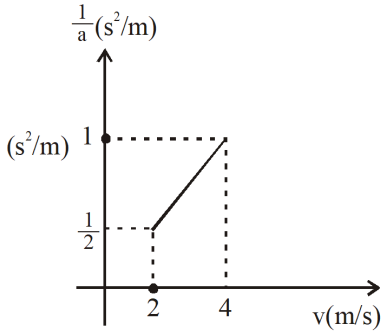
SECTION-I (i) (Maximum Marks:18)

Single Correct Answer Type **6 Q. [3 M (–1)]**

1. Given graph is $\frac{1}{\text{acceleration}}$ vs velocity graph. If the time interval during which velocity changes from 2m/s to 4m/s is given by Δt seconds. Then find the value of $2\Delta t$:-



(A) 3 (B) 4 (C) 5 (D) 6

2. A boatman moves his boat with a velocity 'v' (relative to water) in river and finds to his surprise that velocity of river 'u' (with respect to ground) is more than 'v'. He has to reach a point directly opposite to the starting point on another bank by travelling minimum possible distance. Then

(A) he must steer the boat (with velocity v) at certain angle with river flow so that he can reach the opposite point on other bank directly.

(B) his velocity 'v' must be towards directly opposite point, So, that he can travel rest of distance by walking on other bank to reach the directly opposite point.

(C) boatman should maintain velocity v of boat at certain angle greater than 90° with direction of river flow to minimize drifting and then walk rest of distance on other bank.

(D) boat velocity 'v' should be at an angle less than 90° with direction of river flow to minimize the drift and then walk to the point.

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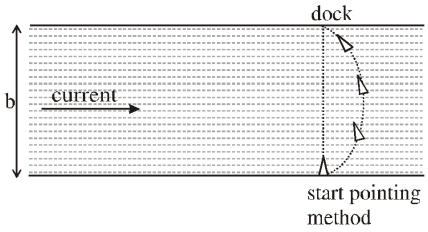
3. Suppose you are anchored near the shore of a channel in which there is steady current, and you are going to run your (motor) boat at constant throttle to a dock directly across the channel on the opposite shore. The way in which one might steer the boat to the dock is pointing method. In pointing method keep the nose of the boat pointed directly at the dock. Velocity of boat with respect to water is 'v' and velocity of water is 'u'. Width of water channel is 'b'. The time taken by boat to reach the dock is :-



(A) $\frac{bv}{v^2-u^2}$ (B) $\frac{b}{\sqrt{v^2-u^2}}$ (C) $\frac{bu}{v^2}$ (D) $\frac{bv}{u^2}$

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4. When a man started from rest with constant acceleration along an incline plane of angle 37°, from horizontal he found rain to be falling vertically. After t seconds, he finds rain to strike him horizontally. Assume that actual velocity of rain is constant with magnitude 4 m/s. Choose the correct option(s) :

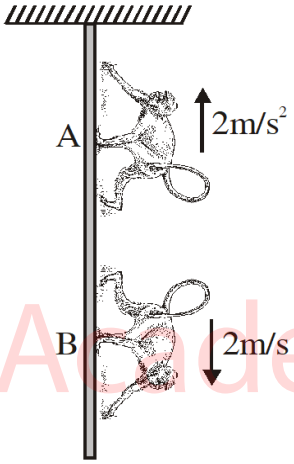
(A) Man is moving up the incline with acceleration of 1 m/s^2 if t = 5 sec.

(B) Man is moving down the incline and if a = 1 m/s^2 then t = 5 sec.

(C) Man is moving down the incline and if t = 20 /3 s then a = 1 m/s^2

(D) Man is moving up the incline and if a = 1 m/s^2 then t = 20/3 s.

5. Two monkeys of masses 10 kg and 8 kg are moving along a vertical light rope, the former climbing up with an acceleration of 2 m/s^2, while the latter coming down with a uniform velocity of 2 m/s. Find tension in the rope at the fixed support.



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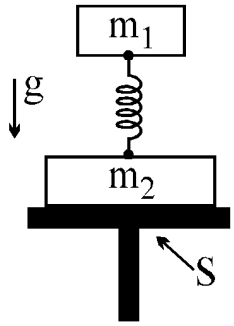
(A) 180 N (B) 200 N (C) 80 N (D) 216 N

6. The system of two weights with masses m_1 and m_2 are connected with weightless spring as shown. The system is resting on the support S. The support S is quickly removed. The accelerations of each of the weights right after the support S is removed are.



(A) $a_1 = 0, a_2 = \frac{(m_1 + m_2)g}{m_2}$

(B) $a_1 = 0, a_2 = \frac{(m_1 + m_2)g}{m_1}$

(C) $a_1 = \frac{(m_1 + m_2)g}{m_1}, a_2 = 0$

(D) $a_1 = 0, a_2 = 0$

SECTION-I (ii) : (Maximum Marks: 32)

Multiple Correct Answer Type 8 Q. [4 M (–2)]

7. In terms of potential difference V, electric current I, permittivity ε_0, permeability μ_0 and speed of light c, the dimensionally correct equation(s) is(are) :-

(A) $\mu_0 I^2 = \varepsilon_0 V^2$ (B) $\varepsilon_0 I = \mu_0 V$ (C) $I = \varepsilon_0 cV$ (D) $\mu_0 cI = \varepsilon_0 V$

8. The motion of a body is given by the equation $\frac{dv(t)}{dt} = 6.0 - 3\,v(t)$; where v (t) is the speed in m/s & t in sec., if the body was at rest at t = 0 :

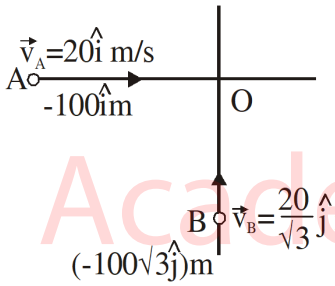
(A) the terminal speed is 2.0 m/s

(B) the magnitude of the initial acceleration is 6.0 m/s²

(C) the speed varies with time as $v(t) = 2\,(1 - e^{-3t})$ m/s

(D) the speed is 1.0 m/s when the acceleration is half the initial value.

9. Positions of two vehicles A and B with reference to origin O and their velocities are as shown.



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(A) they will collide (B) distance of closest approach is 100 m.

(C) their relative speed is $\frac{40}{\sqrt{3}}$ m/s (D) their relative velocity is $\frac{20}{\sqrt{3}}$ m/s

10. River is flowing with a velocity $\vec{v}_R = 4\hat{i}$ ms^{-1}. A boat is moving with a velocity $\left(-2\hat{i} + 4\hat{j}\right) ms^{-1}$ relative to river and width of river is 100 m along y direction then-

(A) The time of crossing is 25 sec. (B) Absolute velocity of boat is $2\sqrt{5}$ ms^{-1}

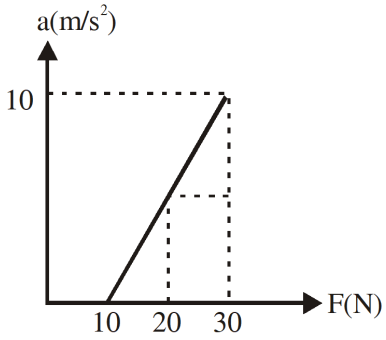
(C) Drift of boat is 50 m. (D) The boat can never cross the river.

11. A carpenter of mass 50 kg is standing on a weighing machine placed in a lift of mass 20 kg. A light string is attached to the lift. The string passes over a smooth pulley and the other end is held by the carpenter as shown. When carpenter keeps the lift moving upward with constant velocity :- ($g = 10$ m/s²)

(A) the reading of weighing machine is 15 kg. (B) the man applies a force of 350 N on the string.

(C) net force on the man is 150 N. (D) Net force on the weighing machine is 150 N.

12. A block placed on a rough horizontal surface is pushed with a force F acting horizontally on the block. The magnitude of F is increased and acceleration produced is plotted in the graph shown.



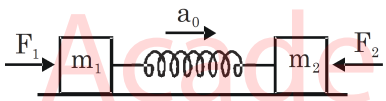
(A) Mass of the block is 2 kg.

(B) Coefficient of friction between block and surface is 0.5.

(C) Limiting friction between block and surface is 10 N.

(D) When F = 8 N, friction between block and surface is 10 N.

13. Two blocks m_1 and m_2 are connected with a compressed spring and placed on a smooth horizontal surface as shown in figure. Force constant of spring is k. Under the influence of forces F_1 and F_2, at an instant blocks move with common acceleration a_0. At that instant force F_2 is suddenly withdrawn. Mark correct option.



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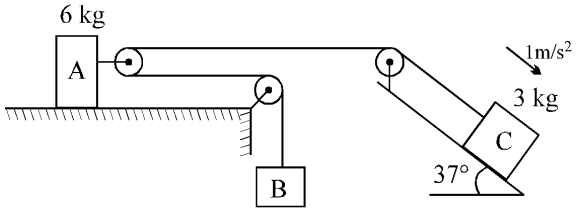
(A) Instantaneous acceleration of m_1 is $a_0 - \frac{F_1}{m_1}$

(B) Instantaneous acceleration of m_2 is $a_2 = a_0 + \frac{F_2}{m_2}$

(C) Instantaneous acceleration of m_1 is $a_1 = a_0$

(D) Spring force is $F_{spring} = m_2a_0 + F_2$

14. Three blocks A, B & C are arranged as shown. Pulleys and strings are ideal. All surfaces are frictionless. If block C is observed moving down along the incline at 1 m/s². Then :- (system is released from rest.)



(A) The mass of block B is 15 kg.

(B) Tension in string is 15 N .

(C) Acceleration of A and B are 5 m/s² and 9 m/s² respectively.

(D) Acceleration of A and B are 9 m/s² and 5 m/s² respectively.

SECTION-I (iii) : (Maximum Marks: 21)

Linked Comprehension Type (Single Option Correct) (1 Para × 3 Q.) (2 Para × 2 Q.) [3 M (–1)]

Paragraph for Questions 15 and 16

At t = 0 on the planet "Gravitus Increasicus", a projectile is fired with speed 6 m/s at an angle 30° above the horizontal. This planet is a strange one, in that the acceleration due to gravity increases linearly with time, starting with a value of zero when the projectile is fired. In other words, g(t) = 2t.

15. The time of flight of the projectile is :-

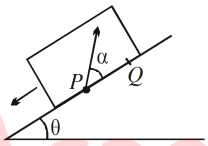
(A) 1 sec (B) 2 sec (C) 3 sec (D) $3\sqrt{3}$ sec

16. Range of the projectile is :-

(A) 9m (B) $\frac{9}{\sqrt{3}}$m (C) $6\sqrt{3}$ m (D) $9\sqrt{3}$ m

Paragraph for Questions 17 and 18

A large heavy box is sliding without friction down a smooth plane of inclination θ. From a point P on the bottom of the box, a particle is projected inside the box. The initial speed of the particle with respect to box is u and the direction of projection makes an angle a with the bottom as shown in figure.



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17. Find the distance along the bottom of the box between the point of projection P and the point Q where the particle lands. (Assume that the particle does not hit any other surface of the box. Neglect air resistance).

(A) $\frac{u^2 \sin 2\theta}{g \cos \alpha}$ (B) $\frac{u^2 \sin 2\alpha}{g \cos \theta}$ (C) $\frac{u^2 \sin 2\theta}{g \sin \alpha}$ (D) $\frac{u^2 \sin 2\alpha}{g \sin \theta}$

18. If the horizontal displacement of the particle as seen by an observer on the ground is zero, find the speed of the box with respect to the ground at the instant when the particle was projected.

(A) $v = \frac{u \sin(\alpha + \theta)}{\cos \theta}$ (B) $v = \frac{u \sin(\alpha + \theta)}{\sin \theta}$ (C) $v = \frac{u \cos(\alpha + \theta)}{\cos \theta}$ (D) $v = \frac{u \cos(\alpha + \theta)}{\sin \theta}$

Paragraph for Questions 19 - 21

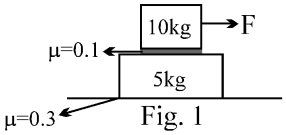


Fig. 1

19. The maximum F which will not cause motion of any of the blocks is

(A) 10 N (B) 15 N (C) data insufficient (D) None

20. The maximum acceleration of 5 kg block is :-

(A) 1 m/s^2 (B) 3 m/s^2 (C) 0 (D) None

21. The acceleration of 10 kg block when F = 30 N is

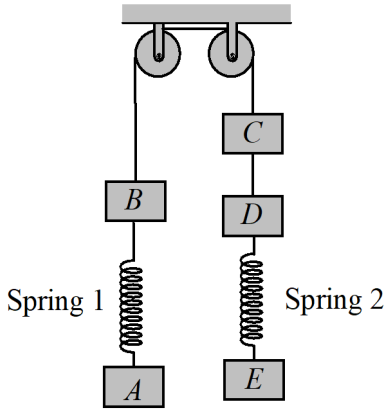
(A) 2 m/s^2 (B) 3 m/s^2 (C) 1 m/s^2 (D) None

SECTION-I (iv) : (Maximum Marks: 16)

Linked Comprehension Type (Multiple Options Correct) 1 Para × 4 Q. [4 M (–1)]

Paragraph for Questions 22 - 25

The system shown in the figure is initially in equilibrium. A is of mass 2m and B, C, D and E are of mass m.



22. When spring 1 is cut, then the acceleration of each blocks will be :-

(A) $a_A = g \downarrow$ (B) $a_D = \frac{2g}{3} \downarrow$ (C) $a_E = 0$ (D) $a_E = g \downarrow$

23. When spring 2 is cut, then the acceleration of each blocks will be :-

(A) $a_A = 0$ (B) $a_B = \frac{g}{3} \downarrow$ (C) $a_B = \frac{2g}{3} \uparrow$ (D) $a_E = g \downarrow$

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24. When string between C & D is cut, then the acceleration of each blocks will be :-

(A) $a_A = 0$ (B) $a_E = 0$ (C) $a_A = g \downarrow$ (D) $a_E = g \downarrow$

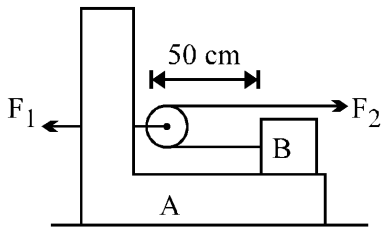
25. When string between B and C is cut, then the acceleration of each blocks will be :-

(A) $a_A = 0$ (B) $a_E = 0$ (C) $a_A = g \downarrow$ (D) $a_E = g \downarrow$

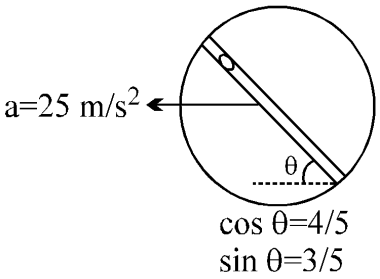
SECTION-II (i) : (Maximum Marks: 8)

Numerical Answer type Questions (Upto Second Decimal Place) 2 Q. [4 M (0)]

1. A 1kg block B rests as shown on a bracket A of same mass. Constant forces $F_1 = 20N$ and $F_2 = 8N$ start to act at time t = 0 when the distance of block B from pulley is 50cm. Time when block B reaches the pulley is _____.



2. A circular disc with a groove along its diameter is placed horizontally. A block of mass 1 kg is placed as shown. The co-efficient of friction between the block and all surfaces of groove in contact is $\mu = 2/5$. The disc has an acceleration of 25 m/s^2. Find the acceleration of the block with respect to disc.



SECTION-II (ii) : (Maximum Marks: 8)

Numerical-Answer-Type questions (Non Negative Integer) 2 Q. [4 M (0)]

3. A bird is at a point P (4, −1, −5) and sees two points P_1 (−1, −1, 0) and P_2 (3, −1, −3). At time t = 0, it starts flying with a constant speed of 10 m/s to be in line with points P_1 and P_2 in minimum possible time t. Find t (in sec), if all coordinates are in kilometers.

4. A bow man is riding on a horse moving with speed of $2\sqrt{2}$ ms^{-1} along a straight road. He aims at his enemy moving perpendicularly to the road at speed of $\sqrt{2}$ ms^{-1}. At the instant when he fires the arrow, the line joining man and his enemy makes an angle of 45° with the road. Find the angle (in degree) with the road at which he should aim to hit his enemy? Muzzle velocity of arrow is 5ms^{-1}. (given that sin 37° = 3/5). Neglect effect of gravity.

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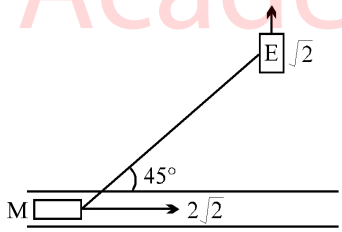
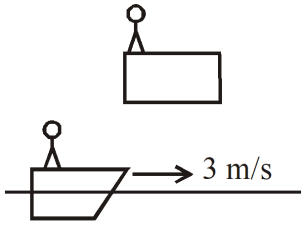


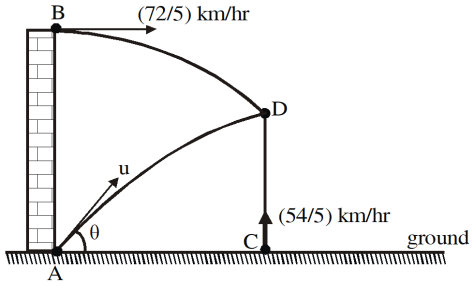
SECTION-III : (Maximum Marks: 24)

Numerical Grid Type (Ranging from 0 to 9) 6 Q. [4 M (0)]

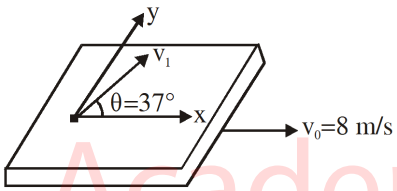
1. You are standing on the Chambal Bridge watching the boats in the river. You see a motorboat pass directly below you, traveling perpendicular to the bridge at a speed of 3 m/s. A person on the boat throws a baseball at an initial speed of v_0 and at an angle of 37° from the vertical (Note: both v_0 and the angle are with respect to the boat). Find the value of v_0 (in m/s) necessary for the ball to travel straight up towards you.



2. In the given figure points A and C are on the horizontal ground & A and B are in same vertical plane. Simultaneously bullets are fired from A, B and C and they collide at D. The bullet at B is fired horizontally with speed of $\frac{72}{5}$ km/hr and the bullet at C is projected vertically upward at velocity of $\frac{54}{5}$ km/hr. Find velocity of the bullet projected from A in m/s.



3. A board is being moved with a cosntant velocity v_0 on a smooth horizontal plane. A small block is projected on the rough boards. Coefficient of friction between board and block is $\mu = 0.3$. Block is projected with velocity $v_1 = 25$ m/s at an angle of $\theta = 37°$ relative to ground. What is the velocity of block (in m/s) relative to ground after time t = 10 sec.



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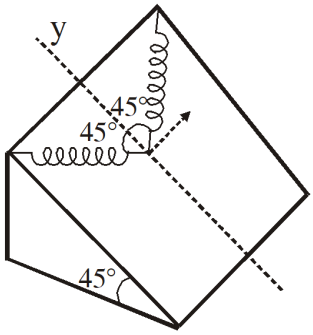
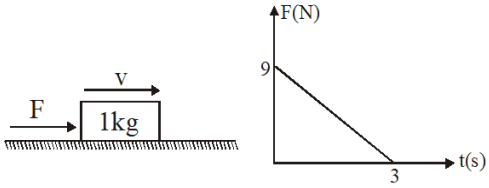
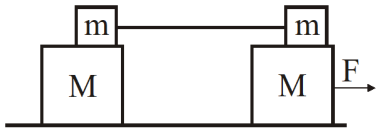
4. Figure shows a ball of mass m connected with two ideal springs of force constant k, kept in equilibrium on a smooth incline, suddenly right spring is cut. What is magnitude of instantaneous acceleration (in m/s2) of ball.



5. A 1 kg block moving on horizontal rough surface of friction coefficient 0.6 is pushed with a force varying with time t in seconds as shown in figure. If initial velocity of block was 4.5 m/s. Find the velocity (in m/s) at t = 3 s.



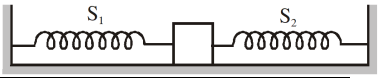
6. Four blocks are arranged on a smooth horizontal surface as shown. The masses of the blocks are given (see the diagram). The coefficient of static friction between the top and the bottom blocks is μ_s. If the maximum value of the horizontal force F, applied to one of the bottom blocks as shown, that makes all four blocks move with the same acceleration is $2\mu_s mg\left(\frac{m+M}{\alpha m+M}\right)$, Then find the value of α.



SECTION-IV : (Maximum Marks: 8)

Matrix Match Type (4 × 5) (One or more options correct) 1 Q. [8 M; (for each entry +2(0)]

1. The figure shows a block B of mass 2 kg kept on a smooth horizontal floor in equilibrium with two identical springs of S_1 & S_2 force constant k = 100 N/m attached to it and to fixed supports as shown. The block is then displaced horizontally from this position by amount x and released. Match the initial equilibrium conditions & subsequent values of x in column-I with corresponding acceleration of the block when released in column II.



Column-I		Column-II	
(A)	S_1 and S_2 are relaxed and x = 2 cm.	(P)	6 m/s^2
(B)	S_1 and S_2 are stretched by 1 cm and x = 2 cm	(Q)	4 m/s^2
(C)	S_1 and S_2 are compressed by 2 cm and x = 4 cm	(R)	3 m/s^2
(D)	S_1 and S_2 are compressed by 4 cm and x = 1 cm	(S)	2 m/s^2
		(T)	1 m/s^2

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ANSWER KEY

GR_Unit and Dimension, Vectors, Kinematics, NLM & Friction

PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A	C	A	C	B	A		
SECTION-I (ii)	Q.	7	8	9	10	11	12	13	14
	A.	A,C	A,B,C,D	B,C	A,B,C	A,B	A,B,C	B,C,D	A,B,C
SECTION-I (iii)	Q.	15	16	17	18	19	20	21	
	A.	C	D	B	C	A	C	A	
SECTION-I (iv)	Q.	22	23	24	25				
	A.	A,B,C	A,B,D	A,B	A,B				
SECTION-II (i)	Q.	1	2						
	A.	0.5	10						
SECTION-II (ii)	Q.	3	4						
	A.	100	82						
SECTION-III	Q.	1	2	3	4	5	6		
	A.	5	5	8	5	0	2		
SECTION-IV	Q.	1							
	A.	A->S,B->S,C->Q,D->T							

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Important Notes

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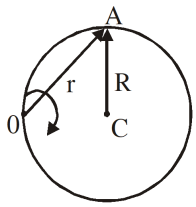
SECTION-I (i) (Maximum Marks:21)

Single Correct Answer Type 7 Q. [3 M (–1)]

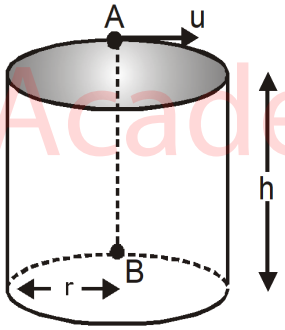
1. A particle A moves along a circle of radius R = 50 cm so that its radius vector r relative to the point O (figure) rotates with the constant angular velocity $\omega = 0.40$ rad/s. Then modulus of the velocity of the particle, and the modulus of its total acceleration will be



(A) v= 0.4 m/s, a = 0.4 m/s^2 (B) v = 0.32 m/s, a = 0.32 m/s^2

(C) v = 0.32 m/s, a = 0.4 m/s^2 (D) v = 0.4 m/s, a = 0.32 m/s^2

2. A hollow vertical cylinder of radius r and height h has a smooth internal surface. A small particle is placed in contact with the inner side of the upper rim, at point A, and given a horizontal speed u, tangential to the rim. It leaves the lower rim at point B, vertically below A. If n is an integer then



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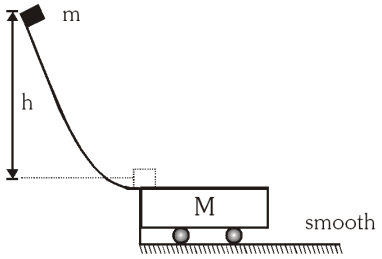
(A) $\frac{u}{2\pi r}\sqrt{\frac{2h}{g}} = n$ (B) $\frac{h}{2\pi r} = n$ (C) $\frac{2\pi r}{h} = n$ (D) $\frac{u}{\sqrt{2gh}} = n$

3. A carriage of mass M and length ℓ is joined to the end of a slope as shown in the figure. A block of mass m is released from the slope from height h. It slides till end of the carriage. The friction between the mass and the slope and also friction between carriage and horizontal floor is negligible. Coefficient of friction between block and carriage is μ. Find h in the given terms.



(A) $\mu\left(1+\frac{M}{m}\right)\ell$ (B) $2\mu\left(1+\frac{M}{m}\right)\ell$ (C) $\mu\left(2+\frac{m}{M}\right)\ell$ (D) $\mu\left(1+\frac{m}{M}\right)\ell$

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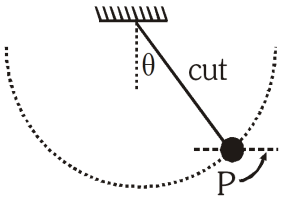
4. Consider a pendulum, consisting of a massless string with a mass on the end. The mass is held with the string horizontal, and then released. The mass swings down, and then on its way back up, the string is cut at point P when it makes an angle of θ with the vertical. What should θ be, so that the mass travels the largest horizontal distance from P by the time it returns to the height it had when the string was cut?

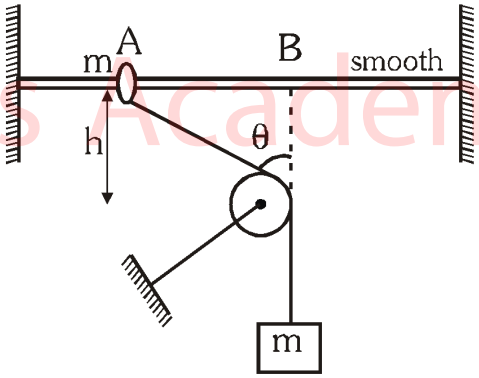


(A) $\tan^{-1}\sqrt{3}$ (B) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (C) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (D) $\tan^{-1}(1)$

5. A uniform chain has a mass M and length L. It is placed on a frictionless table with length ℓ_0 hanging over the edge. The chain begins to slide down. The speed V with which the chain slides away from the edge is given by

(A) $V = \sqrt{\frac{g\ell_0}{L}(L+\ell_0)}$ (B) $V = \sqrt{\frac{g\ell_0}{L}(L-\ell_0)}$ (C) $V = \sqrt{\frac{g}{L}(L^2-\ell_0^2)}$ (D) $V = \sqrt{2g(L-\ell_0)}$

6. A ring of mass m slide from rest on the smooth rod as shown in the figure, due to the block of mass m. Pully and string are massless. Then find the speed of ring when the string become straight. (Given $\theta = 60°$)



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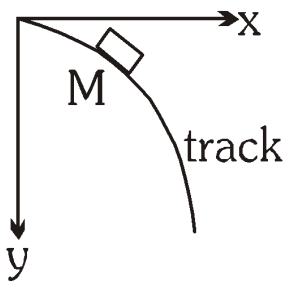
(A) $4\sqrt{gh}$ (B) $\sqrt{2gh}$ (C) $2\sqrt{2gh}$ (D) $8\sqrt{gh}$

7. A man M slides down a curved frictionless track, starting from rest. The curve obeys the equation $y = \frac{x^2}{2}$. The tangential acceleration of man is :



(A) g (B) $\frac{gx}{\sqrt{x^2+4}}$ (C) $\frac{g}{2}$ (D) $\frac{gx}{\sqrt{x^2+1}}$

SECTION-I (ii) : (Maximum Marks: 16)

Multiple Correct Answer Type 4 Q. [4 M (–2)]

8. An ant travels along a long rod with a constant velocity $\vec{u}$ relative to the rod starting from the origin. The rod is kept initially along the positive x-axis. At t = 0, the rod also starts rotating with an angular velocity ω (anticlockwise) in x-y plane about origin. Then

(A) the position of the ant at any time t is $\vec{r} = ut[\cos \omega t\hat{i} + \sin \omega t\hat{j}]$

(B) the speed of the ant at any time t is $u\sqrt{1+\omega^2t^2}$

(C) the magnitude of the tangential acceleration of the ant at any time t is $\dfrac{\omega^2 tu}{\sqrt{1+\omega^2t^2}}$

(D) the speed of the ant at any time t is $\sqrt{1+2\omega^2t^2}u$

9. Which of the following is/are conservative force(s)?

(A) $\vec{F} = 2r^3\hat{r}$ (B) $\vec{F} = -\dfrac{5}{r}\hat{r}$ (C) $\vec{F} = \dfrac{3(x\,i + y\,\hat{j})}{(x^2+y^2)^{3/2}}$ (D) $\vec{F} = \dfrac{3(y\,i + x\,\hat{j})}{(x^2+y^2)^{3/2}}$

10. On a train moving along east with a constant speed v, a boy revolves a bob with string of length ℓ on smooth surface of a train, with equal constant speed v relative to train. Mark the correct option(s).

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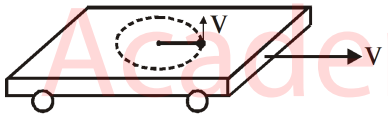


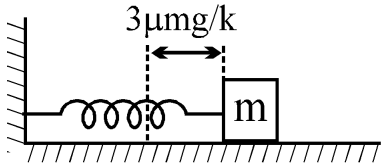
(A) Maximum speed of bob is 2 v in ground frame.

(B) Tension in string connecting bob is $\dfrac{4mv^2}{\ell}$ at an instant.

(C) Tension in string is $\dfrac{mv^2}{\ell}$ at all the moments.

(D) Minimum speed of bob is zero in ground frame.

11. A spring block system is placed on a rough horizontal surface having coefficient of friction m. Spring is given initial elongation $3\mu mg/k$ (where m = mass of block and k = spring constant) and the block is released from rest. For the subsequent motion :-



(A) initial acceleration of block is $2\mu g$.

(B) maximum compression in spring is $\mu mg/k$.

(C) maximum speed of the block is $2\mu g\sqrt{\dfrac{m}{k}}$.

(D) maximum speed of the block is $\mu g\sqrt{\dfrac{m}{k}}$.

SECTION-I (iii) : (Maximum Marks: 9)

Linked Comprehension Type (Single Option Correct) 1 Para × 3 Q. [3 M (–1)]

Paragraph for Questions 12 - 14

The potential energy at a point, relative to the reference point is defined as the negative of work done by the conservative force as the object moves from the reference point to the point considered. The value of potential energy at the reference point itself can be set equal to zero because we are always concerned only with differences of potential energy between two points and the associated change of kinetic energy. A particle A is fixed at origin of a fixed coordinate system. Another particle B which is free to move experiences an force $\vec{F} = \left(-\frac{2\alpha}{r^3} + \frac{\beta}{r^2}\right)\hat{r}$ due to particle A where $\vec{r}$ is the position vector of particle B relative to A. It is given that the force is conservative in nature and potential energy at infinity is zero. If B has to be removed from the influence of A, energy has to be supplied for such a process. The ionization energy E_0 is work that has to be done by an external agent to move the particle from a distance r_0 to infinity slowly. Here r_0 is the equilibrium position of the particle.

12. What is potential energy function of particle as function of r.

(A) $\frac{\alpha}{r^2} - \frac{\beta}{r}$ (B) $-\frac{\alpha}{r^2} + \frac{\beta}{r}$ (C) $-\frac{\alpha}{r^2} - \frac{\beta}{r}$ (D) $\frac{\alpha}{r^2} + \frac{\beta}{r}$

13. Find the ionization energy E_0 of the particle B.

(A) $\frac{\beta^2}{2\alpha}$ (B) $\frac{2\beta^2}{\alpha}$ (C) $\frac{\beta^2}{4\alpha}$ (D) $\frac{\beta^2}{\alpha}$

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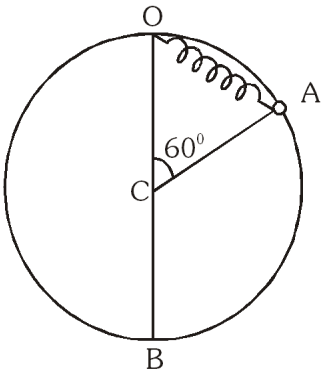
14. If particle B is transfered slowly from point $P_1\left(\sqrt{2}r_0, \sqrt{2}r_0\right)$ to point $P_2\left(\frac{r_0}{\sqrt{2}}, \frac{r_0}{\sqrt{2}}\right)$ in the xy-plane by an external agent, calculate work required to be done by it in the process.

(A) $\frac{9\beta^2}{64\alpha}$ (B) $\frac{\beta^2}{16\alpha}$ (C) $\frac{\beta^2}{64\alpha}$ (D) None of these

SECTION-II : (Maximum Marks: 12)

Numerical-Answer-Type questions (Non Negative Integer) 3 Q. [4 M (0)]

1. A particle of mass 5 kg is free to slide on a smooth ring of radius r = 20 cm fixed in a vertical plane. The particle is attached to one end of a spring whose other end is fixed to the top point O of the ring. Initially the particle is at rest at a point A of the ring such that $\angle OCA = 60°$, C being the centre of the ring. The natural length of the spring is also equal to r = 20 cm. After the particle is released and slides down the ring the contact force between the particle & the ring becomes zero when it reaches the lowest position B. Determine the force constant of the spring.



2. A particle is moving in a circle of radius R in such a way that at any instant the normal and the tangential component of its acceleration are equal. If its speed at t = 0 is v_0 then the time it takes to complete the first revolution is $\frac{R}{\alpha v_0}\left(1-e^{-\beta\pi}\right)$. Find the value of $(\alpha+\beta)^6$.

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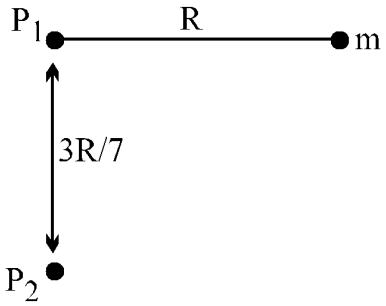
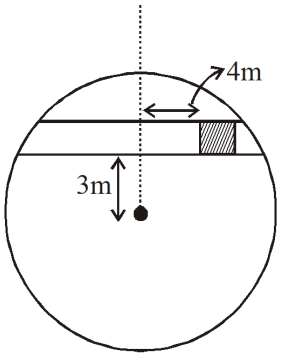
3. A simple pendulum consists of a bob of mass m and a string of length R suspended from a peg P_1 on the wall. A second peg P_2 is fixed vertically below the first one at a distance 3R/7 from it. The pendulum is drawn aside such that the string is horizontal and released. The maximum height (with respect to the lowest point) to which it rises is $\frac{\alpha R}{28}$. Find α.



SECTION-III : (Maximum Marks: 28)

Numerical Grid Type (Ranging from 0 to 9) 7 Q. [4 M (0)]

1. A small cubical block of mass 1 kg is placed inside a rough rectangular groove made in a circular rough table as shown in the figure. Coefficient of friction for all the rough surfaces is $\mu = 0.5$. The table starts rotating clockwise with angular acceleration 1 rad/sec^2 in a horizontal plane about its axis. Find the time (in sec) after which the block will start motion with respect to table. Assume the size of block slightly smaller then the width of groove.



2. A mass m = 1kg, which is free to move on a horizontal frictionless surface, is attached to one end of a massless string that wraps partially around a frictionless vertical pole of radius r (see the top view in figure). You hold on to the other end of the string. At t = 0, the mass has speed v_0 in the tangential direction along the dotted circle of radius R shown. Your task is to pull on the string so that the mass keeps moving along the circular path of radius R. If at any instant 't' force applied by hand on string is $f_0 = 2N$ then find out centripetal acceleration (in m/s^2) of mass. (Given : $R = \frac{2r}{\sqrt{3}}$)

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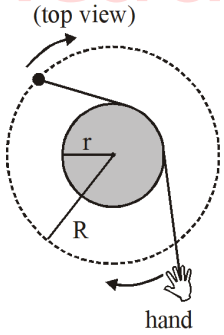
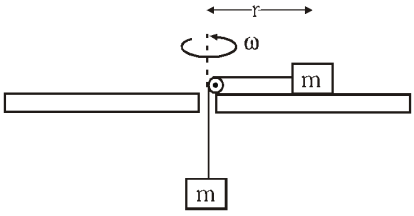
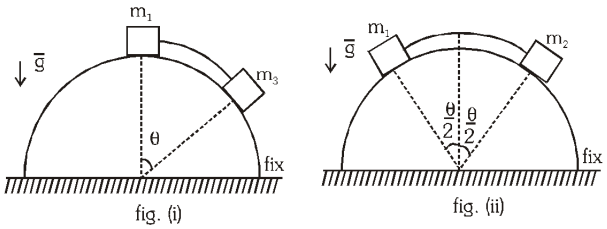


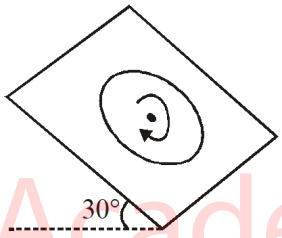
3. Two equal masses m are attached by a string. One mass lies at radial distance r from the center of a horizontal turntable which rotates with constant angular velocity ω, while the second hangs from the strings inside the turntable's hollow spindle (see figure). The coefficient of static friction between the turntable and the mass lying on it is $\mu_s = 0.5$. The maximum and minimum values r_{max}, r_{min} of r such that the mass lying on the turntable does not slide. Find the ratio of r_{max} / r_{min} ?



4. There are three blocks of masses $m_1 = m$, $m_2 = 4m$ and $m_3 = m$. If two block system is kept over a smooth hemisphere with two different arrangement one by one as shown in figure (i) & (ii) respectively are connected with massless & inextensible string. In arrangement figure (i), after releasing the system from rest acceleration of blocks is found to be $\left(a = \frac{5\sqrt{3}}{2}\right)$ m/s^2. In second arrangement block m_1 & m_2 are kept symmetrically. Find the acceleration of block (in m/s^2) after releasing the system from rest according to figure (ii).



5. An old record player of 10 cm radius turns at 10 rad/s while mounted on a 30° incline as shown in the figure. A particle of mass m can be placed any where on the rotating record. If the least possible coefficient of friction m that must exist for no slipping to occur is μ, find $2\sqrt{3}\mu$.



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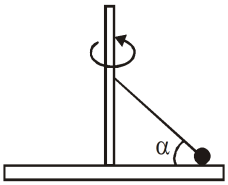
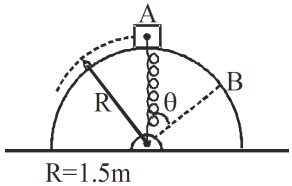
6. A circular platform rotates around a vertical axis with angular velocity $\omega = 10$ rad/s. On the platform is a ball of mass 1 kg, attached to the long axis of the platform by a thin rod of length 10 cm ($\alpha = 30°$). Find normal force exerted by the ball on the platform (in newton). Friction is absent.



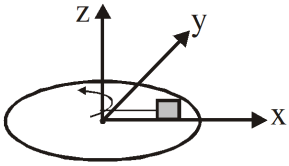
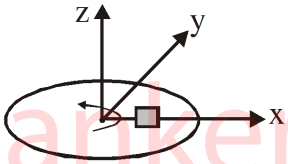
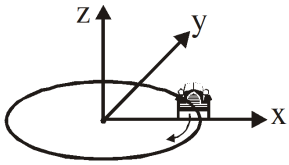
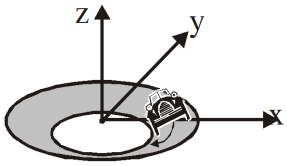
7. A 2 kg block is gently pushed from rest at A and it slides down along the fixed smooth circular surface as shown in figure. If the attached spring has a force constant k = 20 N/m. What is unstretched length of spring (in m) so that it does not allow the block to leave the surface until angle with the vertical is $\theta = 60°$.



SECTION-IV : (Maximum Marks: 8)

Matrix Match Type (4 × 5) (One or more options correct) 1 Q. [8 M; (for each entry +2(0)]

1. Mark the matching options for situations shown in respective figures of column-I at the instant, when particle / car is located on the x-axis as shown :

Column-I		Column-II	
(A)	Block attached to string is moving along a circle on rough surface 	(P)	Force due to friction may have non-zero x-component
(B)	Block is placed on a disc rotating with non-uniform angular velocity as shown below. There is no slipping between block & disc. 	(Q)	Force due to friction may have non-zero y-component.
(C)	Car moving on ground along a circular horizontal track at a constant speed 	(R)	Force due to friction may have non-zero z-component.
(D)	Car moving on ground at constant speed along a circular banked track. 	(S)	Force due to friction may be zero.
		(T)	Centripetal force is only due to friction.

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ANSWER KEY

GR_Circular Motion, Work, Power & Energy

PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7
	A.	D	A	D	C	C	B	D
SECTION-I (ii)	Q.	8	9	10	11			
	A.	A,B,C	A,B,C	A,C,D	A,B,C			
SECTION-I (iii)	Q.	12	13	14				
	A.	B	C	B				
SECTION-II	Q.	1	2	3				
	A.	500	729	27				
SECTION-III	Q.	1	2	3	4	5	6	7
	A.	2	1	3	3	6	5	1
SECTION-IV	Q.	1						
	A.	A->Q,B->PQT,C->PT,D->P						

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Important Notes

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Important Notes

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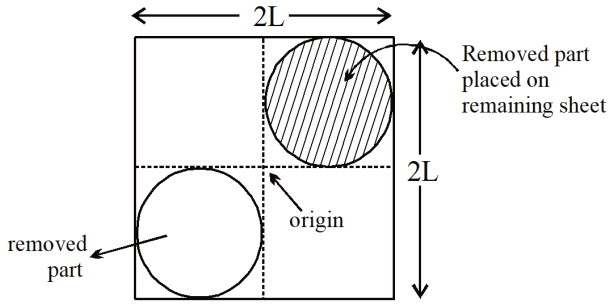
SECTION-I (i) (Maximum Marks:12)

Single Correct Answer Type 4 Q. [3 M (–1)]

1. Figure shows a square lamina with a disc of radius $\frac{L}{2}$ removed from it which is now placed symmetrically over upper right quarter. What is location of centre of mass of system relative to origin shown in figure.



(A) $\frac{\pi L}{12}(\hat{i}+\hat{j})$ (B) $\frac{\pi L}{8}(\hat{i}+\hat{j})$ (C) $\frac{\pi L}{4}(\hat{i}+\hat{j})$ (D) $\frac{\pi L}{16}(\hat{i}+\hat{j})$

2. A fixed wedge ABC is in the shape of an equilateral triangle of side ℓ. Initially, a chain of length 2ℓ and mass m rests on the wedge inside of a tube as shown. The chain is pulled slightly, as a result it starts sliding out of the tube. Work done by gravity till the time, the chain leaves the wedge will be :-

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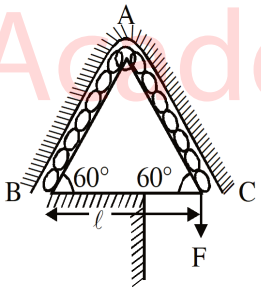
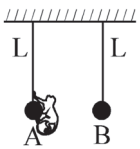


(A) $\left(\frac{(\sqrt{3}+1)mg\ell}{2}\right)$ (B) $\left(\frac{(\sqrt{3}+2)mg\ell}{2}\right)$ (C) $\left(\frac{(\sqrt{3}+2)mg\ell}{4}\right)$ (D) $\left(\frac{(\sqrt{3}+4)mg\ell}{4}\right)$

3. A man weighing 80 kg is standing at the centre of a flat boat and he is 20 m from the shore. He walks 8 m on the boat towards the shore and then halts. The boat weight 200 kg. How far is he from the shore at the end of this time ?

(A) 11.2 m (B) 13.8 m (C) 14.3 m (D) 15.4 m

4. A monkey jumps from ball A onto ball B which are suspended from inextensible light strings each of length L. The mass of each ball & monkey is same. What should be the minimum relative velocity of jump w.r.t. ball, if both the balls manage to complete the circle?



(A) $\sqrt{5gL}$ (B) $\sqrt{20gL}$ (C) $4\sqrt{5gL}$ (D) none

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SECTION-I (ii) : (Maximum Marks: 40)

Multiple Correct Answer Type 10 Q. [4 M (–2)]

5. A man of mass m on an initially stationary boat gets off of the boat by leaping to the left in an exactly horizontal direction. Immediately after the leap, the boat of mass M, is observed to be moving to the right at speed v.

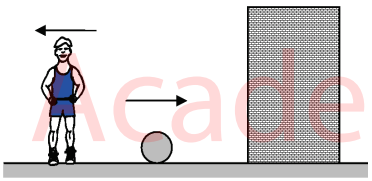
(A) Work done by man on boat is $\frac{1}{2}(m)v^2$.

(B) Increase in the mechanical energy of system of man and boat is $\frac{1}{2}\left(\frac{M^2}{m}+M\right)v^2$.

(C) Velocity of centre of mass of system is v.

(D) Increase in kinetic energy of man is $\frac{1}{2}\frac{M^2}{m}v^2$

6. A man of mass M is carrying a ball of the mass M/2. The man is initially in the state of rest at a distance D from fixed vertical wall. He throws the ball along the floor towards the wall with a velocity v with respect to earth at t = 0. Because of throwing, the man also starts moving backwards. The ball rebounds elastically from the wall. The man finally collects the ball. Assume friction is absent.



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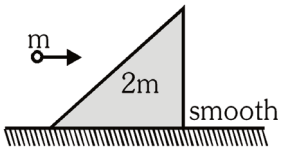
(A) The velocity of the man + ball system after the man has collected the ball is $\frac{2v}{3}$

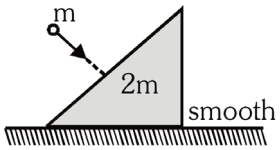
(B) Net linear impulse by ball on man is $\frac{Mv}{3}$

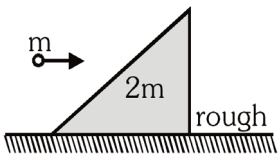
(C) Total linear impulse by ball on man is $\frac{Mv}{6}$

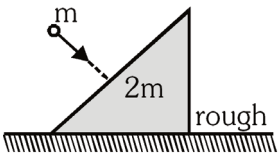
(D) He collects the ball at t = $\frac{4D}{v}$

7. In which of the following system (s) linear momentum cannot be conserved during collision along horizontal line (parallel to base) ?

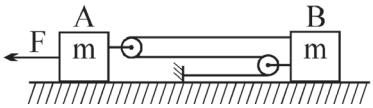
(A) 

(B) 

(C) 

(D) 

8. Two blocks A and B of equal mass are connected by a light inextensible taut string passing over two light smooth pulleys fixed to the blocks. The parts of the string not in contact with the pulleys are horizontal. A horizontal force F is applied to the block A as shown. There is no friction, then



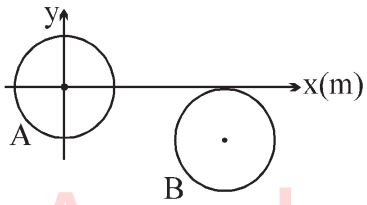
(A) the acceleration of A will be more that of B

(B) the acceleration of A will be less than that of B

(C) the sum of rate of changes of momentum of A and B is greater than the magnitude of F.

(D) the sum of rate of changes of momentum of A and B is equal to the magnitude of F.

9. Two smooth balls A and B, each of mass m and radius R, have their centres at (0, 0, R) and at (5R, –R, R) respectively, in a coordinate system as shown. Ball A, moving along positive x axis, collides with ball B. Just before the collision, speed of ball A is 4 m/s and ball B is stationary. The collision between the balls is elastic.



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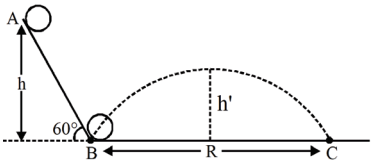
(A) Velocity of the ball A just after the collision is $(\hat{i} + \sqrt{3}\hat{j})$ m/s

(B) Velocity of B just after collision is zero.

(C) Impulse of the force exerted by A on B during the collision, is equal to $(3m\hat{i} - \sqrt{3}m\hat{j})$ kg-m/s

(D) Impulse of the force exerted by A on B is $\left(4m\hat{i}\right)$kg-m/s.

10. In the figure shown, a ball is released from the smooth track. If the ball strikes the horizontal surface of the track and bounces off, it again strikes the horizontal surface at some distance R from B and rises to a height maximum of h' above the surface (If the coefficient of restitution for the collision is e) :-



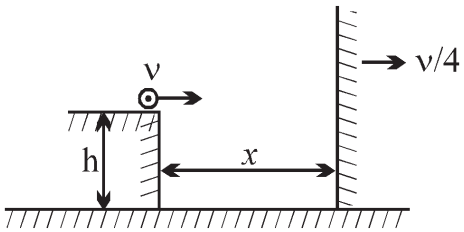
(A) R is maximum for e = $\frac{1}{\sqrt{3}}$

(B) R is maximum for e = 1

(C) $h' \leqslant \frac{3h}{4}$ for all values of e

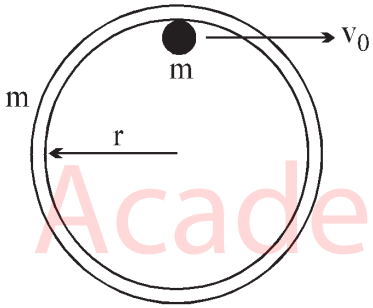
(D) h' = h for e = 1

11. A particle is thrown from a height h horizontally towards a smooth vertical wall moving away with a speed n as shown in the figure. If the particle returns to the point of projection after suffering two elastic collisions, one with the wall and another with the ground, then the total time of flight is T and initial separation between the particle and the wall is x. Then :-



(A) $T = 2\sqrt{\frac{2h}{g}}$ (B) $T = 2\sqrt{\frac{h}{2g}}$ (C) $x = v\sqrt{\frac{h}{2g}}$ (D) $x = v\sqrt{\frac{2h}{g}}$

12. A small body of mass m is at rest inside a thin, narrow tyre of mass m and radius r which is lying on a horizontal table with its plane horizontal. At t = 0 it is given tangential velocity v_0 as shown. After time T velocity of tyre become zero, then :- (Neglect friction everywhere)



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(A) $T = \frac{2\pi r}{v_0}$ (B) $T = \frac{3\pi r}{v_0}$

(C) at $t = \frac{T}{6}$ speed of ball is $\frac{\sqrt{3}v_0}{2}$. (D) at $t = \frac{T}{6}$ speed of ball is $\frac{v_0}{2}$.

13. A 24 kg projectile is fired at an angle of 53° above the horizontal with an initial speed of 50 m/s. At the highest point in its trajectory, the projectile explodes into two fragments of equal mass, the first of which falls vertically with zero initial speed. (Assume the ground is level.)

(ii) How much ?

(A) The second fragment strike the ground from the point of firing is 360 m.

(B) Energy was released during the explosion 10800 J.

(C) The second fragment strike the ground from the point of firing is 240 m.

(D) Energy was released during the explosion 21600 J.

14. A rocket ascends from rest in a uniform gravitational field g by ejecting exhaust with a uniform relative speed u. Assume that the rate at which mass is expelled is given by $\left|\frac{dm}{dt}\right| = km$ where m is the instantaneous mass of the rocket & k is a constant & that the rocket is retarded by air resistance with a force mbv where b is a constant & v is instantaneous velocity of rocket.

(A) its velocity after a long time will be (ku – g)/b.

(B) its velocity after a long time will be ku/b.

(C) The acceleration of rocket as a function of time is $(ku - g)e^{-bt}$.

(D) The velocity of rocket as a function of time is $[\{(ku - g)/b\}(1 - e^{-bt})]$.

SECTION-I (iii) : (Maximum Marks: 9)

Linked Comprehension Type (Single Option Correct) 1 Para × 3 Q. [3 M (–1)]

Paragraph for Questions 15 - 17

A small block of mass M moves on a frictionless surface of an inclined plane, as shown in figure. The angle of the incline suddenly changes from 60° to 30° at point B. The block is initially at rest at A. Assume that collisions between the block and the incline are totally inelastic ($g = 10\ m/s^2$)

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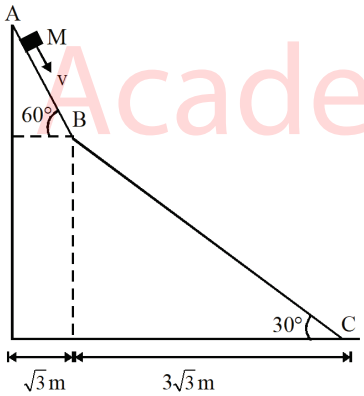


15. The speed of the block at point B immediately after it strikes the second incline is :

(A) $\sqrt{60}$ m/s (B) $\sqrt{45}$ m/s (C) $\sqrt{30}$ m/s (D) $\sqrt{15}$ m/s

16. The speed of the block at point C, immediately before it leaves the second incline is :

(A) $\sqrt{120}$ m/s (B) $\sqrt{105}$ m/s (C) $\sqrt{90}$ m/s (D) $\sqrt{75}$ m/s

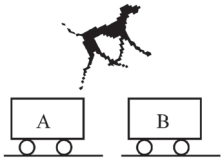
17. If collision between the block and the incline is completely elastic, then the vertical (upward) component of the velocity of the block at point B, immediately after it strikes the second incline is :

(A) $\sqrt{30}$ m/s (B) $\sqrt{15}$ m/s (C) 0 (D) $-\sqrt{15}$ m/s

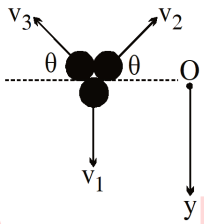
SECTION-II : (Maximum Marks: 20)

Numerical Answer type Questions (Upto Second Decimal Place) 5 Q. [4 M (0)]

1. In a circus act, a 4kg dog is trained to jump from B cart to A and then immediately back to the B cart. The carts each have a mass of 20kg and they are initially at rest. In both cases the dog jumps at 6m/s relative to the cart. If the cart moves along the same line with negligible friction, the final velocity of cart A is v_A and cart B is v_B with respect to the floor. Then find the $v_A : v_B$.



2. A rocket is projected straight up and explodes into three equally massive fragments just as it reaches the top of its flight (refer figure). One of the fragments is observed to come straight down in 2 sec, while the other two take 4 sec to come to ground, after the burst. Find the height h (in m) at which the fragmentation occurred.



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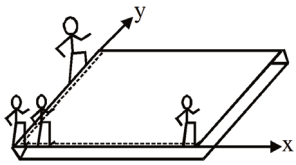
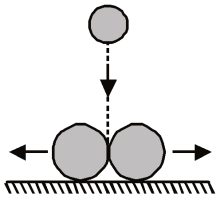
3. A square plank of mass $m_1 = 100$ kg and edge length $L = 20\sqrt{2}$ m is placed on a smooth surface. Two person each of mass $m_2 = m_3 = 50$ kg are at corner of a plank as shown in figure. Two person begin to walk on the plank along two different paths as shown in figure and reach nearest corners. What is magnitude of displacement of plank (in m) in the process.



4. Two bodies of same mass tied with an inelastic string of length $\ell = 5$m lie together. One of them is projected vertically upwards with velocity $\sqrt{6g\ell}$. Find the maximum height (in m) up to which the centre of mass of system of the two masses rises. (Initially string is loose)

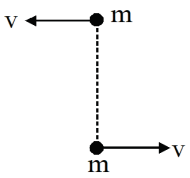
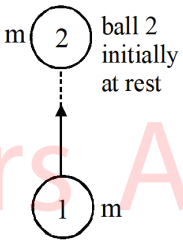
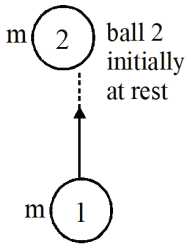
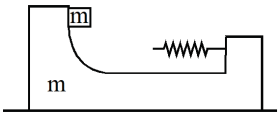
5. Two identical balls of mass M and radius R are placed in contact with each other on a frictionless horizontal surface as shown. The third ball of mass M and radius R/2 moves vertically downward and hits the two balls symmetrically with speed $\sqrt{5}$ m/s and comes to rest. Find the speed of any one of the bigger balls after collision if bigger balls are moving horizontally after collision.



SECTION-IV : (Maximum Marks: 24)

Matrix Match Type (4 × 5) (One or more options correct) 3 Q. [8 M; (for each entry +2(0)]

1. Column I describes the system and initial motion of its particles and column II gives characteristics associated with subsequent motion.

Column-II		**Column-II**	
(A)	Two identical particles under influence of mutual gravitational force only. Particles are given initial velocity perpendicular to line joining as shown. 	(P)	Centre of mass of system does not accelerate.
(B)	Two identical balls on a smooth table. Elastic head on collision takes place. 	(Q)	In centre of mass frame momentum of system is zero.
(C)	Two identical balls on a smooth table. Oblique elastic collision takes place. 	(R)	Momentum of system is conserved.
(D)	Block is released on a wedge as shown in figure. All the surfaces are smooth. 	(S)	Mechanical energy of system is conserved
		(T)	Total kinetic energy is constant

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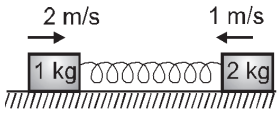
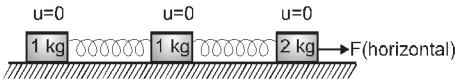
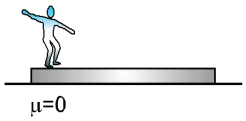
2. Match the column:

In all cases in column–I, the blocks/plank/trolley are placed on the smooth horizontal surface.

Column–I		**Column–I**	
(A)	The initial velocities given to the blocks when spring is relaxed are as shown 	(P)	Centre of mass of the complete system shown will not move horizontally
(B)	A constant force is applied on 2 kg block. Springs are initially relaxed 	(Q)	Centre of mass of the complete system shown will move horizontally
(C)	Initially system is at rest. Man starts moving on a large plank with constant velocity. 	(R)	Mechanical energy of the system will be conserved
(D)	A man standing on one of the trolleys (initially at rest) jumps to the other with relative velocity of 4 m/s horizontally	(S)	Mechanical energy of the system will increase
		(T)	Linear momentum of the complete system will always remain constant

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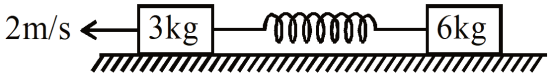
3. Two blocks of masses 3 kg and 6 kg are connected by an ideal spring and are placed on a frictionless horizontal surface. The 3 kg block is imparted a speed of 2 m/s towards left. (Initially spring is relaxed)



	Column-I		**Column-II**
(A)	When the speed of 3 kg block is $\frac{2}{3}$ m/s	(P)	Velocity of centre of mass is $\frac{2}{3}$ m/s
(B)	When the speed of 6 kg block is $\frac{2}{3}$ m/s है तो	(Q)	Deformation of the spring is zero.
(C)	When the speed of 3 kg block is maximum	(R)	Deformation of the spring is maximum
(D)	When the speed of 6 kg block is minimum	(S)	Both the blocks are at rest with respect to each other.
		(T)	Both the blocks are at rest with respect to ground

ANSWER KEY

GR_Centre of Mass & COLLISION

PHYSICS

SECTION-I (i)	Q.	1	2	3	4							
	A.	B	D	C	C							
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12	13	14	
	A.	B,D	A,D	B,C,D	A,C	A,C	B,C	A,C	A,C	A,B	A,C,D	
SECTION-I (iii)	Q.	15	16	17								
	A.	B	B	C								
SECTION-II	Q.	1	2	3	4	5						
	A.	1.2	50	10	5	1						
SECTION-IV	Q.	1	2	3								
	A.	A->PQRS,B->PQRS,C->PQRS,D->QS	A->PRT,B->QS,C->PST,D->PST	A->PQRS,B->PRS,C->PQ,D->PQ								

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Important Notes

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Important Notes

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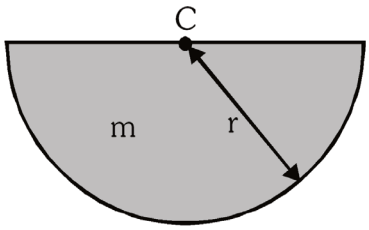
SECTION-I (i) (Maximum Marks:21)

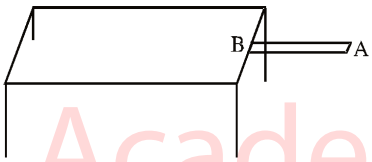
Single Correct Answer Type 7 Q. [3 M (–1)]

1. A semicircular lamina of mass m has radius r and centre C. Its moment of inertia about an axis through its centre of mass and perpendicular to its plane is–



(A) $\frac{1}{2}mr^2$ (B) $\left(\frac{1}{2}-\frac{4}{\pi^2}\right)mr^2$ (C) $\left(\frac{1}{2}-\frac{8}{9\pi^2}\right)mr^2$ (D) $\left(\frac{1}{2}-\frac{16}{9\pi^2}\right)mr^2$

2. A uniform thin stick of length ℓ and mass m is held horizontally with its end B hinged at a point B on the edge of a table. Point A is suddenly released. The acceleration of the centre of mass of the stick at the time of release, is :-



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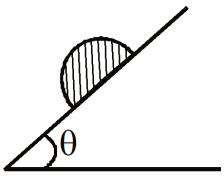
(A) $\frac{3}{4}g$ (B) $\frac{3}{7}g$ (C) $\frac{2}{7}g$ (D) $\frac{1}{7}g$

3. An uniform hemi-solid sphere is placed with flat surface on rough inclined plane as shown in figure. If friction is large for no sliding, then the minimum angle θ at which toppling occur is



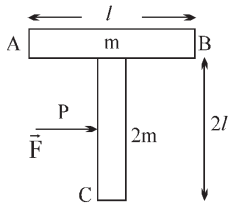
(A) $\tan^{-1}\left(\frac{1}{2}\right)$ (B) 45° (C) $\tan^{-1}\left(\frac{8}{3}\right)$ (D) $\tan^{-1}\left(\frac{4}{3}\right)$

4. A ring of radius R is rolling purely on the outer surface of a pipe of radius 4R. At some instant, the center of the ring has a constant speed = v. Then , the acceleration of the point on the ring which is in contact with the surface of the pipe is :

(A) $4v^2/5R$ (B) $3v^2/5R$ (C) $v^2/4R$ (D) zero

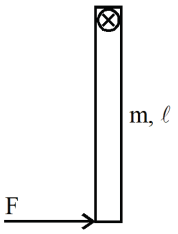
1001CJA00PHEN2500401

5. A T shaped object with dimensions shown in the figure is lying on a smooth floor. A force $\vec{F}$ is applied at the point P parallel to AB such that the object has only the translational motion without rotation. Find the location of P with respect to C.



(A) (3/4)l (B) l (C) (4/3)l (D) (3/2)l

6. As shown in the figure rod of mass m and length ℓ is hinged at O at one end and free to rotate in vertical plane. A force F of magnitude mg is applied on another end of rod which always act perpendicular to the rod. Force is start acting at time t = 0 (as shown). Find vertical component of normal reaction acting on rod when rod become horizontal for 3rd time:- (given m = 2 kg, ℓ = 1m, g = 10 m/s^2)



(A) 5 N (B) 10 N (C) 15 N (D) 20 N

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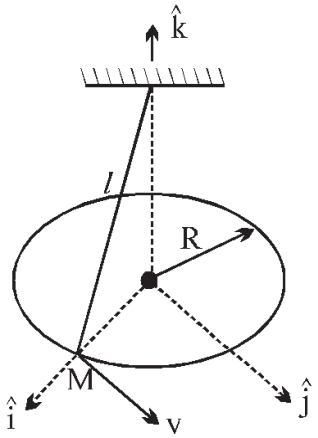
7. A conical pendulum consists of a mass M suspended from a string of length ℓ. The mass executes a circle of radius R in a horizontal plane with speed v. At time t, the mass is at position $R\hat{i}$ and has velocity $v\hat{j}$. At time t, the angular momentum vector of the mass M about the point from which the string suspended is :



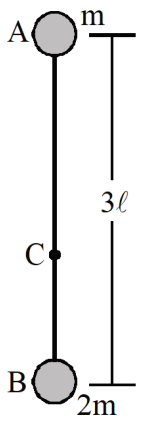
(A) $MvR\hat{k}$ (B) $Mvl\hat{k}$

(C) $Mvl\left[\frac{\sqrt{l^2-R^2}}{l}\hat{i}+\frac{R}{l}\hat{k}\right]$ (D) $-Mvl\left[\frac{\sqrt{l^2-R^2}}{l}\hat{i}+\frac{R}{l}\hat{k}\right]$

SECTION-I (ii) : (Maximum Marks: 24)

Multiple Correct Answer Type 6 Q. [4 M (–2)]

8. A light rigid rod AB of length 3ℓ has a point mass m at end A and a point mass 2m at end B is kept on a smooth horizontal surface. Point C is the center of mass of the system. Initially the system is at rest. The mass 2m is suddenly given a velocity v_0 towards right. Take Z axis to be perpendicular to the plane of the paper.



(A) The minimum moment of inertia (about Z-axis), I_{zz} of the system is $5m\ell^2$.

(B) The magnitude of tension in the rod in subsequent motion is $\frac{2mv_0^2}{9\ell}$

(C) The ratio of moment of inertia about Z-axis at points A and B, $\frac{I_{zz}^A}{I_{zz}^B} = 2$

(D) Point C remains stationary during subsequent motion.

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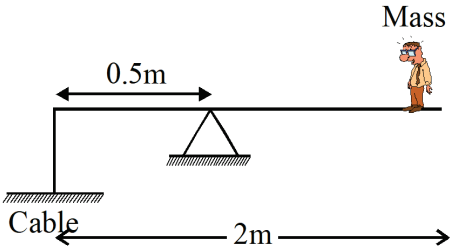
9. A man of mass 85 kg stands on one end of a uniform board of length 2 m and mass 15 kg. The other end is tied to the floor by a steel cable, and the board is supported from below just 0.5 m from the cable end (see figure).



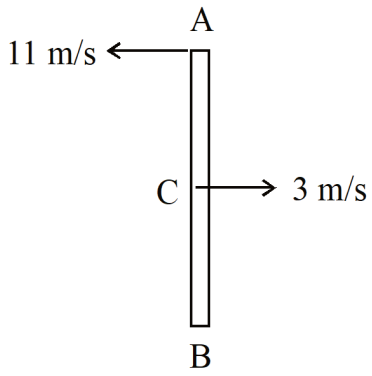
(A) The tension in cable is 1000 N

(B) The tension in cable is 2700 N

(C) The normal reaction at support is 3700 N

(D) The normal reaction at support is 1700 N

10. A uniform rod AB of length 7m is undergoing combined rotational and translational motion on smooth horizontal surface such that, at some instant of time, velocities of its end point A and centre C are both perpendicular to the rod and opposite in direction, having magnitude 11 m/s and 3 m/s respectively as shown in the fig. Now consider that no external force and torque is acting on the rod. Then select correct options.



(A) acceleration of point A is 56 m/s^2

(B) acceleration of point B is 56 m/s^2

(C) at the instant shown in the figure acceleration of point C is zero.

(D) angular velocity of the rod is 4 rad/s

11. A ball moves over a fixed track as shown in the figure. From A to B the ball rolls without slipping. Surface BC is frictionless. K_A, K_B and K_C are kinetic energies of the ball at A, B and C, respectively. Then

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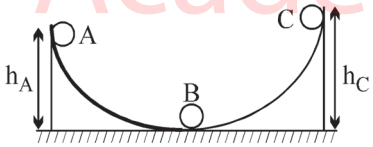


(A) $h_A > h_C$; $K_B > K_C$

(B) $h_A > h_C$; $K_C > K_A$

(C) $h_A = h_C$; $K_B = K_C$

(D) $h_A < h_C$; $K_B > K_C$

12. A uniform solid cylinder of mass m and radius R is released from rest on a sufficiently rough inclined plane. During its downward journey along the incline, the cylinder moves distance ℓ along the incline. The angle of inclination from horizontal is a. Mark correct statements :-

(A) The acceleration of centre of mass of cylinder is $\frac{2g\sin\alpha}{3}$.

(B) The final angular speed of cylinder is $\sqrt{\frac{4}{3}\frac{g\ell\sin\alpha}{R^2}}$.

(C) The minimum coefficient of friction required so that there is no slipping is $\frac{\tan\alpha}{3}$.

(D) If released from rest cylinder will undergo pure rolling even on smooth inclined plane.

13. Suppose a solid disk having radius R is put on a rough horizontal surface and given an angular speed ω_0 about an axis through it's centre and released. If coefficient of friction between disc and surface is μ then:

(A) Time interval before rolling motion starts is $\frac{R\omega_0}{3\mu g}$.

(B) Distance travelled by disc before pure rolling starts is $\frac{R^2\omega_0^2}{18\mu g}$.

(C) Rolling motion doesn't occur

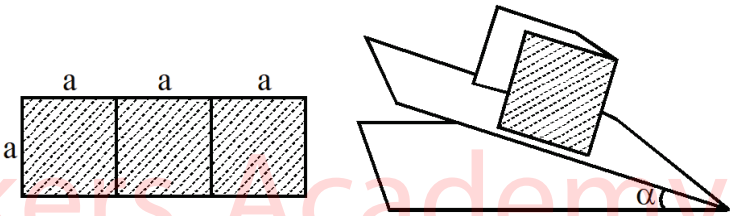
(D) Time interval before rolling motion starts is $\frac{2R\omega_0}{3\mu g}$.

SECTION-I (iii) : (Maximum Marks: 39)

Linked Comprehension Type (Single Option Correct) (3 Para × 3 Q.) (2 Para × 2 Q.) [3 M (–1)]

Paragraph for Questions 14 - 16

A cardboard strip, bent in the shape of the letter C, is put on a rough inclined plane, as shown in the figure.



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14. At what angle of inclination to the horizontal plane will it topple? (assume that it does not slide).

(A) $\tan^{-1}\left(\frac{2}{3}\right)$ (B) $\tan^{-1}\left(\frac{3}{2}\right)$ (C) 45° (D) $\tan^{-1}\left(\frac{1}{3}\right)$

15. What should be the coefficient of friction so that it does not slide before toppling.

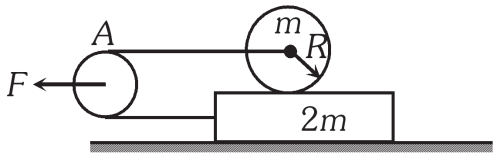
(A) 0.66 (B) 0.75 (C) 1.0 (D) 0.33

16. What should be the time taken to travel a distance a if the inclined plane is made completely smooth? [Take angle of inclined plane as α]

(A) $\sqrt{\frac{a}{g\sin\alpha}}$ (B) $\sqrt{\frac{2a}{g\sin\alpha}}$ (C) $\sqrt{\frac{2a}{3g\sin\alpha}}$ (D) None of these

Paragraph for Questions 17 - 19

A solid cylinder of mass m and radius R is kept at rest on a plank of mass 2m lying on a smooth horizontal surface. Massless and inextensible string connecting cylinder to the plank is passing over a massless pulley. The friction between the cylinder and the plank is sufficient to prevent slipping. Pulley A is pulled with a constant horizontal force F.



17. Acceleration of cylinder with respect to earth is

(A) $\frac{5F}{21m}$ (B) $\frac{F}{7m}$ (C) $\frac{3F}{7m}$ (D) $\frac{2F}{7m}$

18. Acceleration of plank with respect to earth is

(A) $\frac{5F}{21m}$ (B) $\frac{F}{7m}$ (C) $\frac{3F}{7m}$ (D) $\frac{2F}{7m}$

19. Magnitude of friction force acting on the plank is

(A) $\frac{F}{7}$ (B) $\frac{F}{14}$ (C) $\frac{F}{21}$ (D) $\frac{2F}{7}$

Paragraph for Questions 20 - 22

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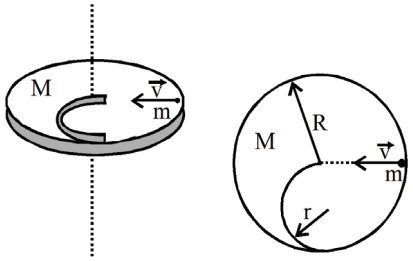
A smooth disc of mass M and radius R can rotate freely around a vertical axis supported by frictionless bearings. A constraining vertical surface of negligible mass, whose shape is a semicircular arc of radius r = $\frac{R}{2}$, is fixed on the disc as shown in the figure. A small ball of mass m is placed on the stationary disc and is bowled at a speed v in such a way that it reaches the internal side of the constraining surface tangentially.



20. Mark the correct statement :-

(A) Momentum of ball & disc system is conserved

(B) Mechanical energy of ball & disc system is not conserved

(C) Angular momentum of ball & disc system is conserved

(D) Mechanical energy of ball & disc system is not conserved due to impulsive reaction from support bearings

21. What is speed of the ball when it leaves the disc :-

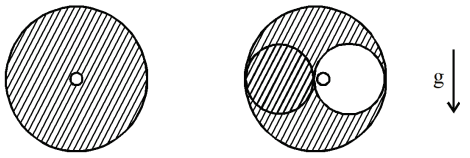
(A) $v\sqrt{\frac{M}{M+2m}}$ (B) $v\sqrt{\frac{m}{M+2m}}$ (C) $v\sqrt{\frac{2M}{M+m}}$ (D) $2v\sqrt{\frac{M}{2M+m}}$

22. What is angular velocity of disc when ball leaves it :-

(A) $2\frac{mv}{MR}\sqrt{\frac{M}{M+2m}}$ (B) $2\frac{mv}{MR}\sqrt{\frac{m}{M+2m}}$ (C) $2\frac{mv}{MR}\sqrt{\frac{2M}{M+m}}$ (D) $2\frac{mv}{MR}\sqrt{\frac{M}{2M+m}}$

Paragraph for Questions 23 and 24

A disk of radius 2R and mass 'M' is hinged from its center. If a disk of radius R been removed such that radius of bigger disk behave as a diameter of the disk and it is welded on the opposite side as shown in diagram.



Case-I Case-II

23. Find the ratio of moment of inertia about point 'O' in Case-I and Case-II :-

(A) $\frac{1}{2}$ (B) $\frac{2}{1}$ (C) $\frac{1}{\sqrt{2}}$ (D) 1

24. Find the angular acceleration of the system in case-II if the system is released from the given position about the axis passing through O and perpendicular to the plane of disc:-

(A) $\frac{g}{R}$ (B) $\frac{g}{2R}$ (C) $\frac{g}{3R}$ (D) $\frac{g}{4R}$

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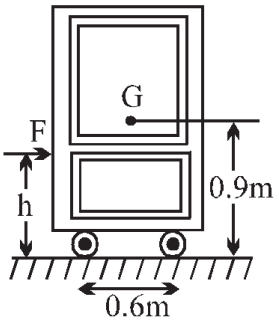
Paragraph for Questions 25 and 26

A 20 kg cabinet is mounted on small casters that allow it to move freely ($\mu = 0$) on the floor. If a 100 N force is applied as shown.



25. The acceleration (in m/s^2) of the cabinet is :-

(A) 5 (B) 0 (C) > 5 (D) < 5

26. The range of values of h for which the cabinet will not tip is :-

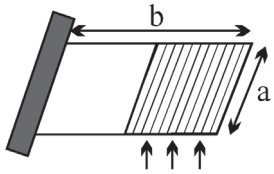
(A) It will topple for all values of h. (B) 0.3 m < h < 1.5 m

(C) h > 0.3 m (D) h < 1.5 m

SECTION-II : (Maximum Marks: 8)

Numerical Answer type Questions (Upto Second Decimal Place) 2 Q. [4 M (0)]

1. There is a rectangular plate of mass M kg of dimensions (a × b). The plate is held in horizontal position by striking n small balls each of mass m per unit area per unit time. These are striking in the shaded half region of the plate. The balls are colliding elastically with velocity v. What is v? It is given n = 100, M = 3 kg, m = 0.01 kg; b = 2 m; a = 1m; g = 10 m/s^2.

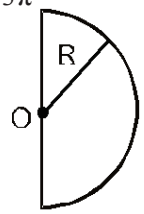


2. I_0 is the moment of inertia of a rod about an axis through its centre of mass and perpendicular to the length whose linear density varies as $\lambda = ax$ where a is a constant and x is the position of an element of the rod relative to its left end. The length of the rod is ℓ. If $I_0 = I_0 = \frac{a\ell^4}{P}$. Find the value of P.

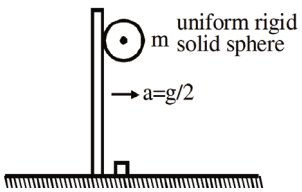
SECTION-III : (Maximum Marks: 20)

Numerical Grid Type (Ranging from 0 to 9) 5 Q. [4 M (0)]

1. A uniform semicircular disc of mass 'M' and radius 'R' hinged at point O a shown in figure is released from rest from a vertical position as shown. If the initial magnitude of angular acceleration of the disc is $\frac{xg}{3\pi R}$ than the value of x is (Center of mass is at a distance $\frac{4R}{3\pi}$ from center of disc):



2. A uniform solid sphere rolls down on a vertical rigid surface without slipping.
The vertical surface moves with an acceleration a = g/2 in horizontal direction as shown. The minimum coefficient of friction between the sphere and vertical surface so as to prevent slipping is m. Write 14 m as answer.



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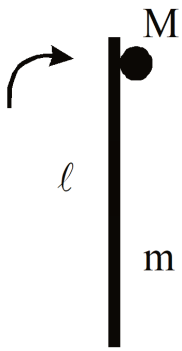
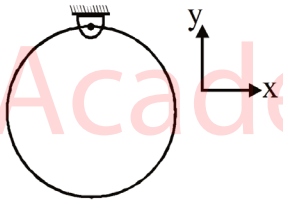
3. A uniform stick of mass 18 kg and length 3 m spins around on a frictionless horizontal plane, with its Centre of Mass stationary. A mass M is placed on the plane, and the stick collides elastically with it, as shown (with the contact point being the end of the stick). What should M (in kg) be so that after the collision the stick has translational motion, but no rotational motion?



4. A bullet of mass m moving with velocity $v_0(-\hat{k})$ strikes the bottom of a stationary vertical uniform ring of same mass m and radius R = 1 m. The ring lies in XY plane with its topmost point hinged on the ceiling. The ring can rotate about X-axis. There is no friction between the hinge and the ring. The bullet gets embedded in the ring immidiately after collision. Find the angular velocity of the system (in radian /s) just after collision. [Take v_0 = 11 m/s]



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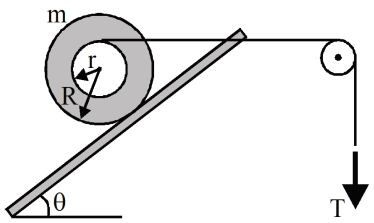
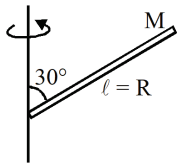
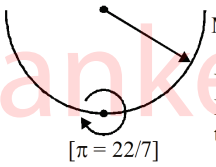
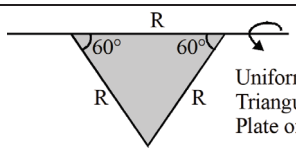
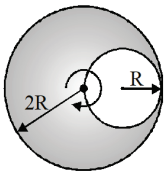
5. A spool, initially at rest, is kept on a frictionless incline making an angle $\theta = 37°$ with the horizontal. The mass of the spool is 2kg and it is pulled by a string as shown with a force of T = 10N. The string connecting the spool and the pulley is initially horizontal. Find the initial acceleration of the spool on the incline. Express your answer in m/s^2.



SECTION-IV : (Maximum Marks: 8)

Matrix Match Type (4 × 5) (One or more options correct) 1 Q. [8 M; (for each entry +2(0))]

1.

Column-I		Column-II	
(A)	 Uniform Rod	(P)	$\frac{8MR^2}{11}$
(B)	 Uniform Semicircular Ring. Axis is perpendicular to plane of ring. [π = 22/7]	(Q)	$\frac{MR^2}{12}$
(C)	 Uniform Triangular Plate of mass M	(R)	$\frac{13MR^2}{8}$
(D)	 Uniform disk of initial mass M from which circular portion of radius R is then removed. M.I. of remaining mass about axis which is perpendicular to plane of plate	(S)	$\frac{MR^2}{8}$

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PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7			
	A.	D	A	C	A	C	C	C			
SECTION-I (ii)	Q.	8	9	10	11	12	13				
	A.	B,C	B,C	A,B,C,D	A,B	A,B,C	A,B				
SECTION-I (iii)	Q.	14	15	16	17	18	19	20	21	22	23
	A.	A	A	B	C	D	B	C	A	A	D
	Q.	24	25	26							
	A.	D	A	B							
SECTION-II	Q.	1	2								
	A.	10	36.00								
SECTION-III	Q.	1	2	3	4	5					
	A.	8	8	9	4	2					
SECTION-IV	Q.	1									
	A.	A->Q,B->P,C->S,D->R									

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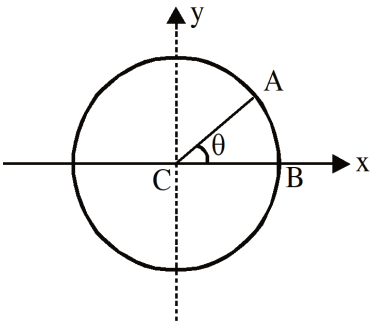
SECTION-I (i) (Maximum Marks:36)

Single Correct Answer Type 12 Q. [3 M (–1)]

1. Two lines AC and BC are drawn on a circular disc as shown in figure. The disc is made of a material with linear expansion coefficients $\alpha_x > \alpha_y$. Here α_x and α_y are linear expansion coefficient for expansion in x-direction and y-direction respectively. C is the centre of disc. If the disc is heated uniformly. Angle ACB (= θ) will :-



(A) increase

(B) decrease

(C) remains constant

(D) may increase or decreases depending on value of temperature increment

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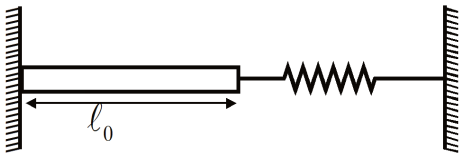
2. A rod of mass M, area of cross section A and length ℓ_0 is connected with a spring as shown in figure. If coefficient of linear expansion of rod is α and initially no extension was there in the spring of spring constant k, then the stress developed in rod when its temperature is increased by ΔT, is

[Young's Modulus of material of the rod is Y]



(A) $\frac{k[Y\alpha\Delta T]\ell_0}{[k\ell_o + YA]}$ (B) $Y\alpha\Delta T$ (C) $\left[\frac{k\ell_o + YA}{k\ell_o}\right] Y\alpha\Delta T$ (D) $\left[\frac{k\ell_o + Y}{k\ell_o}\right] Y\alpha\Delta T$

3. A wire of cross-section A is stretched horizontally between two clamps located 2l meters apart. A weight W kg is suspended from the mid point of the wire. If the mid point sags vertically through a distance x << l the strain produced is :-

(A) $\frac{2x^2}{l^2}$ (B) $\frac{x^2}{l^2}$ (C) $\frac{x^2}{2l^2}$ (D) none of these

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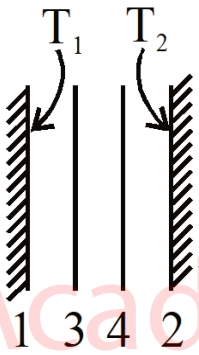
4. In an industrial process 10 kg of water per hour is to be heated from 20°C to 80°C . To do this steam at 150°C is passed from a boiler into a copper coil immersed in water. The steam condenses in the coil and is returned to the boiler as water at 90°C. How many kg of steam is required per hour.
(Specific heat of steam = 1 cal/g°C, Latent heat of vaporisation = 540 cal/g)

(A) 1 g (B) 1 kg (C) 10 g (D) 10 kg

5. Two large black plane surfaces are maintained at constant temperature T_1 and $T_2(T_1 > T_2)$. Two thin black plates are placed between the two surfaces and in parallel to these. After some time, steady conditions are obtained. What is the ratio of heat transfer rate between plate-1 & plate-3 to the ratio of original (when plate-3 & plate-4 was not present) heat tranfer rate between plate-1 & plate-2 (η) in steady state?



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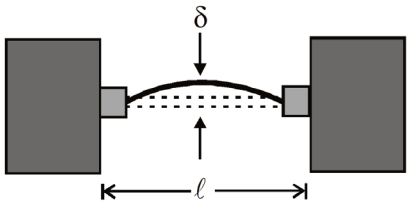
(A) $\eta = \frac{1}{2}$ (B) $\eta = \frac{1}{3}$ (C) $\eta = 1$ (D) $\eta = 0$

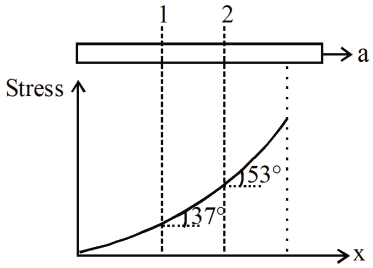
6. A long, thin metal bar of length ℓ is clamped rigidly at its ends at temperature t_0. When the temperature is increased to t, the expanding bar will bow out, as shown below. If the bowing is not too large, a fair first approximation to the shape of the bar is two equal straight segments in the form of a wide V. What is the arch δ of the bow as a function of t? (This is the distance between the corner of the V and the straight line that represents the form of the bar at t_0.)



(A) $\delta = (\ell/2)\sqrt{2\alpha(t - t_0)}$ (B) $\delta = (\ell)\sqrt{\alpha(t - t_0)}$

(C) $\delta = (\ell/2)\sqrt{\alpha(t - t_0)}$ (D) $\delta = (\ell)\sqrt{2\alpha(t - t_0)}$

7. Rod of constant cross-section moves towards right with constant acceleration. Graph of stress and distance from left end is given as in figure. If density of material of rod at cross section 1 is 9 gm/cm^3. Find density at cross section 2.



(A) 16 gm/cm^3 (B) 20 gm/cm^3 (C) 24 gm/cm^3 (D) 12 gm/cm^3

8. 'Gulab Jamuns' (assumed to be spherical) are to be heated in an oven. They are available in two sizes, one twice bigger (in radius) than the other. Pizzas (assumed to be discs) are also to be heated in oven. They are also in two sizes, one twice big (in radius) than the other. All four are put together to be heated to oven temperature. Choose the correct option from the following:

(A) Both size gulab jamuns will get heated in the same time.

(B) Smaller gulab jamuns are heated before bigger ones.

(C) Smaller pizzas are heated before bigger ones.

(D) Bigger pizzas are heated before smaller ones.

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9. Three very large plates of same area are kept parallel and close to each other. They are considered as ideal black surfaces and have very high thermal conductivity. The first and third plates are maintained at temperatures 2T and 3T respectively. The temperature of the middle (i.e. second) plate under steady state condition is

(A) $\left(\frac{65}{2}\right)^{1/4}T$ (B) $\left(\frac{97}{4}\right)^{1/4}T$ (C) $\left(\frac{97}{2}\right)^{1/4}T$ (D) $(97)^{1/4}T$

10. There is formation of layer of ice x cm thick on water, when the temperature of air is $-\theta°C$ (less than freezing point). The thickness of layer increases from x to y in the time t, then the value of t is given by-

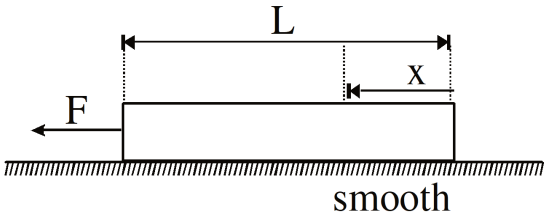
(A) $\frac{(x+y)(x-y)\rho L}{2K\theta}$ (B) $\frac{(x-y)\rho L}{2K\theta}$

(C) $\frac{(x+y)(x-y)\rho L}{K\theta}$ (D) $\frac{(x-y)\rho LK}{2\theta}$

11. Two thin walled spheres of different materials, one with double the radius and one-fourth wall thickness of the other, are filled with ice. If the time taken for complete melting of ice in the sphere of larger radius is 25 minutes and that for smaller one is 16 minutes, the ratio of thermal conductivities of the materials of larger sphere to the smaller sphere is

(A) 4 : 5 (B) 25 : 1 (C) 1 : 25 (D) 8 : 25

12. A rod of mass m, uniform cross sectional area A and length L is accelerated by applying force F as shown in figure on a smooth surface. If Young's modulus of elasticity of the material of rod is Y. (Consider x as measured from the right end)



(A) Tension in rod as a function of distance x is $\frac{Fx}{2L}$.

(B) Strain in rod is $\frac{F}{2AY}$.

(C) Elastic potential energy stored in the rod is $\frac{F^2L}{8AY}$.

(D) There is no stress in rod.

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SECTION-I (ii) : (Maximum Marks: 32)

Multiple Correct Answer Type 8 Q. [4 M (–2)]

13. A metal cylinder of very high conductivity and mass 0.5 kg is heated electrically by a 12 W heater in a room at 15°C. The cylinder temperature rises to 25°C in 5 min and finally becomes constant at 45°C. Assuming that the rate of heat loss is proportional to the excess temperature over the surroundings

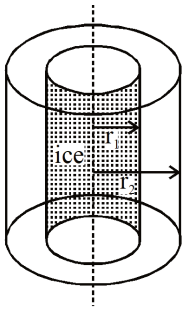
(A) The rate of loss of heat of the cylinder to surrounding at 20°C is 2 W

(B) The rate of loss of heat of the cylinder to surrounding at 45°C is 2 W

(C) Specific heat capacity of metal is $\frac{240}{\ell n(3/2)}$ J/kg°C

(D) None of these

14. A 100 cm long cylindrical flask with inner and outer diameter 2 cm and 4 cm respectively is completely filled with ice as shown in the figure. The constant temperature outside the flask is 40ºC. (Thermal conductivity of the flask is 0.693 W/m °C, L_{ice} = 80 cal/gm & $\ell n2 = 0.693$).



(A) Rate of heat flow from outside to the flask is 80 π J/s.

(B) The rate at which ice melts is $\frac{\pi}{4200}$ kg/s

(C) The rate at which ice melts is 100 π kg/s

(D) Rate of heat flow from outside to flask is 40 π J/s

15. Which of the following statements are true?

(A) Hole in the wall of a cavity radiator behaves like a black body.

(B) Hole in the wall of a cavity radiator does not act like a black body.

(C) When a body is kept in a surrounding of low temperature it does not absorb any energy from the surroundings.

(D) When a body is kept in surrounding of low temperature it simultaneously radiates heat to the surroundings and absorbs heat from the surroundings.

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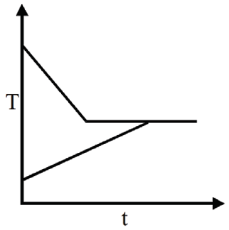
16. A sample A of liquid water and a sample B of ice of identical mass are kept in two neighbouring chambers in an otherwise insulated container. The chambers can exchange heat with each other. The graph of temperatures of the two chambers is plotted with time $S_{ice} = \frac{S_{water}}{2}$:-



(A) Finally the contents in sample A is water.

(B) Equilibrium temperature is freezing point of water

(C) Ice melts partly

(D) Finally the contents in sample B is ice only.

17. 50 gm ice at $-10°C$ is mixed with 10gm steam at 100°C. When the mixture finally reaches its steady state inside a calorimeter of water equivalent 1.5 gm then : [Assume calorimeter was initially at 0°C, Take latent heat of vaporization of water = 540 cal/gm, Latent heat of fusion of water = 80 cal/gm and specific heat capacity of water = 1 cal/gm-°C, specific heat of ice = 0.5 cal/gm°C]

(A) Mass of water remaining is 60 gm

(B) Mass of ice remaining is 3 gm

(C) Mass of steam remaining is 1.20 gm

(D) Final temperature is between 0°C and 100°C

18. The temperature of earth is maintained by a dynamic equilibrium between Sun and Earth. Sun & Earth can be assumed to be black bodies :

(A) If the power output of sun would double with changing the temperature, equilibrium temperature of earth also doubles.

(B) If the radius of sun doubles without changing its' power, its surface temperature would decrease by factor of $\sqrt{2}$.

(C) If the radius of earth doubles without any change in sun, it's equilibrium temperature would increase by factor of $\sqrt{2}$.

(D) If the distance between earth and sun would decrease by a factor of 2, the equilibrium temperature of earth would increase by factor of $\sqrt{2}$.

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19. There are two spheres of same radius and material at same temperature but one being solid while the other hollow. Then :-

(A) They are heated to the same temperature solid sphere will expand more than hollow sphere.

(B) They are heated to the same temperature both sphere will expand equally.

(C) If same amount of heat is given to them then solid sphere will expand more then hollow sphere.

(D) If same amount of heat is given to them then hollow sphere will expand more then solid sphere.

20. A thin uniform metallic rod of length 0.5 m and radius 0.1 m rotates with an angular velocity 400 rad/s in a horizontal plane about a vertical axis passing through one of its ends. The density of material of the rod is 10^4 kg/m^3 and the Young's modulus is 2×10^{11} N/m^2.

(A) The elongation of the rod is $\frac{1}{3} \times 10^{-3}$m.

(B) The elongation of the rod is non-uniform along its length.

(C) Tension at the center of the rod is $\frac{3\pi}{2} \times 10^6$N.

(D) Tension at the center of the rod is $\frac{2\pi}{2} \times 10^6$N.

SECTION-I (iii) : (Maximum Marks: 9)

Linked Comprehension Type (Single Option Correct) 1 Para × 3 Q. [3 M (–1)]

Paragraph for Questions 21 - 23

Multi-layer insulation, or MLI, is thermal insulation composed of multiple layers of thin sheets often used on spacecraft. It is one of the main items of the spacecraft thermal design, primarily intended to reduce heat loss by thermal radiation. In its basic form, it does not appreciably insulate against other thermal losses such as heat conduction or convection. It is therefore commonly used on satellites and other applications in vacuum where conduction and convection are much less significant and radiation dominates. MLI gives many satellites and other space probes the appearance of being covered with gold foil.

The principle behind MLI is radiation balance. To see why it works, start with a concrete example - imagine a square meter of a surface in outer space, at 300 K, with an emissivity of 1, facing away from the sun or other heat sources. From the Stefan-Boltzmann law, this surface will radiate 460 watts. Now imagine we place a thin (but opaque) layer 1 cm away from the plate, thermally insulated from it, and also with an emissivity of 1. This new layer will cool until it is radiating 230 watts from each side, at which point everything is in balance. The new layer receives 460 watts from the original plate. 230watts is radiated back to the original plate, and 230 watts to space. The original surface still radiates 460 watts, but gets 230 back from the new layers, for a net loss of 230 watts. So overall, the radiation losses have been reduced by half by adding the additional layer.

More layers can be added to reduce the loss further. The blanket can be further improved by making the outside surfaces highly reflective to thermal radiation, which reduces both absorption and emission. The performance of a layer stack can be quantified in terms of its overall heat transfer coefficient U, which defines the radiative heat flow rate Q between two parallel surfaces with a temperature difference ΔT and area A as

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$$Q = UA\Delta T$$

Theoretically, the heat transfer coefficient between two layers with emissivities ϵ_1 and ϵ_2, under vacuum, is

$$U = 4\sigma T^3 \frac{1}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1},$$

where T is the mean of the temperatures (in K) of the two layers and $\sigma = 5.7 \times 10^{-8}\ Wm^{-2}K^{-4}$ is the Stefan-Boltzmann Constant. If each layer has the same emissivity ϵ on both sides, then a stack of N layers placed between two high-emissivity surfaces will have an overall heat transfer coefficient

$$U = 4\sigma T^3 \frac{1}{N\left(\frac{2}{\epsilon} - 1\right) + 1}$$

Clearly, increasing the number of layers and decreasing the emissivity, lower the heat transfer coefficient, which is equivalent to a higher insulation value. In this equation, it is assumed that the temperature difference is small compared to the absolute temperature.

21. Main purpose of multilayer insulation is insulation against heat transfer, where heat transfer is mainly due to :-

(A) Conduction
(B) Conduction & radiation
(C) Conduction & convection
(D) Radiation

22. In space there is MLI consisting of 50 layers. Each layer has emissivity ($\in$ = 0.2) both sides & outer most temperature is 550 K and inner most temperature is 100 K less than outer most then find over all heat transfer coefficient (in $Wm^{-2}k^{-1}$) :-

(A) 6.33 (B) 6.33×10^{-2} (C) 12.66×10^{-3} (D) 6.33×10^{-3}

23. For a particular plate heat transfer due to conduction and radiation are equal. Width of a plate is 2mm, mean temperature is 1000 K and emissivity of inner and outer surface are 1 & $\frac{1}{2}$ respectively. Then find conductivity of plate material (in SI) :-

(A) 0.228 (B) 11

(C) 1.11 (D) Data is insufficient

SECTION-II : (Maximum Marks: 16)

Numerical-Answer-Type questions (Non Negative Integer) 4 Q. [4 M (0)]

1. A liquid of volumetric thermal expansion coefficient 200×10^{-6} /°C and bulk modulus = 1.2×10^{9} Pa is filled in a spherical tank of negligible heat expansion coefficient. Its radius is 25 cm and wall thickness is 2 mm. When the temperature of the liquid is raised by 20 °C, find the tensile stress developed (in MPa) in the wall of the tank?

2. A rod of negligible heat capacity has length 40 cm, area of cross-section 1.0 cm^2 and thermal conductivity 100 W/m °C. The temperature of one end is maintained at 0°C and that of the other end is slowly and linearly varied from 0ºC to 60°C in 10 minutes. Assuming no loss of heat through the sides, find the total heat transmitted through the rod in these 10 minutes (in J).

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3. A cylindrical block of length 0.4 m an area of cross-section 0.04m^2 is placed coaxially on a thin metal disc of mass 0.4 kg and of the same cross-section. The upper face of the cylinder is maintained at a constant temperature of 400K and the initial temperature of the disc is 300K. If the thermal conductivity of the material of the cylinder is 10 watt/m-K and the specific heat of the material of the disc in
600 J/kg-K, how long will it take for the temperature of the disc to increase to 350K? Assume, for purposes of calculation, the thermal conductivity of the disc to be very high and the system to be thermally insulated except for the upper face of the cylinder.

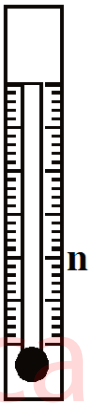
4. An electric heater is used in a room of total wall area 50m^2 to maintain a constant temperature of 15°C inside when the outside temperature is −10°C. The walls have two different layers. The inner layer is of cement of thickness 2 cm, while the outer layer is of brick of thickness 10 cm. Assume that there is no loss of heat from the floor and the ceiling. Take the thermal conductivities of cement and brick to be 0.2 W/m°C and 0.5 W/m°C, respectively. If another identical heater is now switched on, along with the old heater, find the steady state temperature (in °C) inside the room, after a long time, if the outside temperature has now dropped to −15°C.

SECTION-III : (Maximum Marks: 12)

Numerical Grid Type (Ranging from 0 to 9) 3 Q. [4 M (0)]

1. An object can be hanged to a 1.2m long wire of diameter 2 mm, and the wire does not break. However, if the object is slightly displaced and begins to slowly swing, the wire breaks. If another piece of wire having the same length and material but wider diameter is used, the speed of the same swinging object at lowest point can even be 6 m/s. Calculate the least diameter (in mm) of this wider wire.

2. A thermometer has a spherical bulb of volume 1 cm^3 having 1 cm^3 of mercury. A long cylindrical capillary tube is connected to spherical bulb. Volumetric coefficient of expansion of mercury is $1.8 \times 10^{-4}\ K^{-1}$; cross-section area of capillary is $1.8 \times 10^{-4}\ cm^2$. Ignoring expansion of glass, how far apart (in cm) on the stem are marks indicating 1K temperature change.



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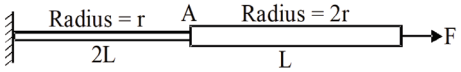
3. Two steel wires of radii r and 2r are connected together end to end and tied to a wall as shown. The force stretches the combination by $\frac{27}{4}$ mm. How far does the junction point A move. (in mm)



ANSWER KEY

GR_Elasticity, Thermal Expansion, Calorimetry & Heat transfer

PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	C	B	B	A	A	B	C	A
	Q.	11	12								
	A.	D	B								
SECTION-I (ii)	Q.	13	14	15	16	17	18	19	20		
	A.	A,C	A,B	A,D	B,D	A,D	B,D	B,D	A,B,C		
SECTION-I (iii)	Q.	21	22	23							
	A.	D	B	A							
SECTION-II	Q.	1	2	3	4						
	A.	300	4500	1663	3500						
SECTION-III	Q.	1	2	3							
	A.	4	1	6							

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Important Notes

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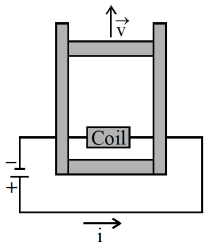
SECTION-I (i) (Maximum Marks:30)

Single Correct Answer Type 10 Q. [3 M (–1)]

1. In figure, a resistance coil, wired to an external battery, is placed inside a thermally insulated cylinder fitted with a frictionless piston and containing an ideal diatomic gas. A current i = 200 mA exists in the coil, which has a resistance R = 350 Ω. If the pressure of the gas remains constant, what should be the speed v (in cm/s) of the piston, of mass m = 10 kg ? (Take $p_{atm} = 10^5$ Pa, A = 10 cm^2)



(A) 1 (B) 2 (C) 3 (D) 4

2. The volume thermal expansion coefficient γ for an ideal gas is computed under constant pressure condition. Expressed in terms of its absolute temperature, the correct expression is :-

(A) $\gamma = \frac{1}{2T}$ (B) $\gamma = \frac{1}{\sqrt{2}T}$ (C) $\gamma = \frac{2}{T}$ (D) $\gamma = \frac{1}{T}$

3. What is fraction of molecule below an altitude h in atmosphere? Assume uniform gravitational field, isothermal conditions, mass of a molecule m, Boltzman constant k, temperature T.

(A) $f = e^{(mgh/kT)}$ (B) $f = e^{-(mgh/kT)}$

(C) $f = 1 - e^{-(mgh/kT)}$ (D) $f = 1 - e^{(mgh/kT)}$

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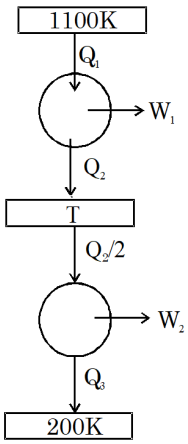
4. A carnot refrigerator operates between −3°C and 27°C. How much work must be done so as to extract 1 kJ of heat from the refrigerator?

(A) 100 J (B) 10^4 J (C) $\frac{1000}{9}$ J (D) 900 J

5. Consider the given series combination of carnot cycles. If $W_1 = W_2$ then the value of T is (all temperatures are maintained at their respective values)



(A) 600 K (B) 800 K (C) 700 K (D) 900 K

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6. In a refrigerator one removes heat from a lower temperature and deposits to the surroundings at a higher temperature. In this process, mechanical work has to be done, which is provided by an electric motor. If the motor is of 1kW power, and heat is transferred from $-3°C$ to $27°C$, find the heat taken out of the refrigerator per second assuming its efficiency is 50% of a perfect engine.

(A) 500 J (B) 5 kJ (C) 12 kJ (D) 19 kJ

7. A household refrigerator with a coefficient of performance 1.2 removes heat from the refrigerated space at the rate of 60kJ/min.what would be cost of running this fridge for one month (30 days) (assuming each day it is used for 4 hours and cost of one electrical unit is 6Rs.):-

(A) 180 Rs. (B) 300 Rs. (C) 480 Rs. (D) 600 Rs.

8. Two Carnot engines A and B are operated in succession. The first one, A receives heat from a source at T_1 = 800 K and rejects to sink at T_2K. The second engine B receives heat rejected by the first engine and rejects to another sink at T_3 = 300K If the work outputs of two engines are equal, then the value of T_2 is :-

(A) 100K (B) 300K (C) 550K (D) 700K

9. An ideal gas is taken from temperature T_i to T_f. $T_f > T_i$ by 3 different reversible processes.

(a) isobaric (b) isochoric (c) adiabatic

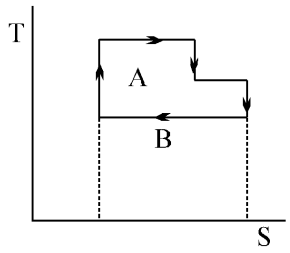
The change in entropy is

ΔS_b, ΔS_u & ΔS_S respectively. Choose the correct relation between them.

(A) $\Delta S_b > \Delta S_u > \Delta S_s$ (B) $\Delta S_s > \Delta S_u > \Delta S_b$

(C) $\Delta S_b > \Delta S_s > \Delta S_u$ (D) $\Delta S_u > \Delta S_b > \Delta S_s$

10. An engine is represented by a cyclic process shown in the T-S (temperature versus entropy)diagram. Here A represents the area of the region enclosed by the cycle and B of the area below the curve. Compute the efficiency cycle.



(A) $\frac{A}{(A+B)}$ (B) $\frac{A}{(A-B)}$ (C) $\frac{B}{(A-B)}$ (D) $\frac{B}{(A+B)}$

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SECTION-I (ii) : (Maximum Marks: 24)

Multiple Correct Answer Type 6 Q. [4 M (–2)]

11. Two moles of O_2 $\left(\gamma = \frac{7}{5}\right)$ at temperature T_0 and 3 mole of CO_2 $\left(\gamma = \frac{4}{3}\right)$ at temperature $2T_0$ are allowed to mix together in a closed adiabatic vessel. The resulting mixture finally comes in thermal equilibrium. Then

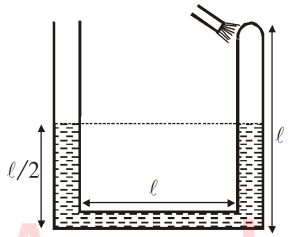
(A) Final temperature of the mixture is $\frac{23T_0}{14}$

(B) Final temperature of the mixture is $\frac{31T_0}{19}$

(C) Adiabatic exponent of the mixture formed is $\frac{14}{5}$

(D) Adiabatic exponent of the mixture formed is $\frac{19}{14}$

12. A rectangular narrow U-tube has equal arm lengths and base length, each equal to ℓ. The vertical arms are filled with mercury up to $\ell/2$ and then one end is sealed. By heating the enclosed gas all the mercury is expelled. If atmospheric pressure is P_0, the density of mercury is ρ and cross-sectional area is S, then [Neglect thermal expansion of glass and mercury]



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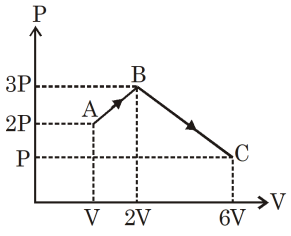
(A) Work done by the gas against the atmospheric pressure is $\frac{5\ell}{2} P_0 S$

(B) Work done by the gas against the gravity is $\frac{7}{4} S\rho g\ell^2$

(C) Work done by the gas against the atmospheric pressure is $P_0 S\ell$

(D) Word done by the gas against the gravity is $S\rho g\ell^2$

13. A monoatomic ideal gas undergoes a process ABC :



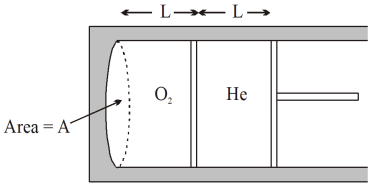
(A) Heat given to the gas is $\frac{33PV}{2}$.

(B) Heat given to the gas is $\frac{35PV}{2}$.

(C) Maximum temperature of the gas during the process is $\frac{8PV}{nR}$.

(D) Maximum temperature of the gas during the process is $\frac{4PV}{nR}$.

14. An insulating cylinder contains equal volumes of He and O_2 separated by a massless freely moving adiabatic piston as shown. The gas is compressed by moving the insulating piston so that volume of He becomes half. Select the correct alternative(s).



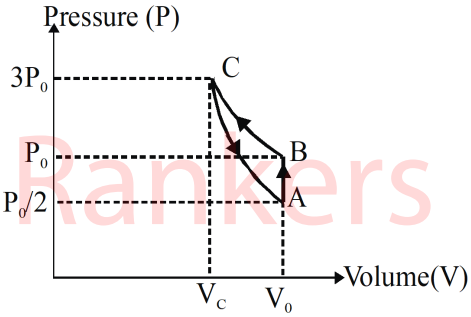
(A) Pressure in He chamber will be equal to pressure in O_2 chamber

(B) Pressure in He chamber will be less then pressure in O_2 chamber

(C) Volume of He chamber will be equal to volume of O_2 chamber

(D) Volume of O_2 chamber will be $\frac{(LA)}{(2)^{25/21}}$.

15. One mole of an ideal gas is carried through a thermodynamic cycle as shown in the figure. The cycle consists of an isochoric, an isothermal and an adiabatic processes. The adiabatic exponent of the gas is γ. Choose the correct option(s).



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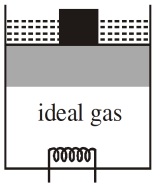
(A) $\gamma = \frac{\ln 6}{\ln 3}$ (B) $\gamma = \frac{\ln 5}{\ln 3}$ (C) BC is adiabatic (D) AC is adiabatic

16. A block of ice of mass 10 gm is in thermal equilibrium in 10 gm of water both are in contact of a conducting movable massless piston on a cylinder containing 3 moles of an ideal diatomic gas. In thermal equilibrium with water, the walls of cylinder are adiabatic and heat lost to surrounding is negligible. The gas is heated slowly by a heater. [Given $L_f = 80$ cal/gm, $R = 25/3$ J/mole – k, $S_W = 1$ cal/gm]



(A) work done by the gas by the time ice melts completely is zero.

(B) Net heat absorbed by the gas by the time ice melts is zero.

(C) Net heat supplied by the heater by the time water starts boiling $\frac{14650}{3}$ cal.

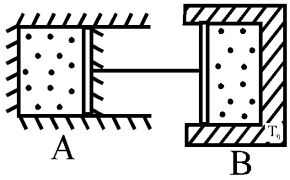
(D) Net heat supplied by the heater by the time water starts boiling 2800 cal.

SECTION-I (iii) : (Maximum Marks: 18)

Linked Comprehension Type (Single Option Correct) (1 Para × 4 Q.) (1 Para × 2 Q.) [3 M (–1)]

Paragraph for Questions 17 - 20

Two cylinder A and B having piston connected by massless rod (as shown in figure). The cross-sectional area of two cylinders are same & equal to 'S'. The cylinder A contains m gm of an ideal gas at Pressure P & temperature T_0. The cylinder B contain identical gas at same temperature T_0 but has different mass. The piston is held at the state in the position so that volume of gas in cylinder A & cylinder B are same & is equal to V_0.



The walls & piston of cylinder A are thermally insulated, whereas cylinder B is maintained at temperature T_0 reservoir. The whole system is in vacuum. Now the piston is slowly released and it moves towards left & mechanical equilibrium is reached at the state when the volume of gas in cylinder A becomes $\frac{V_0}{2}$. Then (here γ for gas = 1.5)

17. The mass of gas in cylinder B

(A) $2\sqrt{2}$ m (B) $3\sqrt{2}$ m (C) $\sqrt{2}$ m (D) none

18. The change in internal energy of gas in cylinder A

(A) $(\sqrt{2}-1)PV_0$ (B) $2(\sqrt{2}-1)PV_0$ (C) $\frac{PV_0}{(\sqrt{2}-1)}$ (D) none

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19. If work done by gas in cylinder B is W_B & work done by gas in cylinder A is W_A then

(A) $W_A = -W_B$ (B) $|W_A| > |W_B|$

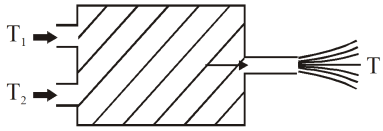
(C) $|W_A| < |W_B|$ (D) we can't say anything

20. What will be the compressive force in connecting rod at equilibrium

(A) PS (B) $\sqrt{2}$PS (C) $2^{3/2}$ PS (D) none

Paragraph for Questions 21 and 22

A self-contained machine only inputs two equal steady streams of hot and cold water at temperatures T_1 and T_2. Its only output is a single high-speed jet of water. The heat capacity per unit mass of water, C, may be assumed to be independent of temperature. The machine is in a steady state and the kinetic energy in the incoming streams is negligible.



21. What is the speed of the jet of water in the jet?

(A) $v = \sqrt{c(T_1 + T_2 - 2T)}$ (B) $v = \sqrt{c(T_1 + T_2 + 2T)}$

(C) $v = 2\sqrt{c(T_1 + T_2 + 2T)}$ (D) $v = 2\sqrt{c(T_1 + T_2 - 2T)}$

22. What is the maximum possible speed of the jet ?

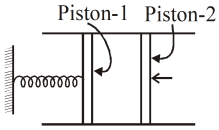
(A) $v_{max} = \sqrt{c}\left(\sqrt{T_1}\,\backslash;+\quad\sqrt{T_2}\right)$ (B) $v_{max} = \sqrt{c}\left(\sqrt{T_1}-\sqrt{T_2}\right)$

(C) $v_{max} = \sqrt{c\,(T_1+T_2)}$ (D) $v_{max} = \sqrt{c\,(T_1-T_2)}$

SECTION-II : (Maximum Marks: 32)

Numerical-Answer-Type questions (Non Negative Integer) 8 Q. [4 M (0)]

1. A long container has air enclosed inside at room temperature and atmospheric pressure (10^5 pa). It has a volume of 20,000 cc. The area of cross section is 100 cm^2 and force constant of spring is k_{spring} = 1000N/m. We push the right piston isothermally and slowly till it reaches the original position of the left piston which is movable. What is the final length of air column in cm. Assume that spring is initially relaxed.



2. Figure shows an ideal gas. Its pressure, volume & temperature are P_0, V_0 & T_0 respectively. Thin U-tube contains mercury. It is seen that when 20 cm of extra mercury is added in limb B of U-tube and temperature of gas is doubled, level of A is maintained at its position. Consider surrounding to be vacuum. Find the initial pressure of gas (in cm of Hg.)

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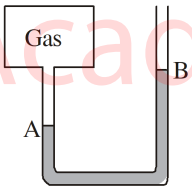
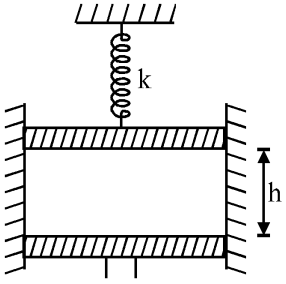
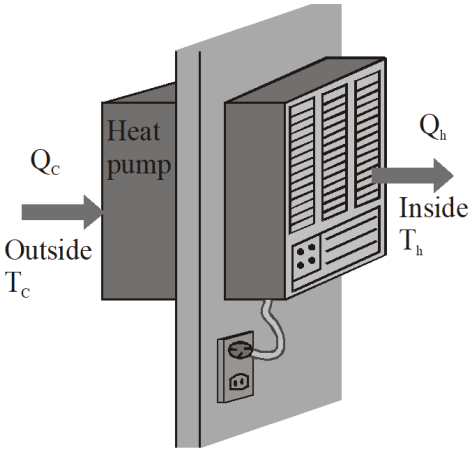


3. A balloon is filled with helium at the atmospheric pressure P_0. The volume of the balloon is V_0. The balloon is made of the material of mass m and density ρ. After being released, the balloon bursts at an altitude where the atmospheric pressure is $(P_0/2)$. Immediately before bursting, the balloon has a volume of $1.25V_0$. The maximum stress that the balloon material can withstand is given by αP_0, find the value of α. Assume that the temperature of helium remains constant, the balloon remains spherical, and the density of the material remains virtually constant. [Take : $V_0 = 8\ m^3$, $r = 1500\ kg/m^3$, $m = 18\ kg$]

4. An ideal gas at NTP is enclosed in a adiabatic vertical cylinder having area of cross section A = $27cm^2$, between two light movable pistons as shown in the figure. Spring with force constant k = 3700 N/m is in a relaxed state initially. Now the lower piston is moved upwards a height h/2, h being the initial length of gas column. It is observed that the upper piston moves up by a distance h/16. (γ = 1.5) The final temperature (in K) of the gas is :-



5. A heat pump used for heating shown in figure is essentially an air conditioner installed backward. It extracts energy from colder air outside and deposits it in a warmer room. Suppose the ratio of the actual energy entering the room to the work done by the device's motor is 20.0% of the theoretical maximum ratio. Determine the energy entering the room per joule of work done by the motor, given that the inside temperature is 27.0°C and the outside temperature is −3°C.



6. An ideal (Carnot) freezer in a kitchen has a constant temperature of 260 K, whereas the air in the kitchen has a constant temperature of 300 K. Suppose the insulation for the freezer is not perfect and conducts energy into the freezer at a rate of 0.150 W. The average power required for the freezer's motor to maintain the constant temperature in the freezer is $\frac{n}{260}$W. Find the value of n.

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7. A carnot engine working between 400 K and 800 K has a work output of 1200 J per cycle. What is the amount of heat energy supplied to the engine from source per cycle (in J) ?

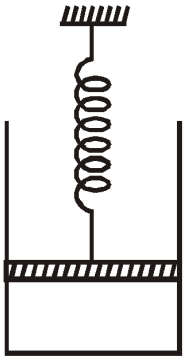
8. An electric freezer is turned on inside a tent for a long time. It is 0 °C outside the tent, +1 °C inside the tent, and −13 °C inside the freezer. T_0 would be the equilibrium temperature inside the tent if another identical freezer is turned on inside the tent ? The outside temperature remains the same. The freezers are identical and follow the Carnot cycle if $T_0 = = \frac{-1767 + \sqrt{1767^2 + 15288\alpha}}{2\alpha}$. Find α.

SECTION-III : (Maximum Marks: 8)

Numerical Grid Type (Ranging from 0 to 9) 2 Q. [4 M (0)]

1. One mole of an ideal monoatomic gas is taken from state A to state B through the process $P = \frac{3}{2}T^{1/2}$. It is found that its temperature increases by 115.2 K in this process. Now it is taken from state B to C through a process for which internal energy is related to volume as $U = \frac{1}{2}V^{1/2}$. The volume at B is 100 m^3 and at C it is 1600 m^3, then the total work performed by the gas is $\alpha \times 10^2$ J. Fill a in OMR sheet. (Use $R = \frac{25}{3}$ J/mol-K)

2. An ideal monoatomic gas is enclosed in a vertical cylinder of cross section area A and length ℓ, under frictionless piston connected to spring of spring constant K. Atmosphere pressure is such that in absence of gas under the piston, equilibrium of piston is achieved when piston just touches the bottom of the cylinder and spring is in stretched configuation. Q amount of heat is supplied to gas slowly to move piston to the upper edge. Initially volume occupied by gas is $\frac{1}{3}$rd of the volume of cylinder. If $Q = 2\lambda$, $\frac{K\ell^2}{9}$ find λ, (through the motion of the piston spring is in stretched configuration)



SECTION-IV : (Maximum Marks: 8)

Matrix Match Type (4 × 5) (One or more options correct) 1 Q. [8 M; (for each entry +2(0)]

1. A monatomic ideal gas at pressure P_0 & volume V_0 can be made to undergo 4 different process on PV diagram as shown. For each of the processes, match with entries in column-II. (Symbols have their usual meanings).

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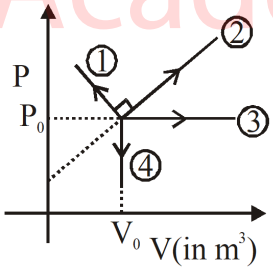

Column-I		Column-II	
(A)	Process−1	(P)	ΔQ + ve
(B)	Process−2	(Q)	ΔU + ve
(C)	Process−3	(R)	W −ve
(D)	Process−4	(S)	Temperature decreases
		(T)	ΔQ − ve

ANSWER KEY

GR_Heat & Thermodynaimcs

PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	C	C	B	D	D	C	A	A
SECTION-I (ii)	Q.	11	12	13	14	15	16				
	A.	A,D	A,B	A,C	A,D	A,D	A,B,C				
SECTION-I (iii)	Q.	17	18	19	20	21	22				
	A.	B	B	C	C	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	100	20	375	364	2	6	2400	137		
SECTION-III	Q.	1	2								
	A.	5	8								
SECTION-IV	Q.	1									
	A.	A->R,B->PQ,C->PQ,D->ST									

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Important Notes

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Important Notes

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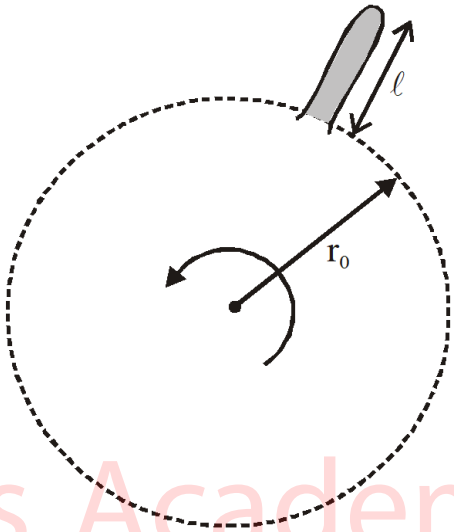
SECTION-I (i) (Maximum Marks:33)

Single Correct Answer Type 11 Q. [3 M (–1)]

1. A test tube filled with water is being spun around in an ultracentrifuge with angular velocity ω. The test tube is lying along a radius and the free surface of the water is at radius r_0 (Figure). The length of the test tube is ℓ and ρ is the density of the water. Ignore gravity and ignore atmospheric pressure. Choose the correct statement.



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(A) The pressure at the mid point of test tube is given by $\frac{1}{8}\rho\ell\omega^2(\ell+4r_0)$.

(B) The pressure at the mid point of test tube is given by $\frac{1}{2}\rho\ell\omega^2\left(\frac{\ell}{2}+r_0\right)$.

(C) The pressure at the mid point of test tube is given by $\frac{1}{4}\rho\ell\omega^2\left(\frac{\ell}{2}+r_0\right)$.

(D) The pressure remains constant through out the tube.

2. A thin uniform cylindrical shell, closed at both ends, is partially filled with water. It is floating vertically in water in half-submerged state. If ρ_C is the relative density of the material of the shell with respect to water, then the correct statement is that the shell is

(A) more than half-filled if ρ_C is less than 0.5

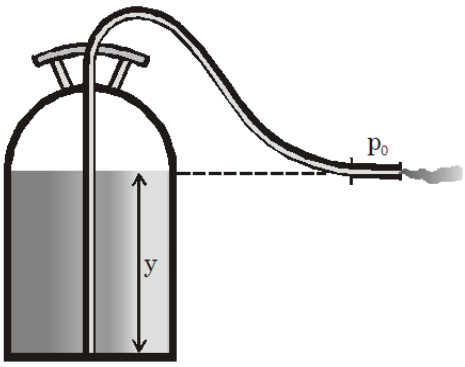
(B) more than half-filled if ρ_C is more than 1.0

(C) half-filled if ρ_C is more than 0.5

(D) less than half-filled if ρ_C is less than 0.5

1001CJA00PHEN2500701

3. A certain fire extinguisher consists of a large sealed tank with a thin tube leading from the bottom of the tank to an exit nozzle; the air above the liquid in the tank is at an overpressure p_{tank} with respect to atmospheric pressure p_0. The exit nozzle is at the same level as the top of the liquid; both are a distance y above the bottom of the tank (see figure). The liquid has density ρ. Find an expression for the speed of the stream of liquid emerging from the nozzle.



(A) $\sqrt{\frac{2(p_{tank}+p_0)}{\rho}}$ (B) $\sqrt{\frac{2p_{tank}}{\rho}}$ (C) $\sqrt{\frac{(p_{tank}+p_0)}{2\rho}}$ (D) $\sqrt{\frac{p_{tank}}{\rho}}$

4. A tank full of water has a small hole at its bottom. Let t_1 be the time taken to empty first one third of the tank and t_2 be the time taken to empty second one third of the tank and t_3 be the time taken to empty rest of the tank then

(A) $t_1 = t_2 = t_3$ (B) $t_1 > t_2 > t_3$ (C) $t_1 < t_2 < t_3$ (D) $t_1 > t_2 < t_3$

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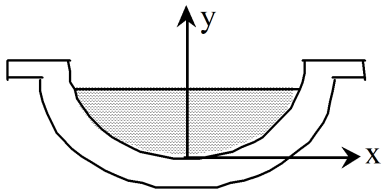
5. A flat plate moves normally with a speed v_1 towards a horizontal jet of water of uniform area of cross-section. The jet discharges water at the rate of volume V per second at a speed of v_2. The density of water is ρ. Assume that water splashes along the surface of the plate at right angles to the original motion. The magnitude of the force acting on the plate due to jet of water is :–

(A) $\rho V v_1$ (B) $\rho\left(\frac{V}{v_2}\right)(v_1 + v_2)^2$

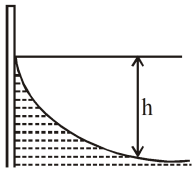
(C) $\frac{\rho V}{v_1 + v_2} v_1^2$ (D) $\rho V(v_1 + v_2)$

6. A small hole is made at the bottom of a symmetrical jar as shown in figure. A liquid is filled into the jar upto a certain height. The rate of descension of liquid is independent of the level of the liquid in the jar. Then the surface of the jar is a surface of revolution of the curve -



(A) $y = kx^4$ (B) $y = kx^2$ (C) $y = kx^3$ (D) $y = kx^5$

7. Water of density ρ in a clean aquarium forms a meniscus, as illustrated in the figure. Calculate the difference in height h between the centre and the edge of the meniscus. The surface tension of water is γ.



(A) $\sqrt{\frac{2\gamma}{\rho g}}$ (B) $\sqrt{\frac{\gamma}{\rho g}}$ (C) $\frac{1}{2}\sqrt{\frac{\gamma}{\rho g}}$ (D) $2\sqrt{\frac{\gamma}{\rho g}}$

8. Figure (a) and (b) show water drop and mercury drop in two identical conical glass pipes, then

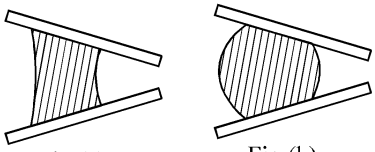


Fig.(a) Fig.(b)

(A) The water drop tends to move towards narrow end and mercury drop towards wide end.

(B) The water drop tends to move towards wide end and mercury drop towards narrow end.

(C) both tend to move towards wide end.

(D) both tend to move towards narrow end.

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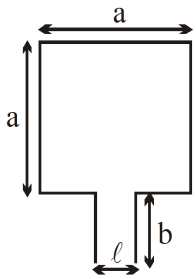
9. A light open rigid wire frame floats on the surface of water as shown in figure. What force will act on the frame, immediately after some soap solution is dropped inside it? α_1 and α_2 are the surface tensions of water and soap respectively ($\alpha_1 > \alpha_2$)



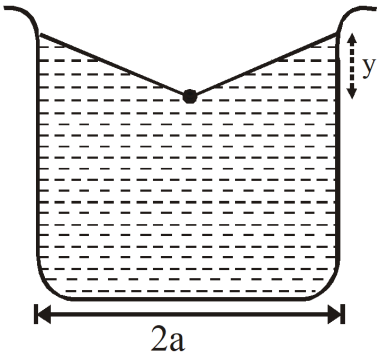
(A) zero (B) $(\alpha_1 - \alpha_2)\,\ell$

(C) $(\alpha_1 + \alpha_2)(4a + 2b + \ell)$ (D) $(\alpha_1 - \alpha_2)(4a + 2b - \ell)$

10. A drop of water volume 0.05 cm^3 is pressed between two glass-plates, as a consequence of which, it spreads between the plates. The area of contact with each plate is 40 cm^2. If the surface tension of water is 70 dyne/cm, the minimum normal force required to seperate out the two glass plates in newton is approximately (assuming angle of contact is zero) :

(A) 45 N (B) 100 N (C) 90 N (D) None of these

11. A container of width 2a is filled with a liquid. A thin wire of weight per unit length λ is gently placed over the liquid surface in the middle of the surface as shown in the figure. As a result, the liquid surface is depressed by a distance y (y << a). Determine the surface tension of the liquid.



(A) $\frac{\lambda ag}{2y}$ (B) $\frac{3\lambda ag}{2y}$ (C) $\frac{2\lambda ag}{3y}$ (D) None of these

SECTION-I (ii) : (Maximum Marks: 16)

Multiple Correct Answer Type 4 Q. [4 M (–2)]

12. A thin walled hemispherical shell of mass m and radius R is pressed against a smooth vertical wall. Through a very small aperture at its top water of density ρ is filled in it completely. Minimum magnitude of force is to be applied to the shell for liquid not to escape from it. Weight of hemispherical shell is mg and total mass of water is $M = \frac{2}{3}\pi R^3\rho$:

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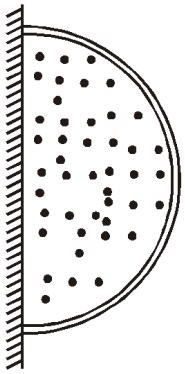


(A) Horizontal component of net force is $\frac{1}{2}\pi R^3\rho g$

(B) Net external force is $Mg\sqrt{\frac{9}{4}+\left(1+\frac{m}{M}\right)^2}$

(C) Vertical component of force is (M + m)g

(D) Direction of net force is given by $\theta = \text{arc tan}\frac{(m+M)}{3M}$, where θ is from horizontal direction.

13. A cubical block is floating in a liquid with one third of its volume immersed in the liquid. When the whole system accelerates upwards with acceleration of g/2 :-

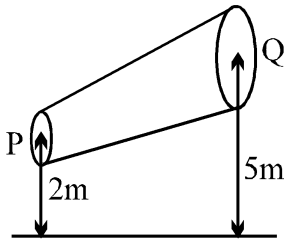
(A) the fraction of volume immersed in the liquid will change.

(B) the buoyancy force on the block will change.

(C) the buoyancy force will increase by 50 percent.

(D) the pressure in the liquid will increased.

14. A non-viscous liquid of constant density 1000 kg/m^3 flows in a streamline motion along a tube of variable cross section. The tube is kept inclined in the vertical plane as shown in the figure. The area of cross section of the tube at two points P and Q at heights of 2 meters and 5 meters are respectively 4×10^{-3}m^2 and 8×10^{-3} m^3. The velocity of the liquid at point P is 1 m/s and fluid flow from P to Q, then



(A) the work done per unit volume by the pressure is + 29625 J/m^3.

(B) the work done per unit volume by gravity forces is – 30000 J/m^3.

(C) Net work done per unit volume is zero

(D) velocity of liquid at point Q is 0.5 m/s

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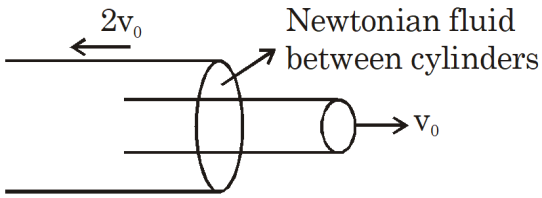
15. Consider a situation in which there is a Newtonian fluid between two long coaxial uniformly moving cylinders of radius R and 2R.



(A) Force per unit length required to keep the inner cylinder moving is $4\pi\eta v_0$

(B) Force per unit length required to keep the inner cylinder moving is $\frac{6\pi\eta v_0}{\ell n 2}$.

(C) Distance of the point from the axis at which speed of fluid is zero is $2^{1/3}$R.

(D) Distance of the point from the axis at which speed of fluid is zero is $\sqrt{2}$R.

SECTION-I (iii) : (Maximum Marks: 21)

Linked Comprehension Type (Single Option Correct) (1 Para × 3 Q.) (2 Para × 2 Q.) [3 M (–1)]

Paragraph for Questions 16 - 18

When liquid medicine of density ρ is to be put in the eye, it is done with the help of a dropper. As the bulb on the top of the dropper is pressed, a drop forms at the opening of the dropper. We wish to estimate the size of the drop. We first assume that the drop formed at the opening is spherical because that requires a minimum increase in its surface energy. To determine the size, we calculate the net vertical force due to the surface tension T when the radius of the drop is R. When this force becomes smaller than the weight of the drop, the drop gets detached from the dropper.

16. If the radius of the opening of the dropper is r, the vertical force due to the surface tension on the drop of radius R (assuming r << R) is

(A) $2\pi rT$ (B) $2\pi RT$ (C) $\frac{2\pi r^2T}{R}$ (D) $\frac{2\pi R^2T}{r}$

17. If $r = 5 \times 10^{-4}$ m, $\rho = 10^3$ kgm^{-3}, $g = 10$ ms^{-2}, $T = 0.11$ Nm^{-1}, the radius of the drop when it detaches from the dropper is approximately

(A) 1.4×10^{-3} m (B) 3.3×10^{-3} m (C) 2.0×10^{-3} m (D) 4.1×10^{-3} m

18. After the drop detaches, its surface energy is

(A) 1.4×10^{-6} J (B) 2.7×10^{-6} J (C) 5.4×10^{-6} J (D) 8.1×10^{-6} J

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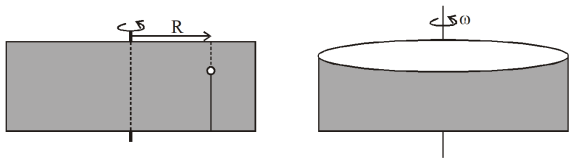
Paragraph for Questions 19 and 20

A closed cylindrical container with a vertical axis is completely filled with liquid (ρ_ω). A plastic bead of density ρ_{body} and volume V is placed at distance R from the axis and is anchored to the bottom of the container by a thin thread of length ℓ. If as a result of the containers revolutions, the beads sink by h and final distance of bead from axis is r. (In the final state the total content of the container rotates at the same angular speed.)



19. What is force exerted by liquid on the bead ?

(A) $\rho_\omega V g$ (B) $(\rho_\omega - r_{body}) V g$

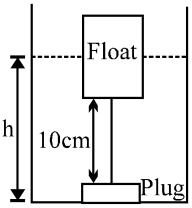
(C) $(\rho_\omega - r_{body}) V\sqrt{g^2 + (\omega^2 r)^2}$ (D) $\rho_\omega V\sqrt{g^2 + (\omega^2 r)^2}$

20. What is ω :-

(A) $\omega = \sqrt{\frac{(R+r)g}{(\ell+h)r}}$ (B) $\omega = \sqrt{\frac{(R-r)g}{(\ell-h)r}}$ (C) $\omega = \sqrt{\frac{(R+r)g}{(\ell-h)r}}$ (D) $\omega = \sqrt{\frac{(R-r)g}{(\ell+h)r}}$

Paragraph for Questions 21 and 22

A level controller is shown in the figure. It consists of a thin circular plug of diameter 10cm and a cylindrical float of diameter 20cm tied together with a light rigid rod of length 10cm. The plug fits in snugly in a drain hole at the bottom of the tank which opens into atmosphere. As water fills up and the level reaches height h, the plug opens. The float has a mass 3kg and the plug may be assumed as massless.



21. Find h (in m) :-

(A) $\frac{2(3+\pi)}{15\pi}$ (B) $\frac{(3+\pi)}{15\pi}$ (C) $\frac{(3+\pi)}{10\pi}$ (D) $\frac{2(3+\pi)}{10\pi}$

22. Determine the level of water (in meter) in the tank when the plug closes again

(A) $\frac{2(3+\pi)}{15\pi}$ (B) $\frac{(3+\pi)}{15\pi}$ (C) $\frac{(3+\pi)}{10\pi}$ (D) $\frac{2(3+\pi)}{10\pi}$

SECTION-II (i) : (Maximum Marks: 8)

Numerical Answer type Questions (Upto Second Decimal Place) 2 Q. [4 M (0)]

1. A U-tube filled with a liquid of volumetric coefficient of $10^{-5}/°C$ lies in a vertical plane. The height of liquid column in the left vertical limb is 100 cm. The liquid in the left vertical limb is maintained at a temperature = 0°C while the liquid in the right limb is maintained at a temperature = 100°C. Find the difference (in cm) in levels in the two limbs.

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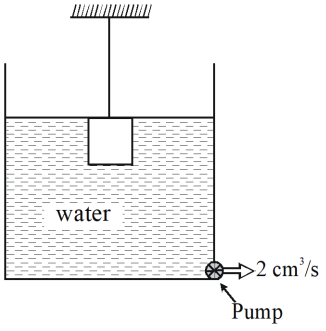
2. An expansible balloon filled with air floats on the surface of a lake with 2/3 of its volume submerged. How deep must it be sunk in the water so that it is just inequilibrium neither sinking further nor rising? It is assumed that the temperature of the water is constant & that the height of the water barometer is 9 meters.

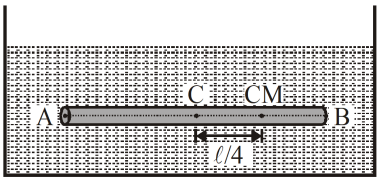
SECTION-II (ii) : (Maximum Marks: 16)

Numerical-Answer-Type questions (Non Negative Integer) 4 Q. [4 M (0)]

3. Figure shows a cubical block of side 10 cm and relative density 1.5 suspended by a wire of cross sectional area 10^{-6} m^2. The breaking stress of the wire is 7×10^6 N/m^2. The block is placed in a beaker of base area 200 cm^2 and initially i.e. at t = 0, the top surface of water & the block coincide. There is a pump at the bottom corner which ejects 2 cm^3 of water per sec. Find the time (in sec) at which the wire will break.



4. A non uniform cylinderical rod of mass 5kg, length 2m and radius 0.1m is having its center of mass at a distance $\frac{1}{2}$m from the centre and lying on the axis of the rod. The rod is kept in a liquid of uniform density 1400 kg/m^3. The moment of inertia of the rod about the centre of mass is 4 kg-m^2. Find the angular acceleration (in SI unit) of point A of the rod just after the rod is released from the position shown in figure.



5. A horizontal pipeline carries water in a streamline flow. At a point along the pipe, where the cross-sectional area is 10 cm^2, the water velocity is 1 m/s and the pressure is 4000 Pa. The pressure of water at another point where the cross-sectional area is 5 cm^2, is..... Pa. (Density of water = 10^3 kg. m^{-3}).

6. A block of wood is floating in water such that 1/3rd of it is submerged in water when the same block is floated in alcohol, 1/2 of it's volume is submerged. Now a mixture of water and alcohol is made taking equal volume of both and block is floated in it. What is the % of it's volume that is now submerged.

SECTION-III : (Maximum Marks: 16)

Numerical Grid Type (Ranging from 0 to 9) 4 Q. [4 M (0)]

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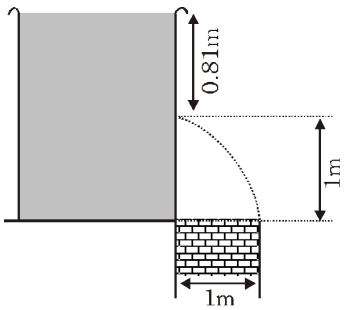
1. For the arrangement shown in the figure. The time interval in seconds after which the water jet ceases to cross the wall is found to be $(\alpha)^3$. Area of the cross section of the tank A = $\sqrt{5}$ m^2 and area of the orifice a = 32 cm^2. [Assume that the container remaining fixed]. Find the value of α. (Take g = 10 m/s^2)



2. A glass capillary sealed at the upper end is of length 0.11 m and internal diameter 2×10^{-5} m. This tube is immersed vertically into a liquid of surface tension 5.06×10^{-2} N/m. When the length $x \times 10^{-2}$ m of the tube is immersed in liquid then the liquid level inside and outside the capillary tube becomes the same, then the value of x is : (Assume atmospheric pressure is $1.01 \times 10^5 \frac{N}{m^2}$)

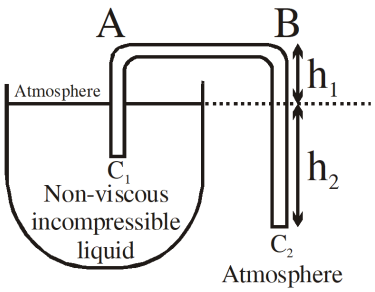
3. Due to the excess pressure of surface tension find the change in the radius (in Å) of a liquid drop of radius 1 mm, surface tension 0.075 N/m and Bulk modulus $1.25 \times 10^{+8}$ N/m^2.

4. The velocity distribution in a viscous flow over a plate is given by $u = 4y - y^2$ for $y \leq 2$m where u is velocity in m/s at a point which is at a distance y from the plate. If the coefficient of dynamic viscosity is 1.5 in SI units. The shear stress at y = 0 in N/m^2 is :-

SECTION-IV : (Maximum Marks: 16)

Matrix Match Type (4 × 5) (One or more options correct) 2 Q. [8 M; (for each entry +2(0)]

1. An evacuated thin tube has two tight caps C_1 and C_2 at its ends. The tube is adjusted as shown in the figure below. Atmospheric pressure = P and density of the liquid = d, Neglect surface tension of liquid.
Match the entities of columns



Column-I		Column-II	
(A)	$h_1 > \frac{P}{dg}, h_2 = 0$	(P)	If C_2 is opened first and then C_1 is opened, liquid level in the left arm of the tube will be on level with liquid in the pot.
(B)	$h_1 > \frac{P}{dg}, h_2 > 0$	(Q)	If C_1 is opened with C_2 closed, liquid will rise in the left arm but tube will not be fully filled.
(C)	$h_1 < \frac{P}{dg}, h_2 > 0$	(R)	If C_1 is opened with C_2 closed, tube will be fully filled with liquid.
(D)	$h_1 + h_2 < \frac{P}{dg}$, $h_2 > 0$	(S)	If C_1 is opened first and then C_2 is opened, liquid will flow down continuously.
		(T)	After any of the above action pressure at the mid point of tube AB is non-zero

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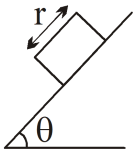
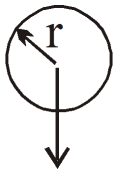
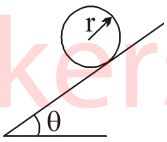
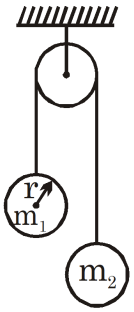
2.

Column I		Column II	
(A)	Body is a block lying on a rough inclined plane. Angle of inclination θ < angle of repose. A is inclined surface 	(P)	Force exerted by A on body can be equal to it's weight in magnitude.
(B)	Body is a ball falling in liquid with terminal velocity. A is liquid. 	(Q)	If density of body is somehow increased without changing r, it's state of motion (constant acceleration / constant velocity) or equilibrium position will be disturbed.
(C)	Body is a spherical ball rolling purely on inclined plane; inclined at an angle equal to angle of repose. The inclined surface is A. 	(R)	Work done by the force exerted by A on the body is zero.
(D)	Body is m_1 hanging on an ideal pulley system. A is string $m_1 < m_2$. 	(S)	Mechanical energy of body is constant as time passes.
		(T)	The kinetic energy of body is constant as time passes

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PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	B	C	B	A	A	A	B	A
	Q.	11									
	A.	A									
SECTION-I (ii)	Q.	12	13	14	15						
	A.	B,C	B,C,D	A,B	B,C						
SECTION-I (iii)	Q.	16	17	18	19	20	21	22			
	A.	C	A	B	D	B	A	C			
SECTION-II (i)	Q.	1	2								
	A.	0.1	4.5								
SECTION-II (ii)	Q.	3	4	5	6						
	A.	100	110	2500	40						
SECTION-III	Q.	1	2	3	4						
	A.	5	1	4	6						
SECTION-IV	Q.	1	2								
	A.	A->PQ,B->PQ,C->PRST,D->PRST	A->PQRST,B->PQT,C->RS,D->Q								

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SECTION-I (i) (Maximum Marks:12)

Single Correct Answer Type 4 Q. [3 M (–1)]

1. A particle of mass 1 kg, executing simple harmonic motion of period 20 s, crosses the mean position at t = 0 with velocity p cm/s.

(A) the maximum acceleration of the particle is 10 cm/s^2

(B) when it is 4 cm from the mean position, its velocity is 6p/10 cm/s

(C) the kinetic energy of the particle when its displacement is 5 cm from the mean position is $\frac{3\pi^2}{8\times10^6}$J

(D) its velocity at displacement 6 cm from the mean position is 8π/10 cm/s

2. A uniform rod of length 2ℓ rocks to and fro on the top of a rough semicircular fixed cylinder of radius a. The period of small oscillations, of rod is

(A) $\frac{2\pi\ell}{\sqrt{3ga}}$ (B) $2\pi\sqrt{\frac{\ell}{3g}}$ (C) $2\pi\sqrt{\frac{a}{3g}}$ (D) $\frac{\pi\ell}{\sqrt{3ga}}$

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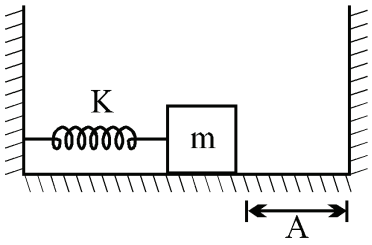
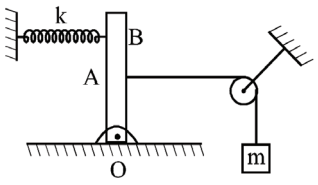
3. In the figure shown the spring is relaxed. The spring is compressed by 2 A and released. Mass m attached with the spring collides with the wall & loses two third of its kinetic energy & returns. Find the time after which the spring will have maximum compression first time after releasing. (Neglect Friction)



(A) $\frac{13\pi}{12}\sqrt{\frac{m}{K}}$ (B) $\frac{17\pi}{12}\sqrt{\frac{m}{K}}$ (C) $\frac{17\pi}{12}\sqrt{\frac{2m}{K}}$ (D) $\frac{7\pi}{12}\sqrt{\frac{2m}{K}}$

4. A massless rod is hinged at O. A string carrying a mass m at one end is attached to point A on the rod so that OA = a. At another point B (OB= b) of the rod , a horizontal spring of force constant k is attached as shown. Find the period of small vertical oscillations of mass m around its equilibrium position.



(A) $\pi\sqrt{\frac{m}{K}}$ (B) $2\pi\sqrt{\frac{m}{K}}$ (C) $\frac{2\pi a}{b}\sqrt{\frac{m}{K}}$ (D) $\frac{2\pi b}{a}\sqrt{\frac{m}{K}}$

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SECTION-I (ii) : (Maximum Marks: 80)

Multiple Correct Answer Type 20 Q. [4 M (–2)]

5. A particle is performing SHM with its position given as $x = 2 + 5\sin\left(\pi t + \frac{\pi}{6}\right)$ where x (in m) & t (in sec). Which of the following is/are correct :-

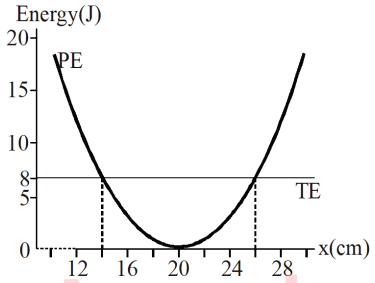
(A) Equilibrium position is at x = 2m

(B) Maximum speed of particle is 5π m/s

(C) At t = 0 particle is 2.5 m away from mean position moving in negative direction

(D) At t = 0, x = 4.5 m; acceleration of particle is $-\pi^2(4.5)$ m/s^2

6. Figure shows the potential-energy diagram and the total energy line of a particle oscillating on a spring.



(A) the amplitude of oscillation of particle is 6 cm.

(B) the spring constant is 4.44 kN/m.

(C) the mass of particle is 1kg.

(D) the maximum kinetic energy of particle is at x = 20 cm.

7. A linear harmonic oscillator of force constant 2×10^6 Nm^{-1} and amplitude 0.01 m has a total mechanical energy of 160 J. Its

(A) maximum potential energy is 100 J

(B) maximum kinetic energy is 100 J

(C) maximum potential energy is 160 J

(D) minimum potential energy is zero.

8. A body is performing S.H.M., then its

(A) average total energy of SHM per cycle is equal to its maximum kinetic energy.

(B) average kinetic energy per cycle is equal to half of its maximum kinetic energy.

(C) mean velocity over a complete cycle is equal to $2/\pi$ times of its maximum velocity.

(D) root mean square velocity is $1/\sqrt{2}$ times of its maximum velocity

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9. The potential energy of a particle of mass 0.1 kg, moving along the x-axis, is given by U = 5x (x − 4)J, where x is in metres. It can be concluded that :

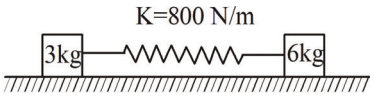
(A) the particle is acted upon by a constant force.

(B) the speed of the particle is maximum at x = 2 m.

(C) the particle executes simple harmonic motion

(D) the period of oscillation of the particle is $\pi/5$ s.

10. The system shown in the figure can move on a smooth surface. The spring is initially compressed by 6 cm and then released .



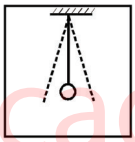
(A) the particles perform SHM with time period $\frac{\pi}{10}$ sec

(B) the block of mass 3 kg perform SHM with amplitude 4 cm

(C) the block of mass 6 kg will have maximum momentum 2.40 kg m/s

(D) none of these

11. A simple pendulum is oscillating in a stationary lift with a time period T and an angular amplitude θ_0. When it is passing through the mean position, the lift starts accelerating upwards.



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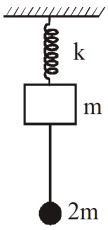
(A) The time period of its oscillation will increase

(B) Its angular amplitude would increase.

(C) The time period of its oscillation would decrease

(D) Its angular amplitude would decrease.

12. A bob of mass 2m hangs by a string attached to the block of mass m of a spring block system. The whole arrangement is in a state of equilibrium. The bob of mass 2m is pulled down slowly by a distance x_0 and released.



(A) For $x_0 = \frac{3mg}{k}$ maximum tension in string is 4mg

(B) For $x_0 > \frac{3mg}{k}$, minimum tension in string is mg

(C) Frequency of oscillation of system is $\frac{1}{2\pi}\sqrt{\frac{k}{3m}}$, for all non-zero values of x_0

(D) The motion will remain simple harmonic for $x_0 \leq \frac{3mg}{k}$

13. Consider a spring that exerts the following restoring force

$F = -kx$ for $x > 0$

$F = -4kx$ for $x < 0$

A mass m on a frictionless surface is attached to the spring displaced to x = A by stretching the spring and released :

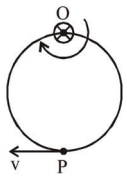
(A) The period of motion will be $T = \frac{3}{2}\pi\sqrt{\frac{m}{k}}$

(B) the most negative value the mass m can reach will be $x = -\frac{A}{2}$

(C) The time taken to move from x = A to $x = -\frac{A}{\sqrt{2}}$, straight away will be equal to $\frac{5\pi}{8}\sqrt{\frac{m}{k}}$

(D) The total energy of oscillations will be $\frac{5}{2}kA^2$

14. A ring of mass m, radius r can oscillate in a vertical plane about is top most point 'O' as shown. Axis is normal to the plane of ring. Maximum speed of lowest point 'P' of ring is v.



(A) Time period of SHM for small amplitudes is $2\pi\sqrt{\frac{2r}{g}}$

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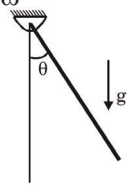
(B) Acceleration of centre of mass of ring at lowest position is zero.

(C) Hinge force when centre of mass is at lowest point is $\frac{mv^2}{4r}$

(D) Mechanical energy of oscillation is $\frac{mv^2}{4}$ (when potential energy at mean position is taken as zero.)

15. An uniform rod of mass m and length ℓ is pivoted at top and it can perform angular SHM in vertical plane according to equation $\theta = \theta_0 \sin \omega t$. At time $t = \frac{2\pi}{\omega}$ a point mass m (at rest) sticks to lowest end of rod.



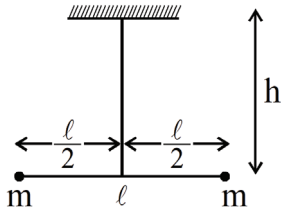
(A) New time period of rod will be times of the $\frac{2}{\sqrt{3}}$ time period before collision.

(B) New angular amplitude is $\frac{\theta_0}{\sqrt{12}}$

(C) New angular amplitude is $\frac{\theta_0}{4}$

(D) There will always be loss of kinetic energy of rod, irrespective of time at which mass sticks to rod.

16. Figure shows two particles connected with a light rod and entire arrangement hangs on a elastic wire. Neglect any gravitational effect, θ_0 = maximum angular displacement of the particles, C = torsional stiffness of wire. Mark the correct statements :-



(A) Time period of oscillation is $\pi \ell \sqrt{\frac{2m}{C}}$

(B) Maximum tension in the rod is $\frac{C\theta_0^2}{\ell}$

(C) The energy of oscillation is $\frac{1}{2}C\theta_0^2$

(D) The maximum tangential acceleration of particle is $\frac{C\theta_0}{m\ell}$

17. A block suspended from a spring at natural length and is free to move vertically in the y-direction. Mark the CORRECT statement (s) :-

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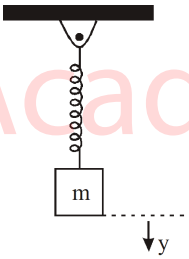


(A) Mass is released when y = 0; the maximum value of y reached by the mass m is 2mg/k

(B) Mass is released when y = 0; the maximum value of y reached by the mass m is mg/k

(C) Due to air resistance the mass settles down into an equilibrium position y_{eq}, the mechanical energy loss is $\frac{1}{2}(m^2g^2/k)$

(D) Due to air resistance the mass settles down into an equilibrium position y_{eq}, the mechanical energy lost is (m^2g^2/k)

18. A particle performing S.H.M. undergoes displacement of A/2 (where A = amplitude of S.H.M.) in one second. At t = 0 the particle was located at either extreme position or mean position. The time period of S.H.M. can be : (consider all possible cases) :-

(A) 12 s (B) 2.4 s (C) 6s (D) 1.2 s

19. Two particles A and B are performing SHM along same straight line with same amplitude a and same angular frequency ω. At t = 0, A is left extreme and B is at a distance a/2 from mean position to the right and moving toward mean position. Then:

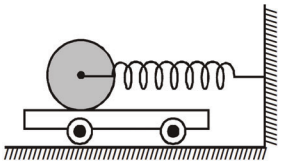
(A) their phase difference is $\frac{5\pi}{3}$

(B) their phase difference is $\frac{2\pi}{3}$

(C) time after which they cross each other is $\frac{2\pi}{3\omega}$

(D) time after which they cross each other is $\frac{\pi}{3\omega}$

20. If the figure shown, the mass of the disc as well as that of the trolley is M. The spring is ideal and has stiffness k. The trolley can move horizontally on the floor without friction, and the disc can roll on the trolley surface without slipping. The spring is compressed and the system released so that oscillations begin. At the instant of release, magntiude of :



(A) Acceleration of center of disc = twice of that of trolley

(B) Acceleration of center of disc = thrice of that of trolley

(C) Spring force is four times of friction.

(D) Spring force is two times of friction.

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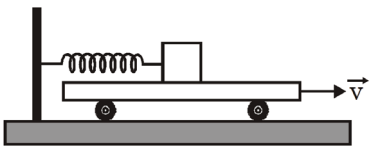
21. A block of mass m is attached to a fixed support by a horizontal spring with force constant k and negligible mass as shown in figure. Hooke's law describes the spring both in extension and in compression. The block placed on a very long horizontal board, with which it has coefficient of static friction μ_s and a smaller coefficient of kinetic friction μ_k. The board moves to the right at constant speed v. Assume the block spends most of its time sticking to the board and moving to the right, so the speed v is very small in comparison to the average speed the block has as it slips back toward the left. Ignore the time when the speed of block changes from v to 0 and from 0 to v. Mark the correct statement(s):



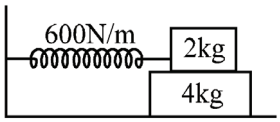
(A) The maximum extension of the spring from its unstretched position is very nearly given by $\frac{\mu_s mg}{k}$

(B) The block oscillates about an equilibrium position at which the spring is stretched by $\frac{\mu_k mg}{k}$

(C) The amplitude of the block's motion is A = $\frac{(\mu_s - \mu_k)mg}{k}$

(D) The period of the block's for a complete oscillation is T = $\frac{2(\mu_s - \mu_k)mg}{vk} + \pi\sqrt{\frac{m}{k}}$

22. In the arrangement shown, the spring of force constant 600N/m is in the unstretched position. The coefficient of friction between the two blocks is 0.4 and that between the lower block & ground surface is zero. If both the blocks are displaced slightly and released, the system executes SHM.



(A) Time period of their oscillation if they do not slip w.r.t. each other is $\pi/5$ sec.

(B) Time period of their oscillation if they do not slip w.r.t. each other is $\pi/3$ sec.

(C) The maximum amplitude of the oscillation for which sliding between them does not occur is 5 cm.

(D) The maximum amplitude of the oscillation for which sliding between them does not occur is 2 cm.

Paragraph for Questions 23 and 24

A solid disk of mass M and radius R on a vertical shaft. The shaft is attached to a coil spring which exerts a linear restoring torque of magnitude $C\theta$ where θ is the angle measured from the static equilibrium position and C is a constant. Neglect the mass of the shaft and the spring and assume the bearings to be frictionless.

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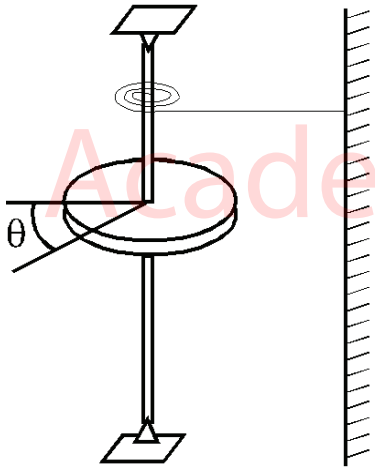


23. If the disc is rotated by θ and released :-

(A) The disks will execute SHM.

(B) The disks will not execute SHM.

(C) The frequency of motion is $\frac{1}{2\pi}\sqrt{\frac{2C}{MR^2}}$.

(D) The frequency of motion is $\frac{1}{2\pi}\sqrt{\frac{C}{3MR^2}}$.

24. Suppose that the disk is moving according to the $\theta = \theta_0 \sin(\omega t)$ where ω is the frequency found in part (a) At time $t_1 = \pi/\omega$ a ring of stickly putty of mass M and radius R is dropped concentrically on the disk,

(A) The new frequency of motion is $\frac{1}{2\pi}\sqrt{\frac{2C}{3MR^2}}$

(B) The new frequency of motion is $\frac{1}{2\pi}\sqrt{\frac{2C}{MR^2}}$.

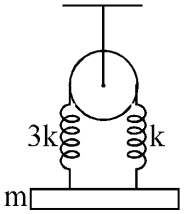
(C) The new amplitude of motion is $\frac{\theta_0}{2}$.

(D) The new amplitude of motion is $\frac{\theta_0}{\sqrt{3}}$.

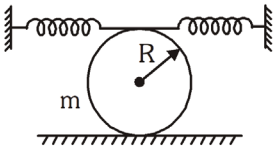
SECTION-II : (Maximum Marks: 24)

Numerical-Answer-Type questions (Non Negative Integer) 6 Q. [4 M (0)]

1. A uniform rod of mass m is suspended on two springs at equal distance from ends as shown. The force constants of the springs are k and 3k. During the vertical oscillations of the system, the rod remains horizontal. Find the angular frequency (in rad./s) of these osicllations. The strings and the pulley are ideal. (Given m = 3 kg, k = 676 N/m)



2. A solid uniform cylinder of mass m performs small oscillations due to the action of two springs combined stiffness equal to k (figure). The period of these oscillations in the absence of sliding is $T = \pi\sqrt{\frac{xm}{2k}}$. Then find x :



3. A car accelerates uniformly from rest. Initially, its door is slightly ajar (open). Calculate how far (in m) the car travels before the door slams shut. Assume the door has a frictionless hinge, a uniform mass distribution and a length of 1.2 m from front to back.

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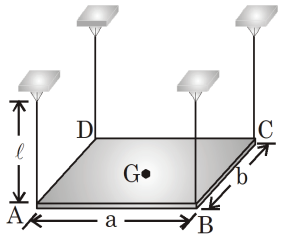
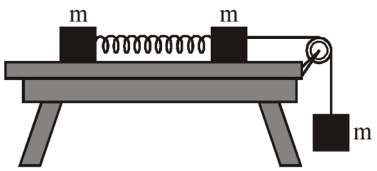
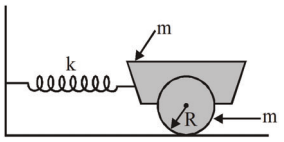
4. A thin rectangular plate of mass M, sides a and b is suspended from four vertical wires of the same length ℓ. Determine the period of small oscillations of the plate when it is given a small horizontal displacement in a direction perpendicular to AB for a = b = ℓ. If your answer is $T = 2\pi\sqrt{\frac{x\ell}{yg}}$ fill value of x + y.



5. At a smooth horizontal table are two identical cubes of mass m, connected by a spring of rigidity k. The length of spring in the unstretched state is ℓ_0. The right cube is linked to the load mass m at the end. At some time the system is released and the system moves without initial velocity. Find the maximum distance (in cm) between blocks during the motion of the system. [ℓ_0 = 1 cm, m = 3 kg, k = 1000 N/m]



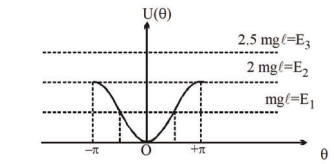
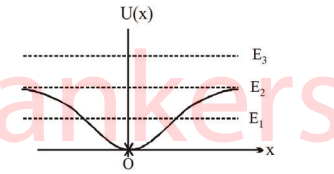
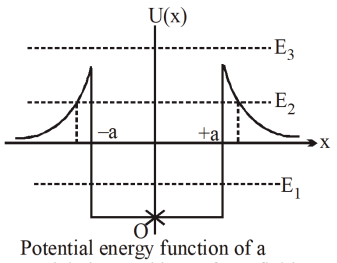
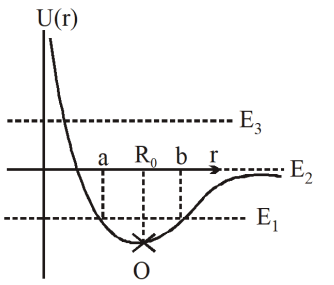
6. A cart consists of a body of mass m and two wheels, each of mass m and radius R. The cart is attached to a spring of constant k. The other end of the spring is fixed to a wall as shown in figure. If time period of oscillation is $\beta \times 2\pi\sqrt{\frac{m}{k}}$ then find the value of β



SECTION-IV : (Maximum Marks: 4)

Matrix Match Type (4 × 5) (One or more options correct) 1 Q. [8 M; (for each entry +2(0)]

1. Column-I represents potential energy graph for certain system. Column-II gives statements related to graphs.

	Column-I		**Column-II**
(A)	 U vs θ graph for a bob hanging vertically from a string with its lowest position as reference level and θ is angle of string from vertical line	(P)	If total energy is E_3, it is not possible for the body to have any turning point in its motion.
(B)	 A particle moving along x axis with potential energy function as $U(x) = [1-e^{-x^2}]$	(Q)	For a small displacement about point O potential energy function is quadratic in variable plotted on x-axis.
(C)	 Potential energy function of a particle in an arbitrary force field	(R)	For a small displacement about position O motion is simple harmonic.
(D)	 Graph represents potential energy for a particle	(S)	If total energy is $E_{total} < E_2$ particle executes periodic and oscillatory motion for all energy values greater than energy at O.
		(T)	Point O is position of stable equilibrium

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ANSWER KEY

GR_Simple Harmonic Motion

PHYSICS

SECTION-I (i)	Q.	1	2	3	4						
	A.	D	A	B	C						
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12	13	14
	A.	A,B	A,B,D	B,C	A,B,D	B,C,D	A,B,C	C,D	A,D	A,B	A,D
	Q.	15	16	17	18	19	20	21	22	23	24
	A.	A,B	A,B,C,D	A,C	A,B,C,D	B,D	B,C	A,B,C,D	A,D	A,C	A,D
SECTION-II	Q.	1	2	3	4	5	6				
	A.	26.00	3	1	2	3	2				
SECTION-IV	Q.	1									
	A.	A->PQRT,B->PQRST,C->P,D->PRST									

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Important Notes

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SECTION-I (i) (Maximum Marks:33)

Single Correct Answer Type 11 Q. [3 M (–1)]

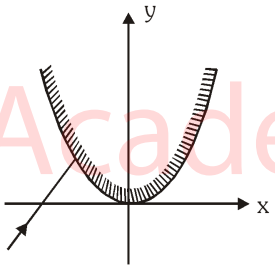
1. The apparent depth of water in cylindrical water tank of diameter 2R cm is reducing at the rate of x cm/minute when water is being drained out at a constant rate. The amount of water drained in c.c. per minute is (n_1 = refractive index of air, n_2 = refractive index of water) :-

(A) $x\,\pi\,R^2\,n_1/n_2$ (B) $x\,\pi\,R^2\,n_2/n_1$ (C) $2\,\pi\,R\,n_1/n_2$ (D) $\pi\,R^2x$

2. A light ray incident along vector $2\hat{i}+4\hat{j}+\sqrt{5}\hat{k}$ strikes on the x-z plane from medium-I of refractive index 2 and enters intomedium-II of refractive index μ_2. The value of μ_2 for which the ray is just totally reflected from the boundary, is

(A) $\frac{5}{4}$ (B) $\frac{6}{5}$ (C) $\frac{3\sqrt{3}}{5}$ (D) 8

3. You are given a parabolic mirror whose inner surface is silvered. The equation of the curve formed by its intersection with x–y plane is given by $y= x^2/4$. A ray travelling in x–y plane along line y= x+3 hits the mirror in second quadrant and gets reflected. The unit vector in the direction of reflected ray is :-



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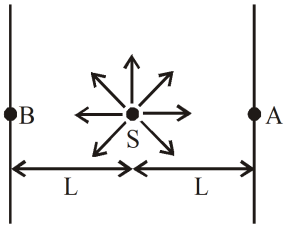
(A) $\frac{1}{\sqrt{2}}\left(-\hat{i}-\hat{j}\right)$ (B) $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)$ (C) $\frac{1}{\sqrt{2}}\left(-\hat{i}+\hat{j}\right)$ (D) None of these

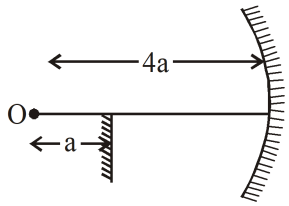
4. The intensity of light from a source decreases with the distance x from the source, as $\frac{1}{x^2}$. In the following picture, the intensity of light from the source S at point A on a curtain is 8.1 U, where U is some unit. We then add a big mirror parallel to the curtain at point B. The distances AS and SB are equal, and the points A, S and B are in line, perpendicular to the curtain and the mirror. What is the intensity in U of light reaching point A now ?



(A) 9 (B) 15 (C) 12 (D) 18

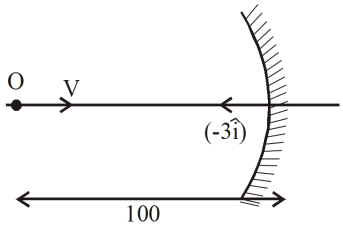
1001CJA00PHEN2500901

5. The diagram shows a mirror system. If after two reflections the image and the object O coincide with each other, then the value of radius of curvature of the concave mirror is : (Take a = 3m)



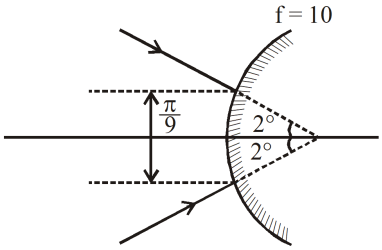
(A) 4 m (B) 6 m (C) 8 m (D) 10 m

6. Radius of curvature of a concave mirror varies as $R = \left|\frac{6}{t^2 - 5t + 6}\right|$ cm. Initially the object was at a distance 100 cm from the pole as shown. At t = 0 the object started moving with velocity $v = 2t\hat{i}$ (cm/s). Find the speed of image at time t = 3 sec. Mirror is moving with constant velocity $\left(-3\hat{i}\right)$.



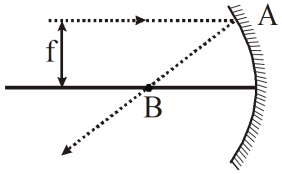
(A) 12 cm/s (B) 20 cm/s (C) 5 cm/s (D) 24 cm/s

7. A converging beam of light having angle of convergence 4° is incident upon a convex mirror as shown. Find the angle of convergence after reflection. Focal length of mirror is 10 cm.



(A) 0.5° (B) 1° (C) 1.5° (D) 2°

8. A ray of light is incident on a concave mirror. It is parallel to the principal axis and its height from principal axis is equal to the focal length of the mirror. The ratio of the distance of point B to the distance of the focus from the centre of curvature is (AB is the reflected ray) :-



(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{2}$

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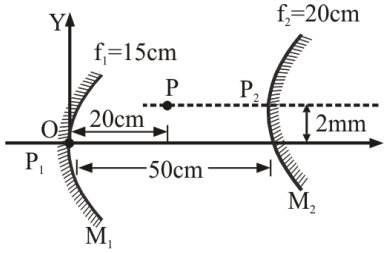
9. A person has D cm wide face and his two eyes are separated by d cm. The minimum width of a mirror required for the person to view his complete face is

(A) $\frac{D+d}{2}$ (B) $\frac{D-d}{4}$ (C) $\frac{D+d}{4}$ (D) $\frac{D-d}{2}$

10. Find the co-ordinates of image of point object P formed after two successive reflection in situation as shown in figure considering first reflection at concave mirror and then at convex.



(A) 60 cm , – 16 mm (B) 30 cm , – 14 mm

(C) 30 cm , – 16 mm (D) 60 cm , – 14 mm

11. The x-y plane is the boundary between two transparent media. Medium-1 with z > 0 has refractive index $\sqrt{2}$ and medium – 2 with z < 0 has a refractive index $\sqrt{3}$. A ray of light in medium–1 given by the vector $A = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$ is incident on the plane of separation. Find the unit vector in the direction of refracted ray in medium-2.

(A) $\frac{1}{\sqrt{3}}\left(\hat{i}-\hat{j}-\hat{k}\right)$ (B) $\frac{1}{\sqrt{3}}\left(\hat{i}+\hat{j}+\hat{k}\right)$ (C) $\frac{1}{5\sqrt{2}}\left(3\hat{i}+4\hat{j}+5\hat{k}\right)$ (D) $\frac{1}{5\sqrt{2}}\left(3\hat{i}+4\hat{j}-5\hat{k}\right)$

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SECTION-I (ii) : (Maximum Marks: 32)

Multiple Correct Answer Type 8 Q. [4 M (–2)]

12. Two mirror's are inclined at an angle of 30° one of the ray is incident parallel to the one of the mirror. Then select the correct statement(s)-

(A) Ray will undergo five reflections from the system

(B) Total deviation from the system will 120°

(C) Ray will retrace it path finally

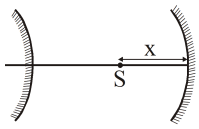
(D) Total deviation suffered by the ray will be 90°

13. A ray of light travelling in the direction $\frac{1}{5}(3\hat{i}+4\hat{j})$ is incident on a plane mirror. After reflection, it travel along the direction $\frac{1}{5}(3\hat{i}-4\hat{j})$. Then choose the correct statement(s).

(A) Mirror is kept in x-y plane (B) Normal of the mirror is along $-\hat{j}$

(C) Angle of incidence is 37° (D) Mirror is kept in x-z plane

14. A convex mirror and a concave mirror are placed on the same optic axis, separated by distance L = 20cm. The radius of curvature of each mirror has a magnitude of 12 cm. A light source is located at a distance x from the concave mirror as shown in figure.



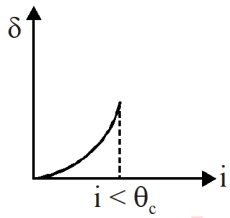
(A) The rays from the source return to the source after reflecting first from the convex mirror and then from the concave mirror for x = 8 cm.

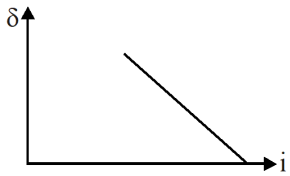
(B) The rays from the source return to the source after reflecting first from the convex mirror and then from the concave mirror for x = 16 cm.

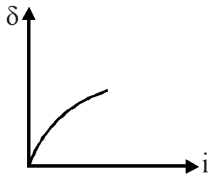
(C) If the rays from the source return to the source after first reflecting from the concave mirror and then from the convex mirror for x = 8 cm.

(D) If the rays from the source return to the source after first reflecting from the concave mirror and then from the convex mirror for x = 16 cm.

15. Following are graphs of angle of deivation versus angle of incidence.

(a) 

(b) 

(c) 

Based on the above graphs mark the correct options.

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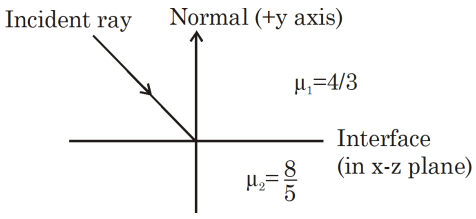
(A) Graph-a may be a part of the graph for ray of light that travels from denser to rarer medium.

(B) Graph-b may be for ray of light that is totally internally reflected from a denser to rarer medium boundary.

(C) Graph-c may be a part of the graph for ray of light that travels from rarer to denser medium.

(D) Graph-b may be a part of the graph for ray of light that is reflected from a plane mirror.

16. A light ray incident in a medium of refractive index $\frac{4}{3}$ is shown in diagram. Interface seperating the two mediums lies in xz plane. Vector along incident ray is $3\hat{i} - 4\hat{j}$.



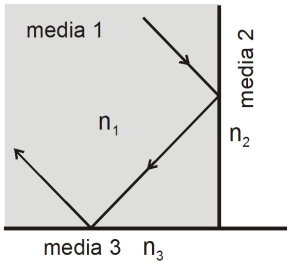
(A) Angle of incidence is 37°.

(B) The angle of refraction is 60º.

(C) The angle of refraction is 30°

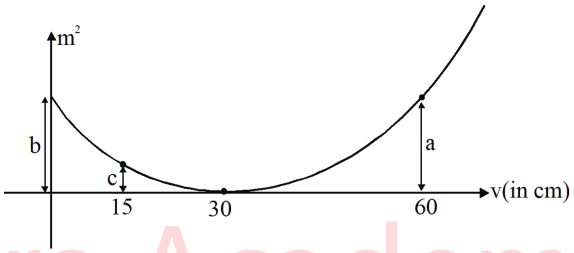
(D) If angle of incidence of the incident ray is greater than the critical angle then total internal reflection takes place.

17. In the diagram shown, light is incident on the interface between media 1 (refractive index n_1) and 2 (refractive index n_2) at angle slightly greater than the critical angle, and is totally reflected. The light is then also totally reflected at the interface between media 1 and 3 (refractive index n_3), after which it travels in a direction opposite to its initial direction. The media must have a refractive indices such that



(A) $n_1 < n_2 < n_3$ (B) $n_1^2 - n_3^2 > n_2^2$ (C) $n_1^2 - n_2^2 < n_3^2$ (D) $n_1^2 + n_2^2 > n_3^2$

18. For a concave mirror, graph of square of magnification and image distance from pole is given for real object. Choose the correct statement(s).



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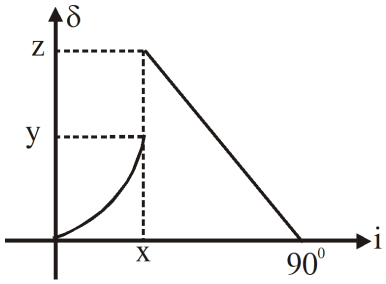
(A) This graph is hyperbola. (B) length a must be equal to length b.

(C) radius of concave mirror is 60 cm (D) the length c will be half of length a.

19. Light is incident from glass to air. The variation of the angle of deviation d with the angle of incident i for $0 < i < 90°$ is shown. The values of x,y and z, in terms of critical angle C, shown in the figure are

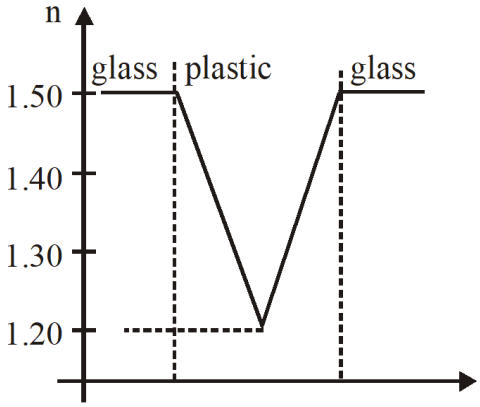


(A) $x = C$ (B) $y = 90° - C$ (C) $z = 180° - 2C$ (D) $z = 180° - C$

SECTION-II : (Maximum Marks: 28)

Numerical-Answer-Type questions (Non Negative Integer) 7 Q. [4 M (0)]

1. A thin plate of transparent plastic is embedded in a thick slab of glass. The index of refraction of the glass is n = 1.50 ; the index of refraction of the plate changes as shown in the diagram. A beam of light passes through glass and strikes the surface of the plastic plate. The maximum angle of incidence θ_{max} enables the beam to pass through the plate. Find θ_{max} (in °).



2. A ray of light is falling on a glass sphere of $\mu = \sqrt{3}$ such that the incident ray and the emergent ray, when produced, intersect at a point on the surface of the sphere. The angle of incidence in degrees is found to be $\frac{2\pi}{k}$. Find the value of k.

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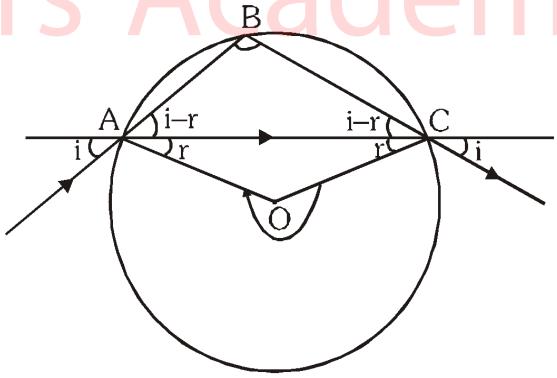
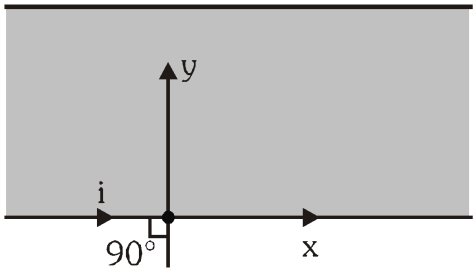
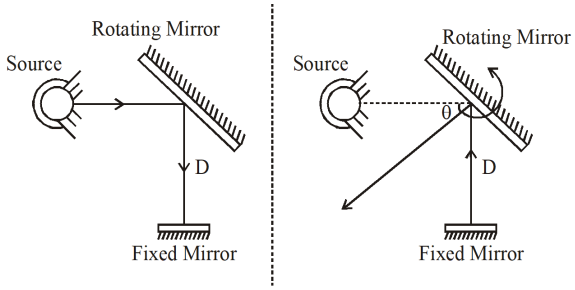


3. In direction of y, medium become denser as $\mu = 1 + y^2$ above x–axis. A light ray enters at an angle almost at 90° at origin. At what value of y, the slope of path of light is $\sqrt{3}$.



4. In the mid-19th century, French physicist Foucault made the most accurate measurement of the speed of light using a laboratory-sized apparatus consisting of rotating and fixed mirrors. Consider a source of light emitting a beam which bounces off a rotating mirror and then reflects off a small fixed mirror. By the time it reaches back to the rotating mirror, the mirror has rotated by a small angle. This rotation will deflect the beam through a small angle 30 minute of an arc from its original path. If the mirror rotates with a frequency f ((5/3) x 10^4 Hz) and the speed of light is given as 3×10^8 m/sec , the value of D (where D is distance between rotating mirror and fixed mirror) is $\frac{5}{4}\alpha$ m then the value of a is



5. A long rectangular slab of transparent medium of thickness d is placed on table with length parallel to x-axis and width parallel to the y-axis. A ray of light travelling along y-axis enters at origin such that angle of incidence θ (θ ⟶ 90°). The refractive index μ of the medium varies as $\mu = \sqrt{1 + e^{x/d}}$. The refractive index of the air is 1. If the x-coordinate of the point A, where the ray intersects the upper surface of the slab-air boundary is (kd) ℓn2, then find the value of k.

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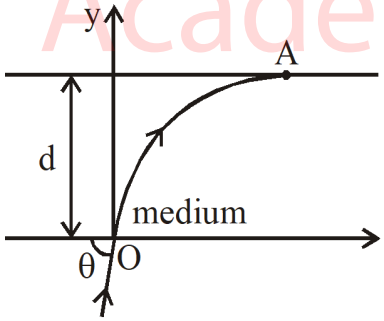
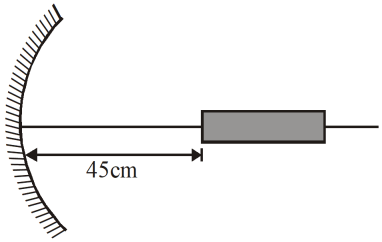
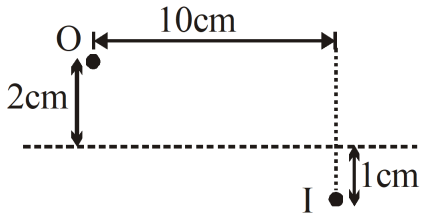


6. An object of length 30 cm is placed on principal axis of a concave mirror of focal length 30 cm. Its one end at a distance of 45 cm as shown. If length of image is 10x (in cm) find the value of x.



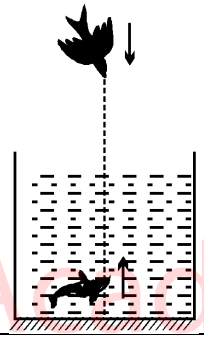
7. The principal axis of a spherical mirror is shown by dotted line. O is the point object whose real image is I. The distance of the pole and centre of curvature of the mirror from object measured along principal axis by drawing ray diagram is x & y cm, then find x/y.



SECTION-IV : (Maximum Marks: 8)

Matrix Match Type (4 × 5) (One or more options correct) 1 Q. [8 M; (for each entry +2(0)]

1. A bird in air is diving vertically over a tank with speed 6 cm/s. Base of the tank is silvered. A fish in the tank is rising upward along the same line with speed 4 cm/s. [Take : $\mu_{water} = 4/3$]



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Column I		Column II	
(A)	Speed of the image of fish as seen by the bird directly	(P)	12
(B)	Speed of the image of fish formed after reflection from the mirror as seen by the bird	(Q)	4
(C)	Speed of image of bird relative to the fish looking upwards	(R)	9
(D)	Speed of image of bird relative to the fish looking downwards in the mirror	(S)	3

PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	A	A	C	A	D	A	D	B
	Q.	11									
	A.	D									
SECTION-I (ii)	Q.	12	13	14	15	16	17	18	19		
	A.	A,C	B,C,D	A,C	A,B,D	A,C	B,D	B,C	A,B,C		
SECTION-II	Q.	1	2	3	4	5	6	7			
	A.	53	6	1	5	2	4	3			
SECTION-IV	Q.	1									
	A.	A->R,B->S,C->P,D->Q									

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Important Notes

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Important Notes

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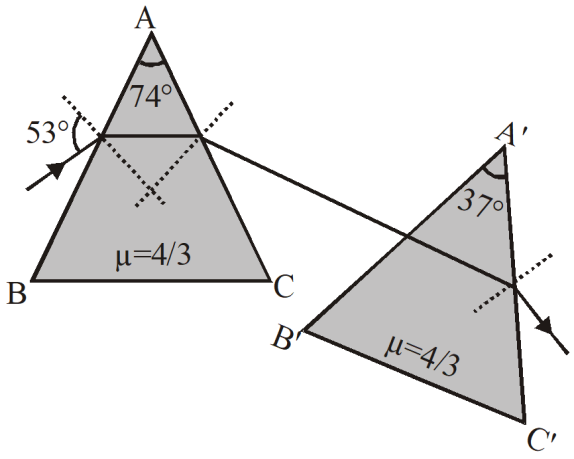
SECTION-I (i) (Maximum Marks:12)

Single Correct Answer Type 4 Q. [3 M (–1)]

1. A light ray is incident on face AB of prism ABC as shown in figure. The second prism is kept in such a manner that emergent ray from prism ABC is falling normally on face A'B' of prism A'B'C'. The net deviation produced by the optical system of the two prisms is :-



(A) 48° (B) 112° (C) 32° (D) 18°

2. Figure shows a thin converging lens for which the focal length is 5 cm. The lens is in front of a concave spherical mirror of radius R = 30 cm. If the lens and mirror are 20 cm apart and an object is placed 15 cm to the left of the lens, determine the approximate distance of the final image from the lens (in cm)

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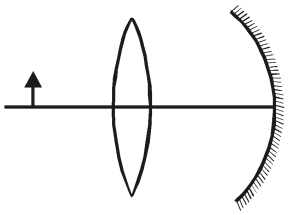
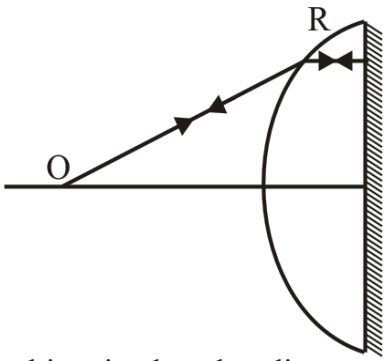


(A) 5.3 cm (B) 4.6 cm (C) 6.1 cm (D) 12.7 cm

1001CJA00PHEN2501001

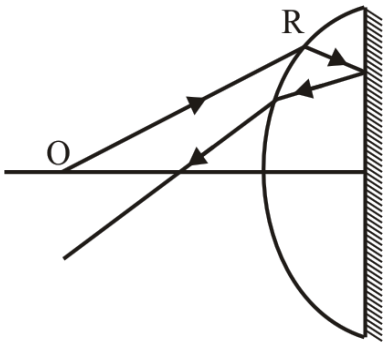
3. A thin plano-convex glass lens ($\mu = 1.5$) has its plane surface reflecting and R is the radius of curvature of curved part, then which of the following ray diagram is true for an object placed at O?

(A)



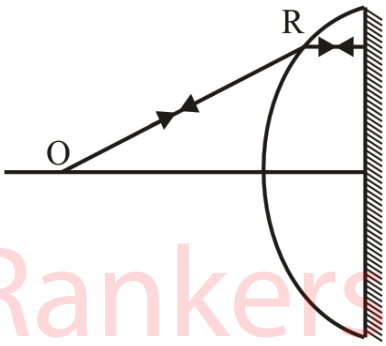
object is placed at distance 2R from lens

(B)



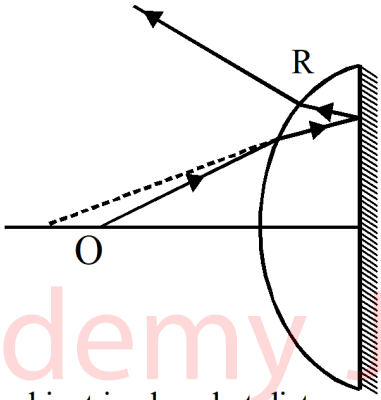
object is placed at distance 2R from lens

(C)



object is placed at distance 3R from lens

(D)



object is placed at distance 2R from lens

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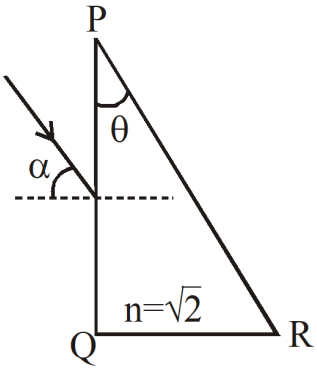
4. A parallel beam of light is incident from air at an angle α on the side PQ of a right angled triangular prism of refractive index $n = \sqrt{2}$. Light undergoes total internal reflection in the prism at the face PR when α has a minimum value of 45°. The angle θ of the prism is :

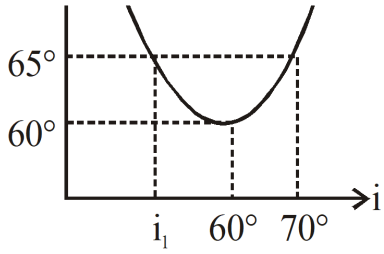


(A) 15° (B) 22.5° (C) 30° (D) 45°

SECTION-I (ii) : (Maximum Marks: 52)

Multiple Correct Answer Type 13 Q. [4 M (–2)]

5. The angle of deviation (δ) vs angle of incidence (i) is plotted for a prism.
Choose the correct statement(s).



(A) The angle of prism is 60°

(B) The refractive index of the prism is n = $\sqrt{3}$

(C) For deviation to be 65° the angle of incidence i_1 = 55°

(D) for minimum deviation, the angle of emergence e = 60°.

6. A concave spherical surface of radius of curvature 10 cm separates two mediums X and Y of refractive indices 4/3 and 3/2 respectively. Centre of curvature of the surface lies in the medium X. An object is placed in medium X. Choose incorrect option(s) :-

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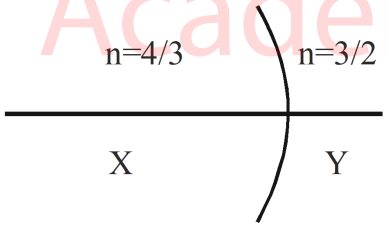


(A) Image is always real

(B) Image is real if the object distance is greater than 90 cm.

(C) Image is always virtual

(D) Image is virtual only if the object distance is less than 90 cm.

7. The principal axis of an optical device is along y = −1. If the image of a small body placed at (−30,3) is formed at a point (60, −3), then the optical device may be :-

(A) A convex lens of focal length 20 cm

(B) A concave mirror of focal length 60 cm

(C) A concave lens of focal length 20 cm

(D) A convex mirror of focal length 60 cm

8. In displacement method, the distance between object and the screen is 96 cm. The ratio of length of two images formed by a converging lens placed between them is 4. Then

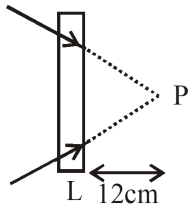
(A) ratio of the length of object to the length of shorter image is 2.

(B) ratio of the length of object to the length of shorter image is 2.

(C) focal length of the lens is 64/3 cm.

(D) when the shorter image is formed on screen, distance of the lens from the screen is 32 cm.

9. A beam of light converges to a point P. A device L is placed in the path of the convergent beam 12 cm from P as shown. Choose the correct option(s) :-



(A) If L is a convex lens of focal length 20 cm, the beam converges at a point at 4.5 cm distance from P

(B) If L is a convex lens of focal length 20 cm, the beam converges at a point at 10 cm distance from lens

(C) If L is a concave lens of focal length 16 cm, the beam converges at a point at 48 cm distance from lens

(D) If L is a plane mirror the beam converges at a point at 24 cm distance from P.

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10. Variation of magnification (m) produced by a thin converging lens versus distance (v) of image from pole of the lens is plotted. Which of the following statements is/are correct?

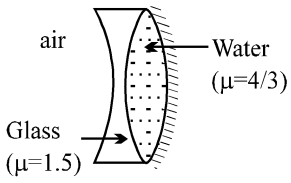
(A) Focal length of the lens is equal to intercept on v-axis.

(B) Focal length of the lens is equal to inverse of the magnitude slope of the line.

(C) Magnitude of intercept on m-axis is equal to unity.

(D) Magnitude of intercept on v-axis is equal to unity.

11. The radius of curvature of the left and right surface of the concave lens are 10cm and 15cm respectively. The radius of curvature of the mirror is 15cm.



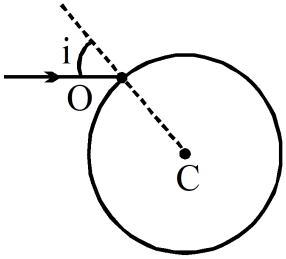
(A) equivalent focal length of the combination is −18cm.

(B) equivalent focal length of the combination is +36cm.

(C) the system behaves like a concave mirror.

(D) the system behaves like a convex mirror.

12. A spherical transparent medium of refractive index n is placed in air. A thin parallel beam of light of monochromatic wavelength l is incident at a point O of the sphere at an angle of incidence i. Then which of the following statements will be true for this optic system?



(A) For the ray to come out of the sphere after refraction through it, the incident angle i should be less than the critical angle C for the medium of the sphere.

(B) The angle of emergence q (made with the normal) for the ray will be equal to the incident angle i for all values of i.

(C) If angle i ≃ 90° (grazing incidence) the emergent ray will come out grazing the surface.

(D) If the angle of incidence i is such that the angle of refraction at O is equal to the critical angle C for the medium of sphere, the emergent ray will be parallel to the incident ray.

13. A prism of refractive index $\sqrt{2}$ and apex angle A is shown. Light is incident from PQ side at angle of incidence i ($0 < i \leq 90°$)

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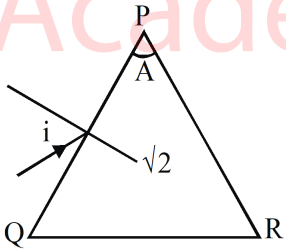


(A) if A = 40° then light incident at all angles will emerge from surface PR.

(B) if A = 80° then light incident at some angles will emerge from surface PR and at some other angles light will suffer TIR at surface PR.

(C) if A = 100° then light incident at all angles will be reflected back from surface PR.

(D) whatever is the value of A, light will emerge from the surface PR for some value of i.

14. Optical axis of a thin equi-convex lens is the X-axis. The co-ordinate of a point object and its image are (−20 cm, 1 cm) and (25 cm, −2 cm) respectively

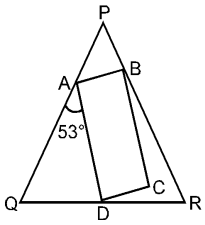
(A) the lens is located at x = 5 cm

(B) the lens is located at x = −5 cm

(C) the focal length of the lens is 10 cm

(D) the focal length of the lens is 15 cm

15. A rectangular cavity ABCD is carved inside an equilateral prism PQR of refractive index $\sqrt{2}$shown in the figure. For a ray entering at face PQ and emerging at face PR, without passing through AB and CD :



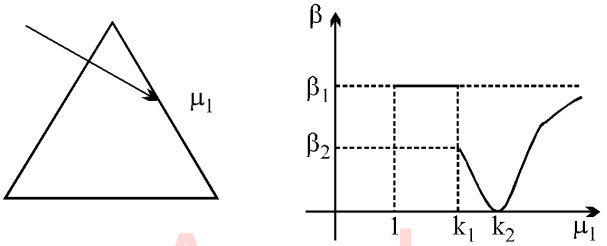
(A) there will not be any effect of cavity on δ

(B) there may be effect of cavity on δ

(C) for cavity filled with water $\delta_{min} = 30^o$

(D) for cavity filled with air $\delta_{min} = 30^o$

16. The diagram shows an equilateral prism. The medium on one side of the prism is μ_1. The refractive index of the prism is $\mu = 4/\sqrt{3}$. The diagram shows the variation of the magnitude of angle of deviation with respect to μ_1? Consider the light ray to be normally incident on the first face. Then :-



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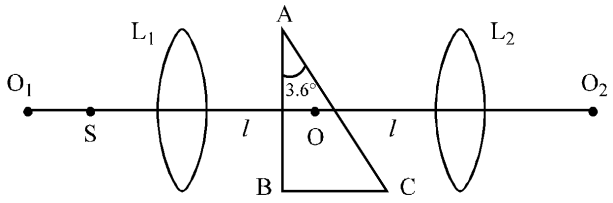
(A) The value of β_1 is 60°.

(B) The value of β_2 is 30°.

(C) The value of k_1 is 2.

(D) The value of k_2 is $\frac{4}{\sqrt{3}}$.

17. A thin right angled glass prism of cross-section ABC with a very small refracting angle $\alpha = 3.6°$ is placed between the convex lenses along a common axis O_1O_2 (as shown in figure)



The lenses are placed at equal distances l = 30 cm from the prism. The refractive index of the prism glass is μ = 1.5. The surface AB does not reflect any light energy while the surface AC is partially reflecting and partially refracting. A point source of light S is placed at a distance of 20 cm to the left of the lens L_1 as shown. Considering the rays that are closed to the axis O_1O_2, The focal lengths of the lenses L_1 and L_2 are 20 cm each. The face AB of the prism is normal to the axis O_1O_2.

(A) The number of the images(s) formed by this system are 2.

(B) The position of one of the image formed by this system from principle axis is $\frac{\pi}{5}$ cm.

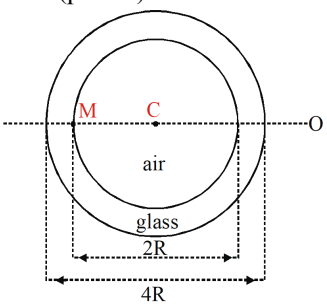
(C) The position of one of the image formed by this system from principle axis is 1.2π cm.

(D) The number of the image formed by this system is 1.

SECTION-II : (Maximum Marks: 16)

Numerical-Answer-Type questions (Non Negative Integer) 4 Q. [4 M (0)]

1. A hollow sphere of glass of R.I. n has a small mark M on its interior surface which is observed by an observer O from a point outside the sphere. C is the centre of the sphere. The inner cavity (air) is concentric with the external surface and thickness of the glass is every where equal to the radius of the inner surface. The distance by which the mark will appear nearer than is $\frac{(n-1)R}{(pn-1)}$. Find p. (assuming paraxial rays.)



2. A point object is placed at a distance of 0.3m from a convex lens (focal length 0.2 m) cut into two halves each of which is displaced by 0.0005 m as shown in the figure. If number of images are x and the distance between them is y mm, then find xy.

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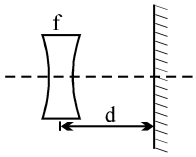
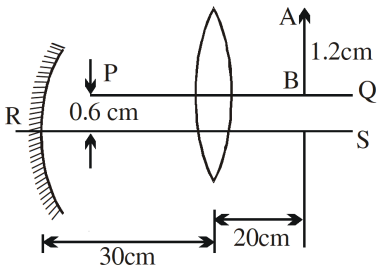
3. A diverging lens of focal length 10 cm is placed 10 cm in front of a plane mirror as shown in the figure. Light from a very far away source falls on the lens. Find the distance (in cm) of the image of source due to plane mirror (before hitting lens again) from mirror?



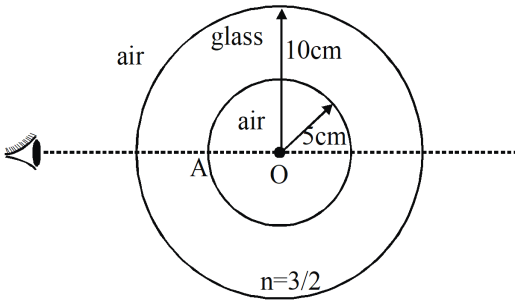
4. A convex lens of focal length 15 cm and a concave mirror of focal length 30 cm are kept with their optic axes PQ and RS parallel but separated in vertical direction by 0.6 cm as shown. The distance between the lens and mirror is 30 cm. An upright object AB of height 1.2 cm is placed on the optic axis PQ of the lens at a distance of 20 cm from the lens . If A' B' is the image after refraction from the lens and reflection from the mirror, find the distance of A'B' (in cm) from the pole of the mirror.



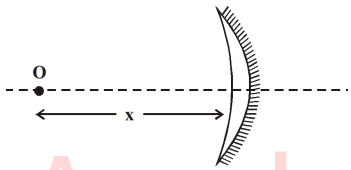
SECTION-III : (Maximum Marks: 16)

Numerical Grid Type (Ranging from 0 to 9) 4 Q. [4 M (0)]

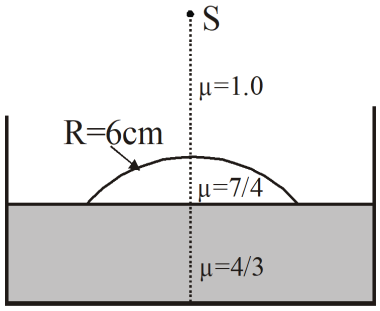
1. A hollow glass sphere of inner radius 5 cm & outer radius 10 cm is viewed as shown in figure. If refractive index of glass is 3/2, find distance (in cm) between A & O as seen by the observer?



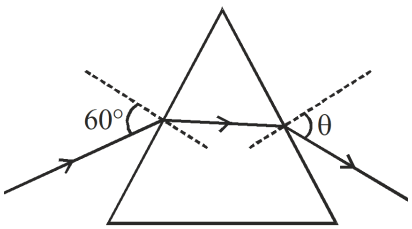
2. Radii of curvature of a concavo-convex lens (refractive index = 1.5) are 40 cm and 20 cm as shown. The convex side is silvered. The distance x (in cm) on the principal axis where an object is placed so that its image is created on the object itself, is given as 4β. Find the value of β.



3. Water (with refractive index = $\frac{4}{3}$) in a tank is 18 cm deep. Oil of refractive index $\frac{7}{4}$ lies on water making a convex surface of radius of curvature 'R = 6 cm' as shown. Consider oil to act as a thin lens. An object 'S' is placed 24 cm above water surface. The location of its image is at 'x' cm above the bottom of the tank. Then 'x' is



4. A monochromatic beam of light is incident at 60° on one face of an equilateral prism of refractive index n and emerges from the opposite face making an angle θ(n) with the normal (see the figure). For n = $\sqrt{3}$ the value of θ is 60° and $\frac{d\theta}{dn}$ = m. The value of m is



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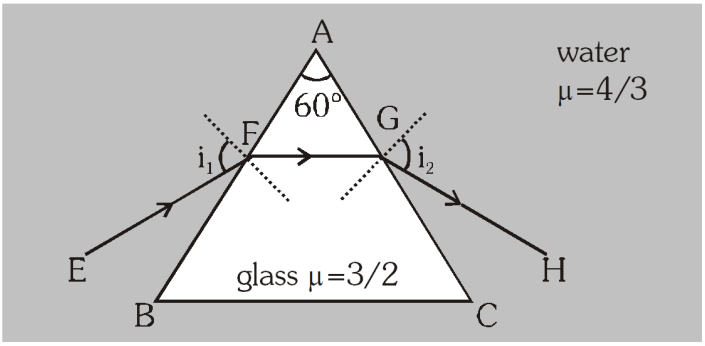
SECTION-IV : (Maximum Marks: 24)

Matrix Match Type (4 × 5) (One or more options correct) 3 Q. [8 M; (for each entry +2(0))]

1. Consider an equilateral prism ABC of glass $\left(\mu = \frac{3}{2}\right)$ placed in water $\left(\mu = \frac{4}{3}\right)$



	Column–I		Column–II
(A)	FG is parallel to BC	(P)	maximum deviation
(B)	$i_1 = 90°$	(Q)	minimum deviation
(C)	$i_1 = i_2 = \sin^{-1}\left(\frac{9}{16}\right)$	(R)	TIR will take place at surface AC
(D)	EF is perpendicular to AB	(S)	No TIR will take place at surface BC
		(T)	Can't say anything

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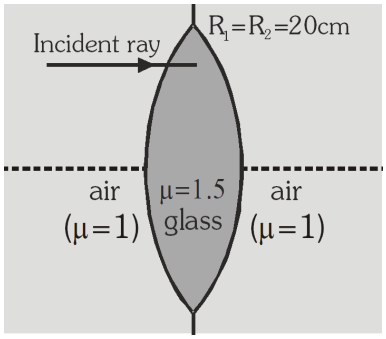
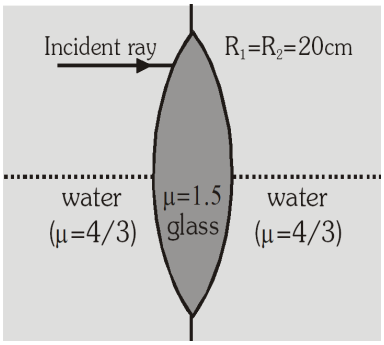
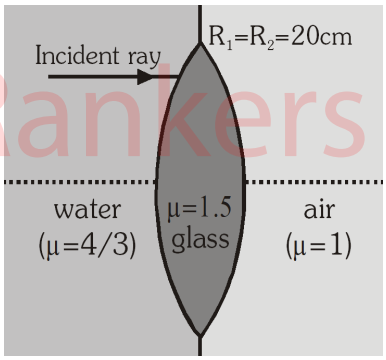
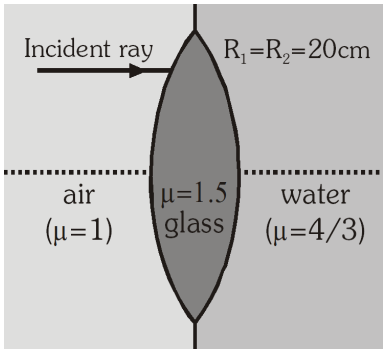
2.

Column I (optical system)		Column II (focal length)	
(A)		(P)	80 cm
(B)		(Q)	40 cm
(C)		(R)	30 cm
(D)		(S)	20 cm

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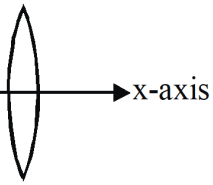
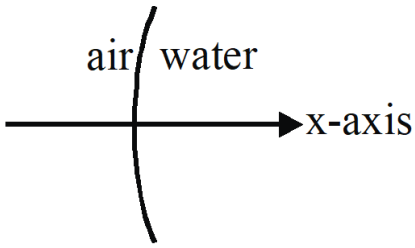
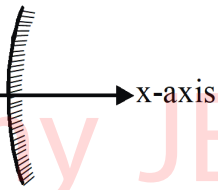
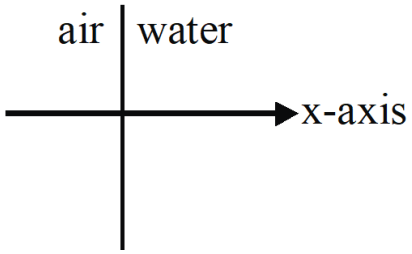
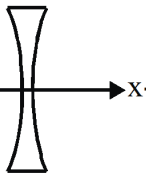
3. In column I certain combination of small object and image are given. Match them with possible refracting / reflecting body in column II. In all cases, object is perpendicular to x-axis. Object can't be at zero distance from the interface.

	Column I		Column II
(A)	Virtual image of virtual object	(P)	Converging lens  x-axis
(B)	Real image of real object	(Q)	Light is incident from air on the spherical surface 
(C)	Erect real image of same size as object	(R)	Convex mirror  x-axis
(D)	Erect and virtual image of larger size	(S)	Light is incident from right to left. 
		(T)	Diverging lens  x-axis

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PHYSICS

SECTION-I (i)	Q.	1	2	3	4							
	A.	A	A	A	A							
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12	13	14	
	A.	A,B,C,D	A,B,D	A,B	A,B,C,D	B,D	A,B,C	A,C	B,C	A,B,C	B,C	
	Q.	15	16	17								
	A.	A,C,D	A,B,C,D	A,B,C								
SECTION-II	Q.	1	2	3	4							
	A.	3	6	20	15							
SECTION-III	Q.	1	2	3	4							
	A.	6	4	2	2							
SECTION-IV	Q.	1	2	3								
	A.	A->QS,B->PS,C->QS,D->S	A->S,B->P,C->R,D->Q	A->PRST,B->PQ,C->S,D->PQ								

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SECTION-I (i) (Maximum Marks:30)

Single Correct Answer Type 10 Q. [3 M (–1)]

1. A ring of radius R is made out of a thin wire of area of cross section A. The ring has a uniform charge Q distributed on it. A charge q_0 is placed at the centre of the ring. If Y is the young's modulus for the material of the ring and ΔR is the change in the radius of the ring then,

(A) $\Delta R = \frac{q_0 Q}{4\pi\varepsilon_0 RAY}$ (B) $\Delta R = \frac{q_0 Q}{4\pi^2\varepsilon_0 RAY}$ (C) $\Delta R = \frac{q_0 Q}{8\pi^2\varepsilon_0^2 RAY}$ (D) $\Delta R = \frac{q_0 Q}{8\pi^2\varepsilon_0 RAY}$

2. The electrostatic potential V at a point on the circumference of a thin non−conducting disk of radius r and uniform charge density σ is given by equation $V = 4\sigma r$. Which of the following expression correctly represents electrostatic energy stored in the electric field of a similar charged disk of radius R?

(A) $U = \frac{8}{3}\pi\sigma^2 R^3$ (B) $U = \frac{4}{3}\pi\sigma^2 R^3$ (C) $U = \frac{2}{3}\pi\sigma^2 R^3$ (D) None of these

3. Two smooth spherical non conducting shells each of radius R having uniformly distributed charge Q and −Q on their surfaces are released on a smooth non-conducting surface when the distance between their centres is 5R. The mass of A is m and that of B is 2m. The speed of A just before A and B collide is :- [Neglect gravitational interaction] $\left(K = \frac{1}{4\pi\varepsilon_0}\right)$

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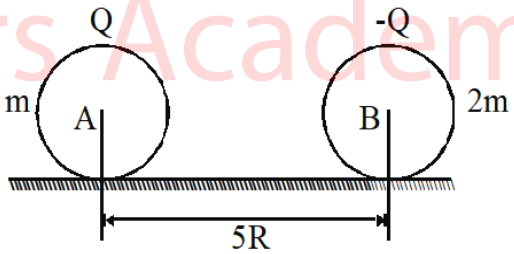
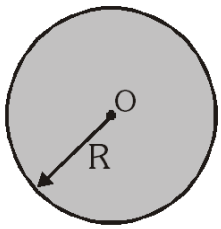


(A) $\sqrt{\frac{2KQ^2}{5mR}}$ (B) $\sqrt{\frac{4KQ^2}{5mR}}$ (C) $\sqrt{\frac{8KQ^2}{5mR}}$ (D) $\sqrt{\frac{16KQ^2}{5mR}}$

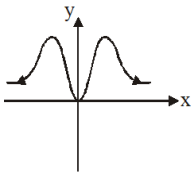
4. Volume charge density of a non−conducting sphere is varying as $\rho = \rho_0 r$ where r0 is a constant and r is the distance from centre of sphere.If potential at r=2R is zero then potential at infinity will be

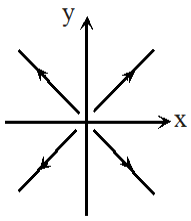


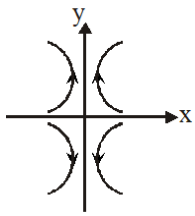
(A) $\frac{\rho_0 R^3}{8\epsilon_0}$ (B) $-\frac{\rho_0 R^3}{8\epsilon_0}$ (C) $-\frac{\rho_0 R^3}{4\epsilon_0}$ (D) Data insufficient

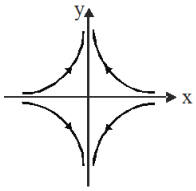
1001CJA00PHEN2501101

5. In a certain region of space, the potential field depends on x and y coordinates as $V = (x^2 - y^2)$. The corresponding electric field lines in x-y plane are correctly represented by :

(A) 

(B) 

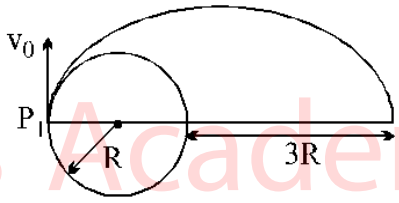
(C) 

(D) 

6. An electric field $\vec{E} = E_0\hat{i} + E_0\hat{j}$ exists in a space. Find the flux through a triangular loop with vertices at $\left(\frac{a}{2}, 0, 0\right)$, $\left(\frac{a}{2}, 0, a\right)$ and $\left(\frac{a}{2}, a, \frac{a}{2}\right)$

(A) $\sqrt{2}\,E_0a^2$ (B) $\frac{E_0a^2}{2}$ (C) $\frac{\sqrt{3}}{2}E_0a^2$ (D) $\frac{\sqrt{3}}{4}E_0a^2$

7. The acceleration due to gravity on the surface of earth is g & radius R. A ball is projected horizontally from point P_1 on the surface of earth (as shown in the figure) If the maximum height attained by the ball above the earth surface is 3R then speed of projection V_0 is (neglect cosmic dust resistance & earth rotation about its axis)



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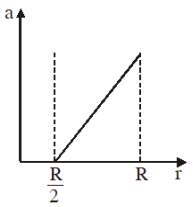
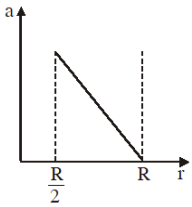
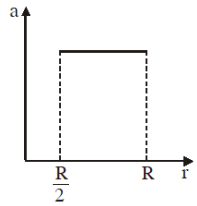
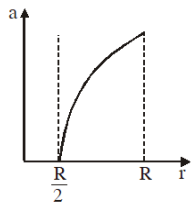
(A) $\sqrt{\frac{8}{5}gR}$ (B) $\sqrt{2gR}$ (C) $\sqrt{gR}$ (D) $\sqrt{\frac{6}{5}gR}$

8. A frictionless tunnel is dug along a chord of the earth at a perpendicular distance $\frac{R}{2}$ from the centre of earth (where R is radius of earth). An object is released from one end of the tunnel. The correct graph, showing the variation of acceleration of particle with its distance r from centre of earth is :

(A) 

(B) 

(C) 

(D) 

9. A satellite of mass m is orbiting a planet of mass M at a radial distance r_0 from the centre of the planet. The satellite expels a small mass in a direction opposite to its orbital velocity. The immediate recoil velocity of the satellite exceeds the initial orbital velocity by Δv. The largest possible value of Δv for which the satellite remains within the gravitational field of the planet is

(A) $\sqrt{\frac{2GM}{r_0}}$ (B) $\sqrt{\frac{GM}{2r_0}}$ (C) $(\sqrt{2}-1)\sqrt{\frac{GM}{r_0}}$ (D) None of these

10. With what minimum speed should m be projected from point P in presence of two fixed spherical masses M each at A and B is shown in figure such that mass m should escape the gravitational attraction of A and B.

(A) $\frac{6}{5}\sqrt{\frac{GM}{a}}$ (B) $\frac{8}{5}\sqrt{\frac{GM}{a}}$ (C) $\sqrt{\frac{2GM}{a}}$ (D) $2\sqrt{\frac{GM}{a}}$

SECTION-I (ii) : (Maximum Marks: 32)

Multiple Correct Answer Type 8 Q. [4 M (–2)]

11. Electric field in a region is given as $\vec{E} = (10 - 5x)\hat{i}$. A charge particle of mass 5kg and charge Q (= 1C) is situated at origin and free to move in given electric field. Then choose the correct options (Neglect gravity) :-

(A) Motion of charge particle is Oscillatory

(B) Maximum displacement of charge particle from origin is 4 SI units

(C) Maximum velocity gain by charge particle is 2 SI units

(D) The position of charge particle, when velocity gained by particle is maximum, is 2 SI units

12. A stationary electric dipole $\vec{\phi} = p\hat{k}$ is situated at origin. A positive charge q, mass m executes circular motion of radius s at constant speed in the field of the dipole. Characterize the plane of orbit is :-

(A) The speed of charge particle is $v = \frac{1}{s}\sqrt{\frac{2qp}{\sqrt{3}\pi\epsilon_0 m}}$

(B) The speed of charge particle is $v = \frac{1}{s}\sqrt{\frac{qp}{3\sqrt{3}\pi\epsilon_0 m}}$

(C) Angular momentum of charge is $L = \sqrt{\frac{2qpm}{\sqrt{3}\pi\epsilon_0}}$

(D) Angular momentum of charge is $L = \sqrt{\frac{qpm}{3\sqrt{3}\pi\epsilon_0}}$

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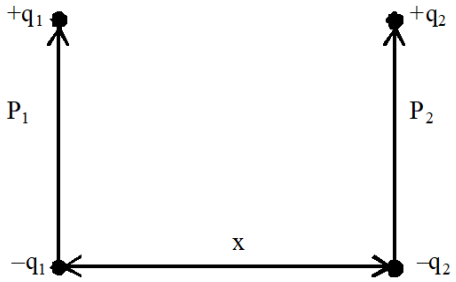
13. Figure shows two dipole moments parallel to each other and placed at a distance x apart x is very large, then :



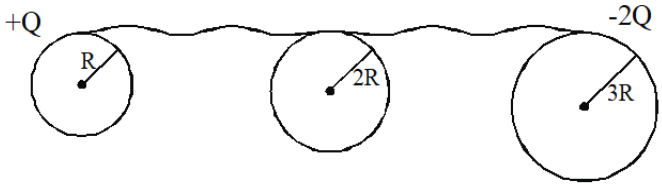
(A) they will repel each other

(B) they will attract each other

(C) force of interaction is of magnitude of $\frac{3P_1P_2}{4\pi\varepsilon_0 x^4}$

(D) force of interaction is of magnitude of $\frac{6P_1P_2}{4\pi\varepsilon_0 x^4}$

14. Two conducting spheres of radii R and 3R carry charges Q and $-2Q$. Between these spheres a neutral conducting sphere of radius 2R is connected. The separation between the sphere is considerably large then :-



(A) The final charge on initially neutral conducting sphere is $-\frac{Q}{3}$

(B) The decrease in electric potential energy of sphere of radius R is $\frac{35KQ^2}{72R}$

(C) The decrease in electric potential energy of sphere of radius R is $\frac{37KQ^2}{72R}$

(D) The final electric potential of sphere of radius 3R will be $-\frac{KQ}{6R}$

15. Two point objects of masses m and 4m are at rest at an infinite separation. They move towards each other under mutual gravitational attraction. If G is the universal gravitational constant, then at a separation r

(A) the total mechanical energy of the two objects is zero

(B) their relative velocity is $\sqrt{\frac{10Gm}{r}}$

(C) the total kinetic energy of the objects is $\frac{4Gm^2}{r}$

(D) their relative velocity is zero

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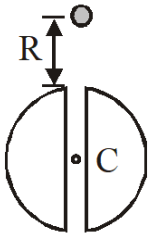
16. A particle is dropped from a height equal to the radius of the earth above the tunnel dug through the earth as shown in the figure.

R : Radius of earth.

M : Mass of earth



(A) Particle will oscillate through the earth to a height R on both sides

(B) Particle will execute simple harmonic motion

(C) Motion of the particle is periodic

(D) Particle passes the centre of earth with a speed = $\sqrt{\frac{2GM}{R}}$

17. Two spherical bodies of masses M and 2M and radii R and 2R, respectively, start approaching each other at time t = 0 from rest, due to mutual gravitational attraction. They were initially very far away from each other. They collide at time t = T.

(A) Distance travelled by their COM till time t = T is zero (COM–centre of mass)

(B) Their relative velocity at time t = T is $\frac{1}{3}\sqrt{\frac{2GM}{R}}$

(C) PE of the system decreases as the bodies approach each other

(D) In COM frame, speeds of the bodies are always equal

18. A particle at a distance r from the centre of a uniform spherical planet of mass M radius R (< r) has a velocity of magnitude v.

(A) for $0 < v < \sqrt{\frac{GM}{r}}$ trajectory may be ellipse

(B) for $v = \sqrt{\frac{GM}{r}}$ trajectory may be ellipse

(C) for $\sqrt{\frac{GM}{r}} < v < \sqrt{\frac{2GM}{r}}$ trajectory may be ellipse.

(D) for $v = \sqrt{\frac{GM}{r}}$ trajectory may be circle

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SECTION-I (iii) : (Maximum Marks: 3)

Matching list Type (Single Option Correct) 1 Q. [3 M (–1)]

19. A planet of mass M, has two natural satellites with masses m_1 and m_2. The radii of their circular orbits are R_1 and R_2 respectively. Ignore the gravitational force between the satellites. Define v_1, L_1, K_1 and T_1 to be, respectively, the orbital speed, angular momentum, kinetic energy and time period of revolution of satellite 1 ; and v_2, L_2, K_2 and T_2 to be the corresponding quantities of satellite 2. Given $m_1/m_2 = 2$ and $R_1/R_2 = 1/4$, match the ratios in List-I to the numbers in List-II.

	List-I		List-II
(P)	$\frac{v_1}{v_2}$	(1)	$\frac{1}{8}$
(Q)	$\frac{L_1}{L_2}$	(2)	1
(R)	$\frac{K_1}{K_2}$	(3)	2
(S)	$\frac{T_1}{T_2}$	(4)	8

(A) P ⟶ 4 ; Q ⟶ 2 ; R ⟶ 1 ; S ⟶ 3

(B) P ⟶ 3 ; Q ⟶ 2 ; R ⟶ 4 ; S ⟶ 1

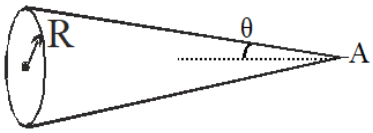
(C) P ⟶ 2 ; Q ⟶ 3 ; R ⟶ 1 ; S ⟶ 4

(D) P ⟶ 2 ; Q ⟶ 3 ; R ⟶ 4 ; S ⟶1

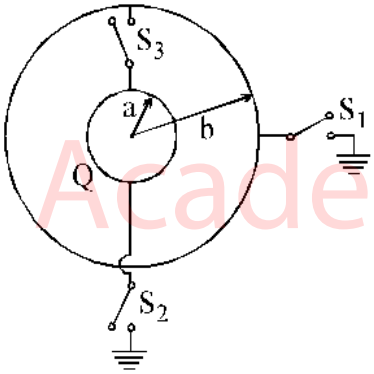
SECTION-II : (Maximum Marks: 8)

Numerical-Answer-Type questions (Non Negative Integer) 2 Q. [4 M (0)]

1. A cone made of insulating material has a total charge Q = 3mC spread uniformly over its sloping surface. Calculate the energy (in Joule) required to take a charge q = 5μC from infinity to apex A of cone. (Given : half cone angle θ = 37°, R = 2m)



2. The figure shows a conducting sphere 'A' of radius 'a' which is surrounded by a neutral conducting spherical shell B of radius 'b' (>a). Initially switches S_1, S_2 and S_3 are open and sphere 'A' carries a charge Q. First the switch 'S_1' is closed to connect the shell B with the ground and then opened. Now the switch 'S_2' is closed so that the sphere 'A' is grounded and then S_2 is opened. Finally, the switch 'S_3' is closed to connect the spheres together. Find the heat (in joule) which is produced after closing the switch S_3. [Consider b = 4 cm, a = 2 cm and Q = 80 μC]



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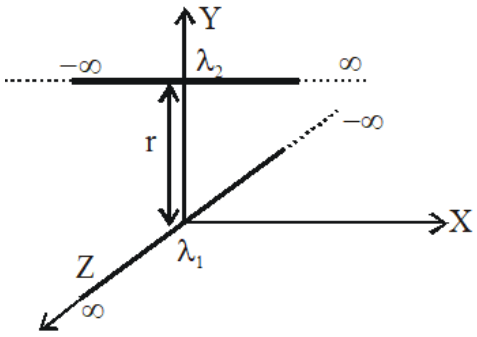
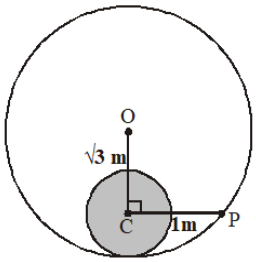
SECTION-III : (Maximum Marks: 36)

Numerical Grid Type (Ranging from 0 to 9) 9 Q. [4 M (0)]

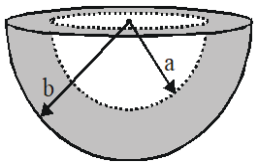
1. AB and CD are uniform line charges of infinite length having charge density λ_1 and λ_2 and lying parallel to the Z−axis and X−axis respectively. The force between them depends on the perpendicular distance between them given by 'r' as shown in figure. The value of $F = \frac{\lambda_1\lambda_2}{2\varepsilon_0 r^n}$, then write the value of n in OMR sheet.



2. A uniformly charged sphere is placed inside a charged hollow sphere as shown in figure. O is the centre of hollow sphere and C is the centre of solid sphere. The magnitude of charge on both the spheres is $4\left(\sqrt{\frac{3}{7}}\right)$ μC. The electric field at point 'P' which lie just outside the hollow sphere is given by 3α kN/C. Find the value of α.



3. A uniformly charged hemisphere of radius b and charge density ρ has a hemispherical cavity of radius a $\left(a = \frac{b}{2}\right)$ cut from its centre. If the potential at the centre of the cavity is $\frac{n\rho b^2}{16\epsilon_0}$ then n = ?



4. Three small dipoles of dipole moment P each are placed at the three vertices of an equilateral triangle of side a as shown. If total potential at the centroid G is given by $V = \frac{n \times 3\sqrt{3}kp}{2a^2}$. Find n $\left(k = \frac{1}{4\pi\epsilon_0}\right)$.

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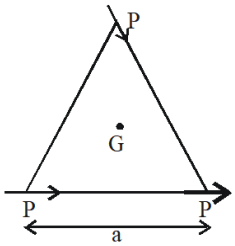

5. A small point charge (+q = 1μC, m = 3 kg) is started from a very long distance towards a fixed point charge (+Q = 1 mC). The initial line of motion of q is at a ^ distance of d = 1m from the line passing from the centre of Q, (as shown in figure). Take V_0 = 2 m/s. Determine the least distance between the two point charges (in m)

6. The electric potential in a region is given by V(x, y, z) = ax2 + ay2 + abz2. 'a' is a positive constant of appropriate dimensions and b, a positive constant such that V is volts when x, y, z are in m. Let b = 2. The work done by the electric field when a point charge +4μC moves from the point (0, 0, 0.1m) to the origin is 50 μJ. The radius of the circle of the equipotential curve corresponding to V = 6250 volts and z = $\sqrt{2}$ m is α m. Fill α^2 in OMR sheet.

7. The electric field strength depends only on the x, y and z coordinates according to the law $E = \frac{a(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}}$, where a = 122.5 SI unit and is a constant. Find the potential difference (in volt) between (3, 2, 6) and (0, 3, 4).

8. A particle of mass m was transferred from the centre of the base of a uniform hemisphere of mass M and radius R into infinity. What work was performed in the process by the gravitational force exerted on the particle by the hemisphere is $\frac{-n}{2}\frac{GMm}{R}$, Then find the value of n :-

9. A binary star consists of two stars A (mass 2.2 MS) and B (mass 11 MS), where MS is the mass of the sun. They are separated by distance d and are rotating about their centre of mass, which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of mass is:-

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ANSWER KEY

GR_Electrostatics & Gravitation-1

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	A	B	D	B	A	D	C	D
SECTION-I (ii)	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,C,D	B,D	A,C	A,B,D	A,B,C	A,C,D	A,C	A,B,C,D		
SECTION-I (iii)	Q.	19									
	A.	B									
SECTION-II	Q.	1	2								
	A.	81	180								
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	
	A.	0	9	3	1	2	6	7	3	6	

SECTION-I (i) (Maximum Marks:18)

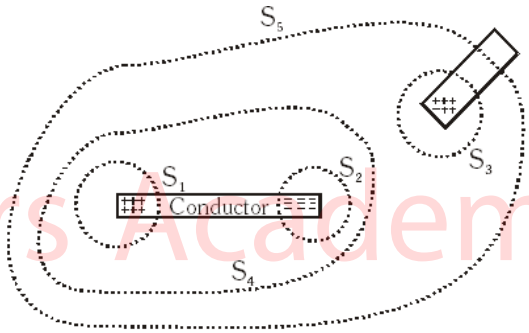
Single Correct Answer Type 6 Q. [3 M (–1)]

1. A positive point charge Q is placed at a point P, which is at a distance L from the centre 'C' of an uncharged metallic ball of radius R (L > R). There is a point G on the line PC, at a distance R/2 from C. Then, the electric potential at G is :-

(A) $\frac{Q}{4\pi\varepsilon_0 L}$ (B) $\frac{Q}{4\pi\varepsilon_0}\times\left(\frac{1}{L}-\frac{1}{R}\right)$

(C) $\frac{Q}{4\pi\varepsilon_0}\times\left(\frac{1}{L}-\frac{2}{R}\right)$ (D) None

2. A rod containing charge +Q is brought near an initially uncharged isolated conducting rod as shown. Regions with total surface charge +Q and −Q are induced in the conductor as shown in the figure. The only regions where the net charge in this configuration is non−zero are indicated by the "+" and "−" signs. Let us denote the total flux of electric field outward through closed surface S_1 as ϕ_1, through S_2 as ϕ_2, etc. Which of the following is necessarily false.

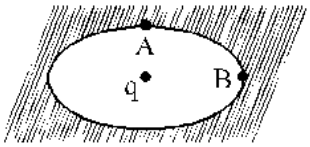


(A) $\phi_1 > 0$ (B) $\phi_2 = \phi_1$ (C) $\phi_3 = \phi_1$ (D) $\phi_4 = 0$

3. A sphere of radius R carries charge such that its volume charge density is proportional to the square of the distance from the centre. What is the ratio of the magnitude of the electric field at a distance 2R from the centre to the magnitude of the electric field at a distance of R/2 from the centre (i.e. $E_{r=2R}/E_{r=R/2}$)?

(A) 1 (B) 2 (C) 4 (D) 8

4. An ellipsoidal cavity is carved within a perfect conductor. A positive charge q is placed at the center of the cavity. The points A & B are on the cavity surface as shown in the figure. Then :



(A) electric field near A in the cavity = electric field near B in the cavity

(B) charge density at A = charge density at B

(C) potential at A = potential at B

(D) total electric field flux through the surface of the cavity is q/ε_0.

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5. A small satellite of mass 'm' is revolving around earth in a circular orbit of radius r_0 with speed v_0. At certain point of its orbit, the direction of motion of satellite is suddenly changed by angle $\theta = \cos^{-1}(3/5)$ by turning its velocity vector in the same plane of motion, such that speed remains constant. The satellite, consequently goes to elliptical orbit around earth. The ratio of speed at perigee to speed at apogee is :-

(A) 3 (B) 9 (C) 1/3 (D) 1/9

6. Given a thin homogeneous disc of radius a and mass m_1. A particle of mass m_2 is placed at a distance ℓ from the disc on its axis of symmetry. Initially both are motionless in free space but they ultimately collide because of gravitational attraction. Assuming $a << \ell$, find the relative velocity at the time of collision.

(A) $\left[2G(m_1+m_2)\left(\frac{2}{a}+\frac{1}{\ell}\right)\right]^{1/2}$ (B) $\left[2G(m_1+m_2)\left(\frac{2}{a}-\frac{1}{\ell}\right)\right]^{1/2}$

(C) $\left[G(m_1+m_2)\left(\frac{2}{a}-\frac{1}{\ell}\right)\right]^{1/2}$ (D) None of these

SECTION-I (ii) : (Maximum Marks: 56)

Multiple Correct Answer Type 14 Q. [4 M (–2)]

7. An infinite plane in the xz plane carries a uniform surface charge density $\sigma_1 = 8.85\ nC/m^2$. A second infinite plane carrying a uniform charge density $\sigma_2 = 17.7\ nC/m^2$ intersects the xz plane at the z-axis and makes an angle of 60° with the xz plane as shown in figure. The electric field in the xy plane.

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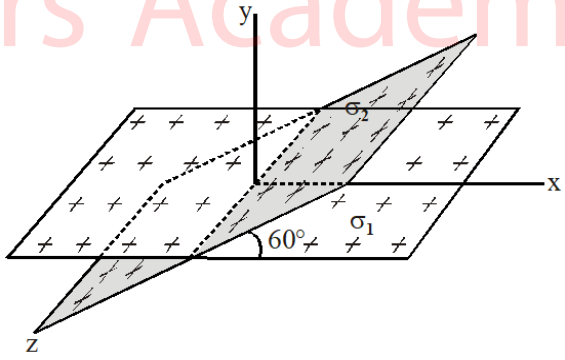


(A) at x = 6m, y = 2m is $500\sqrt{3}$ N/C (B) at x = −5m, y = 0 is $500\sqrt{3}$ N/C

(C) at x = 2m, y = 6 m is $500\sqrt{7}$ N/C (D) at x = −1m, y = −1m is $500\sqrt{7}$ N/C

8. $\vec{E} = x\hat{i} + y\hat{j} + z\hat{k}$ exists in space then choose incorrect statement(s) :

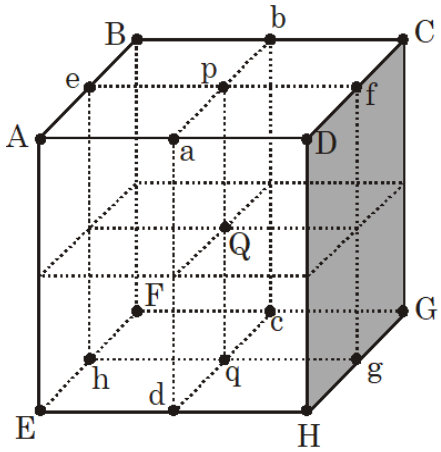
(A) This electric field is not possible

(B) Volume charge density is non uniform.

(C) If a charge particle goes from (3, 4, 5) to (5, 3, 4) then work done by electric force on charge will be positive

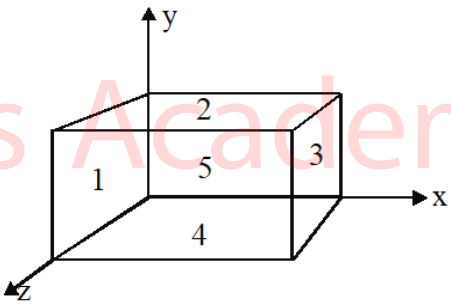
(D) Charge enclosed in sphere of unit radius and centered at origin is $4\pi\in_0$.

9. The following figure shows a charge Q kept at the centre of a cube. Let ϕ represent the flux of field due to the charge Q, the correct options are :-



(A) $\phi_{ABCD} = \frac{Q}{6\varepsilon_0}$ (B) $\phi_{Cbpf} = \frac{Q}{24\varepsilon_0}$ (C) $\phi_{abcd} = 0$ (D) $\phi_{abcd} = \frac{Q}{2\varepsilon_0}$

10. In a region of space, the electric field $\vec{E} = E_0x\hat{i} + E_0y\hat{j}$. Consider an imaginary cubical volume of edge 'a' with its edges parallel to the axes of coordinates. Now,



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(A) the total electric flux through the faces 1 and 3 is E_0a^3

(B) the charge inside the cubical volume is $2\varepsilon_0E_0a^3$

(C) the total electric flux through the faces 2 and 4 is $2E_0a^3$

(D) the charge inside the cubical volume is $\varepsilon_0E_0a^3$

11. A long thin straight wire with linear charge density λ runs along axis of a thin hollow metal cylinder of radius R. The cylinder has a net linear charge density 2λ. Assume λ is positive. Mark correct options:-

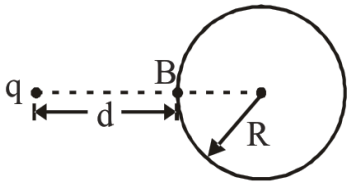
(A) $\vec{E}(r > R) = \frac{3\lambda}{2\pi\epsilon_0}\frac{\hat{r}}{r}$

(B) $\vec{E}(r < R) = \frac{3\lambda}{2\pi\epsilon_0}\frac{\hat{r}}{r}$

(C) Linear charge density on inner surface of cylinder is $-\lambda$

(D) Linear charge density on outer surface of cylinder is 3λ

12. For the situation shown in the figure below, mark out the correct statement(s).



Hollow neutral conductor

(A) Potential of the conductor is $\frac{q}{4\pi\varepsilon_0(d+R)}$.

(B) Potential of the conductor is $\frac{q}{4\pi\varepsilon_0 d}$.

(C) Potential of the conductor can't be determined as nature of distribution of induced charges is not known.

(D) Potential at point B due to induced charges is $\frac{-qR}{4\pi\varepsilon_0(d+R)d}$.

13. A spherical soap bubble of radius R has uniformly distributed charge over its surface with surface charge density σ then [T = surface tension of the soap solution]

(A) excess pressure inside the bubble is $\frac{4T}{R}-\frac{\sigma^2}{2\varepsilon_0}$ (B) excess pressure inside the bubble is $\frac{4T}{R}+\frac{\sigma^2}{2\varepsilon_0}$

(C) excess pressure inside the bubble is $\frac{4T}{R}$ (D) electrostatic pressure is $\frac{\sigma^2}{2\varepsilon_0}$

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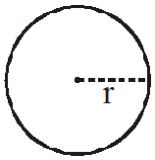
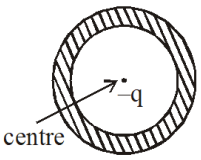
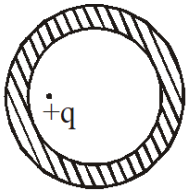
14. In which of the cases we will get uniform charge distribution of (+q) on external spherical surface. Given every object is a conductor :

(A) 

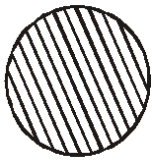
thin shell having net charge +q

(B) 

thick shell having net charge zero

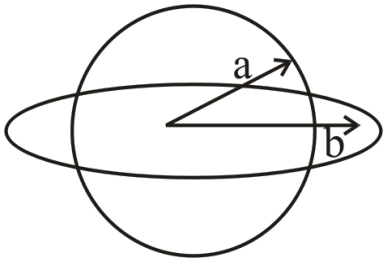
(C) 

thick shell having net charge zero

(D) 

solid sphere having net charge +q

15. An insulating ring of charge $2\pi\lambda b$, radius b is concentric with a charged conducting solid sphere of charge Q and radius a (b > a) as shown in the figure. Then :-



(A) Potential of the sphere is $\frac{Q}{4\pi\epsilon_0 a}+\frac{\lambda}{2\epsilon_0}$

(B) If sphere is grounded then charge on the sphere will be $\frac{-a\lambda}{8\pi^2\epsilon_0}$

(C) Electric field inside the sphere is non-zero

(D) The sphere is equipotential body.

16. Two large thin conducting plates with small gap in between are placed in a uniform electric field E (perpendicular to the plates). Area of each plate is A and charges +Q and −Q are given to these plates as shown in the figure. If points R,S and T as shown in the figure are three points in space, then the

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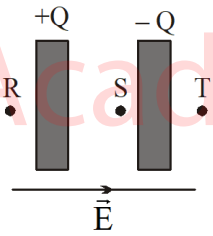

(A) field at point R is E

(B) field at point S is E

(C) field at point T is $\left(E+\frac{Q}{\varepsilon_0 A}\right)$

(D) field at point S is $\left(E+\frac{Q}{A\varepsilon_0}\right)$

17. Two tunnels are dug from one side of the earth's surface to the other side, first along a diameter and the second along a chord. Now two particles are dropped from one end of each of the tunnels. Both the particles oscillate simple harmonically along the tunnels. Let T_1 and T_2 be the time period v_1 and v_2 be the maximum speed of the particle in the two tunnels. Then

(A) $T_1 = T_2$ (B) $T_1 > T_2$ (C) $v_1 = v_2$ (D) $v_1 > v_2$

18. Satellite A is in a circular orbit of radius r and satellite B in another circular orbit of radius 4r, both revolving around the earth. The masses of the satellites A and B are in the ratio 3 : 1. If the symbols V, E, u and T represent the speed, total energy, escape velocity and period of revolution of satellite, then

(A) $V_A/V_B = 2$ (B) $E_A/E_B = 12$ (C) $u_A/u_B = 2$ (D) $T_A/T_B = \frac{1}{8}$

19. A particle of mass m is thrown in a parabolic path towards earth having mass M and radius R. It comes to a greatest approach distance R/2 from surface of earth. Magnitude of total energy change required to make the particle moving in an elliptical path having eccentricity 1/2, with the greatest approach point either aphelion or perihelion are :

(A) $\frac{1}{3}\frac{GMm}{R}$ (B) $\frac{1}{6}\frac{GMm}{R}$ (C) $\frac{1}{2}\frac{GMm}{R}$ (D) $\frac{GMm}{R}$

20. A small body of mass m is projected with a velocity just sufficient to make it reach from the surface of a planet (of radius 2R and mass 3M) to the surface of another planet (of radius R and mass M). The distance between the centers of the two spherical planets is 6R. The distance of the body from the center of bigger planet is 'x' at any moment. During the journey, find the distance x where the speed of the body is maximum & minimum. Assume motion of body along the line joining centres of planets.

(A) 2R (B) $3\sqrt{3}\left(\sqrt{3}-1\right)R$ (C) $3\sqrt{3}\left(\sqrt{3}+1\right)R$ (D) 4R

SECTION-II : (Maximum Marks: 12)

Linked Comprehension Type (Upto Second Decimal Place) 1 Para × 3 Q. [4 M (0)]

Paragraph for Questions 21 - 23

A pair of stars rotates about a common center of mass. One of the stars has a mass M which is twice as large as the mass m of the other. Their centres are at a distance d apart, d being large compared to the size of either star.

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1. The period of rotation of the stars about their common centre of mass in terms of d, m, G is $T = \frac{2\pi d^{x/2}}{\sqrt{yGm}}$. Then x + y is :-

2. Compare the angular momentum of the two stars about their common centre of mass by calculating the ratio L_m/L_M is :-

3. Compare the kinetic energies of the two stars by calculating the ratio K_m/K_M is :-

SECTION-III : (Maximum Marks: 20)

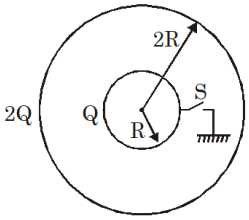
Numerical Grid Type (Ranging from 0 to 9) 5 Q. [4 M (0)]

1. Electric dipole of moment P = Pî is kept at a point (x,y) in an electric field, $E = 4xy^2\hat{i} + 4x^2y\hat{j}$. Find the forces on the dipole is $npy\left(y^2+4x^2\right)^{\frac{1}{2}}$. Find the value of n.

2. Four point charges $+8\mu C$, $-1\mu C$, $-1\mu C$ and $+8\mu C$ are fixed at the points, $-\sqrt{\frac{27}{2}}m$, $-\sqrt{\frac{3}{2}}m$, $+\sqrt{\frac{3}{2}}m$ and $+\sqrt{\frac{27}{2}}m$ respectively on the y-axis. A particle of mass 6×10^{-4} kg and of charge $+0.1\mu C$ moves along the $-x$ direction. Its speed at $x = +\infty$ is v_0. Find the least value of v_0 (in m/s) for which the particle will cross the origin. Assume that space is gravity free. (Given : $1/(4\pi\varepsilon_0) = 9 \times 10^9$ Nm^2/C^2)

3. A non-uniform electric field in x-direction is increasing uniformly from 2N/C at x = 1m to 8 N/C at x = 7m. The center of cube is at x = 2m. If the charge enclosed inside a small cube of side length 10 cm is 8.85×10^{-5n} C. The value of n will be.

4. Two conducting spherical shell of radii R & 2R given charges Q and 2Q respectively. Inner shell is provided with a switch which can ground the inner shell, as shown. Switch is initially open and energy stored in the system is U_1. After the switch is closed, energy stored in the system is found to be U_2. Find $\frac{U_1}{U_2}$.



5. A cube is made from six thin non-conducting square faces of size a each. Each face carries a uniformly distributed charge Q. If electrostatic force acting on each face is $\frac{Q^2}{N \in_0 a^2}$, then find N.

SECTION-IV : (Maximum Marks: 16)

Matrix Match Type (4 × 5) (One or more options correct) 2 Q. [8 M; (for each entry +2(0)]

1. Four concentric hollow spheres of radii R, 2R, 3R and 4R have given the charges as shown. Now, the conductors 1 and 3, 2 and 4 are connected by conducting wires at the same time, then match the entries of column I with entries of column II.

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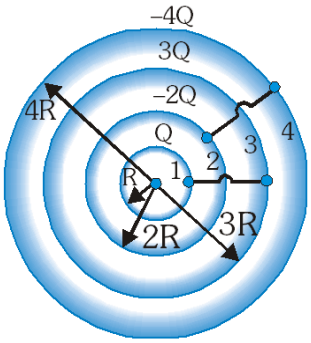
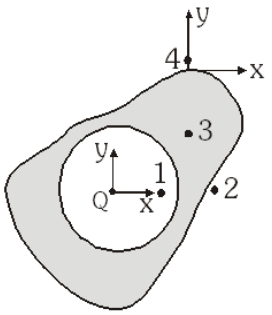


	Column-I		Column-II
(A)	The charge an inner surface of 3rd conductor	(P)	$-\frac{22Q}{5}$
(B)	The charge on 4th conductor	(Q)	$-\frac{Q}{8\pi \in_0 R}$
(C)	The potential of conductor 1	(R)	$\frac{6Q}{5}$
(D)	The potential of conductor 2	(S)	$\frac{-3Q}{40\pi \in_0 R}$
		(T)	None of these

2. Figure shows an uncharged conducting body having a spherical cavity. Charge Q is placed at the centre of the cavity.



	Column-I (Point)		Column-II (Components of electric field)
(A)	1 (inside cavity near surface)	(P)	$E_x = 0$
(B)	2 (near but outside conductor surface)	(Q)	$E_y = 0$
(C)	3 (inside bulk of conductor)	(R)	$E_x \neq 0$
(D)	4 (near but outside conductor surface)	(S)	$E_y \neq 0$
		(T)	None of these

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ANSWER KEY

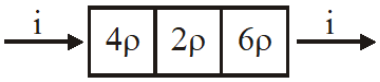
GR_Electrostatics & Gravitation-2

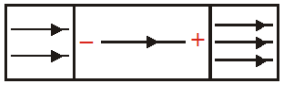
SECTION-I (i)	Q.	1	2	3	4	5	6				
	A.	A	B	B	C	B	B				
SECTION-I (ii)	Q.	7	8	9	10	11	12	13	14	15	16
	A.	A,C	A,B,C	A,B,C	A,B	A,C,D	A,D	A,D	A,C	A,D	A,D
	Q.	17	18	19	20						
	A.	A,D	A,B,C,D	B,C	A,B						
SECTION-II	Q.	1	2	3							
	A.	6.00	2.00	2.00							
SECTION-III	Q.	1	2	3	4	5					
	A.	4	3	3	5	2					
SECTION-IV	Q.	1	2								
	A.	A->R,B->P,C->S,D->Q	A->QR,B->RS,C->PQ,D->PS								

SECTION-I (i) (Maximum Marks:6)

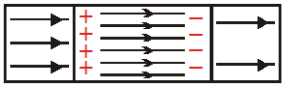
Single Correct Answer Type 2 Q. [3 M (–1)]

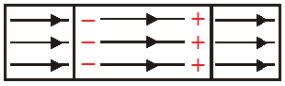
1. Which of the following electric field pattern is correct inside the conductor having three materials of resistivity ?



(A) 

(B)

(C) 

(D) 

2. A constant current I flows through a manganin wire of resistance R, length ℓ, radius of cross-section r, density d and specific heat C. The resistance of manganin wire is independent of the temperature. The amount of heat given off by unit surface area of the wire in unit time is given according to Newton's law of cooling as $\frac{dQ}{dt} = k(T - T_0)$ where T_0 is the temperature of the surrounding. Find the expression of temperature of wire at any instant t.

(A) $T_0 + \frac{I^2R}{2\pi r\ell k}\left(1 - e^{\frac{2kt}{rCd}}\right)$

(B) $T_0 + \frac{I^2R}{\pi r\ell k}\left(1 - e^{-\frac{2kt}{rCd}}\right)$

(C) $T_0 + \frac{I^2R}{4\pi r\ell k}\left(1 - e^{-\frac{2kt}{rCd}}\right)$

(D) $T_0 + \frac{I^2R}{2\pi r\ell k}\left(1 - e^{-\frac{2kt}{rCd}}\right)$

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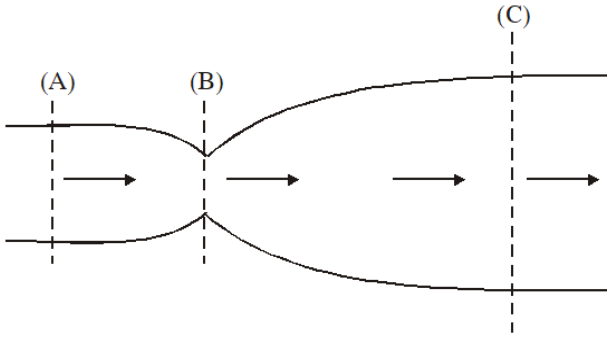
SECTION-I (ii) : (Maximum Marks: 40)

Multiple Correct Answer Type 10 Q. [4 M (–2)]

3. Figure shows a copper wire of nonuniform cross section carrying current towards right for sectors A, B and C.



(A) $i_B > i_A > i_C$ (i = current)

(B) $E_B > E_A > E_C$ (E = electric field)

(C) $V_B > V_A > V_C$ (V = drift velocity)

(D) $J_B > J_A > J_C$ (J = current density)

1001CJA00PHEN2501301

4. A metal sphere of radius a is surrounded by a concentric metal sphere of inner radius of b, where b > a. The space between the spheres is filled with a material whose electrical conductivity σ varies with the electric field strength E as $\sigma = kE$ where k is a constant. A potential difference V is maintained between spheres.

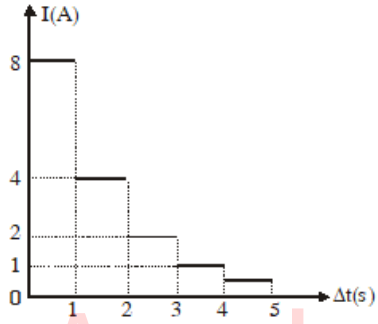
(A) Current is $4\pi r^2 kE^2$, where $(a < r < b)$

(B) Current is $2\pi r^2 kE^2$, where $(a < r < b)$

(C) Potential difference between spheres is $V = \sqrt{\frac{I}{4\pi k}}\ell n\left(\frac{b}{a}\right)$ where I is total current

(D) Potential difference between spheres is $V = \sqrt{\frac{I}{2\pi k}}\ell n\left(\frac{b}{a}\right)$ where I is total current

5. The current in a wire of resistance 10Ω varies as shown. Select correct alternative(s).



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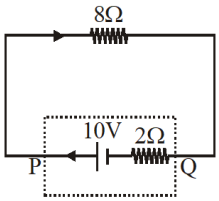
(A) The heat produced in the wire in 4 s is 852.5 J

(B) The heat produced in the wire in 4 s is 850 J

(C) The charge flows across any section of the wire in 4 s is 15 C

(D) The charge flows across any section of the wire in 4 s is 15.5 C

6. A battery of emf 10 volt and internal resistance 2Ω is connected to an external resistance 8W as shown in the figure :-



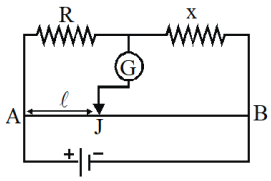
(A) Work done due to conservative electric field while a unit positive charge passes through battery from Q to P (along the arrow) is 8 Joule.

(B) Work done due to conservative electric field while a unit positive charge passes through battery from Q to P (along the arrow) is − 8 Joule.

(C) Work done due to conservative electric field while a unit positive charge passes through 8W along the arrow is −8 Joule.

(D) Work done due to non conservative electric field while a unit positive charge moves from Q to P (along the arrow) is 10 Joule.

7. In the meter bridge experiments initially balance point is J, where AJ= ℓ. One wants to make some changes in the shown setup. After any change balance length being AJ' = ℓ', then choose the correct statement(s). [Here R $\neq$ x]



(A) If in shown set-up the galvanometer and battery are inter changed current through galvanometer is zero.

(B) radius of wire AB is double then $\ell' = 2\ell$

(C) if length of wire AB is doubled then $\ell' = 2\ell$

(D) If the values of R and x both are double and then interchanged then $\ell' = \ell$

8. A 20 m long potentiometer wire has a resistance of 20 ohm. It is connected in series with a battery of emf 3V and a resistance of 10W. The internal resistance of cell is negligible. If the length can be read accurately up to 1 mm, the potentiometer can read voltage

(A) up to minimum of 0.2 mV

(B) with an accuracy of 0.2 mV

(C) with an accuracy of 0.1 mV

(D) upto maximum of 2V

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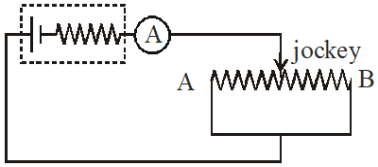
9. The circuit diagram shows a source of emf 10 V having an internal resistance of 2.5 Ω connected to an ideal ammeter. The jockey can be slided to different points on the Rheostat whose total resistance is 10 Ω. What can be the reading of the ammeter.



(A) 1A (B) 2A (C) 3A (D) 4A

10. Two identical moving coil galvanometer have 10 Ω resistance and full scale deflection at 2 μA current. One of them is converted into a voltmeter of 100 mV full scale reading and the other into an Ammeter of 1 mA full scale current using appropriate resistors. These are then used to measure the voltage and current in the Ohm's law experiment with R = 1000 Ω resistor by using an ideal cell. Which of the following statement(s) is/are correct ?

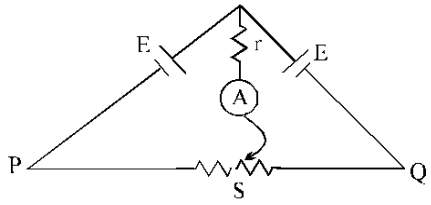
(A) The measured value of R will be 978 Ω < R < 982 Ω.

(B) The resistance of the Voltmeter will be 100 kΩ.

(C) The resistance of the Ammeter will be 0.02 Ω (round off to 2nd decimal place)

(D) If the ideal cell is replaced by a cell having internal resistance of 5 W then the measured value of R will be more than 1000 Ω.

11. In the figure. PQ is a wire of uniform cross-section and of resistance R_0. A is an ideal ammeter and the cells are of negligible resistance. The jockey J can freely slide over the wire PQ making contact on it at S. If the length of the wire PS is $f = 1/n^{th}$ of PQ, find the value of 'f' for maximum and minimum reading on the ammeter.



(A) $f = 0,1$ for I_{max} (B) $f = 0,1$ for I_{min} (C) $f = 1/2$ for I_{max} (D) $f = 1/2$ for I_{min}

12. A galvanometer (coil resistance 99Ω) is converted into a ammeter using a shunt of 1 Ω and connected as shown in the figure (i). The ammeter reads 3A. The same galvanometer is converted into a voltmeter by connected a resistance of 101 Ω in series. This voltmeter is connected as shown in figure (ii). Its reading is found to be 4/5 of the full scale reading. Then :

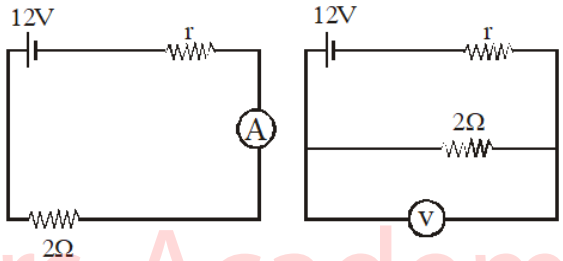


Fig. (i) Fig. (ii)

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(A) Internal resistance r of the cell is 1.01Ω.

(B) Range of the ammeter is 0-5 A and voltmeter is 0-10 V.

(C) Full scale deflection current of the galvanometer is 0.05 A

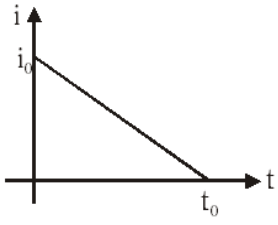
(D) Full scale deflection current of the galvanometer is 0.1 A.

SECTION-I (iii) : (Maximum Marks: 24)

Linked Comprehension Type (Single Option Correct) (2 Para × 3 Q.) (1 Para × 2 Q.) [3 M (–1)]

Paragraph for Questions 13 and 14

Relation between current in conductor and time is shown in figure then determine.



13. Total charge flow through the conductor is :-

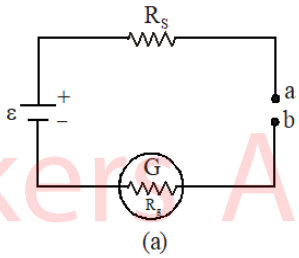
(A) $\frac{1}{2}i_0t_0$ (B) $\frac{1}{4}i_0t_0$ (C) i_0t_0 (D) None of these

14. If resistance of conductor is R then total heat dissipated across resistance R is :-

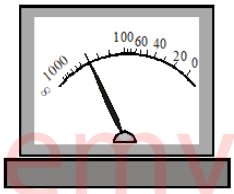
(A) $\frac{Rt_0i_0^2}{2}$ (B) $\frac{Rt_0i_0^2}{6}$ (C) $\frac{Rt_0i_0^2}{3}$ (D) $\frac{Rt_0i_0^2}{4}$

Paragraph for Questions 15 - 17

A simple ohmmeter consists of battery connected in series with a galvanometer and a resistor, as shown in figure. The resistance R_s is chosen such that when the terminals a and b are shorted (put in electrical contact, with negligible resistance between them), the current through the galvanometer gives a full-scale deflection. Thus, a full scale deflection indicates no resistance between terminals a and b. A zero deflection indicates an infinite resistance between the terminals. When the terminals are connected across an unknown resistance R, the current through the galvanometer depends on R, so the scale can be calibrated to give a direct reading of R, as shown in figure. Because an ohmmeter sends a current through the resistance to be measured, some caution must be exercised when using this instrument. For example, you would not want to try to measure the resistance of sensitive ammeter with an ohmmeter, because the current provided by the battery in the ohmmeter would probably damage the ammeter. Let us use a galvanometer with resistance of 20 Ω and maximum current of 10 mA, ε = 1 V. For a current more that 10 mA, galvanometer would be damaged.



(a)



(b)

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15. The scale is non linear because

(A) the resistance changes with change in temperature.

(B) of internal resistance of the battery.

(C) the current in circuit decreases with increase in resistance.

(D) of internal resistance of galvanometer.

16. The minimum resistance required for rest of the circuit (other than galvanometer) is

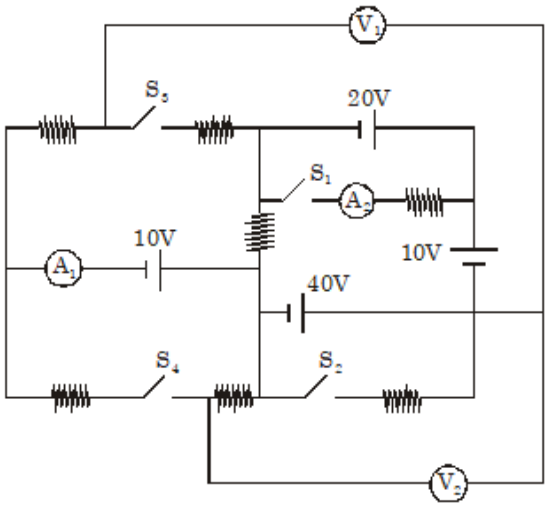
(A) 20Ω (B) 30Ω (C) 80Ω (D) 100Ω

17. The reading in galvanometer when resistance connected between a and b is 100 Ω is

(A) 5 mA (B) 3.3 mA (C) 2 mA (D) 8 mA

Paragraph for Questions 18 - 20

In given circuit, all batteries, voltmeters and ammeters are ideal. Voltmeters and ammeters have only +ive scale for reading. Resistance of each resistor is 5Ω. Reading of voltmeters are V_1 and V_2 and of ammeters are A_1 and A_2. Initially all switches are closed.



Coloumn–I $\lvert A_1 - A_2 \rvert$		Column-II V_1		Column-III V_2	
(I)	3A	(i)	20 V	(P)	45 V
(II)	0A	(ii)	30 V	(Q)	35 V
(III)	5A	(iii)	50 V	(R)	40 V
(IV)	1A	(iv)	40 V	(S)	25 V

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18. Only switch S_2 is open

(A) (I) (iii) (S)
(B) (IV) (ii) (P)
(C) (I) (iii) (P)
(D) (IV) (ii) (S)

19. Only switch S_3 is open :

(A) (III) (iii) (R)
(B) (IV) (ii) (S)
(C) (I) (iii) (P)
(D) (II) (iv) (Q)

20. Only switch S_1 is open :

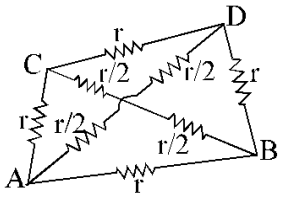
(A) (III) (iii) (P)
(B) (I) (iii) (R)
(C) (IV) (ii) (P)
(D) (III) (ii) (P)

SECTION-I (iv) : (Maximum Marks: 12)

Linked Comprehension Type (Multiple Options Correct) (1 Para × 3 Q.) [4 M (–1)]

Paragraph for Questions 21 - 23

The figure shows a tetrahedron, each side of which has a resistance r



21. Choose the correct statement(s) related to the resistance between any two points.

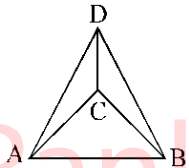
(A) $R_{AB} = R_{BD} = R_{BC} = R_{CD} = R_{CA} = R_{AD}$

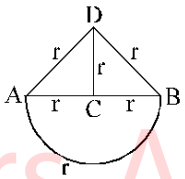
(B) $R_{AB} = R_{AC} = R_{AD} = R_{BD} = R_{BC} \neq R_{CD}$

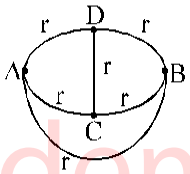
(C) R_{CD} is the least

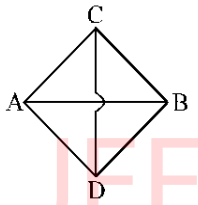
(D) $R_{AB} = R_{AC} = R_{BC}$ and $R_{CD} = R_{AD} = R_{BD}$

22. Choose the correct diagram(s), which show two-dimensional equivalent of the tetrahedron.

(A)


(B)


(C)


(D)


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23. If a battery is connected between any two points of the tetrahedron, then identify the correct statement(s).

(A) The potentials of the other two points are always equal.

(B) There always exists a branch through which no current flows.

(C) The current coming out of the battery in each case is same.

(D) None of these

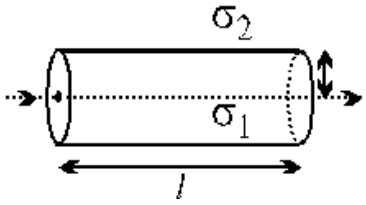
SECTION-II : (Maximum Marks: 24)

Numerical-Answer-Type questions (Non Negative Integer) 6 Q. [4 M (0)]

1. On quadrupling the resistance 'R' connected to a battery the power dissipated in the resistance surprisingly does not change. What is the internal resistance of the battery in Ohm? (Take R = 10Ω)

2. One and a half litres of water at a temperature of 20°C is heated for 15 minutes on an electric table stove burner having two sections with the same resistance. When the sections are connected in parallel, the water begins to boil and 100 grams of it is converted into steam. The latent heat of vaporization is 540 cal per gram. Starting from same initial conditions, the sections are connected in series. In what time (in min) will 100 gm of water get converted into steam. Neglect all heat loses and heat capacity of the container. Density of water 1 gm/cc and specific heat capacity of water = 4200 J/kg°C.

3. A long conductor of circular cross-section has radius r and lenght ℓ as shown in the figure. The conductivity of the material near the axis is σ_1. and increases linearly with the distance from axis and becomes σ_2 near the surface. If the inverse of resistance of the conductor when the current enters from the one end and leaves from the other end is given by $\frac{1}{R} = \frac{\pi r^2}{n\ell}(2\sigma_2 + \sigma_1)$. Find the value of n.



4. In circuit (a) the instruments read $U_1 = 100$ V, $I_1 = 5\mu$A. In circuit (b) we have $U_2 = 25$ mV, $I_2 = 2.5$ A. What is the internal resistance of the cell (in Ohm). Round off to nearest integer. Consider both instruments to be regular laboratory instruments.

5. The circuit diagram given in the figure shows the experimental setup for the measurement of unknown resistance by using a meter bridge. The wire connected between the points P & Q has non-uniform resistance such that resistance per unit length varies directly as the distance from the point P. Null point is obtained with the jockey J with R_1 and R_2 in the given position. On interchanging the positions R_1 and R_2 in the gaps the jockey has to be displaced through a distance Δ from the previous position along the wire to establish the null point. If the ratio of $\frac{R_1}{R_2} = 3$, find the value of Δ (in cm). Ignore any end corrections. [Take $\sqrt{3} = 1.7$]

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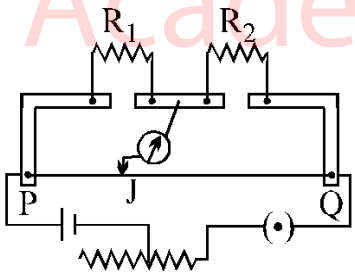
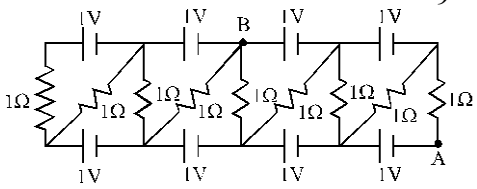


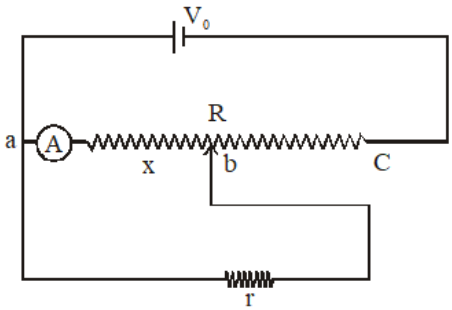
6. The potential difference $V_A - V_B$ for the circuit shown in the figure is $-\frac{n}{9}$V . Find the value of n.



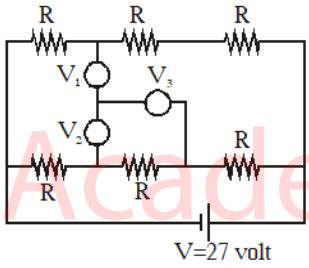
SECTION-III : (Maximum Marks: 12)

Numerical Grid Type (Ranging from 0 to 9) 3 Q. [4 M (0)]

1. A constant voltage V_0 (= 12V) is applied to a potential divider of resistance R (= 4Ω), connected to an ideal ammeter. A constant resistor r (= 2 Ω) is connected to the sliding contact of the potential divider (as shown). Find the minimum current (in A) measured by ammeter.



2. In the circuit shown below, all the voltmeter identical and have very high resistance. Each resistor has the same resistance. The voltage of the ideal battery shown is 27 V. Find the reading of voltmeter V_3 (in volts).



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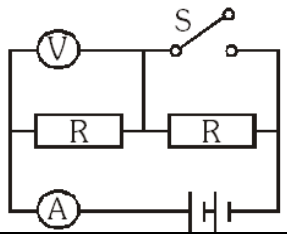
3. The coil of a heater has resistance of 5Ω and is powered from a source of internal resistance 20Ω. With what resistance (in Ω) should the coil be shunted to reduce the amount of power dissipated in it to one ninth of the value without shunt resistance ?

SECTION-IV : (Maximum Marks: 8)

Matrix Match Type (4 × 5) (One or more options correct) 1 Q. [8 M; (for each entry +2(0)]

1. The diagram shows a circuit with two identical resistors. The battery has a negligible internal resistance. What will the effect on the ammeter and voltmeter be if the switch S is closed ?



Column-I		Column-II	
(A)	Ammeter reading	(P)	Increases
(B)	Voltmeter reading	(Q)	Decreases
(C)	Equivalent resistance of circuit	(R)	Does not change
(D)	Power dissipated across R in right branch	(S)	Becomes zero
		(T)	Cannot be determined

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SECTION-V : (Maximum Marks: 8)

SUBJECTIVE TYPE QUESTIONS 2 Q. (4 M, 0)

1. A long conductor of a circular cross-section of radius R is made of a material whose resistivity depends only on the distance 'r' from the conductor's central axis as $\rho = a / r^2$, where a is a constant. The conductor has length L (>>R). Its ends are maintained at a potential difference = ΔV. Find

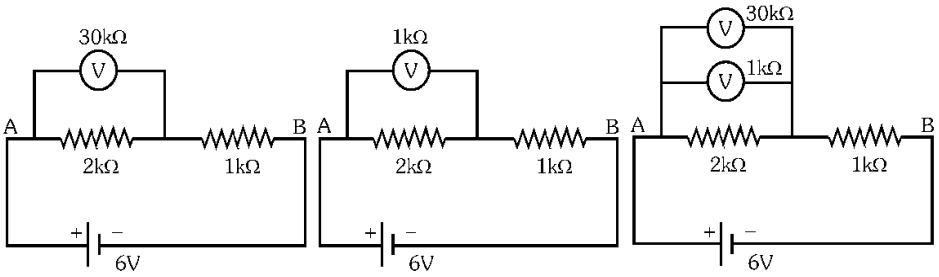
(a) the electric field and current density in the conductor at any distance 'r' from the central axis.

(b) total rate of joule-heat production. The rate of loss of heat from the surface of the conductor is $\lambda(T_S - T_A)$, where 'T_S' is the surface temperature of the conductor, 'T_A' the ambient temperature and 'λ' is some constant. Find

(c) surface temperature (T_S) in steady state. If the thermal conductivity of the conductor's material is 'k' everywhere find

(d) the temperature in the conductor as a function of 'r'. Assuming steady state.

2. A series combination of a 2kΩ resistor and a 1kΩ resistor, is connected across a battery of emf 6V and negligible internal resistance. The potential drop, across the 2kΩ resistor, is measured by (a) a 30kΩ voltmeter (b) a 1kΩ voltmeter and (c) both these voltmeters connected across it. If the voltmeter readings in the three cases are V_1, V_2 and V_3 respectively, arrange these readings in descending other.



How will the three readings compare with one another if the potential drop were measured across the series combination of the 2kΩ and the 1kΩ resistor i.e., across the points A and B ?

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ANSWER KEY

GR_CURRENT ELECTRICITY

PHYSICS

SECTION-I (i)	Q.	1	2								
	A.	A	D								
SECTION-I (ii)	Q.	3	4	5	6	7	8	9	10	11	12
	A.	B,C,D	A,C	B,C	B,D	A,C	C,D	B,C,D	A,C	A,D	A,B,C
SECTION-I (iii)	Q.	13	14	15	16	17	18	19	20		
	A.	A	C	C	C	A	B	C	D		
SECTION-I (iv)	Q.	21	22	23							
	A.	A,D	A,B,C,D	A,B,C							
SECTION-II	Q.	1	2	3	4	5	6				
	A.	20	60	3	40	35	22.00				
SECTION-III	Q.	1	2	3							
	A.	2	6	2							
SECTION-IV	Q.	1									
	A.	A->P,B->P,C->Q,D->QS									
SECTION-V	Q.	1	2								
	A.										

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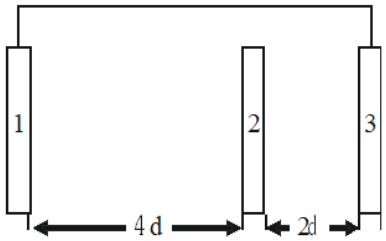
SECTION-I (i) (Maximum Marks:27)

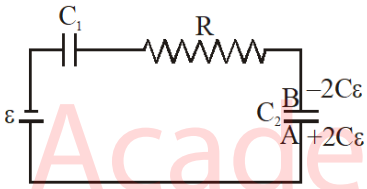
Single Correct Answer Type 9 Q. [3 M (–1)]

1. Three identical metal plates of large area A numbered 1, 2,3 are arranged as shown in the diagram. The plates-1 and 3 are connected by a conducting wire. The charge given to plate 2 is $2q_0$ and no other plate is given any charge. Find the potential difference between the plate 2 and 3.



(A) $\frac{4}{3}\frac{q_0 d}{A\epsilon_0}$ (B) $\frac{8}{3}\frac{q_0 d}{A\epsilon_0}$ (C) $\frac{12}{3}\frac{q_0 d}{A\epsilon_0}$ (D) $\frac{2}{3}\frac{q_0 d}{A\epsilon_0}$

2. The capacitance of C_1 and C_2 shown in the diagram is C. C_1 is initially uncharged and C_2 is given a charge 2Ce as shown. Which of the following graph represents charge on plate B of capacitor C_2 as a function of time ?



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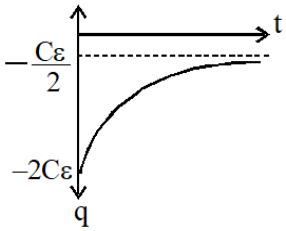
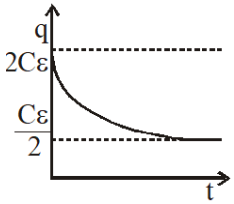
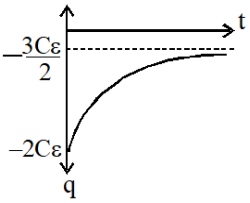
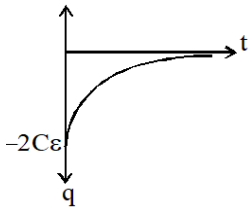
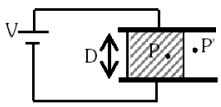
(A)


(B)


(C)


(D)


3. A parallel plate capacitor, partially filled with a dielectric slab of dielectric constant K, is connected with a cell of emf V volt, as shown in the figure. Separation between the plates is D. Then



(A) electric field at point P is less than that at point P′.

(B) electric field at point P′ is less than that at point P.

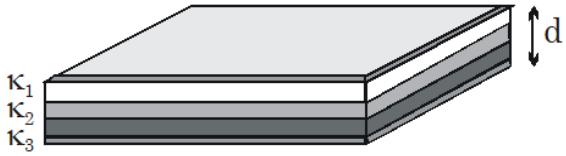
(C) electric fields at points P and P′ are equal

(D) electric field at point P is $E = \frac{V}{KD}$.

4. A metal ball of radius R is surrounded by a homogeneous medium of dielectric constant K and resistivity ρ. The ball is given a charge Q at t = 0, which starts leaking through the medium towards infinity. After how much time will the charge on the ball become Q/2?

(A) $[2K\rho\varepsilon_0].\ln 2$ (B) $[K\rho\varepsilon_0].\ln 2$ (C) $[K\rho\varepsilon_0/2].\ln 2$ (D) None

5. A parallel plate capacitor is filled with 3 dielectric materials of same thickness, as shown in the sketch. The dielectric constants are such that $\kappa_3 > \kappa_2 > \kappa_1$. Let the magnitudes of the electric field in and potential drops across each dielectric be E_3, E_2, E_1, ΔV_3, ΔV_2, and ΔV_1, respectively. Which one of the following statements is true?



(A) $E_3 < E_2 < E_1$ and $\Delta V_3 < \Delta V_2 < \Delta V_1$ (B) $E_3 > E_2 > E_1$ and $\Delta V_3 > \Delta V_2 > \Delta V_1$

(C) $E_3 < E_2 < E_1$ and $\Delta V_3 > \Delta V_2 > \Delta V_1$ (D) $E_3 > E_2 > E_1$ and $\Delta V_3 < \Delta V_2 < \Delta V_1$

6. A dielectric slab is introduced between the plates of a capacitor. If the charge on the capacitor before the slab is introduced is q and the magnitude of the induced charge on the dielectric surface is q', then

(A) q' < q (always) (B) q' > q (always) (C) q' = q (always) (D) q' = 0

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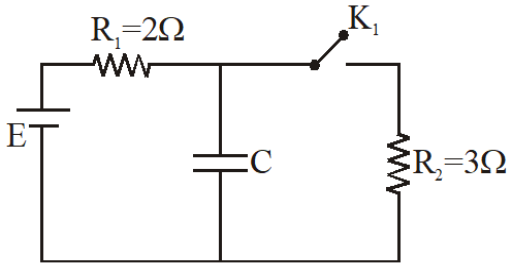
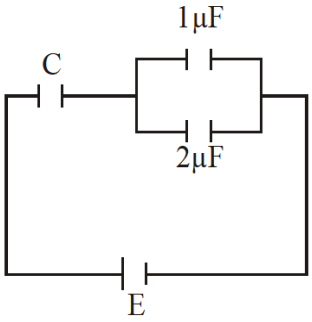
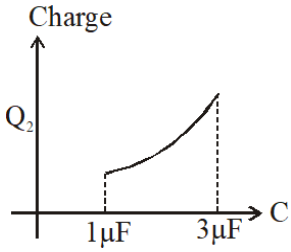
7. In the circuit shown in the figure K_1 is open. The charge on capacitor C in steady state is q_1. Now key is closed and at steady state charge on C is q_2. The ratio of charges q_1/q_2 is :

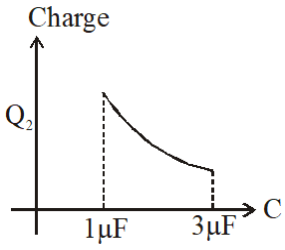


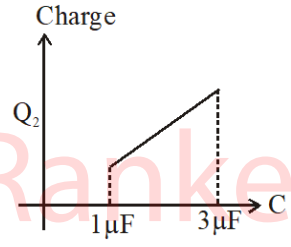
(A) 5/3 (B) 3/5 (C) 1 (D) 2/3

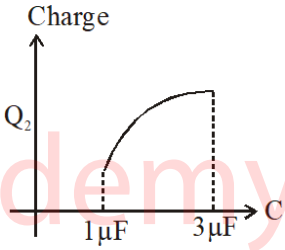
8. In the given circuit, charge Q_2 on the 2μF capacitor changes as C is varied from 1μF to 3μF. Q_2 as a function of 'C' is given properly by : (figures are drawn schematically and are not to scale) :-



(A) 

(B) 

(C) 

(D) 

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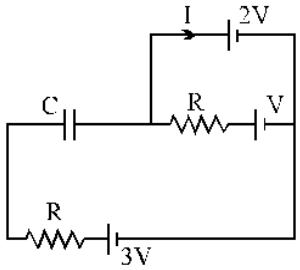
9. In the RC circuit shown, what is the variation of the current I as a function of time? The capacitor is initially uncharged and the switch is closed at time t = 0 sec.

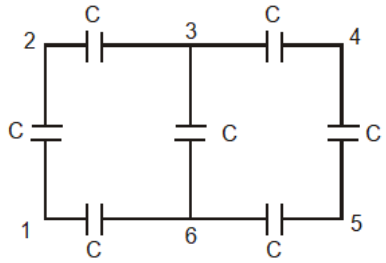


(A) $\frac{V}{R}\left(e^{\frac{-t}{RC}}-1\right)$ (B) $\frac{V}{R}\left(e^{\frac{t}{RC}}-1\right)$ (C) $\frac{V}{R}\left(e^{\frac{-t}{RC}}+1\right)$ (D) None of these

SECTION-I (ii) : (Maximum Marks: 32)

Multiple Correct Answer Type 8 Q. [4 M (–2)]

10. For the given circuit, select the correct alternative(s)



(A) The equivalent capacitance between points 1 & 2 is $\frac{15C}{11}$

(B) The equivalent capacitance between points 3 & 6 is $\frac{5C}{3}$

(C) The equivalent capacitance between points 1 & 3 is $\frac{15C}{14}$

(D) The equivalent capacitance between points 3 & 5 is $\frac{14C}{15}$

11. A parallel plate air capacitor is connected to a battery. If plates of the capacitor are slowly pulled apart, then which of the following statements is/are incorrect ?

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(A) Strength of electric field inside the capacitor remains unchanged, if battery is disconnected before pulling the plates.

(B) During the process, negative work is done by an external force applied to pull the plates whether battery is disconnected or it remains connected.

(C) Potential energy in the capacitor decreases if the battery remains connected during pulling plates apart.

(D) Potential energy in the capacitor decreases if the battery is disconnected before pulling plates apart.

12. We have a parallel plate capacitor made of two conducting, circular sheets of radius 2R each. They are kept at a small separation d between them. Suppose, we insert a circular shape dielectric of radius R and thickness d, between the gap with its centre coinciding with the plate centres the dielectric constant is k. The correct statement is/are :

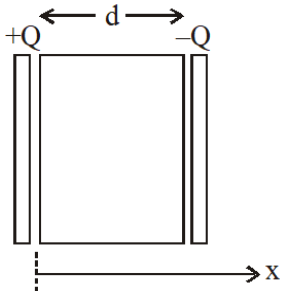
(A) Capacitance of the capacitor is $\frac{(k+3)\in_0 \pi R^2}{d}$

(B) Capacitance of the capacitor is $\frac{k\in_0 \pi\,(4R^2)}{d}$

(C) Ratio of electric field in the region with dielectric & without dielectric is 1

(D) Ratio of electric field in the region with dielectric & without dielectric is 1 : k

13. A dielectric slab is filled in between the plates of parallel plate charged capacitor. Di-electric constant of the capacitor varies with x as $k = \frac{x}{d}k_0$. Capacitor is not connected with the battery. Then choose the correct statement(s) :-



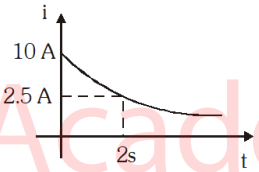
(A) Net electric field inside the di-electric is uniform throughout the capacitor.

(B) Net electric field inside the dielectric decreases on moving along x-direction inside the capacitor.

(C) Bound charge density inside the di-electric is zero.

(D) Bound charge density inside the dielectric is non-uniform.

14. The figure, a graph of the current in a discharging circuit of a capacitor through a resistor of resistance $10\,\Omega$.



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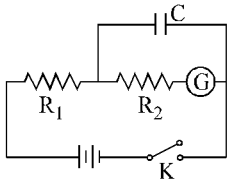
(A) The initial potential difference across the capactitor is 100 volt.

(B) The capacitance of the capacitor is $\frac{1}{10\ell n2}$F.

(C) The total heat produced in the circuit will be $\frac{500}{\ell n2}$ joules.

(D) The thermal power in the resistor will decrease with a time constant $\frac{1}{2\ell n2}$ second.

15. In the connection shown in the figure the switch K is open and the capacitor is uncharged. Then we close the switch and let the capacitor charge upto the maximum and open the switch again. The values indicated by the galvanometer.



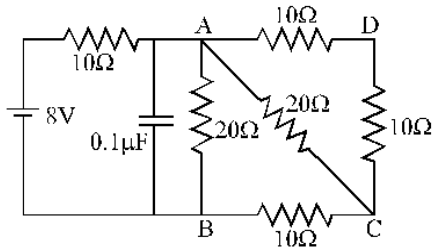
(A) increases after closing the switch and then decreases after opening the switch

(B) direction of current is same through the galvanometer before and after opening the switch again

(C) after opening the switch again energy dissipation in R_1 s zero

(D) after opening the switch again energy dissipation in R_2 is zero

16. A capacitor of capacitance 0.1 μF is connected to a battery of emf 8V as shown in the fig. Under steady state condition



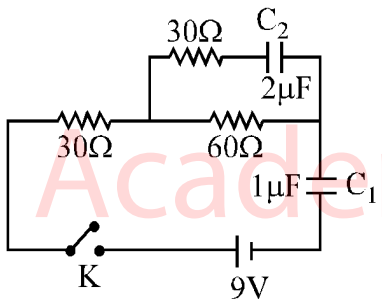
(A) charge on the capacitor is 0.4 μC.

(B) charge on the capacitor is 0.2 μC

(C) current in the resistor between point A and B is 0.2 A

(D) current in the resistor between point A and B is 0.4 A

17. For the arrangement shown in the figure, the key is closed at t = 0. C_2 is initially uncharged while C_1 has a charge of 2μC.



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(A) The current coming out of the battery just after switch is closed $\frac{7}{50}$A.

(B) The current coming out of the battery just after switch is closed $\frac{11}{50}$A.

(C) The charge on the capacitor Q_1 in the steady state condition is 9μC.

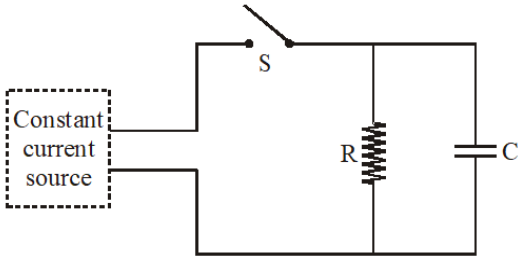
(D) The charge on the capacitor Q_2 in the steady state condition is 0.

SECTION-I (iii) : (Maximum Marks: 12)

Linked Comprehension Type (Single Option Correct) (2 Para × 2 Q.) [3 M (–1)]

Paragraph for Questions 18 and 19

In the circuit shown in figure, switch S is closed at time t = 0. Thereafter the constant current source by varying its emf, maintains a constant current I out of its upper terminal.



18. Derive an expression for the charge on the capacitor as a function of time :-

(A) $q = iRC\left(e^{-\frac{t}{RC}}\right)$ (B) $q = iRC\left(1-e^{-\frac{t}{RC}}\right)$ (C) $q = it\left(1-e^{-\frac{t}{RC}}\right)$ (D) $q = \frac{i}{RC}\left(1-e^{-\frac{t}{RC}}\right)$

19. Find the time at which the rate of energy storage on capacitor is equal to the rate of heat dissipation through resistor :-

(A) $t = \frac{RC\,\ell n2}{2}$ (B) $t = 2RC\,\ell n\,2$ (C) $t = RC\,\ell n\,2$ (D) Not possible

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Paragraph for Questions 20 and 21

Consider a simple RC circuit as shown in figure 1.

Process 1 : In the circuit the switch S is closed at t = 0 and the capacitor is fully charged to voltage V_0 (i.e., charging continues for time T >> RC). In the process some dissipation (E_D) occurs across the resistance R. The amount of energy finally stored in the fully charged capacitor is E_C.

Process 2 : In a different process the voltage is first set to $\frac{v_0}{3}$ and maintained for a charging time T >> RC. Then the voltage is raised to $\frac{2v_0}{3}$ without discharging the capacitor and again maintained for a time T >> RC. The process is repeated one more time by raising the voltage to V_0 and the capacitor is charged to the same final voltage V_0 as in Process 1.

These two processes are depicted in Figure 2.

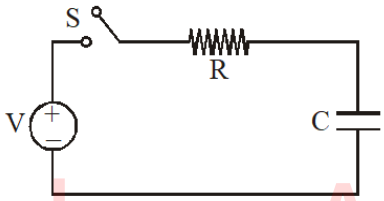

Figure 1

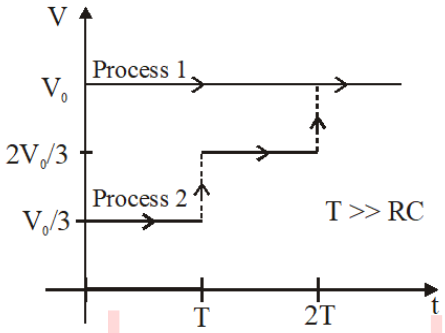

Figure 2

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20. In Process 1, the energy stored in the capacitor E_C and heat dissipated across resistance E_D are related by:-

(A) $E_C = E_D$ (B) $E_C = 2E_D$ (C) $E_C = E_D$ (D) $E_C = E_D \ln 2$

21. In Process 2, total energy dissipated across the resistance E_D is :-

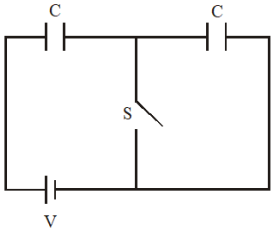
(A) $E_D = \frac{1}{3}\left(\frac{1}{2}CV_0^2\right)$ (B) $E_D = 3\left(\frac{1}{2}CV_0^2\right)$

(C) $E_D = \frac{1}{2}CV_0^2$ (D) $E_D = 3CV_0^2$

SECTION-II : (Maximum Marks: 16)

Numerical Answer type Questions (Upto Second Decimal Place) 4 Q. [4 M (0)]

1. In the given circuit switch s is open initially. If C = 2 μF and v = 3 volt. Then find heat produced (in μJ) in the circuit after closing the switch.



2. An electron enters the region between the plates of a parallel plate capacitor at a point equidistant from either plate. The capacitor plates are 2×10^{-2} m apart & 10^{-1} m long . A potential difference of 300 volt is kept across the plates. Assuming that the initial velocity of the electron is parallel to the capacitor plates. The largest value of the velocity of the electron so that it does not fly out of the capacitor at the other end is $\frac{\sqrt{x}}{2\sqrt{9.1}} \times 10^8$ m/s. Find the value of x.

3. Two square metallic plates of 1 m side are kept 0.01 m apart, like a parallel plate capacitor, in air in such a way that one of their edges is perpendicular, to an oil surface in a tank filled with an insulating oil. The plates are connected to a battery of e.m.f. 500 volt. The plates are then lowered vertically into the oil at a speed of 0.001 m/s. The current drawn from the battery during the process is $n \times 10^{-9}$ Ampere. Find the value of 2n. [di-electric constant of oil = 11, $\in_0 = 8.85 \times 10^{-12}$ $C^2/N^2\ m^2$]

4. A certain series RC circuit is formed using a resistance R, a capacitor without dielectric having a capacitance C = 2F and a battery of emf E = 3V. The circuit is completed and it is allowed to attain the steady state. After this, at t = 0, half the thickness of the capacitor is filled with a dielectric of constant K = 2 as shown in the figure. The system is again allowed to attain a steady state. What will be the heat generated (in joule) in the circuit between t = 0 and t = ∞ ?

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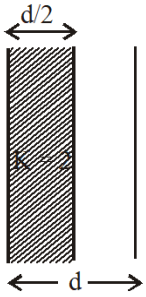
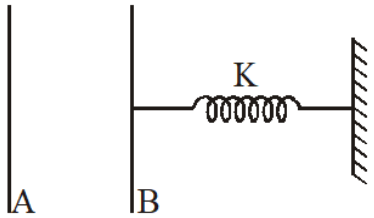


SECTION-III : (Maximum Marks: 8)

Numerical Grid Type (Ranging from 0 to 9) 2 Q. [4 M (0)]

1. Initially the plates of the parallel plate capacitor shown in the figure are A and B containing charge $+2\mu C$ and $-2\mu C$ respectively. Negatively charged plate B is connected to a spring of spring constant 5000N/m at one end and the other end of the spring is connected to a rigid support. Now the plate A is fixed and the system is released from rest. The maximum elongation produced in the spring is $\frac{\alpha \times 10^{-12}}{\epsilon_0}$m. The area of the plates is 2 cm^2, the medium between the plates is air and Electric permittivity of air is ϵ_0 (in SI). Then the value of α.



2. Two square plates with side length ℓ are arranged parallel to each other at distance d. They are charged to a potential difference V_0 and isolated from the source. A dielectric with a permittivity $\varepsilon_r = 2$ whose thickness is d and width is equal to that of the plates is drawn into the space between the plates. The length of the dielectric is greater than ℓ (figure). The resulting force F acting on the dielectric at distance $x = \ell/2$ is $\frac{\alpha C V_0^2}{\beta \ell}$ where $C = \frac{\varepsilon_0 \ell^2}{d}$. Take α and β as minimum integers. Find $\beta - \alpha$.

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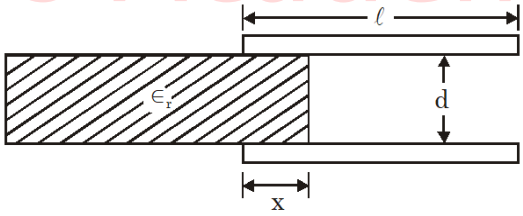
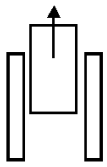
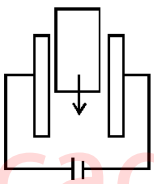
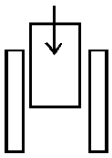
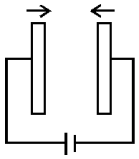
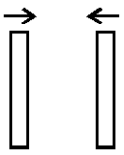


SECTION-IV : (Maximum Marks: 24)

Matrix Match Type (4 × 5) (One or more options correct) 3 Q. [8 M; (for each entry +2(0)]

1. Column–I shows some parameters after completion of event and Column–II shows tha arrangements to which these parameters belong.

	Column-I		Column-II
(A)	Work done by external agent is negative	(P)	Dielectric is being removed slowly from a charged capacitor 
(B)	Electric field between plates increases	(Q)	Dielectric being introduced slowly while battery is connected 
(C)	Energy of capacitor decreases	(R)	Dielectric is being introudced slowly in a charged capacitor 
(D)	Force between plates remains constant	(S)	Plates of capacitor are being slowly moved closer 
		(T)	Plates of capacitor are being moved slowly in a charged capacitor 

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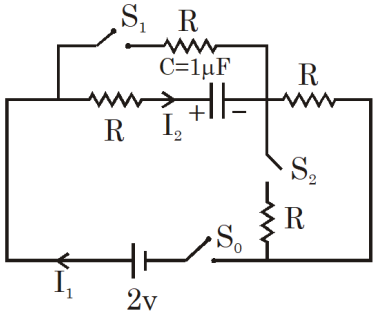
2. In the given figure initially all switches are open and the capacitor has charge q. At t = 0, switch S_0 is closed along with S_1 or S_2 or both. Column I has the entries for the actions on switches and column II has the values of currents I_1 & I_2 at t = 0. (R = 1Ω)



	Column-I		Column-II
(A)	S_0 and S_1 closed and q = 1 μC	(P)	$I_1 < 2I_2$
(B)	S_0 and S_1 closed and q = −1 μC	(Q)	$I_2 = 0$
(C)	S_0, S_1 and S_2 closed and q = 1 μC	(R)	$I_1 > 4I_2$
(D)	S_0, S_1 and S_2 closed and q = −1 μC	(S)	$I_1 > 1A$
		(T)	$I_1 \leq 1A$

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3.

	Column-I		Column-II
(A)	Plates of an isolated, charged, parallel plate, air core capacitor are slowly pulled apart.	(P)	Electric energy stored inside capacitor increases in the process.
(B)	A dielectric is slowly inserted inside an isolated and charged parallel plate air cored capacitor to completely fill the space between plates.	(Q)	Force between the two plates of the capacitor remain unchanged.
(C)	Plates of a parallel plate capacitor connected across a battery are slowly pulled apart.	(R)	Electric field in the region between plates remain unchanged.
(D)	A dielectric slab is slowly inserted inside a parallel plate capacitor connected across a battery to completely fill the space between plates.	(S)	Total electric energy stored inside capacitor decreases in the process.
		(T)	Electric field in the region decreases.

ANSWER KEY

GR_CAPACITOR

PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8	9
	A.	B	C	C	B	A	A	A	D	A
SECTION-I (ii)	Q.	10	11	12	13	14	15	16	17	
	A.	A,B,C	B,D	A,C	B,D	A,B,C,D	A,B,C	A,C	A,B,C,D	
SECTION-I (iii)	Q.	18	19	20	21					
	A.	B	C	A	A					
SECTION-II	Q.	1	2	3	4					
	A.	4.50	4.8	8.85	3.00					
SECTION-III	Q.	1	2							
	A.	4	7							
SECTION-IV	Q.	1	2	3						
	A.	A->QRST,B->PS,C->RT,D->PRT	A->QRT,B->PS,C->RS,D->PS	A->PQR,B->QST,C->ST,D->PR						

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Important Notes

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Important Notes

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SECTION-I (i) (Maximum Marks:21)

Single Correct Answer Type 7 Q. [3 M (–1)]

1. Current I flowing along the path ABCD, along the four edges of the cube (figure-a), creates a magnetic field in the centre of the cube of B_0. Find the magnetic field B created at the center of the cube by a current I flowing along the path of the six edges ABCGHEA (figure-b)

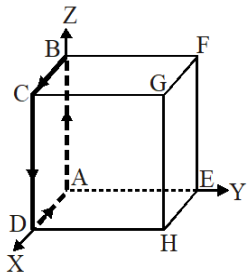

Figure-(a)

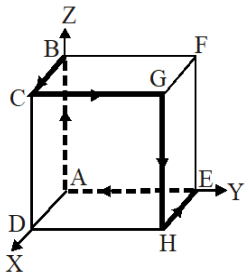

Figure-(b)

(A) $\sqrt{\frac{3}{2}}B_0$ Towards corner G

(B) $\sqrt{3}B_0$ Towards corner E

(C) $\sqrt{\frac{3}{2}}B_0$ Towards corner H

(D) $\sqrt{3}B_0$ Towards corner F

2. A thin uniform rod with negligible mass and length ℓ is attached to the floor by a frictionless hinge at point P. A horizontal spring with force constant k connects the other end to wall. The rod is in a uniform magnetic field B directed into the plane of paper. What is extension in spring in equilibrium when a current i is passed through the rod in direction shown. Assuming spring to be in natural length initially.

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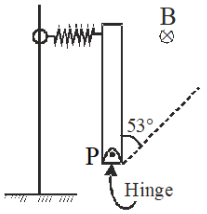
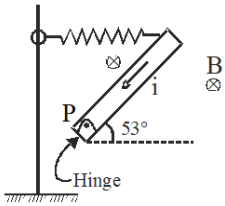


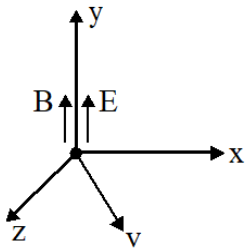
Spring in relaxed position



In equilibrium position

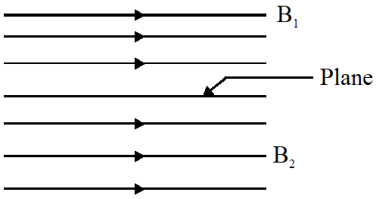
(A) $\frac{5i\ell B}{8k}$ (B) $\frac{3i\ell B}{8k}$ (C) $\frac{5i\ell B}{4k}$ (D) $\frac{5i\ell B}{6k}$

3. A positive charge particle having charge q and mass m has velocity $\vec{v} = v\left(\frac{\hat{i}+\hat{k}}{\sqrt{2}}\right)$ in the magnetic field at the origin. Its speed as the function of y is :-



(A) $\sqrt{v^2+\frac{qE}{2m}y}$ (B) $\sqrt{\left(\frac{B}{E}\right)^2+v^2+\frac{qE}{2m}y}$ (C) $\sqrt{v^2+\frac{2qE}{m}y}$ (D) None of the above

4. Two uniform magnetic fields B_1 and B_2 exist above and below the plane. The magnetic force acting on unit area of the plane is :-



(A) $\frac{B_2^2 - B_1^2}{2\mu_0}\uparrow$ (B) $\frac{|B_1 - B_2|^2}{2\mu_0}\downarrow$ (C) $\frac{\left|\frac{B_1+B_2}{2}\right|^2}{2\mu_0}\uparrow$ (D) None of the above

5. To increase the current sensitivity of a moving coil galvanometer, we should :–

(A) decrease number of turns in coil (B) decrease area of cross section of the coil

(C) increase torsional constant of spiral springs (D) None of the above

6. A straight conductor of mass 100gm and length 20cm is placed on the xy plane in a gravity free space. A uniform and constant magnetic field of 1000T in positive z direction is applied on it. A charge of 1mC is passed through the rod in a very short time duration and the conductor begins to move just after all the charge has passed through it. Assuming no friction, the velocity of the conductor is :

(A) 2cm/s (B) 2 m/s (C) 0 (D) 20 cm/s

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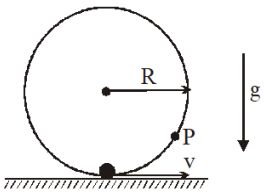
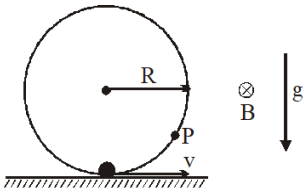
7. In case figure (a) and (b) a positive charge particle of mass m and charge q is given velocity 'v' at the lowest point of a smooth fixed sphere of radius R. In both cases circular motion is in the plane of the paper. In case (b) magnetic field is also switched on in inward direction. If at point P, v_1 and v_2 , N_1 and N_2 , are velocities and normal forces in two cases respectively then.



(a) $v = \sqrt{2gR}$



(b) $v = \sqrt{2gR}$

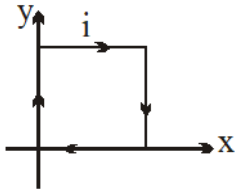
(A) $v_1 = v_2$ and $N_1 > N_2$ (B) $v_1 < v_2$ and $N_1 > N_2$

(C) $v_1 > v_2$ and $N_1 > N_2$ (D) $v_1 = v_2$ and $N_1 = N_2$

SECTION-I (ii) : (Maximum Marks: 40)

Multiple Correct Answer Type 10 Q. [4 M (–2)]

8. Figure shows a square current carrying coil of edge length L. The magnetic field on the coil is given by $\vec{B} = \frac{B_0 y}{L}\hat{i} + \frac{B_0 x}{L}\hat{j}$ where B_0 is a positive constant.



(A) If coil is free to rotate about x axis torque on the coil is given by $\frac{1}{2}iAB_0\hat{i}$.

(B) If coil is free to rotate about y - axis torque on coil is given by $-\frac{1}{2}iAB_0\hat{j}$.

(C) Resultant force on coil is zero.

(D) Equation for the torque $\vec{\mu} \times \vec{B}$ where μ is magnetic moment of coil is not valid on the coil.

9. Consider a cube of side 'a' as shown. Eight point charges are placed at the corners. The cube is rotated about the axis with constant angular velocity 'ω':

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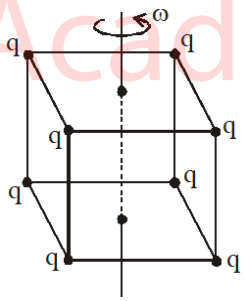


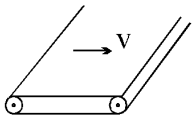
(A) Net magnetic field at the centre of cube is zero

(B) Net magnetic field at the centre of cube is $\frac{\sqrt{2}\mu_0 q\omega}{\pi a}$

(C) Net magnetic field at the centre of cube is $\frac{8}{3\sqrt{3}}\frac{\mu_0 q\omega}{\pi a}$

(D) If polarity of any four charges are reversed, then magnetic field at the centre of cube will be zero.

10. Charge is sprayed onto a large non conducting belt above the left hand roller. The belt carries charge with a uniform surface charge density σ, as it moves with a speed v between the rollers as shown. The charge is removed by a wiper at right hand roller. For a point just above the sheet mark the correct option.



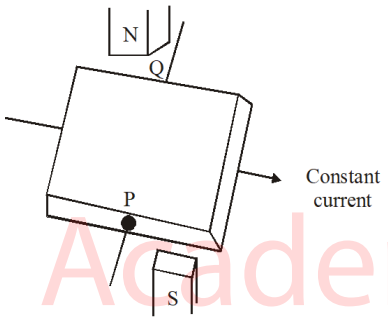
(A) magnetic field is $\frac{\mu_0 \sigma v}{2}$, out of the plane of the page, parallel to axis of roller.

(B) magnetic field is $\mu_0\sigma$, out of the plane of the page, perpendicular to axis

(C) electic field is $\frac{\varepsilon_0 \sigma}{2}$ perpendicular to the plane of sheet

(D) If an electron moves parallel to V just above the sheet it will experience an upward magnetic force.

11. Figure shows the essential parts of an apparatus to demonstrate the Hall effect. Which of the following statement (s) is/are correct ?



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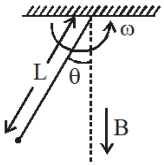
(A) In the arrangement above, the Hall potential difference is developed across PQ

(B) The magnitude of the Hall potential is greater if the applied magnetic flux density is increased

(C) The magnitude of the Hall potential is less if the width PQ of the specimen is decreased

(D) Hall potential is independent of material used for conductor

12. A charged particle having its charge to mass ratio as β goes in a conical pendulum of length L making an angle θ with vertical and angular velocity ω. If a magnetic field B is directed vertically downwards (see figure) :



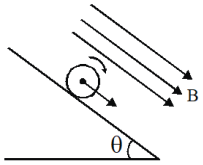
(A) $B = \frac{1}{\beta}\left[\omega - \frac{g}{\omega L \cos\theta}\right]$

(B) Angular momentum of the particle about the point of suspension remains constant.

(C) If the direction of B were reversed maintaining same ω and L, then θ will remain unchanged.

(D) Rate of change of angular momentum of the particle about the point of suspension is not a constant vector.

13. A cylindrical wire is rolling on rough inclined plane. There exist a magnetic field along the plane as shown. At a certain instant, when its angular velocity is ω, a current i is switched on across the length of cylinder in such a way that its angular velocity becomes constant. Find the correct statement(s) :-



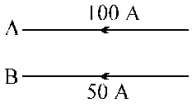
(A) translational velocity will also become constant

(B) normal reaction will become zero

(C) friction force will become zero

(D) force on the cylinder due to magnetic field will be $mg\cos\theta$.

14. Two long parallel conductors are carrying currents in the same direction as shown in the figure. The upper conductor (A) carries a current of 100 A and is held firmly in position. The lower conductor (B) carries a current of 50 A and is free to slide up and down. The linear mass density of the lower conductor is 0.01 kg m^{-1}.



(A) Conductor B will be in equilibrium if the distance between the conductors is 0.01 m^{-1}.

(B) Conductor B performs SHM with period $\frac{\pi}{50}$ s.

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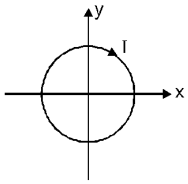
(C) Equilibrium of conductor B is stable.

(D) Equilibrium of conductor B is unstable.

15. A uniform circular loop of mass 'm' and radius R is lying in space where a uniform magnetic field $\vec{B} = B_0(\hat{i} + \sqrt{3}\hat{j})$. The loop carries constant current I in clockwise direction and lies in a gravity free region at rest in xy−plane with its centre at origin. Then choose the CORRECT statement(s) :-



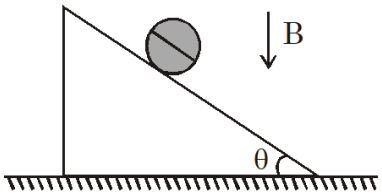
(A) The initial angular acceleration of the loop has magnitude $\frac{4I\pi B_0}{m}$.

(B) The initial angular acceleration of the loop has magnitude $\frac{2I\pi B_0}{m}$

(C) The unit vector in the direction of initial angular acceleration is $\left(+\frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j}\right)$

(D) The initial acceleration of the point with coordinate $\left(\frac{\sqrt{3}}{2}R, \frac{-R}{2}\right)$ is zero

16. In the figure shown a coil of single turn is wound on a sphere of radius R and mass M. The plane of the coil is parallel to the inclined plane and lies in the equatorial plane of the sphere. Current in the coil is i. The sphere is in equilibrium.



(A) The value of B is $\frac{mg\tan\theta}{\pi iR}$

(B) The current in coil is clockwise as seen from top.

(C) Minimum friction coefficient required is $\tan\theta$.

(D) If the surface is smooth, sphere will slide down without rotation.

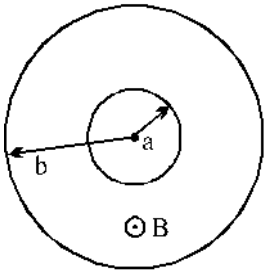
17. Outer cylinder of the coaxial conductor of radius 'b' is given a positive potential V relative to the inner cylinder of radius 'a' as shown in the fig. A static magnetic field B parallel to the axis of the cylinder, directed out of the plane of the paper is also present. Choose the correct statement from the following statements.

(i) An electron (mass = m_e) is set free with negligible velocity at the surface of the inner cylinder under condition $V = V_0$ and $B = 0$. The velocity of electron will be $v = \sqrt{\frac{2eV_0}{m_e}}$, when it hits the outer cylinder.

(ii) An electron (mass = m_e) is set free with negligible velocity at the surface of the inner cylinder under condition $V = V_0$ and $B = 0$. The velocity of electron will be $v = \sqrt{\frac{eV_0}{m_e}}$, when it hits the outer cylinder.

(iii) An electron (mass = m_e) starts out with an initial velocity u_0 in the radial direction under condition $V = 0$ and $B \neq 0$. The critical field will be $B_C = \frac{2b\,m_e\,u_0}{e(b^2 - a^2)}$ for which electron will not reach the outer cylinder.

(iv) An electron (mass = m_e) starts out with an initial velocity u_0 in the radial direction under condition $V = 0$ and $B \neq 0$. The critical field will be $B_C = \frac{4b\,m_e\,u_0}{e(b^2 - a^2)}$ for which electron will not reach the outer cylinder.



(A) (i) (B) (ii) (C) (iii) (D) (iv)

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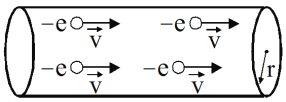
SECTION-I (iii) : (Maximum Marks: 15)

Linked Comprehension Type (Single Option Correct) (1 Para × 3 Q & 1 Para × 2 Q.)[3 M (–1)]

Paragraph for Questions 18 - 20

A beam of electrons has radius r and contains n electrons per cubic meter moving with velocity $\vec{v}$ along the beam (figure). Assume that a beam is much longer than its diameter forms a cylindrically symmetric distribution of charge and current. The beam expand if the electric force exceeds the magnetic force. So long as the expansion is slow, the approximation of cylindrical symmetry and static fields remain valid.



18. What is electric field produced on the edge of the beam?

(A) $\vec{E} = \frac{ner}{4\varepsilon_0}\hat{r}$ (B) $\vec{E} = \frac{ner}{2\varepsilon_0}\hat{r}$ (C) $\vec{E} = \frac{-ner}{4\varepsilon_0}\hat{r}$ (D) $\vec{E} = \frac{-ner}{2\varepsilon_0}\hat{r}$

19. What is the magnetic field produced on the edge of the beam?

(A) $\frac{ner}{4}\mu_0 v$ (B) $\frac{ner}{2}\mu_0 v$ (C) $ner\mu_0 v$ (D) $\frac{ner}{6}\mu_0 v$

20. What is net force on the electron at the edge of the beam?

(A) $\frac{2ne^2r}{\varepsilon_0}\left(1-\mu_0\varepsilon_0 v^2\right)\hat{r}$ (B) $\frac{ne^2r}{2\varepsilon_0}\left(1-\mu_0\varepsilon_0 v^2\right)\hat{r}$

(C) $\frac{4ne^2r}{\varepsilon_0}\left(1-\mu_0\varepsilon_0 v^2\right)\hat{r}$ (D) $\frac{ne^2r}{4\varepsilon_0}\left(1-\mu_0\varepsilon_0 v^2\right)\hat{r}$

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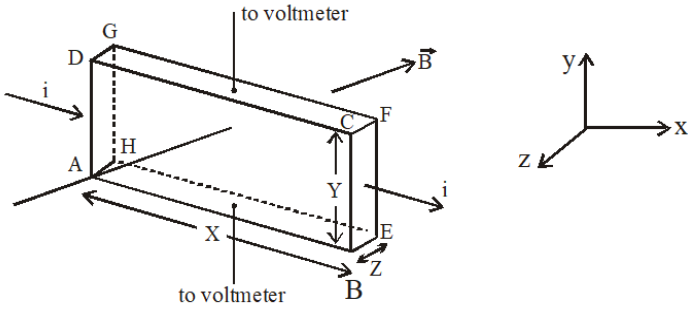
Paragraph for Questions 21 and 22

When a conductor carries a current i the charge carriers move with a drift velocity v_d directed along the length of conductor. A uniform magnetic field B is applied perpendicular to current as shown in figure. Magnetic force deflects charge carriers. As a resultant of magnetic deflection a potential difference called Hall voltage is established in transverse direction to current. In usual notations n = number of electron per unit volume, σ = conductivity of conductor, q = magnitude of charge carrier.



21. Mark the correct option :

(A) In steady state face CFGD becomes positively charged

(B) Hall voltage developed across conductor has magnitude $\frac{1}{nq}\frac{iB}{Y}$

(C) An electric field is established from face ABCD towards EFGH

(D) By decreasing Z, voltage developed can be increased.

22. Direction of resultant electric field inside conductor from x-axis is given by :

(A) $\tan\alpha = \frac{\sigma B}{nq}$ (B) $\tan\alpha = \frac{iB}{\sigma n}$ (C) $\tan\alpha = \frac{\sigma n}{iB}$ (D) $\tan\alpha = \frac{ni}{\sigma B}$

SECTION-II : (Maximum Marks: 8)

Numerical Answer type Questions (Upto Second Decimal Place) 2 Q. [4 M (0)]

1. Two coaxial long solenoids having number of turns 3000 & 1500 per metre, radii 2.0 m & 1.0 m and carry current of 2A & 4A in opposite sense respectively. Find the magnetic energy stored per unit length (in J/m). [Take $\pi^2 = 10$]

2. A 1.5 m diameter cyclotron is to accelerate protons to an energy of 8 MeV. The magnetic field strength for this cyclotron required is given by B tesla. Fill 300 B in OMR sheet. (Given mass of proton = 1.6×10^{-27} kg)

SECTION-III : (Maximum Marks: 20)

Numerical Grid Type (Ranging from 0 to 9) 5 Q. [4 M (0)]

1. A slab is inserted between two parallel plates of a capacitor charged to Q_0. The capacitor is discharged through the slab. A magnetic field is present throughout and is out of the plane of the paper. The total momentum given to the slab after complete discharge is $xdBQ_0$. Find the value of x.

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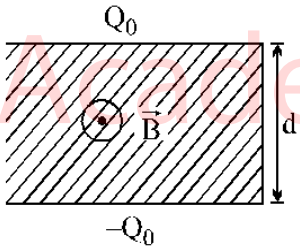
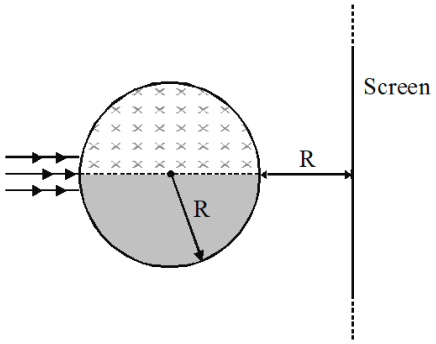
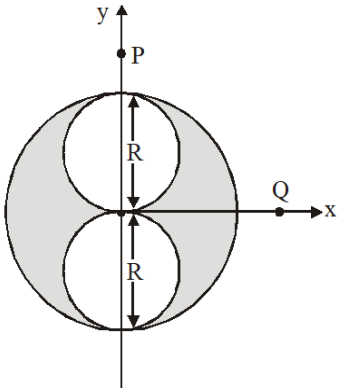


2. Figure shows circular region of radius R = $\sqrt{3}$m in which upper half has uniform magnetic field $\vec{B} = 0.2\left(-\hat{k}\right)$ T and lower half has uniform magnetic field $\vec{B} = 0.2\hat{k}$T. A very thin parallel beam of point charges each having mass m = 2 gm, speed v = 0.3 m/sec and charge q = +1 mC are projected along the diameter as shown in figure. A screen is placed perpendicular to initial velocity of charges as shown. If the distance between the points on screen where charges will strikes after being deflected by the magnetic field is 4X meters, then calculate X.



3. A long, cylindrical conductor of radius R has two cylindrical cavities of diameter R running parallel to its axis as shown in figure. The conductor carries a uniform current density J. The field strength B at position Q (x = 2R) along the x-axis is $\frac{\alpha}{68}\mu_0 JR$ then find α.



4. The current density inside a long solid cylindrical wire of radius a is in the direction of the central axis and varies linearly with radial distance r from the axis according to $J = J_0\frac{r}{a}$. If the magnetic field inside wire is given by $\frac{\mu_0 J_0 r^x}{ya}$ then find value of x + y.

5. An infinite hollow cylinder of radius R having uniform surface charge density σ is rotated with constant angular velocity ω about about its axis. Magnetic field at the centre of cylinder is $B = \mu_0\sigma R^x\omega$. Find x

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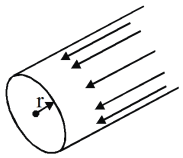
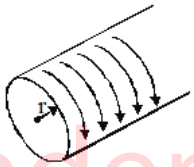
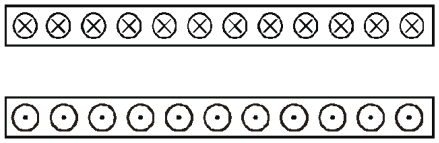
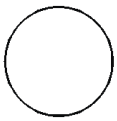
SECTION-IV : (Maximum Marks: 8)

Matrix Match Type (4 × 5) (One or more options correct) 1 Q. [8 M; (for each entry +2(0)]

1. Match the column.

Column-I		Column-II	
(A)	The pressure on walls due to quantity 'X' is tending to contract them	(P)	Current (X) is flowing axially on the surface of a long thin hollow pipe of radius r.  Energy : Magnetic field energy
(B)	The pressure on walls due to quantity 'X' is tending to expand them	(Q)	Current (X) is flowing tangentially on the surface of a long thin hollow pipe of radius r.  Energy : Magnetic field energy
(C)	The energy/volume inside the walls is more than outside.	(R)	Two current (X) carrying infinite thin sheets as shown in figure.  Energy : Magnetic field energy.
(D)	The energy/volume outside the walls is more than inside.	(S)	A soap bubble present in air containing air.The temperature inside and outside air is the same. Surface tension is (X)  Energy : internal energy of air
		(T)	Point source of light (X) is present at the centre of a spherical shell with both surfaces perfectly black. Energy : light energy.

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PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7			
	A.	D	A	C	A	D	B	A			
SECTION-I (ii)	Q.	8	9	10	11	12	13	14	15	16	17
	A.	A,B,C,D	C,D	A,D	A,B	A,D	B,C,D	A,D	A,C,D	B,C	A,C
SECTION-I (iii)	Q.	18	19	20	21	22					
	A.	D	B	B	D	A					
SECTION-II	Q.	1	2								
	A.	216.00	160.00								
SECTION-III	Q.	1	2	3	4	5					
	A.	1	3	9	5	1					
SECTION-IV	Q.	1									
	A.	A->PS,B->QRT,C->QRT,D->P									

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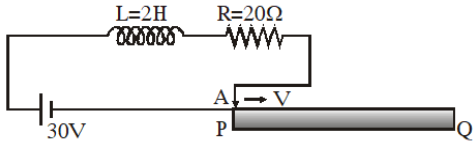
SECTION-I(i) (Maximum Marks:24)

Single Correct Answer Type 8 Q. [3 M (–1)]

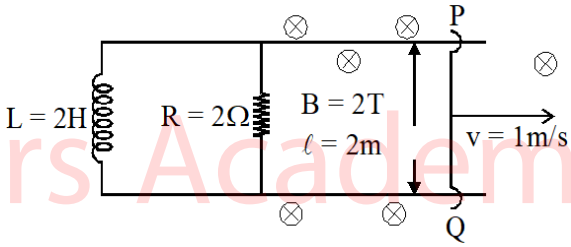
1. In the circuit shown in figure, A is a sliding contact which can move over a smooth rod PQ. Resistance per unit length of the rod PQ is 1 ohm/m. Initially slider is just left to the point P and circuit is in the steady state. At t = 0 slider starts moving with constant velocity v = 5 m/s towards right. Current in the circuit at t = 2 sec is :-



(A) 1 amp (B) less than 0.5 amp

(C) more than 1 amp (D) in between 0.5 to 1.0 amp

2. The given figure shows an inductor and resistor fixed on a conducting wire. A movable conducting wire PQ starts moving on the fixed rails from t = 0 with constant velocity 1 m/s. The work done by the external force on the wire PQ in 2 seconds is :-



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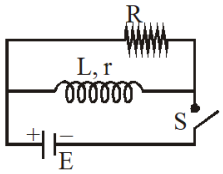
(A) 16 J (B) 32 J (C) 48 J (D) 64 J

3. A solenoid of inductance L and resistance r is connected in parallel to a resistance R. A battery of emf E and of negligible internal resistance is connected across this parallel combination as shown in figure. The circuit is in steady state. At time t = 0, switch S is opened. Choose the correct option(s).



(A) Current in the inductor just after opening of the switch is = $\frac{E(r+R)}{rR}$

(B) Total energy dissipated in the solenoid and the resistor long time after opening of the switch is $\frac{1}{2}L\frac{E^2(R+r)^2}{r^2R^2}$

(C) The amount of heat generated in the solenoid after opening of switch is $\frac{E^2L}{2r(r+R)}$

(D) The amount of heat generated in the solenoid after opening of the switch is $\frac{E^2L}{2R(r+R)}$

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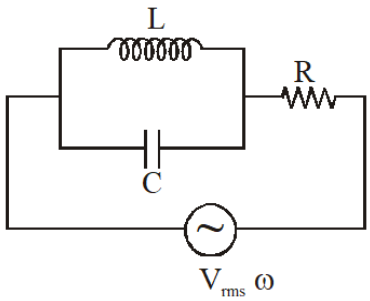
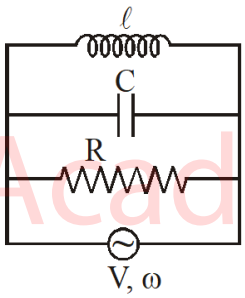
4. In the given L-C-R circuit the value of rms current through the a.c. source is



(A) $\dfrac{V_{rms}}{\sqrt{R^2+\left(\omega L-\frac{1}{\omega C}\right)^2}}$ (B) $\dfrac{V_{rms}}{\sqrt{R^2+\left(\frac{\omega L}{1-\omega^2 LC}\right)^2}}$

(C) $\dfrac{V_{rms}}{\sqrt{R^2+(\omega^2 LC-\omega L)^2}}$ (D) $\dfrac{V_{rms}}{\sqrt{(R+\omega L)^2+\frac{1}{\omega^2 C^2}}}$

5. Consider L, C, R circuit as shown in figure, with a.c. source of peak value V and angular frequency ω. Then the peak value of current through the ac source



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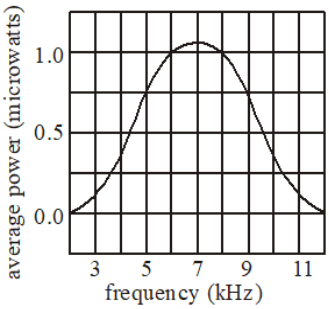
(A) $\dfrac{V}{\sqrt{\frac{1}{R^2}+\left(\omega L-\frac{1}{\omega C}\right)^2}}$ (B) $V\sqrt{\frac{1}{R^2}+\left(\frac{1}{\omega L}-\omega C\right)^2}$

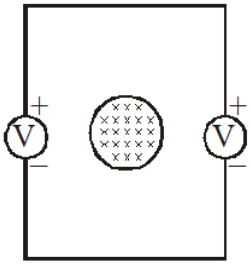
(C) $\dfrac{V}{\sqrt{R^2+\left(\omega L-\frac{1}{\omega C}\right)^2}}$ (D) $\dfrac{V R\omega C}{\sqrt{\omega^2 C^2+R(\omega^2 C^2-1)^2}}$

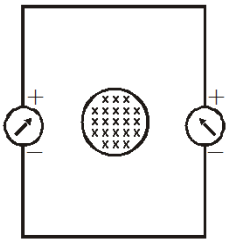
6. The plot given below is of the average power delivered to an LRC circuit versus frequency. The quality factor of the circuit is :

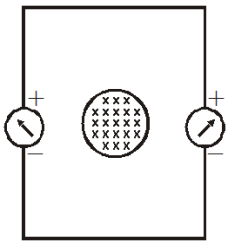


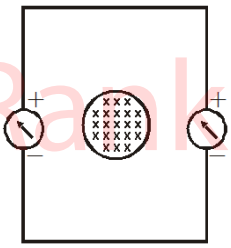
(A) 5 (B) 2.4 (C) 2.8 (D) 1.4

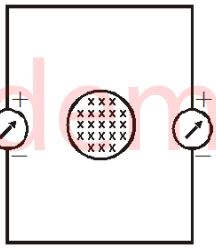
7. Two similar voltmeters with large resistance are connected in a loop as shown. Inside the loop, there is a cylindrical region through which magnetic flux is distributed uniformly. The magnetic field increases uniformly with time. The voltmeters are bidirectional i.e. the needle deflects to right if current in it flows from + terminal to − terminal and needle deflects to left if current in it flows from − terminal to + terminal. Which of the following figures show the readings of voltmeters correctly :-



(A)


(B)


(C)


(D)


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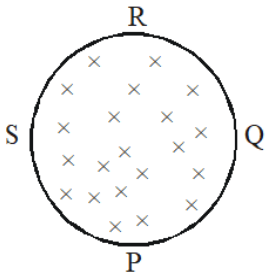
8. A long solenoid contains another coaxial solenoid (whose radius R is half of its own). Their coils have same numbers of turns per unit length and initially both carry no current. At any moment the current flowing in the inner coil is twice as large as that in the outer one and their directions are the same. As a result of the increasing current a charged particle initially at rest between the solenoids, starts moving along a circular trajectory. The radius of the circular trajectory is :

(A) $\frac{5}{4}R$ (B) $\sqrt{2}R$ (C) $\frac{7}{6}R$ (D) $\sqrt{3}R$

SECTION-I(ii) : (Maximum Marks: 24)

Multiple Correct Answer Type 6 Q. [4 M (–2)]

9. A conducting ring is placed in a uniform external magnetic field present in the space within the ring and perpendicular to plane of ring. The magnetic field changes at a constant rate due to which a current of magnitude 4 amp flows in the ring. The resistance of parts PQR and PSR are 4Ω and 8Ω respectively. Then :-



(A) The emf induced in the ring 48 V

(B) Potential difference between P & R is 24 V

(C) Potential difference between P & R is 8V

(D) Potential difference between P & R is zero as conservative electric field in the ring wire is zero

10. In figure the switch is closed at t = 0, with the capacitor of capacity 1 μF and having initial charge of 20 μC (the polarity shown). The left square loop has very large dimensions as compared to the distance between loops and a resistance of 10 Ω is connected in series with capacitor as shown. A wire of length 8 mm and resistance per unit length of 0.5 Ω/mm is bent in the form of square loop A and placed at a distance x = 1 m from left loop.

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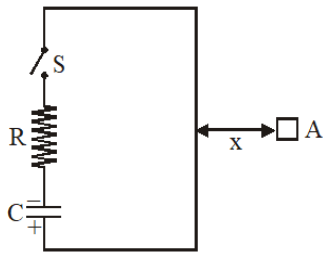


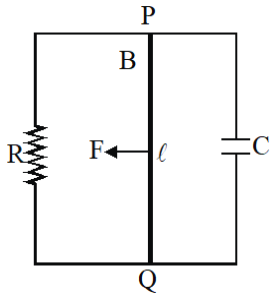
(A) The mutual inductance of the system will be 8×10^{-13} H

(B) The induced current in the loop A will be clockwise

(C) The induced current in the loop A will be anticlockwise

(D) The induced current in the loop A at time $t = 2RC\ell n2$ will be 10^{-8} A

11. A conducting rod PQ of length ℓ is dogged with a constant force F along two smooth parallel rails separated by a distance ℓ as shown in the adjacent figure. Then choose the correct statement(s).



(A) terminal velocity of the rod, $v_t = \dfrac{2FR}{B^2\ell^2}$

(B) terminal velocity of the rod, $v_t = \dfrac{FR}{B^2\ell^2}$

(C) maximum charge on the capacitor, $q_{max} = \dfrac{FCR}{B\ell}$

(D) maximum charge on the capacitor, $q_{max} = \dfrac{2FCR}{B\ell}$

12. Growth of current in two different L-R circuits are depicted by the i-t graphs shown. Angle subtended by the curves with time axis at time t = 0 are also shown in the graph. τ_1 and τ_2 are time constants for the circuits 1 and 2 respectively. Choose the correct alternative

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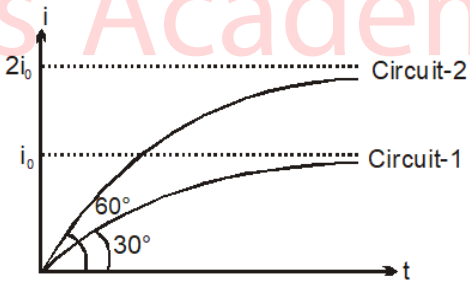


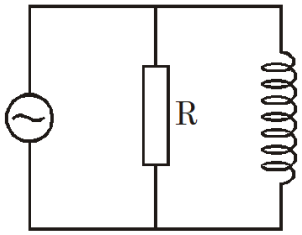
(A) $\dfrac{\tau_1}{\tau_2} = \dfrac{2}{3}$

(B) $\dfrac{\tau_1}{\tau_2} = \dfrac{3}{2}$

(C) Initial rate of change of current for circuit-2 is 2 times that of circuit-1

(D) Initial rate of change of current for circuit-2 is 3 times that of circuit-1

13. In a circuit, alternating voltage with amplitude U_o = 100 V. is applied. The phase difference between current and voltage through the source is equal $\varphi = \pi/4$. R = 100 Ohm and the inductance is resistance free. Take ω = 100 rad/s



(A) The amplitude of current is $\sqrt{2}$ amp.

(B) The average power dissipated is 50 W

(C) The inductance is 1H

(D) The amplitude of current through the inductance is 0.5 A

14. A very small circular loop of radius a is initially coplanar and concentric with a much larger circular loop of radius b (>>a). A constant current I is passed in the large loop which is kept fixed in space and the small loop is rotated with angular velocity ω about a diameter. The resistance of the small loop is R & its self inductance is negligible. The current in the larger loop is clockwise. (Here θ is to be measured from axis of large ring)

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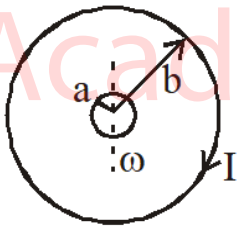


(A) The current in the small loop as a function of time is $\frac{\pi a^2 \mu_0 I \omega \sin \omega t}{2bR}$

(B) The current in the small loop as a function of time is $\frac{\pi a^2 \mu_0 I \omega \cos \omega t}{2bR}$

(C) Torque that must be exerted on the small loop to rotate it, is $\frac{\omega}{R}\left(\frac{\pi b^2 \mu_0 I \sin \omega t}{2b}\right)^2$

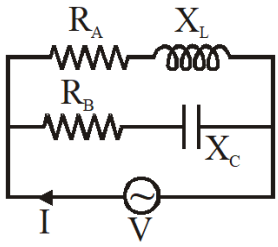
(D) Torque that must be exerted on the small loop to rotate it, is $\frac{\omega}{R}\left(\frac{\pi a^2 \mu_0 I \sin \omega t}{2b}\right)^2$

SECTION-I(iii) : (Maximum Marks: 36)

Linked Comprehension Type (Single Option Correct) (3 Para × 2 Q.) (2 Para × 3 Q.) [3 M (–1)]

Paragraph for Questions 15 and 16

Figure shows a two branched parallel circuit with $R_A = 20\Omega$, $X_L = 15\Omega$, $R_B = 6\Omega$ and $X_C = 8\Omega$. The total current supplied by alternating source V is $I = \sqrt{29}A$.



15. Current passing through the inductor (X_L) is :-

(A) Zero (B) 4A (C) 2A (D) 5A

16. Power dissipated in the circuit is :-

(A) Zero (B) 380 W (C) 920 W (D) 1400 W

Paragraph for Questions 17 and 18

In the figure a small dielectric body of mass m carries charge q. The body lies at the end of a rod of length b, having negligible mass. The other end of the rod is connected to a vertical shaft mounted on frictonless bearings so that the rod is perpendicular to the shaft and thus free to swing in a horizontal plane. At time t = 0, the rod is at rest, and a uniform magnetic field is turned on whose magnitude increases linearly with time : B = kt, where k is a constant. The direction of the field is parallel to the shaft.

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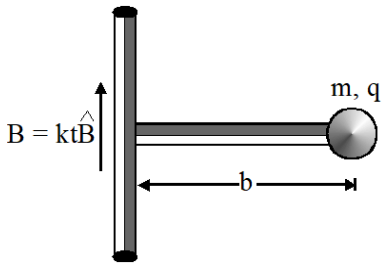


17. What is the angular acceleration of the body about the shaft.

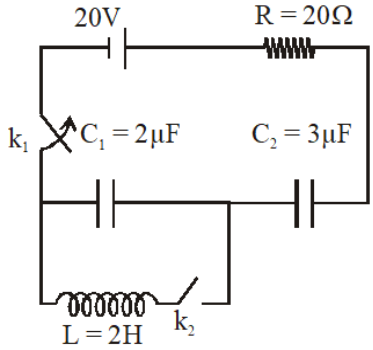
(A) $\alpha = \frac{qk}{2m}$ (B) $\alpha = \frac{2qk}{m}$ (C) $\alpha = \frac{qk}{m}$ (D) $\alpha = \frac{2qk}{3m}$

18. What is the speed v of the body as a function of time ?

(A) $v = \frac{2qkbt}{2m}$ (B) $v = \frac{qkbt}{2m}$ (C) $v = \frac{2qkbt}{m}$ (D) $v = \frac{2qkbt}{3m}$

Paragraph for Questions 19 - 21

A circuit shown in the figure in which k_1 is closed and k_2 is open. Inductor L can be connected to capacitor C_1 by closing switch k_2 and opening k_1.



19. Let switch k_1 is closed and k_2 is opened for long time. The charge on capacitor C_2 will be :-

(A) 60 μC (B) 12 μC (C) 24 μC (D) 8 μC

20. At t = 0, when capacitors are fully charged, switch k_1 is opened and switch k_2 is closed, so that inductor is connected in series with capacitor C_1. The maximum charge will appear on capacitor C_1 for first time after time t = 0 is :-

(A) $\frac{\pi}{2000}$sec (B) $\frac{\pi}{1000}$sec (C) $\frac{\pi}{500}$sec (D) $\frac{\pi}{250}$sec

21. The maximum energy across inductor will be :-

(A) 0.144 mJ (B) 0.288 mJ (C) 0.072 mJ (D) None of these

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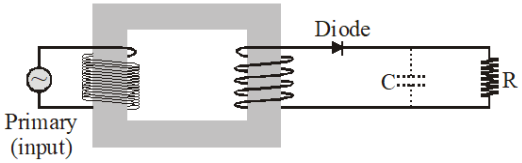
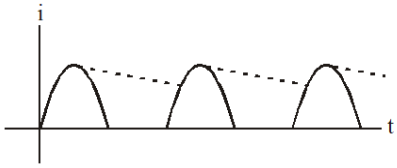
Paragraph for Questions 22 - 24

Portable electronic devices such as radios and compact disc players are often powered by direct current supplied by batteries. Many devices come with AC-DC converters such as that shown in figure (i). Such a converter contains a transformer that steps the voltage down from 120 V to, typically , 9V and a circuit that converts alternating current to direct current. The AC-DC converting process is called rectification, and the converting device is called a rectifier.

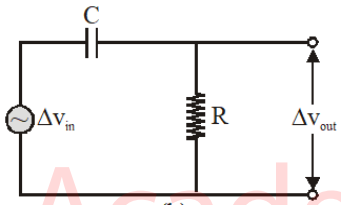


(i) A half-wave rectifier with an optional filter capacitor



(ii) Current versus time in the resistor. The solid curve represents the current with no filter capacitor, and the dashed curve is the current when the circuit includes the capacitor.

We can also design filters that respond differently to different frequencies. Consider the simple series RC circuit shown in figure. The input voltage is across the series combination of the two elements. The output is the voltage across the resistor.



(b)

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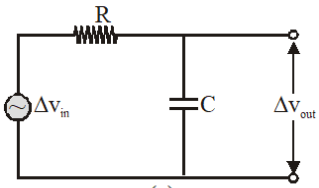
Now consider the circuit shown in figure, where we have interchanged the resistor and capacitor and where the output voltage is taken across the capacitor. At low frequencies, the reactance of the capacitor and the voltage across the capacitor is high. As the frequency increases, the voltage across the capacitor drops. Therefore, this filter is an RC low-pass filter.



(c)

22. Consider the filter circuit shown in Figure (b), the ratio of the input, output voltage is :-

(A) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{R}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

(B) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{R}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

(C) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{R}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

(D) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{R}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

23. What value of $\frac{\Delta v_{out}}{\Delta v_{in}}$ does approach as the frequency decreases toward zero ?

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) Zero

24. Consider the filter circuit shown in figure (c). The ratio of the input, output voltage is :-

(A) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{1/\omega C}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

(B) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{1/\omega C}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

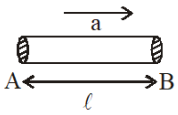
(C) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{1/\omega C}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

(D) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{1/\omega C}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

Paragraph for Questions 25 and 26

If a conductor is accelerated, initially the free electrons lag behind, thus setting up an electric field in the conductor. In steady state this field exerts force on free electrons and make them accelerate along with the conductor. (Historically this was used by Tolman-Stewant in 1916 for their experiments to measure charge to mass ratio).

25. If the above rod accelerates with acceleration along rod length and v_A and v_B are potential of end A and B respectively, then



(A) Potential of point A is less than that of B

(B) Potential of point B is less than that of A

(C) A constant electric field appears in the rod from B to A

(D) A constant electric field appears in the rod from A to B

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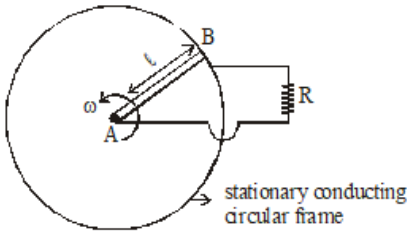
26. Figure shows a rod AB that can rotate with uniform angular velocity ω over a stationary conducting circular frame and resistance R is connected between centre A and frame.



(A) If the current was found to be I then the charge to mass ratio of electrons would be $\frac{\omega^2 \ell^2}{2IR}$

(B) If the current was found to be I then the charge to mass ratio of electrons would be $\frac{2\omega^2 \ell^2}{IR}$

(C) If a magnetic field B is applied perpendicular to plane of the paper then induce current (due to motional emf only) in rod will be $\frac{1}{2}\frac{B\ell^2\omega}{R}$

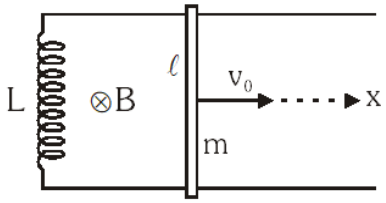
(D) If a magnetic field B is applied perpendicular to plane of the paper then induce current in rod will be radially outward

SECTION-III : (Maximum Marks: 16)

Numerical Grid Type (Ranging from 0 to 9) 4 Q. [4 M (0)]

1. Two coaxial long solenoids of equal lengths have current i_1, i_2, number of turns per unit length n_1, n_2 and radius 3m, 5m respectively. If $n_1i_1 = n_2i_2 = 250$ (in SI unit) and the two solenoids carry current in opposite sense. Find the magnetic energy stored per unit length (in SI unit). (Take $\pi^2 = 10$)

2. A loop is formed by two parallel conductors connected by a solenoid with inductance L = 2H and a conducting rod of mass m = 80 g, which can freely (without friction) slide over the conductors. The conductors are located in a horizontal plane in a uniform vertical magnetic field with induction B = 4T. The distance between the conductors is equal to ℓ = 4cm. At the moment t = 0 the rod is imparted an initial velocity v_0 = 2 cm/s directed to the right. Find the amplitude (in cm) for the oscillations of conducting rod, if the electric resistance of the loop is negligible.



3. A disc of radius r is made of a material of negligible resistance and can rotate about a horizontal shaft. A smaller disc of radius ρ is fixed onto the same shaft and has a massless cord wrapped around it, which is attached to a small object of mass m as shown. Two ends of a resistor of resistance R are connected to the perimeter of the disc and to the shaft by wiping contacts. The system is then placed into a uniform horizontal magnetic field B and mass m is released. Find the constant angular velocity (in rad/s) with which the disc will rotate after a certain time. If your answer is ω give value of $\frac{\omega}{20}$. Data : r = 10 cm, ρ = 2 cm, R = 0.01 Ω, B = 0.2 T, m = 50 g.

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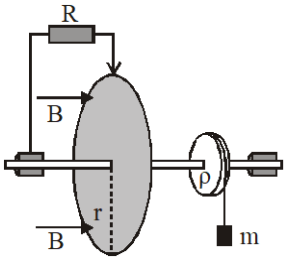
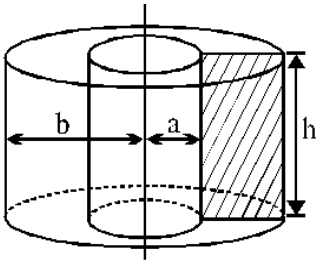


4. A ring of a rectangular cross section (see figure) is made of a material whose resistivity is ρ. The ring is placed in a homogeneous magnetic field. The intensity of the magnetic field is directed along the axis of the ring and increases directly with times, B = kt. The current induced in the ring is $\frac{xkh}{8\rho}(b^2 - a^2)$. Find the value of x.



ANSWER KEY

GR_EMI & AC

PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5	6	7	8		
	A.	C	B	C	B	B	D	A	B		
SECTION-I(ii)	Q.	9	10	11	12	13	14				
	A.	A,C	A,B,D	B,C	B,D	A,B,C	A,D				
SECTION-I(iii)	Q.	15	16	17	18	19	20	21	22	23	24
	A.	C	B	A	B	C	C	A	A	D	A
	Q.	25	26								
	A.	A,C	A,C								
SECTION-III	Q.	1	2	3	4						
	A.	2	5	5	2						

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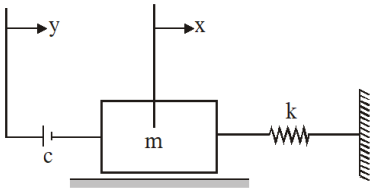
SECTION-I(i) (Maximum Marks:45)

Single Correct Answer Type 15 Q. [3 M (–1)]

1. The differential equation governing the vibrating system is shown in diagram where c is a viscous damper with damping coefficient c. Select the correct equation of motion.



(A) $m\ddot{x} + c\dot{x} + k(x - y) = 0$

(B) $m(\ddot{x} - \ddot{y}) + c(\dot{x} - \dot{y}) + kx = 0$

(C) $m\ddot{x} + c(\dot{x} - \dot{y}) + kx = 0$

(D) $m(\ddot{x} - \ddot{y}) + c(\dot{x} - \dot{y})k(x - y) = 0$

2. In a forced vibration with viscous damping, maximum amplitude occurs when forced frequency is:

(A) equal to natural frequency.

(B) slightly less than natural frequency.

(C) slightly greater than natural frequency.

(D) zero

3. A vibrating system consists of a mass 12.5 kg, a spring with stiffness 1000 N/m and a dashpot with damping factor of 15 N-s/m. The value of critical damping of the system is:

(A) 0.223 Ns/m (B) 17.88 Ns/m (C) 71.4 Ns/m (D) 223.6 Ns/m

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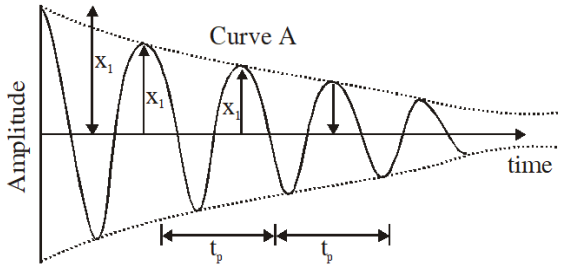
4. The amplitude versus time curve of as damped-free vibration is shown in the below figure. Curve labelled 'A' is :



(A) a longarithmic decrement curve.

(B) an exponentially decreasing curve.

(C) a hyperbolic curve.

(D) a linear curve.

5. When the mass of a critically damped single degree of freedom system is deflected from its equilibrium position and released, it will

(A) return to equilibrium position without oscillation.

(B) oscillate with increasing time period.

(C) oscillate with decreasing amplitude.

(D) oscillate with constant amplitude.

6. If a spring-mass-dashpot system is subjected to excitation by a constant harmonic force, then at resonance, its amplitude of vibration will be:

(A) infinity

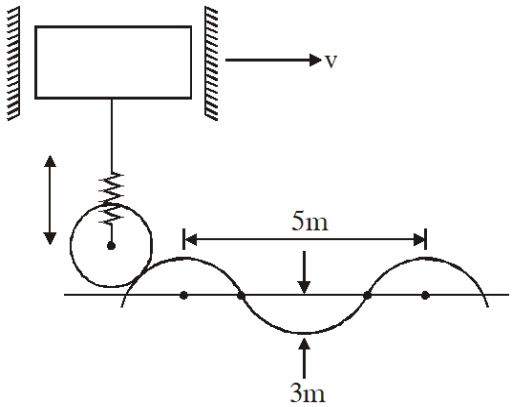
(B) inversely proportion to damping

(C) directly proportional to damping

(D) decreasing exponentially with time.

7. In a spring-mass system, the mass 0.1 kg and the stiffness of the spring is 1 kN/m. By introducing a damper, the frquency of oscillation is found to be 90% of the original value. What is the damping factor of the damper?
(A) 1.2 N.s/m (B) 3.4 N.s/m (C) 8.7 N.s/m (D) 12.0 N.s/m

8. The figure below shows the schematic of an automobile having a mass of 900 kg and the suspension spring constant of 81×10^4 N/m. If it travels at a speed of 72 km/hr on a rough road with periodic waviness as shown, what is the forcing frequency of the road on the wheel?



(A) 10 Hz (B) 4 Hz (C) 1.5 Hz (D) 20 Hz

9. An LCR circuit having a resistor of 20 Ω and inductive reactance is 200 H. After how many oscillations the ratio of current to its maximum value will become $\frac{1}{e}$?
(A) 10 (B) $\frac{10}{\pi}$ (C) 100 (D) $\frac{100}{\pi}$

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10. A spring mass system executes damped harmonic oscillations given by the equation $y = Ae^{-\frac{bt}{2m}} \sin(\omega' t + \phi)$ where the symbols have their usual meanings.
If a 2 kg mass (m) is attached to a spring of force constant (K) 1250 N/m, the period of the oscillation is $(\pi/12)$s. The damping constant 'b' has the value.
(A) 9.8 kg/s (B) 2.8 kg/s (C) 98 kg/s (D) 28 kg/s

11. A simple pendulum of length 1m (consider $\pi^2 = g$) has a time period T when there is no air resistance. If a small air resistance is present during oscillation than :-
(A) The time period will be initially more than 2s and decreases with time.
(B) The time period will be less than 4s initially and increases with time
(C) The time period will be greater than 4s and decrease with time
(D) The time period will be more than 2s and remains constant.

12. A small ball connected to a thread is submerged in a liquid and made to oscillate like a simple pendulum. The time period is T_1 if we consider buoyant force as well as viscous force . If there is no viscous force but there is a buoyant force, the time period is T_2 :
(A) T_1 can be equal to T_2 (B) T_1 is always equal to T_2
(C) $T_1 > T_2$ (D) $T_1 < T_2$

13. Critical damping is the

(A) largest amount of damping for which no oscillation occurs in free vibration

(B) smallest amount of damping for which no oscillation occurs in free vibration

(C) largest amount of damping for which the motion is simple harmonic in free vibration.

(D) smallest amount of damping for which the motion is simple harmonic in free vibration

14. Which of the following statements are TRUE for damped vibrations?

(P) For a system having critical damping, the system does not undergo a vibratory motion.

(Q) Logarithmic decrement method is used to determine the amount do damping in a physical system.

(R) In case of damping due to dry friction between moving surfaces resisting force of constant magnitude acts opposite to the relative motion.

(S) For the case of viscous damping, drag force is directly proportional to the square of relative velocity.

(A) P and Q only (B) P and S only (C) P, Q and R only (D) Q and S only

15. A thin strip 10 cm long is on a U shaped wire of negligible resistance and it is connected to a spring of spring constant 0.5 Nm^{-1} (see figure). The assembly is kept in a uniform magnetic field of 0.1 T. If the strip is pulled from its equilibrium position and released, the number of oscillation it performs before its amplitude decreases by a factor of e is N. If the mass of the strip is 50 grams, its resistance 10 Ω and air drag negligible, N will be close to :

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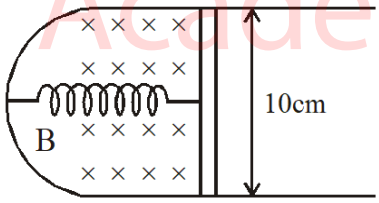


(A) 50000 (B) 5000 (C) 10000 (D) 1000

SECTION-I(ii) : (Maximum Marks: 8)

Multiple Correct Answer Type 2 Q. [4 M (–2)]

16. A Harmonic oscillator is represented by the equation $m\frac{d^2x}{dt^2} + \gamma\frac{dx}{dt} + kx = 0$ With m = 0.25 kg, γ = 0.070 kg/s and k = 85 N/m. For the given harmonic oscillator.

(A) The time period of oscillation is nearly 0.34 sec.

(B) The number of oscillations in which its mechanical energy will drop to one-half of its initial value is approximately 7.

(C) The number of oscillations in which its mechanical energy will drop to one-half of its initial value is approximately 14.

(D) Quality factor of the oscillator is close to 66.

17. A block of mass 1 kg is connected to a horizontal spring of spring constant 100 N/m. The system is made to oscillate in the horizontal direction by oscillating the support point of the spring with an angular frequency of 20 rad/sec and an amplitude of 10 cm. (There is negligible damping)

(A) The amplitude of oscillation of the mass is 3.33 cm

(B) The amplitude of oscillation of the mass is 5 cm

(C) The block oscillates in the same phase as the support point.

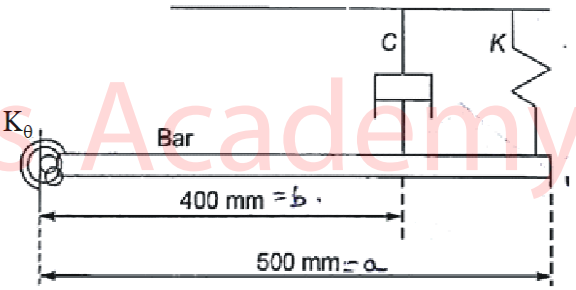
(D) The block oscillates with a phase difference of 180° with the support point.

SECTION-I(iii) : (Maximum Marks: 6)

Linked Comprehension Type (Single Option Correct) 1 Para × 2 Q. [3 M (–1)]

Paragraph for Questions 18 and 19

A uniform rigid slender bar of mass 10 kg, hinged at the left end is suspended with the help of spring and damper arrangement as shown in the figure where K = 2 kN/m, C = 500 Ns/m and the stiffness of the torsional spring K_e is 1 kN /m/rad. Ignore the hinge dimensions.



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18. The undamped natural frequency of oscillators of the bar about the hinge point is

(A) 42.43 rad/s (B) 30 rad/s (C) 17.32 rad/s (D) 14.14 rad/s

19. The damping coefficient in the vibration equation is given by

(A) 500 Nms/rad (B) 500 N/(m/s) (C) 80 Nms/rad (D) 80 N/(m/s)

SECTION-II : (Maximum Marks: 12)

Linked Comprehension Type (Upto Second Decimal Place) 1 Para × 2 Q. [4 M (0)]

Paragraph for Questions 1 and 2

The energy of recoil of a rocket launcher of mass m = 4500 kg is absorbed in a recoil spring. At the end of the recoil, a damping dashpoot is arranged in such a way that the launcher returns to the firing positin without any oscillation (critical damping). The launcher recoils by a distance of 3m with an initial speed of 10 ms^{-1}.

1. Find the recoil's spring constant (in kN/m)

2. Find the dashpot's coefficient of critical damping. (in kNs m^{-1}) [Given kN = 1000 N]

3. In a spring-mass system, the mass is m and the spring constant is k. The critical damping coefficient of the system is 0.1 kg/s. In another spring-mass system, the mass is 2 m and the spring constant is 8 k. The critical damping coefficient (in kg/s) of this system is

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ANSWER KEY

GR_Damped & Forced Oscillations

PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	D	B	A	B	C	B	B	D
	Q.	11	12	13	14	15					
	A.	D	C	B	C	B					
SECTION-I(ii)	Q.	16	17								
	A.	A,B,D	A,D								
SECTION-I(iii)	Q.	18	19								
	A.	A	D								
SECTION-II	Q.	1	2	3							
	A.	50.00	30.00	0.4							

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Important Notes

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SECTION-I(i) (Maximum Marks:27)

Single Correct Answer Type 9 Q. [3 M (–1)]

1. A traveling wave is of the form y (x,t) = A cos (kx − ωt) + B sin (kx − ωt), which can also be written as y (x,t) = D sin (kx − ωt − ϕ) where

(A) D = A + B (B) D = |A + B| (C) $D^2 = A^2 + B^2$ (D) D = A − B

2. A thin string with linear density μ is joined to a thick string with linear density 2μ. A incident pulse is sent down the thin string toward the thick string and eventually creates reflected and transmitted pulses. Which of the following is true ?

(A) The reflected and transmitted pulses are both inverted.

(B) Neither the reflected nor transmitted pulses are inverted.

(C) The reflected pulse is inverted, but the transmitted pulse is not inverted.

(D) The transmitted pulse is inverted, but the reflected pulse is not inverted.

3. Here given snap shot of a progressive wave at t = 0 with time period = T. Then the equation of the wave if wave is going in +ve x-direction and if wave is going in −ve x-direction will be respectively. $\left(\text{Here, } T = \frac{2\pi}{\omega}\right)$

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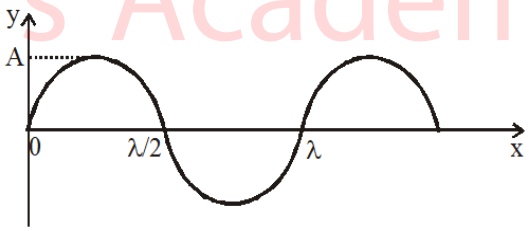
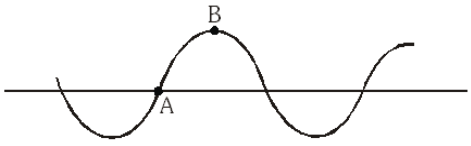


(A) y = A sin (kx + ωt), y = A sin (kx − ωt) (B) y = A cos (kx + ωt), y = A cos (kx − ωt)

(C) y = A sin (ωt − kx), y = A sin (ωt + kx) (D) y = A sin (kx − ωt), y = A sin (kx + ωt)

4. A progressive wave is travelling in a string as shown. Then which of the following statement about KE and potential energy of the elements A and B is true?



(A) For point A : kinetic energy is maximum and potential energy is min.

(B) For point B : kinetic energy is minimum and potential energy is min.

(C) For point A : kinetic energy is minimum and potential energy is max.

(D) For point B : kinetic energy is minimum and potential energy is max.

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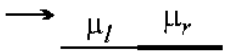
5. A string consists of two parts attached at x = 0. The right part of the string (x > 0) has mass μ_r per unit length and the left part of the string (x < 0) has mass μ_ℓ per unit length. The string tension is T. If a wave of unit amplitude travels along the left part of the string, as shown in the figure, what is the amplitude of the wave that is transmitted to the right part of the string ?



(A) 1 (B) $\frac{2}{1+\sqrt{\mu\ell/\mu_r}}$ (C) $\frac{2\sqrt{\mu\ell/\mu_r}}{1+\sqrt{\mu\ell/\mu_r}}$ (D) $\frac{\sqrt{\mu\ell/\mu_r}-1}{\sqrt{\mu\ell/\mu_r}+1}$

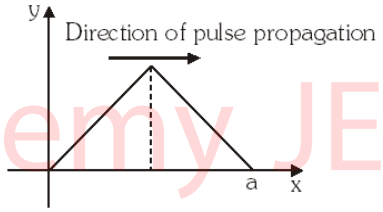
6. Five waveforms moving with equal speeds on the x-axis $y_1 = 8\sin(\omega t + kx)$; $y_2 = 6\sin(\omega t + \frac{\pi}{2} + kx)$; $y_3 = 4\sin(\omega t + \pi + kx)$; $y_4 = 2\sin(\omega t + \frac{3\pi}{2} + kx)$; $y_5 = 4\sqrt{2}\sin(\omega t - kx + \frac{\pi}{4})$ are superimposed on each other. The resulting wave is :

(A) $8\sqrt{2}\cos kx \sin(\omega t + \frac{\pi}{4})$ (B) $8\sqrt{2}\sin(\omega t - kx + \frac{\pi}{4})$

(C) $8\sqrt{2}\sin kx \cos(\omega t + \frac{\pi}{4})$ (D) $8\sin(\omega t + kx)$

7. The wave function of a triangular wave pulse is defined by the relation below at time t = 0 sec.

$$y = \begin{cases} mx & for\ 0 \le x \le \frac{a}{2} \\ -m(x-a) & for\ \frac{a}{2} \le x \le a \\ 0 & every\ where\ else,\ \text{where } m<<1 \end{cases}$$



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The wave pulse is moving in the +X direction in a string having tension T and mass per unit length μ. The total energy present with the wave pulse is :-

(A) $\frac{m^2Ta}{2}$ (B) m^2Ta (C) mTa (D) $\frac{mTa}{2}$

8. A plane progressive harmonic wave is given by the equation : $\varphi = \varphi_m\sin(2t - 3x + 4y + \pi/3)$, where x and y are in meters, and t is in seconds. Let $\hat{n}$ is the unit vector in the direction of wave propagation, and v is the speed of wave w.r.t. the wave medium, then :-

(A) $\hat{n} = -\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$; v = 0.4 m/s (B) $\hat{n} = \frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}$; $v = \frac{2}{3}$ m/s

(C) $\hat{n} = \frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}$; v = 0.4 m/s (D) $\hat{n} = -\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}$; v = 0.5 m/s

9. The vibrations of a string of length 600 cm fixed at both ends are represented by the equation $y = 4\sin\left(\frac{\pi x}{15}\right)\cos(96\pi t)$, where x and y are in cm and t in second. What is the maximum displacement at a point at x=5 cm?

(A) $\sqrt{3}$ cm (B) $2\sqrt{3}$ cm (C) $3\sqrt{3}$ cm (D) $4\sqrt{3}$ cm

SECTION-I(ii) : (Maximum Marks: 36)

Multiple Correct Answer Type 9 Q. [4 M (–2)]

10. The figures represent two snaps of a travelling wave on a string of mass per unit length, $\mu = 0.25$ kg/m. The two snaps are taken at time t = 0 and at t = $\frac{1}{24}$s. Then the possible solution for wave are :

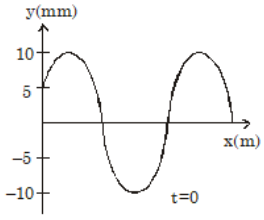

Figure-1

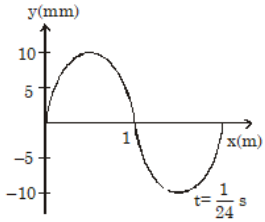

Figure-2

(A) speed of wave is 4 m/s.

(B) the tension in the string is 4 N

(C) the equation of the wave is $y = 10 \sin\left(\pi x - 4\pi t + \frac{\pi}{6}\right)$

(D) the maximum velocity of the particle = $\frac{\pi}{25}$m/s

11. $y(x, t) = 0.8/[(4x + 5t)^2 + 5]$ represents a moving pulse, where x & y are in meter and t in second. Then:

(A) pulse is moving in +x direction

(B) in 2s it will travel a distance of 2.5 m

(C) its maximum displacement is 0.16 m

(D) it is a symmetric pulse.

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12. A string of mass 0.2 kg and length 2m is tied at two ends to fixed supports under a tension of 10 N. A point P on the string is found to travel from one extreme to other in 0.1s. Taking one end as x = 0 and the other end x = 2m and t = 0 as the time when P is at rest. (Position of P is x). The correct statements will be

(A) For $0 < t < 0.1$ s, energy flows across P in positive x-direction for $0 < x < 1$ m

(B) For $0 < t < 0.05$ s, energy flows across P in negative x-direction for $0 < x < 1$ m

(C) At t = 0.05 s, rate of energy flow through P is zero for x = 0.5 m

(D) At t = 0.1s, rate of energy flow through P for all values of x is zero

13. String of length L whose one end is at x = 0, vibrates according to the relations given by different equations. Choose the correct statement(s).

(A) $y = A \sin\frac{\pi x}{L}\sin \omega t$ has 1 antinodes, 2 nodes

(B) $y = A \cos\frac{\pi x}{L}\sin \omega t$ has 2 antinodes, 1 nodes

(C) $y = A \sin\frac{2\pi x}{L}\sin \omega t$ has 3 nodes, 2 antinodes

(D) $y = A \cos\frac{2\pi x}{L}\sin \omega t$ has 3 antinodes, 2 nodes

14. A wave disturbance in a medium is described by y (x, t) = 0.02 cos (50 πt + π/2) cos (10 πx), where x and y are in metres and t in seconds.

(A) A node occurs at x = 0.15 m

(B) An antinode occurs at x = 0.3 m

(C) The speed of the wave is 5.0 m/s

(D) The wavelength is 0.2 m

15. A waveform : $y_1 = A \sin\left(2x - 4t + \frac{\pi}{3}\right)$ is superposed with a second waveform, to produce a standing wave with a node at x = 0. The equation of the second waveform can be :-

(A) $y_2 = A \sin\left(2x + 4t + \frac{5\pi}{3}\right)$

(B) $y_2 = A \sin\left(2x - 4t + \frac{\pi}{3}\right)$

(C) $y_2 = A \sin\left(2x + 4t - \frac{\pi}{3}\right)$

(D) $y_2 = A \sin\left(2x + 4t + \frac{\pi}{3}\right)$

16. A string fixed at both ends and under tension T vibrates in its 1st overtone with an amplitude A at the antinodes. The total energy of the string is E and the maximum possible speed of a particle of the string is v. If the same string were to vibrate in its fundamental mode under a tension 4T and with an amplitude A at the antinode then :-

(A) The total energy of the string will be E

(B) The total energy of the string will be 2E

(C) The maximum possible speed of a particle on the string is v

(D) The maximum possible speed of a particle on the string is 2v

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17. A long wire ABC is made by joining two wires AB and BC of equal cross-sectional area. AB has length 4.80 m and mass 0.12 kg. BC has length 2.56 m and mass 0.4 kg. The wire ABC is under a tension of 160 N. A sinusoidal wave y = 5.6 (cm) sin (ωt − kx) is sent along the wire ABC from the end A. No power dissipates during the propagation of the wave.

(A) The amplitude of the reflected wave is 2.4 cm

(B) The amplitude of transmitted wave is 3.2. cm

(C) The maximum displacement of the nodes of the stationary wave in the wire AB is 3.2 cm

(D) The fraction of power transmitted from the junction B is approximately 0.816

18. In a travelling one dimensional mechanical sinusoidal wave

(A) potential energy and kinetic energy of an element become maximum simultaneously.

(B) all particles oscillate with the same frequency and the same amplitude

(C) all particles may come to rest simultaneously

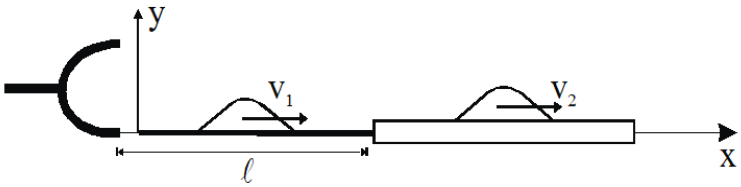
(D) we can find two particles, in a length equal to half of a wavelength, which have the same non zero acceleration simultaneously.

SECTION-I(iii) : (Maximum Marks: 24)

Linked Comprehension Type (Single Option Correct) (2 Para × 3 Q.) (1 Para × 2 Q.) [3 M (–1)]

Paragraph for Questions 19 - 21

A harmonic oscillator at x = 0, oscillates with a frequency $\frac{\omega}{2\pi}$ and amplitude a. It is generating waves at end of a thin string in which velocity of wave is v_1 and which is connected to another heavier string in which velocity of wave is v_2 as shown, length of first string is ℓ.



19. If harmonic oscillator oscillates by an equation $y = a\sin\omega t$. The equation of incident wave in first string is

(A) $y = a\sin\omega\left(t - \frac{x}{v_1}\right)$ (B) $y = a\sin\omega\left(t + \frac{x}{v_1}\right)$

(C) $y = a\sin\left[\omega\left(t - \frac{x}{v_1}\right) + \pi\right]$ (D) $y = a\sin\left[\omega\left(t + \frac{x}{v_1}\right) + \pi\right]$

20. Equation of transmitted wave in second string if its amplitude is a_t is

(A) $y = a_t\sin\omega\left(t - \frac{x}{v_2}\right)$ (B) $y = a_t\sin\omega\left(t - \frac{\ell}{v_1}\right)$

(C) $y = a_t\sin\omega\left(t - \frac{\ell}{v_1} - \frac{x-\ell}{v_2}\right)$ (D) $y = a_t\sin\omega\left(t - \frac{x}{v_2}\right)$

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21. Equation of reflected wave, if it is reflecting at the joint and amplitude of reflected wave is a_R

(A) $y = a_R\sin\omega\left(t - \frac{x}{v_2}\right)$ (B) $y = a_R\sin\left[\omega\left(t - \frac{\ell}{v_1} - \frac{\ell - x}{v_1}\right) + \pi\right]$

(C) $y = a_R\sin\left[\omega\left(t + \frac{x}{v_1}\right) + \pi\right]$ (D) $y = a_R\sin\left[\omega\left(t + \frac{2\ell + x}{v_1}\right) + \pi\right]$

Paragraph for Questions 22 - 24

A 2m string has tension 1N is fixed at both end and its is vibrating in its third harmonic with antinode amplitude 3 cm and frequency 100 Hz, then

22. Possible stationary wave equation for the vibration of the string will be (assume origin at left end of the string and x is measured in meters towards right and t is measured in seconds)

(A) $y = (0.06\text{ m})\sin\left(\frac{3\pi}{2}x\right)\cos(200\,\pi t)$ (B) $y = (0.03\text{ m})\sin\left(\frac{3\pi}{4}x\right)\cos(200\,\pi t)$

(C) $y = (0.06\text{ m})\sin\left(\frac{3\pi}{4}x\right)\cos(200\,\pi t)$ (D) $y = (0.03\text{ m})\sin\left(\frac{3\pi}{2}x\right)\cos(200\,\pi t)$

23. Total wave energy on the string will be nearly equal to

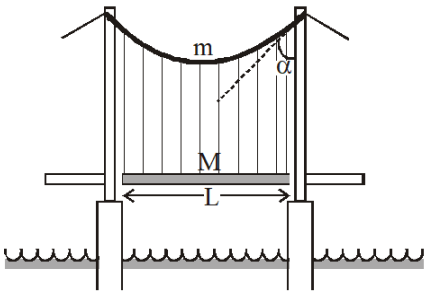
(A) 40 mJ (B) 10 mJ (C) 30 mJ (D) 20 mJ

24. At what time from the start (by the answer of first question in this paragraph) string will have maximum kinetic energy first time (second)

(A) $\frac{1}{200}$ (B) $\frac{1}{100}$ (C) $\frac{1}{400}$ (D) $\frac{1}{800}$

Paragraph for Questions 25 and 26

A suspension bridge consists of a pair of cables hung between two towers with the roadway suspended from these cables by means of closely spaced vertical wires of negligible mass as shown in figure. Each flexible cable of length ℓ (> L) and mass m hangs between two poles and mass of roadway section is M distributed uniformly along length L. At the ends, the cable makes an angle of α with the walls. (Assume each cable bears equal load and M>>m)



Hanging Bridge

25. The shape of any cable is best describe by equation assuming center of a cable is origin, vertical direction is taken as positive y-axis and horizontal right taken as positive x-axis?

(A) $y = \frac{\cot\alpha}{2L}x^2$ (B) $y = \frac{\tan\alpha}{2L}x^2$ (C) $y = \frac{\tan\alpha}{L}x^2$ (D) $y = \frac{\cot\alpha}{L}x^2$

26. If a small amplitude transverse pulse is passed through the cable, what is the speed of transverse waves at the middle ?

(A) $\frac{1}{2}\sqrt{\frac{Mg\ell\tan\alpha}{m}}$ (B) $\sqrt{\frac{Mg\ell\tan\alpha}{m}}$ (C) $\frac{1}{2}\sqrt{\frac{Mg\ell\cot\alpha}{m}}$ (D) $\sqrt{\frac{Mg\ell\cot\alpha}{m}}$

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SECTION-II : (Maximum Marks: 12)

Numerical Answer type Questions (Upto Second Decimal Place) 3 Q. [4 M (0)]

1. A steel of wire of length 25 cm is fixed at its ends to rigid walls. Young's modulus of steel = 200 GPa, coefficient of linear thermal expansion = 10^{-5} / °C. Initially, the wire is just taut at 20°C temperature. The density of steel = 8.0 g/cc. A tuning fork of frequency 200 Hz is touched to the wire, to execute oscillations. Simultaneously, the temperature is slowly lowered. At what temperature (in °C) will resonance occur corresponding to the third overtone?

2. A string of length ℓ is fixed at both ends. It is vibrating in its 3^{rd} overtone with maximum amplitude a = $2\sqrt{3}$ mm. Find the square of amplitude (in mm^2) at a distance $\frac{\ell}{3}$ from one end.

3. A string will break apart if it is placed under too much tensile stress. One type of steel has density. $\rho_{steel} = 10^4$ kg/m^3 and breaking stress $\sigma = 9 \times 10^8$ N/m^2. We make a guitar string from (4π) gram of this type of steel. It should be able to withstand $(900\ \pi)$N without breaking. What is highest possible fundamental frequency (in Hz) of standing waves on the string if the entire length of the string vibrates.

ANSWER KEY

GR_WAVE ON A STRING

PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5	6	7	8	9
	A.	C	C	D	B	C	A	B	C	B
SECTION-I(ii)	Q.	10	11	12	13	14	15	16	17	18
	A.	A,B,C,D	B,C,D	C,D	A,B,C,D	A,B,C,D	A,C	A,C	A,B,C,D	A,B,D
SECTION-I(iii)	Q.	19	20	21	22	23	24	25	26	
	A.	A	C	B	D	B	C	D	A	
SECTION-II	Q.	1	2	3						
	A.	17.5	9.00	375.00						

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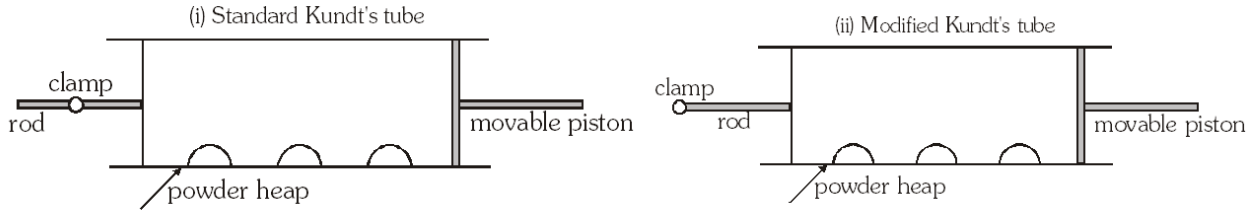
SECTION-I(i) (Maximum Marks:27)

Single Correct Answer Type 9 Q. [3 M (–1)]

1. The speed of sound in neon (Ne) at a certain temperature is V ms^{-1}. The speed of sound in hydrogen (H_2) at the same temperature will be (assume both gases to be ideal)

(A) $V = \sqrt{\frac{42}{5}}\ ms^{-1}$ (B) $V\sqrt{\frac{5}{42}}\ ms^{-1}$ (C) $V\sqrt{5}\ ms^{-1}$ (D) $\frac{V}{\sqrt{5}}\ ms^{-1}$

2. In the Kundts tube experiment (shown in fig. (i)), the rod is clamped at the end instead of clamping it at the center as shown in fig. (ii). It is known that speed of sound in air is 330 m/s, powder piles up at successive distance of 0.6 m and length of rod used is 1m, speed of sound in rod is



(A) 550 m/s (B) 1100 m/s (C) 1200 m/s (D) 600 m/s

3. Two identical wires under the same tension have a fundamental frequency of 500 Hz. The fractional increase in the tension of one wire will give 5 beats per second, is :-

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(A) 0.01 (B) 0.02 (C) 0.03 (D) 0.04

4. A sonometer has two strings stretched side by side parallel to each other. The two strings are made of the same metal and have uniform cross-sections of radii r_1 and r_2 respectively with $r_1 : r_2 :: 2 : 3$. The two strings are each of length 1m and are loaded by identical weights. The first string is made to oscillate with six loops in its length. How many loops will be seen on the second string when it resonates with the first string?

(A) 2 (B) 3 (C) 9 (D) 6

5. **Statement–1 :** When a closed organ pipe vibrates, the pressure of the gas at the closed end remains constant.

Statement–2 : In a stationary–wave system, displacement nodes are pressure antinodes, and displacement antinodes are pressure nodes.

(A) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

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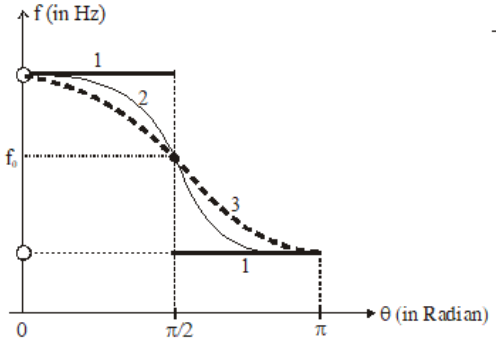
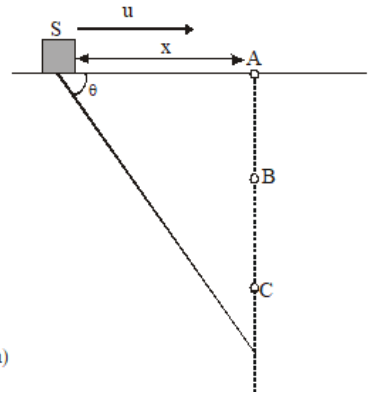
6. A bird is singing on a tree and a man is hearing at a distance 'r' from the bird. Calculate the displacement of the man towards the bird so that the loudness heard by man increases by 20 dB.
[Assume that the motion of man is along the line joining the bird and the man]

(A) $\frac{9r}{10}$ (B) $\frac{r}{10}$ (C) $\frac{3r}{5}$ (D) $\frac{4r}{5}$

7. A source is moving with a constant speed u on a straight line, emitting a sound of frequency f_0. There are three observers A, B and C. A on track, B at a perpendicular distance of d from the track and C at a perpendicular distance of 2d form the track as shown in the figure. The variation of the observed frequency with respect to the position x,





(A) A – 3, B – 2, C – 1 (B) A – 2, B – 3, C– 1
(C) A – 1, B – 2, C – 3 (D) A – 1, B – 3, C – 2

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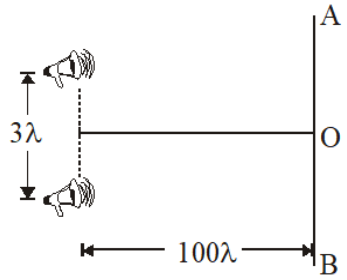
8. If $y_1 = 5$ (mm) $\sin\pi t$ is equation of oscillation of source S_1 and $y_2 = 5$ (mm) $\sin(\pi t + \pi/6)$ be that of S_2 and it takes 1 sec and ½ sec for the transverse waves to reach point A from sources S_1 and S_2 respectively then the resulting amplitude at point A, is

(A) $5\sqrt{2+\sqrt{3}}$ mm (B) $5\sqrt{3}$ mm (C) 5 mm (D) $5\sqrt{2}$ mm

9. Two loudspeakers are emitting sound waves of wavelength λ with an initial phase difference of $\frac{\pi}{2}$. At what minimum distance from O on line AB will one hear a maxima ?



(A) 25 λ (B) $\frac{100\lambda}{\sqrt{15}}$ (C) $\frac{25\lambda}{3}$ (D) 50 λ

SECTION-I(ii) : (Maximum Marks: 32)

Multiple Correct Answer Type 8 Q. [4 M (–2)]

10. A small source of sound operates at 1kHz, 24πW emitting sound uniformly in all directions. The speed of sound is 360m/s in air and density of air is 1.2kg/m^3.

(A) The intensity of sound at a distance 10m from the source is 0.06W/m^2

(B) The intensity of sound at a distance 10m from the source is 0.24W/m^2

(C) The pressure amplitude at distance 10m is 7.2 Pa

(D) The pressure amplitude at distance 10m is 28.8 Pa

11. In a resonance column experiment, the length of air column at 1st resonance is measured to be 13.1 cm with a possible indeterminate error of 0.1 cm. With the same scale, the 2nd resonance occurs at 39.2 cm. The possible wavelength of the sound can be :

(A) 52.2 cm (B) 52.4 cm (C) 52.5 cm (D) 51. 9 cm

12. A narrow steel rod of length 'L' is rigidly clamped at its mid-point and transverse standing waves of frequency 'f' are set up in it. The speed of transverse waves in the rod is 'c'.Then,

(A) The free ends of the rod must be antinodes.

(B) The fundamental frequency of the rod is c/(L)

(C) The second overtone frequency of the rod is 5c/(2L)

(D) 'f' can be any integral multiple of the fundamental frequency.

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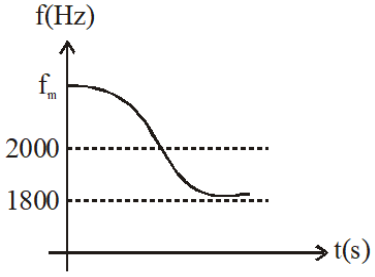
13. A stationary observer receives a sound of frequency f_0 = 2000 Hz when the source is at rest. When the source starts moving with a constant velocity starting from a large distance, the apparent frequency f varies with time as shown in figure. Speed of sound = 300 m/s. Choose the correct alternative (s):-



(A) Speed of source is 66.7 m/s.

(B) f_m shown in figure can be 2500 Hz.

(C) Speed of source is 33.33 m/s.

(D) f_m shown in figure cannot be greater than 2250 Hz.

14. Choose the incorrect one

(A) When an ultrasonic wave travels from air into water, it bends towards the normal to the air water interface

(B) Any function of the form $y(x, t) = f(vt + x)$ represents a travelling wave

(C) The velocity, wavelength and frequency of wave undergo change when it is reflected from a surface

(D) None of the above

15. A person blows into open-end of a long pipe. As a result, a high-pressure pulse of air travels down the pipe. When this pulse reaches the other end of the pipe.

(A) a high-pressure pulse starts travelling up the pipe, if the other end of the pipe is open

(B) a low -pressure pulse starts travelling up the pipe, if the other end of the pipe is open

(C) a low pressure pulse starts travelling up the pipe, if the other end of the pipe is closed

(D) a high-pressure pulse starts travelling up the pipe, if the other end of the pipe is closed

16. An open organ pipe of length 24 cm and an organ pipe closed at one end of length 38 cm have the same diameter. It is found that the 1st overtone of closed pipe is in unison with the fundamental of the open pipe. The velocity of sound in air is 340 ms^{-1} . Choose the CORRECT statement(s)

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(A) The end-correction at each open end is 1cm

(B) the fundamental frequency of the open pipe is nearly 654 Hz

(C) The frequency of 2^{nd} harmonic of open pipe will be twice its fundamental frequency

(D) The frequency of 2^{nd} harmonic of open pipe will be less than twice its fundamental frequency.

17. Mark the CORRECT statement (s) :-

(A) When two waves of different frequency f_1 and f_2 are superimposed at a point, the variation of intensity takes place with frequency $f_1 - f_2$

(B) When an observer is moving, relative speed of the sound waves with respect to the observer is changed, while due to motion of source, the sound wavelength of the wave changes.

(C) When a high pressure compression pulse reaches water surface in a resonance column, it reflects as a low pressure rarefraction pulse due to a phase change of π

(D) When a tuning fork is struck by a hammer a transverse wave sets in the prong while a longitudinal wave sets in the stem (handle) of tuning fork

SECTION-I(iii) : (Maximum Marks: 21)

Linked Comprehension Type (Single Option Correct) (1 Para × 4 Q.) (1 Para × 3 Q.) [3 M (–1)]

Paragraph for Questions 18 - 21

Imagine that you are in Indian army and you have been given charge of commanding parade on 26th January. The soldiers have done extensive training on march past with a band which strikes 120 times/min. The soldier should strike left foot or right foot at each drum beat so that parade marches at 120 steps/min.

On the day of parade, there are 2 parts. In 1st part, the band marches ahead of the soldiers with same speed as that of soldiers. In this march, you notice that the soldiers in the first row strike left foot but the soldiers in last row strike right foot at the same time. Unfortunately, the general also notices this.

In the 2nd part, you hope & pray that everything goes well. In this part, the band is stationary but the soldiers advance towards the band at same constant speed. Here again, you notice that the soldiers in the parade are doing 121 steps/min. As before, the general notices this.

A court martial is due for you. You frantically try to look for an explanation. One of your friends reminds you of your old physics teacher at Allen. You turn to him for solace. He suggests two things for defence.

(a) Sound takes time to reach from one place to another.

(b) doppler's effect

Assume that it was chilly winter & speed of sound = 320 m/sec.

18. Select the correct explanation for mistakes in each part.

(A) part I ⟶ a, part II ⟶ b (B) part I ⟶ a, part II ⟶ a

(C) part I ⟶ b, part II ⟶ a (D) part I ⟶ b, part II ⟶ b

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19. What can be the length of the column of soldiers marching past.

(A) 320 m (B) 160 m (C) 80 m (D) 40 m

20. What is the speed of the soldiers.

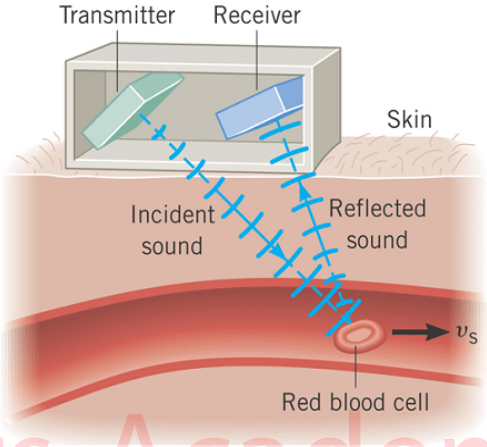
(A) 1.0 m/s (B) 2.0 m/s (C) 1.5 m/s (D) 2.5 m/s

21. What will be the marching speed of the soldiers when they turn around & march away from the band in part 2 ?

(A) 120 steps/min. (B) 119 steps/min. (C) 121 steps/min. (D) 110 steps/min.

Paragraph for Questions 22 - 24

The Doppler flow meter is a particularly interesting medical application of the Doppler effect. This device measures the speed of blood flow, using transmitting and receiving elements that are placed directly on the skin, as in Figure. The transmitter emits a continuous sound whose frequency is typically about 5 MHz. When the sound is reflected from the red blood cells, its frequency is changed in a kind of Doppler effect because the cells are moving with the same velocity as the blood. The receiving element detects the reflected sound, and an electronic counter measures its frequency, which is Doppler-shifted relative to the transmitter frequency. From the change in frequency the speed of the blood flow can be determined. Typically, the change in frequency is around 600 Hz for flow speeds of about 0.1 m/s.



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22. Assume that the red blood cell is directly moving away from the source and the receiver. What is the (approx) speed of the sound wave in the blood?

(A) 1700 m/s (B) 330 m/s (C) 5000 m/s (D) 3000 m/s

23. An abnormal segment of the artery is narrowed down by an arteriosclerotic plaque to one-fourth the normal cross-sectional area. What will be the change in frequency due to reflection from the red blood cell in that region?

(A) 150 Hz (B) 300 Hz (C) 600 Hz (D) 2400 Hz

24. At what extra rate does the heart have to work due to this narrowing down of the artery? Assume the density to be 1.5 gm/cc and the area of the normal artery to be 0.1 cm^2.

(A) 1.125×10^{-4} W (B) 2.5×10^{-4} W (C) 6.25×10^{-5} W (D) 5.625×10^{5} W

SECTION-II (i) : (Maximum Marks: 8)

Linked Comprehension Type (Upto Second Decimal Place) 1 Para × 2 Q. [4 M (0)]

Paragraph for Questions 1 and 2

A point isotropic sonic source of sound power 1 milli watts emits sound of frequency 170 Hz in all directions.

1. The distance of a point from the source where the loudness level is 60 dB is $\sqrt{\frac{50x}{\pi}}$. Find the value of x. [Take v_s = 340 m/s]

2. It is known that at some moment t = 0, the displacement of air particles at a certain point 4 m away from the source is A. The amplitude of oscillation at this point is known to be 2A. The displacement of air particles in terms of A, at a point 55 m away from the source at the moment t = 0 is $-\frac{xA}{55}$. Find the value of x. Consider displacement of particles to be positive if it is away from the source.

SECTION-II (ii) : (Maximum Marks: 4)

Numerical Answer type Questions (Upto Second Decimal Place) 1 Q. [4 M (0)]

3. AB is a cylinder of length 1m fitted a thin flexible diaphragm C at the middle and other thin flexible diaphragms A and B at the ends. The portions AC and BC contain hydrogen and oxygen gases respectively. The diaphragms A and B are set into vibrations of same frequency. What is the minimum frequency of these vibration for which diaphragm C is a node ? (Under the conditions of experiment v_{H_2} = 1100 m/s, v_{O_2} = 300 m/s).

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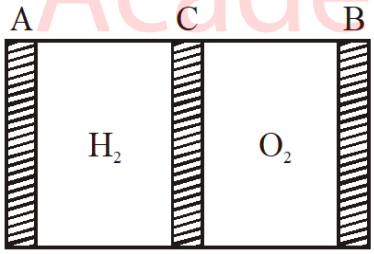


SECTION-III : (Maximum Marks: 20)

Numerical Grid Type (Ranging from 0 to 9) 5 Q. [4 M (0)]

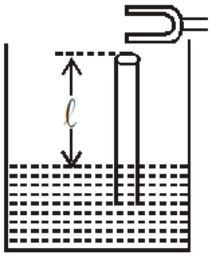
1. A super car is moving with velocity 150 m/sec towards a fixed wall producing a sound of frequency 75 Hz. Speed of sound is 300 m/sec then find the wavelength (in m) of reflected sound.

2. An open organ pipe containing air resonates in fundamental mode due to a tuning fork. The measured values of length ℓ (in cm) of the pipe and radius r (in cm) of the pipe are $\ell = 94 \pm 0.1$, $r = 5 \pm 0.05$. The velocity of the sound in air is accurately known. The maximum percentage error in the measurement of the frequency of that tuning fork by this experiment is given by α^2%. Find the value of 10α.

3. A plane longitudinal wave having angular frequency ω = 500 rad/s is travelling in positive X direction in a medium of density ρ = 1 kg/m^3 and bulk modulus 4×10^4 N/m^2 . The loudness at a point in the medium is observed to be 20dB. The pressure amplitude of the pressure wave is given has $\alpha \times 10^{-4}$ N/m^2. Find the value of α.

4. A ball of radius R = 100 cm is located in the way of propagation of a plane sound wave. The sonic wavelength is $\lambda = 2$ cm, the frequency is $\nu = 100\pi$ Hz, the pressure oscillation amplitude in air is $(\Delta p)_m = 4$ Pa. Find the mean energy flow rate (in watt), averaged over an oscillation period, reaching the surface of the ball. Take density of air 1 kg/m^3.

5. A tuning fork of known frequency is held at the open end of a long tube which is dipped into water as shown in the figure. The tuning fork of frequency 165 Hz resonates when air columns have length $\ell_1 = 50 \pm 0.5$ cm and $\ell_2 = 150 \pm 0.1$ cm. If the speed of sound in air is $[330 \pm \alpha\ (0.99)]$ m/s then the value of α is _____



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ANSWER KEY

GR_SOUND WAVE

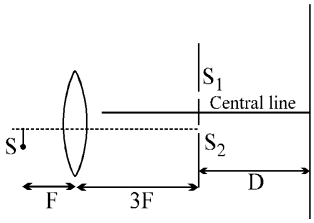
PHYSICS

SECTION-I(i)	Q.	1	2	3	4	5	6	7	8	9
	A.	A	B	B	C	D	A	C	C	C
SECTION-I(ii)	Q.	10	11	12	13	14	15	16	17	
	A.	A,C	A,B,C,D	A,C	C,D	A,C	B,D	A,B,C	A,B,D	
SECTION-I(iii)	Q.	18	19	20	21	22	23	24		
	A.	A	B	D	B	A	D	A		
SECTION-II (i)	Q.	1	2							
	A.	5.00	4.00							
SECTION-II (ii)	Q.	3								
	A.	1650								
SECTION-III	Q.	1	2	3	4	5				
	A.	2	4	2	4	2				

SECTION-I (i) (Maximum Marks:51)

Single Correct Answer Type 17 Q. [3 M (–1)]

1. Observe the set up shown in the figure. What will be the position of the central maxima on the screen? Source S is d/2 distance below the optical axis and the optical axis is equal distance below the central line. (The separation between the slits is d and separation between the plane of slit and the screen is D and focal length of lens is F.) Also F >> d, D >>d



(A) $\frac{Dd}{2F}$ (B) $\frac{Dd}{F}$ (C) $\frac{Dd}{3F}$ (D) None

2. A monochromatic light source of wavelength λ is placed at S. Three slits S_1, S_2 and S_3, source S and point P is shown in the figure. If $S_1P - S_2P = \lambda/6$ and $S_1P - S_3P = 2\lambda/3$ and I be the intensity at P when only one slit is open, the intensity at P when all the three slits are open is :-

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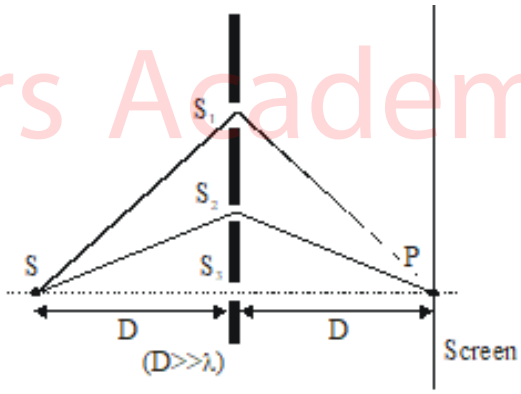
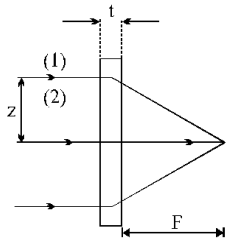


(A) 3I (B) 5I (C) 8I (D) 0

3. Plane wavefronts are incident on a glass slab which has refractive index as a function of distance Z, according to the relation $\mu = \mu_0^- (1-Z^2 / Z_0^2)$, where μ_0 is the refractive index along the axis and Z_0 is a constant. This glass slab can act as lens of focal length F. By using the concept of optical path length calculate the focal length of the slab. Consider t to be very small as compared to F.

[**Hint :** Equate the OPL of ray (1) and ray (2)]



(A) $Z_0^2/(2\mu_0 t)$ (B) $Z_0^2/(\mu_0 t)$ (C) $\mu_0 Z_0^2/(2t)$ (D) None

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4. One requires 11eV of energy to dissociate a carbon monoxide molecule into carbon and oxygen atoms. The minimum frequency of the appropriate electromagnetic radiation to achieve the dissociation lies in

(A) visible region (B) infrared region

(C) ultraviolet region (D) microwave region

5. A linearly polarized electromagnetic wave given as $\vec{E} = E_0\hat{i}\cos(kz - \omega t)$ is incident normally on a perfectly reflecting infinite wall at z = 0. Assuming that the material of the wall is optically inactive, the reflected wave will be given as

(A) $\vec{E}_r = -E_0\hat{i}\cos(kz - \omega t)$ (B) $\vec{E}_r = E_0\hat{i}\cos(kz + \omega t)$

(C) $\vec{E}_r = -E_0\hat{i}\cos(kz + \omega t)$ (D) $\vec{E}_r = E_0\hat{i}\sin(kz - \omega t)$

6. The electric field intensity produced by the radiations coming from 100 W bulb at a 3m distance is E. The electric field intensity produced by the radiations coming from 50 W bulb at the same distance is

(A) $\frac{E}{2}$ (B) 2E (C) $\frac{E}{\sqrt{2}}$ (D) $\sqrt{2}E$

7. Electromagnetic wave consists of periodically oscillating electric and corresponding magnetic vectors .....

(A) in randomly oriented planes but vibrating in phase.

(B) in mutually perpendicular planes but vibrating in phase.

(C) in mutually perpendicular planes but vibrating with a phase difference of π.

(D) in mutually perpendicular planes but vibrating with a phase difference of $\pi/2$.

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8. A flood light is covered with a filter that transmits red light. The electric field of the emerging beam is represented by a sinusoidal plane wave $E_x = 36 \sin(1.20 \times 10^7 z - 3.6 \times 10^{15} t)$ V/m. The average intensity of the beam will be-

(A) 0.86 W/m^2 (B) 1.72 W/m^2 (C) 3.44 W/m^2 (D) 6.88 W/m^2

9. An electromagnetic wave of frequency 1 x 10^{14} hertz is propagating along z-axis. The amplitude of electric field is 4 V/m. If $\varepsilon_0 = 8.8 \times 10^{-12} \frac{C^2}{N - m^2}$, then average energy density of electric field will be:

(A) 35.2×10^{-10} J/m^3 (B) 35.2×10^{-11} J/m^3 (C) 35.2×10^{-12} J/m^3 (D) 35.2×10^{-13} J/m^3

10. A plane electromagnetic wave of angular frequency ω propagates in a poorly conducting medium of conductivity σ and relative permittivity ε. Find the ratio of conduction current density and displacement current density in the medium.

(A) $\frac{\varepsilon\varepsilon_0\omega}{\sigma}$ (B) $\frac{\sigma}{\varepsilon\,\varepsilon_0\omega}$ (C) $\frac{\omega}{\sigma\,\varepsilon_0\varepsilon}$ (D) $\frac{\omega\sigma}{\varepsilon_0\varepsilon}$

11. The electric field of a plane polarized electromagnetic wave in free space at time t = 0 is given by an expression $\vec{E}(x, y) = 10\hat{j}, \cos[(6x + 8z)]$ The magnetic field $\vec{B}(x, z, t)$ is given by : (c is the velocity of light)

(A) $\frac{1}{c}\left(6\hat{k} + 8\hat{i}\right)\cos[(6x - 8z + 10ct)]$

(B) $\frac{1}{c}\left(6\hat{k} - 8\hat{i}\right)\cos[(6x + 8z - 10ct)]$

(C) $\frac{1}{c}\left(6\hat{k} + 8\hat{i}\right)\cos[(6x + 8z - 10ct)]$

(D) $\frac{1}{c}\left(6\hat{k} - 8\hat{i}\right)\cos[(6x + 8z + 10ct)]$

12. An observer is standing 90 cm away from a point source of light of power 250W. RMS value of electric field due to the source at the position of the observer is :-

(A) 48V/m (B) 96V/m (C) 40V/m (D) 80V/m

13. The Sun is directly overhead and you are facing toward the north. Light coming to your eyes from the sky just above the horizon is :-

(A) Partially polarized north-south

(B) Partially polarized east,west

(C) Partially polarized up-down

(D) Randomly polarized

14. The angle of incidence of light is equal to Brewster's angle, then

(a) reflected ray is perpendicular to refracted ray

(b) refracted ray is parallel to reflected ray

(c) reflected light is polarized having its magnetic field vector in the plane of incidence

(d) refracted light is polarized

(A) (a) and (d) are true

(B) (a) and (b) are true

(C) (a) and (c) are true

(D) (b) and (c) are true

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15. The resolving power of a lens can be improved by increasing :

(A) Diameter of the lens

(B) The object distance

(C) The wavelength of light

(D) The brightness of light

16. In single slit fraunhoffer diffraction pattern, resultant ampitude at the centre of central maxima is A_0 and it is A at an angluar position of 30º. Then $\frac{A}{A_0}$ will be - [Slit width a = λ]

(A) $\frac{\pi}{2}$ (B) $\frac{2}{\pi}$ (C) $\frac{\pi}{4}$ (D) $\frac{4}{\pi}$

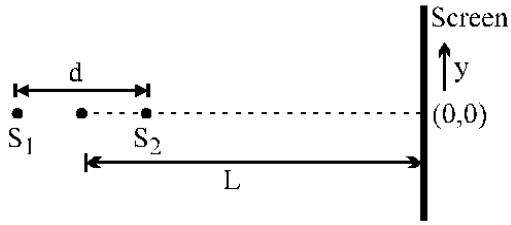
17. An electromagnetic wave with a frequency of v = 3.0 MHz passes from vacuum to a non-magnetic medium with a dielectric constant $\varepsilon_r = 4.0$. The change in its wavelength is :-

(A) 50 (B) 60 (C) 70 (D) 80

SECTION-I (ii) : (Maximum Marks: 52)

Multiple Correct Answer Type 13 Q. [4 M (–2)]

18. The figure shows two points source which emit light of wavelength λ in phase with each other and are at a distance d = 5.5 λ apart along a line which is perpendicular to a large screen at a distance L from the centre of the source. Assume that d is much less than L. Which of the following statement is (are) correct?



(A) Five bright fringes appear on the screen

(B) Six bright fringes appear on the screen

(C) Point y = 0 corresponds to bright fringe

(D) Point y = 0 corresponds to dark fringe.

19. A lens of focal length f = 40 cm is cut along the diameter into two equal halves. In this process, a layer of thickness t = 1mm is lost, then halves are put together to form a composite lens. In between focal plane and the composite lens a narrow slit is placed very close to the focal plane |u| < |f|. The slit is emitting monochromatic light of wavelength 0.6 μm. Behind the lens a screen is located at a distance L = 0.5 m from it as shown :-

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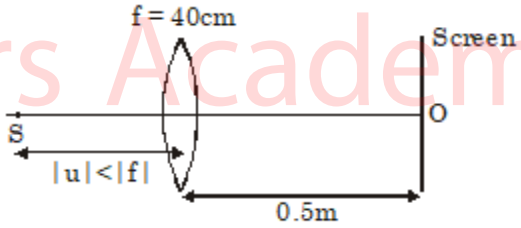


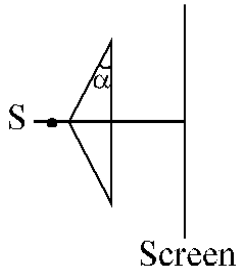
(A) Fringe width is 0.12 mm

(B) Fringe width is 0.24 mm.

(C) Length of interference pattern is 1/8 cm

(D) Length of interference pattern is 1/16 cm

20. For the biprism experiment shown in the figure, the fringe width increases when



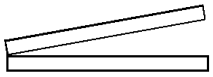
(A) biprism is moved towards the slit

(B) entire apparatus is submerged in a liquid having R.I. less than that of prism

(C) a biprism having smaller angle α is used

(D) the slit width is reduced

21. When monochromatic light is incident normally on a wedge-shaped thin air film, refer figure, an interference pattern may be seen by reflection. Which of the following is/are correct?



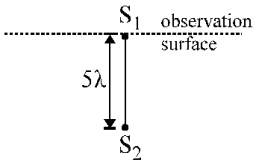
(A) Parallel fringes are observed

(B) If water is introduced into the region between the plates, the fringe separation decreases

(C) If the angle of the wedge is increased, the fringe separation decreases

(D) When white light is used there will not be a completely dark fringe

22. Two microwave coherent point sources emitting waves of wavelength λ are placed at 5λ distance apart. The interference is being observed on a flat non-reflecting surface along a line passing through one source, in a direction perpendicular to the line joining the two sources (refer figure). Considering λ as 4 mm, the positions of maxima. (Take initial phase difference between the two sources to be zero.)



(A) 48 (B) 21 (C) $\frac{32}{3}$ (D) $\frac{9}{4}$

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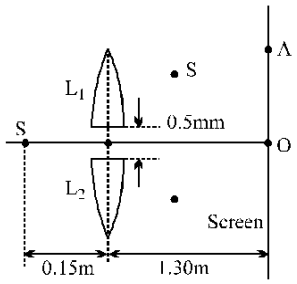
23. In the figure shown S is a monochromatic point source emitting light of wavelength λ = 500 nm. A thin lens of circular shape and focal length 0.10 m is cut into two identical halves L_1 and L_2 by a plane passing through a diameter. The two halves are placed symmetrically about the central axis SO with a gap of 0.5 mm. The distance along the axis from S to L_1 and L_2 is 0.15 m, while that from L_1 & L_2 to O is 1.30 m. The screen at O is normal to SO.



(A) If the third intensity maximum occurs at the point A on the screen, find the distance OA is 1mm.

(B) If the gap between L_1 & L_2 is reduced from its original value of 0.5 mm, will the distance OA increase

(C) If the gap between L_1 & L_2 is reduced from its original value of 0.5 mm, will the distance OA decrease

(D) If the gap between L_1 & L_2 is reduced from its original value of 0.5 mm, will the distance OA remain the same

24. An electromagnetic wave travelling along z-axis is given as: $\vec{E} = E_0 \cos(kz - \omega t)$. Choose the correct options from the following;

(A) The associated magnetic field is given as $\vec{B} = \frac{1}{c}\hat{k} \times \vec{E} = \frac{1}{\omega}\left(\hat{k} \times \vec{E}\right)$

(B) The electromagnetic field can be written in terms of the associated magnetic field as $\vec{E} = c\left(\vec{B} \times \hat{k}\right)$

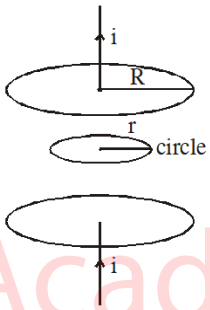
(C) $\hat{k}.\vec{E} = 0, \hat{k}.\vec{B} = 0$

(D) $\hat{k} \times \vec{E} = 0, \hat{k} \times \vec{B} = 0$

25. A plane electromagnetic wave propagating along x direction can have the following pairs of E and B

(A) E_x, B_y (B) E_y, B_z (C) B_x, E_y (D) E_z, B_y

26. A capacitor with circular plates of radius R is charged with a constant current I as shown. Which of the following are correct?



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(A) Poynting vector points radially inward everywhere on the circle.

(B) Magnetic field is clockwise as seen from above.

(C) Magnetic energy density is increasing with time.

(D) Poynting vector flux is equal to the rate of change of stored energy.

27. A long straight cable of length ℓ is placed symmetrically along z-axis and has radius a($<<$ I). The cable consists of a thin wire and a co-axial conducting tube. An alternating current $I(t) = I_0 \sin(2\pi\nu t)$ flows down the central thin wire and returns along the co-axial conducting tube. The induced electric field at a distances from the wire inside the cable is $\vec{E}(s,t) = \mu_0 I_0 \nu \cos(2\pi\nu t) \ln\left(\frac{s}{a}\right)\hat{k}$

(A) The displacement current density inside the cable is $\frac{2\pi}{\lambda^2} I_0 \ln\left(\frac{a}{s}\right) \sin(2\pi\nu t)\hat{k}$.

(B) The displacement current density inside the cable is $\frac{2\pi\nu^2}{C^2} I_0 \ln\left(\frac{a}{s}\right) \sin(2\pi\nu t)\hat{k}$.

(C) the total displacement current integrated over the cross section of the cable is $\left(\frac{a}{2}\right)^2 \left(\frac{2\pi}{\lambda}\right)^2 I_0 \sin 2\pi\nu t$

(D) the total displacement current integrated over the cross section of the cable is $\left(\frac{a}{2}\right)^2 \left(\frac{2\pi\nu}{C}\right)^2 I_0 \sin 2\pi\nu t$

28. A plane electromagnetic wave propagates in a homogeneous and isotropic medium with $\varepsilon = 3.00$ and $\mu = 1.00$. The amplitude of the wave's electric field strength is $E_m = 10.0$ V/m. Then correct option are :-

(A) The amplitude of the wave's magnetic field strength H_m is 46 mA/m.

(B) The amplitude of the wave's magnetic field strength H_m is 36 mA/m.

(C) The phase speed v of the wave $\sqrt{3} \times 10^8$ m/s

(D) The phase speed v of the wave $\sqrt{2} \times 10^8$ m/s

29. A plane electromagnetic wave propagate along the x-axis in vacuum. The magnitude of the wave's magnetic field strength is $H_m = 0.0500$ A/m.

(A) The amplitude of the wave's electric field strength $E_m = 18.84$ V/m

(B) The amplitude of the wave's electric field strength $E_m = 16.84$ V/m

(C) The time-averaged density <w> of the wave's energy is 1.57×10^{-9}.

(D) The time-averaged density <w> of the wave's energy is 1.57×10^{9}.

30. A leaky capacitor with circular plates of radius R = 12 cm has a capacitance of 5 μF and an effective resistance of 4×10^5 Ω. At t = 0, the potential difference across the plates is zero but is increasing at the constant rate of 2000 V/s.

(A) the displacement current I_D is 0.01 A

(B) the displacement current I_D is 0.02 A

(C) Displacement current equals to the conduction current at t = 2s.

(D) Displacement current equals to the conduction current at t = 4s.

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SECTION-I (iii) : (Maximum Marks: 18)

Linked Comprehension Type (Single Option Correct) (3 Para × 2 Q.) [3 M (–1)]

Paragraph for Questions 31 and 32

It is known that accelerating charges radiate electromagnetic waves. This is the method of power being radiated by a cellphone. A cell phone has an antenna in which we can assume that charges undergo oscillation with an amplitude x_0 and angular frequency ω. This average power can be assumed to depend on charge Q, x_0, ω, t_0 and velocity of light c.

31. The formula for power P is (a is dimensionless constant) :-

(A) $\frac{aq^2\omega^4x_0^2}{\epsilon_0 c^2}$ (B) $\frac{aq^2\omega^4x_0^2}{\epsilon_0 c^3}$ (C) $\frac{aq^4\omega^2x_0}{\epsilon_0^2 c^6}$ (D) $\frac{aq^4\omega^4x_0^2}{\epsilon_0^2 c^4}$

32. A cell phone radiates 3W power at 300 MHz. If $a = \frac{1}{3}$, find the number of photons being emitted per unit time ?

(A) 1.5×10^{25} / sec (B) 8×10^{23} / sec

(C) 4×10^{19} / sec (D) Information insufficient

Paragraph for Questions 33 and 34

A parallel-plate capacitor with round plates is charged with steady current during the time τ to a voltage of U. The separation distance is d. Imagining that there is a cylindrical surface between the plates coaxial to them whose radius r is much smaller than the radius of the plates.

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33. The magnitude of the Poynting vector at points of the surface is :-

(A) $\frac{r\varepsilon_0u^2}{2\tau^2d^2}t$ (B) $\frac{r\varepsilon_0u^2}{\tau^2d^2}t$ (C) $\frac{r^2\varepsilon_0u^2}{2\tau^2d^3}t$ (D) $\frac{r\varepsilon_0u^2}{2\tau^2d}t$

34. the amount of energy W flowing through the surface during the time t. Compare W with the energy of the electric field contained in the volume V confined by the surface after completion of the charging process.

(A) $\frac{3\varepsilon_0u^2\pi r^2}{2d}$ (B) $\frac{\varepsilon_0u^2\pi r^2}{2d}$ (C) $\frac{\varepsilon_0u^2\pi r^3}{2d^2}$ (D) $\frac{\varepsilon_0u^2\pi d^2}{2r}$

Paragraph for Questions 35 and 36

The current in a very long solenoid increases uniformly form zero to I during the time t. The number of solenoid turns per unit length is n. Imagining inside the solenoid in its middle part a closed surface coaxial to it of length ℓ and radius r.

35. the magnitude of the Poynting vector at points of the surface;

(A) $\frac{\mu_0n^2I^2r}{2\tau^2}t$ (B) $\frac{3\mu_0n^2I^2r}{2\tau^2}t$ (C) $\frac{3\mu_0n^2I^2r}{2\tau}t$ (D) $\frac{\mu_0n^2I^2r^2}{2\tau^2}t$

36. The amount of energy W flowing through the surface during the time t. Compare W with the energy of the magnetic field contained in the volume V confined by the surface after the current I has set in.

(A) $\mu_0n^2I^2\pi r^2\ell$ (B) $\frac{\mu_0n^2I^2\pi r^2\ell}{4}$ (C) $\frac{\mu_0n^2I^2\pi r^2\ell}{2}$ (D) $\frac{\mu_0n^2I^2\pi r^2\ell}{8}$

SECTION-II : (Maximum Marks: 16)

Numerical Answer type Questions (Upto Second Decimal Place) 4 Q. [4 M (0)]

1. In a Young's experiment, the upper slit is covered by a thin glass plate of refractive index 1.4 while the lower slit is covered by another glass plate having the same thickness as the first one but having refractive index 1.7. Interference pattern is observed using light of wavelength 5400 Å. It is found that the point P on the screen where the central maximum (n = 0) fell before the glass plates were inserted now has 3/4 the original intensity. It is further observed that what used to be the 5th maximum earlier, lies below the point P while the 6th minimum lies above P. Calculate the thickness of the glass plate (in μm). (Absorption of light by glass plate may be neglected)

2. A thin film of a specific material can be used to decrease the intensity of reflected light. There is destructive interference of waves reflected from upper and lower surfaces of the film. These films are called anti-reflection coating. Magnesium fluoride (MgF_2) is used as anti-reflecting coating on plane glass surface for light having wavelength 500 nm and normal incidence. The minimum thickness of film required is 100 nm. Find refractive index of magnesium fluoride. It is known that refractive index of glass is more than refractive index of magnesium fluoride.

3. Magnitude of average poynting vector at the surface of sun is $n \times 10^7 \frac{W}{m^2}$. Find n. (Given that power radiated by sun is 3.8×10^{26} W and radius of sun is 7×10^8 m)

4. Earth recieves $2\frac{cal}{min\ cm^2}$ solar energy the amplitude of electric field at Earth's surface is.

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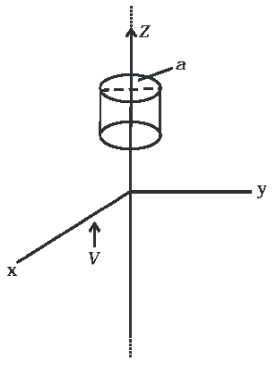
SECTION-III : (Maximum Marks: 8)

Numerical Grid Type (Ranging from 0 to 9) 2 Q. [4 M (0)]

1. An infinitely long thin wire carrying a uniform linear static charge density λ is placed along the z-axis figure. The wire is set into motion along its length with a uniform velocity $\vec{v} = v\hat{k}_z$. the pointing vector at point (a, 0, 0) is $\frac{n\lambda^2 v}{8\pi^2\varepsilon_0 a^2}\hat{k}$. Fill n in OMR Sheet.

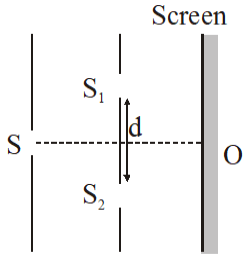
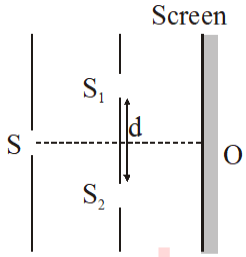
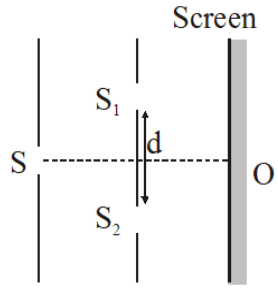
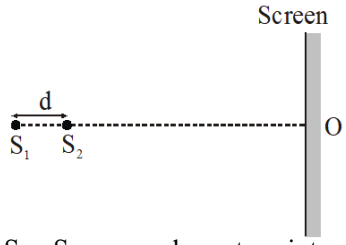


2. A plane electromagnetic wave with ω of the order of 10^{10} s^{-1} propagates in a vacuum. The amplitude of the wave's electric vector is $E_m = 0.775$ V/m. A wave-absorbing surface having the shape of hemisphere of radius r = 0.632 m is placed in the wave's path with its vertex facing the direction of propagation of the wave. What energy W (in mJ) will this surface absorb during the time t = 1.00 sec.

SECTION-IV : (Maximum Marks: 24)

Matrix Match Type (4 × 5) (One or more options correct) 3 Q. [8 M; (for each entry +2(0)]

1. Column-I describes various arrangements to obtain interference pattern on screen.

Column-I		Column-II	
(A)	 S_1, S_2 are pin holes. $d = 100\ \lambda$	(P)	Centre of screen 'O' is dark.
(B)	 S_1, S_2 are slits. $d = 100.5\ \lambda$	(Q)	First order maxima will not be seen.
(C)	 S_1, S_2 are pin holes. $d = 0.5\ \lambda$	(R)	Shape of fringes is hyperbolic
(D)	 S_1, S_2 are coherent point sources with zero initial phase difference. $d = 2.5\ \lambda$	(S)	Shape of fringes is circular.
		(T)	Shape of fringes is linear.

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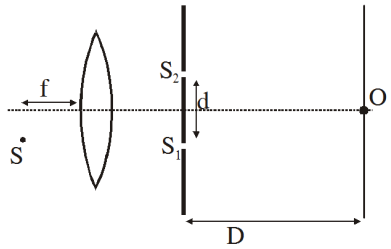
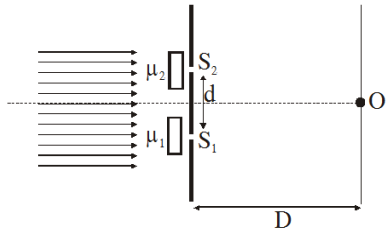
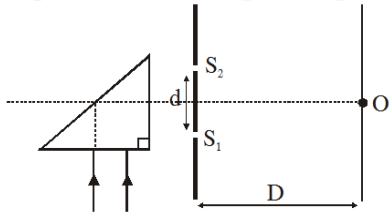
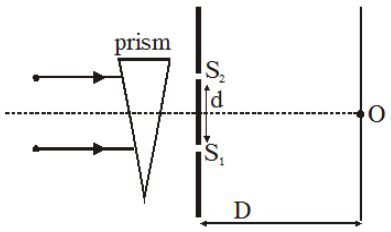
2. Column-I shows some modifications in a standard YDSE setup. Column-II shows the associated characteristics.

Column-I		Column-II	
(A)	 monochromatic point source S placed in focal plane.	(P)	Zero order maxima lies above O.
(B)	 monochromatic parallel beam incident on S_1S_2 through transparent slabs of same thickness but $\mu_1 > \mu_2$	(Q)	If a transparent mica sheet is introduced infront of S_2 central bright fringe can be obtained at O.
(C)	 monochromatic parallel beam incident on a right angled isosceles prism of refractive index 1.50.	(R)	Fringe width $\beta = \frac{\lambda D}{d}$
(D)	 monochromatic parallel beam incident on thin prism	(S)	Point O can be a minima
		(T)	Point O can be a least order minima.

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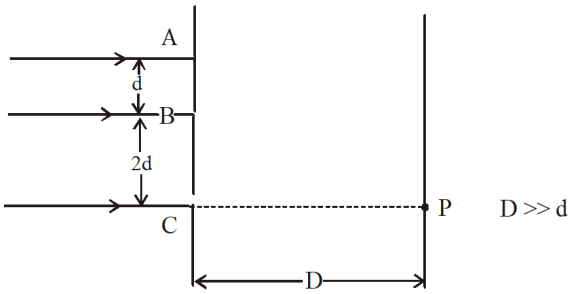
3. In the figure A, B and C are three slits. Intensity of light from each slit is I_0 at point P. Wavelength of light is λ. Match the conditions of column-I to resultant intensity at point P given in column-II.



Column-I		Column-II	
(A)	If $\frac{d^2}{D} = \lambda$	(P)	$I_p = I_o$
(B)	If $\frac{d^2}{D} = \frac{\lambda}{2}$	(Q)	$I_p = 4I_o$
(C)	If $\frac{d^2}{D} = \frac{\lambda}{4}$	(R)	$I_p = 5I_o$
(D)	If $\frac{d^2}{D} = 2\lambda$	(S)	$I_p = 9I_o$
		(T)	$I_p = 3I_o$

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ANSWER KEY

GR_WAVE OPTICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	A	C	B	C	B	B	C	B
	Q.	11	12	13	14	15	16	17			
	A.	B	B	B	C	A	B	A			
SECTION-I (ii)	Q.	18	19	20	21	22	23	24	25	26	27
	A.	A,D	B,C	A,B,C	A,B,C	A,B,C	A,B	A,B,C	B,D	A,D	A,B,C,D
	Q.	28	29	30							
	A.	A,C	A,C	A,C							
SECTION-I (iii)	Q.	31	32	33	34	35	36				
	A.	B	A	A	B	A	C				
SECTION-II	Q.	1	2	3	4						
	A.	9.3	1.25	6.17	1026.94						
SECTION-III	Q.	1	2								
	A.	2	1								
SECTION-IV	Q.	1	2	3							
	A.	A->R,B->T,C->QR,D->PS	A->PRST,B->QRST,C->R,D->PRST	A->P,B->R,C->P,D->S							

SECTION-I (i) (Maximum Marks:33)

Single Correct Answer Type 11 Q. [3 M (–1)]

1. The maximum kinetic energy of photo-electron liberated from the surface of lithium (work function = 2.35 eV) by electromagnetic radiation whose electric component varies with time as $E = a[1 + \cos(2\pi f_1 t)]\cos(2\pi f_2 t)$ where 'a' is a constant, $f_1 = 3.6 \times 1015$ Hz and $f_2 = 1.2 \times 10^{15}$ Hz is : [Take : $h = 6.6 \times 10^{-34}$ J-s]

(A) 2.6 eV (B) 7.55 eV (C) 12.5 eV (D) 17.45 eV

2. Electrons with de-Broglie wavelength λ fall on the target in an X-ray tube. The cut-off wavelength of the emitted X-rays is

(A) $\lambda_0 = \frac{2mc\lambda^2}{h}$ (B) $\lambda_0 = \frac{2h}{mc}$ (C) $\lambda_0 = \frac{2m^2c^2\lambda^3}{h^2}$ (D) $\lambda_0 = \lambda$

3. A stationary hydrogen atom of mass M emits a photon corresponding to the longest wavelength of Balmer series. The recoil velocity acquired by the atom is (R = Rydberg constant and h = plank's constant)

(A) $\frac{Rh}{M}$ (B) $\frac{Rh}{4M}$ (C) $\frac{3}{4}\frac{Rh}{M}$ (D) $\frac{5}{36}\frac{Rh}{M}$

4. In an X-ray experiment target is made up of copper (Z = 29) having some impurity. The $K\alpha$ line of copper have wavelength λ_0. It was observed that another K_α line due to impurity have wavelength $\frac{784}{625}\lambda_0$. The atomic number of the impurity element is

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(A) 22 (B) 23 (C) 24 (D) 26

5. Which one of the following statements is wrong in the context of X-rays generated from a X-ray tube ?

(A) Wavelength of characteristic X-rays decreases when the atomic number of the target increases

(B) Cut-off wavelength of the continuous X-rays depends on the atomic number of the target

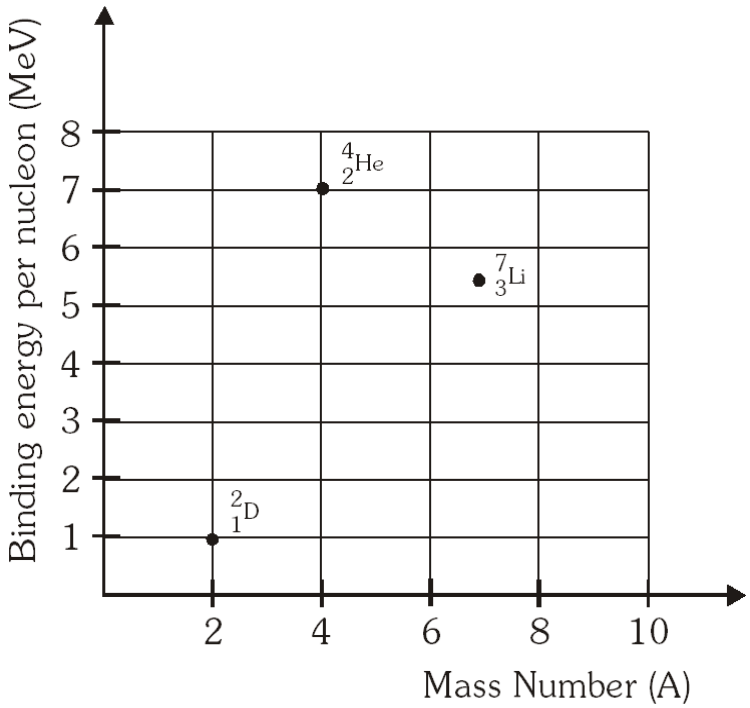
(C) Intensity of the characteristic X-rays depends on the electrical power given to the X-rays tube

(D) Cut-off wavelength of the continuous X-rays depends on the energy of the electrons in the X-ray tube

6. A parallel beam of photons of intensity I strike a plane surface at angle 45° to the surface. If plane surface have reflection coefficient 0.5 then pressure exerted by the beam on the surface (C is speed of light) is :

(A) $\frac{3I}{4C}$ (B) $\frac{3I}{2C}$ (C) $\frac{I}{2\sqrt{2}C}$ (D) $\frac{3I}{2\sqrt{2}C}$

7. The positions of ${}_1^2D$, ${}_2^4He$ and ${}_3^7Li$ are shown on the binding energy curve as shown in figure.
The energy released in the fusion reaction. ${}_1^2D + {}_3^7Li \rightarrow 2\,{}_2^4He + {}_0^1n$



(A) 20 MeV (B) 16 MeV (C) 8 MeV (D) 1.6 MeV

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8. The radionuclide ${}_6^{11}C$ decays by β^+ emission.
Given that $m({}_6^{11}C) = 11.011434$ u
$m({}_5^{11}B) = 11.009305$ u
$m_e = 0.000548$ u, $1u = 931.5$ MeV/c^2
The Q-value of this decay process is :-

(A) 0.962 MeV (B) 0.962×10^3 MeV
(C) 0.962 eV (D) 0

9. For a substance the average life for α-emission is 1620 years and for β emission is 405 years. After how much time the 1/4 of the material remains by simultaneous emission :-

(A) 648 years (B) 324 years (C) 449 years (D) 810 years

10. Tritium (${}_1^3H$), an isotope of hydrogen with a half life of 12.3 year, is produced in the upper atmosphere by cosmic rays and is intimately mixed with the hydrogen in air. In order to determine the age of a bottle of wine found in an ancient cave, the tritium in the wine is measured and found to be 6.9% of that found in new wine. How old is the wine in bottle?

(A) 59.20 years (B) 47.5 years (C) 37.5 years (D) 53.2 years

11. The electrostatic energy of Z protons uniformly distributed throughout a spherical nucleus of radius R is given by $E = \frac{3}{5}\frac{Z(Z-1)e^2}{4\pi\varepsilon_0 R}$

The measured masses of the neutron, ^1_1H, $^{15}_7N$ and $^{15}_8O$ are 1.008665 u, 1.007825 u, 15.000109 u and 15.003065 u respectively. Given that the radii of both the $^{15}_7N$ and $^{15}_8O$ nuclei are same, 1u = 931.5 MeV/c^2 (c is the speed of light) and $\frac{e^2}{(4\pi\varepsilon_0)}$ = 1.44 MeV fm. Assuming that the difference between the binding energies of $^{15}_7N$ and $^{15}_8O$ is purely due to the electrostatic energy, the radius of either of the nuclei is (1fm = 10^{-15}m)

(A) 2.85 fm (B) 3.03 fm (C) 3.42 fm (D) 3.80 fm

SECTION-I (ii) : (Maximum Marks: 12)

Multiple Correct Answer Type 3 Q. [4 M (–2)]

12. A moving neutron collides with stationary H-atom in ground state. As a result it excites and then de excites. The corresponding radiation fall on a surface, having work function σ. The minimum value of required kinetic energy for neutron is E_0 and possible minimum value of de broglie wavelength of emitted photoelectrons is λ_0. If neutron hits stationary He+ ion instead of stationary H atom, then minimum value of kinetic energy for neutron is E_1. Then

(A) The value of energy transferred from neutron to H-atom is $\frac{3E_0}{4}$.

(B) The value of λ_0 is $\frac{h}{\sqrt{m_e(E_0-2\sigma)}}$.

(C) The value of E_1 is $4E_0$

(D) The value of E_1 is $\frac{5E_0}{2}$

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13. A radioactive sample has a half life of 40 seconds. When its activity is measured 80 seconds after the begining, it is found to be 6.932×10^{18} dps. During this time total energy released is 6×10^8 joule ($\ell n2 = 0.6932$) :-

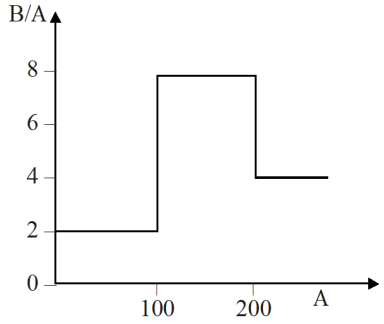
(A) The initial number of atoms in the sample is 1.6×10^{20}

(B) The initial number of atoms in the sample is 1.6×10^{21}

(C) Energy released per fission is 5×10^{-13} J

(D) Energy released per fission is $\frac{5}{3} \times 10^{-13}$ J

14. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below :



(A) Fusion of two nuclei with mass numbers lying in the range of $1 < A < 50$ will release energy

(B) Fusion of two nuclei with mass numbers lying in the range of $51 < A < 100$ will release energy

(C) Fission of a nucleus lying in the mass range of $100 < A < 200$ will release energy when broken into two equal fragments

(D) Fission of a nucleus lying in the mass range of $200 < A < 260$ will release energy when broken into two equal fragments

SECTION-I (iii) : (Maximum Marks: 24)

Linked Comprehension Type (Single Option Correct) (2 Para × 3 Q.) (1 Para × 2 Q.) [3 M (–1)]

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Paragraph for Questions 15 - 17

A light of wavelength λ is incident on a metal sheet of work function $\phi = 2$ eV. The wavelength l varies with time as $\lambda = 3000 + 40t$, where λ is in Å and t is in second. The power incident on metal sheet is constant at 100 W. This signal is switched on and off for time intervals of 2 minutes and 1 minute respectively. Each time the signal is switched on, the λ start from initial value of 3000 Å. The metal plate is grounded and electron clouding is negligible. The efficiency of photoemission is 1%

(hc = 12400 eV Å, 1eV = 1.6×10^{-19} J)

15. The time after which photoemission will stop is :-

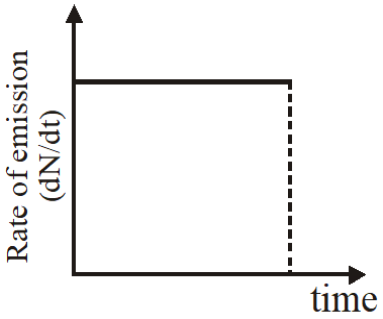
(A) 120 s (B) 80 s (C) 60 s (D) 180 s

16. The total number of photoelectrons ejected in one hour is :-

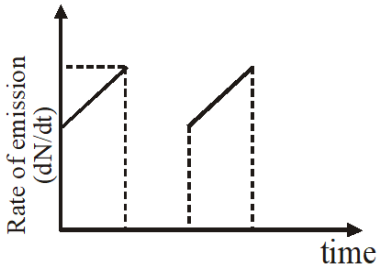
(A) 3.71×10^{17} (B) 3.71×10^{21} (C) 3.71×10^{25} (D) 3.71×10^{7}

17. The variation of rate of emission vs time is :-

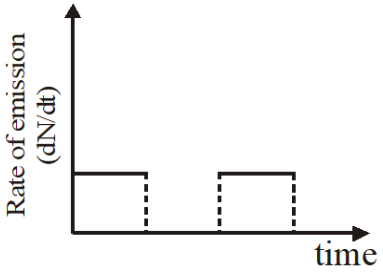
(A)



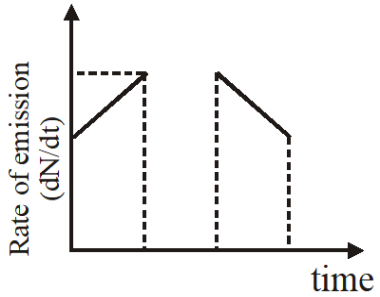
(B)



(C)



(D)



Paragraph for Questions 18 - 20

Polonium (P_0^{210}) emits ${}_2\alpha^4$ particles and is converted into lead (${}_{82}Pb^{206}$). This reaction is used for producing electric power in a space mission: P_0^{210} has half life of 138.6 days. (Given masses of the nuclei) P_0^{210} = 209.98264 amu, Pb^{206} = 205.97440, ${}_2\alpha^4$ = 4.00260 amu)

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18. Assuming an efficiency of 10% of the thermoelectric machine, how much P_0^{210} is required to produce 1.2×10^7 J of electric energy per day at the end of 693 days?

(A) 1.0 gm (B) 2.0 gm (C) 3.0 gm (D) 0.5 gm

19. What is the initial amount of P_0^{210} ?

(A) 16 gm (B) 8 gm (C) 32 gm (D) 64 gm

20. Find the initial activity of the material.

(A) 4.57×10^{21} per day (B) 3.57×10^{21} per day

(C) 2.28×10^{21} per day (D) 1.785×10^{21} per day

Paragraph for Questions 21 and 22

The β-decay process, discovered around 1900, is basically the decay of a neutron (n). In the laboratory, a proton (p) and an electron (e^-) are observed as the decay products of the neutron. Therefore, considering the decay of a neutron as a two-body decay process, it was predicted theoretically that the kinetic energy of the electron should be a constant. But experimentally, it was observed that the electron kinetic energy has a continuous spectrum. Considering a three-body decay process, i.e. $n \longrightarrow p + e^- + \vec{\nu}_e$, around 1930, Pauli explained the observed electron energy spectrum. Assuming the anti-neutrino $(\vec{\nu}_e)$ to be massless and possessing negligible energy, and the neutron to be at rest, momentum and energy conservation principles are applied. From this calculation, the maximum kinetic energy of the electron is 0.8×10^6 eV. The kinetic energy carried by the proton is only the recoil energy.

21. If the anti-neutrino had a mass of 3 eV/c^2 (where c is the speed of light) instead of zero mass, what should be the range of the kinetic energy, K, of the electron?

(A) $0 \leqslant K \leqslant 0.8 \times 10^6$eV

(B) $3.0\text{eV} \leqslant K \leqslant 0.8 \times 10^6$eV

(C) $3.0\text{eV} \leqslant K < 0.8 \times 10^6$eV

(D) $0 \leqslant K < 0.8 \times 10^6$eV

22. What is the maximum energy of the anti-neutrino?

(A) zero

(B) much less than 0.8×10^6 eV

(C) Nearly 0.8×10^6 eV

(D) Much larger than 0.8×10^6 eV

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SECTION-I (v) : (Maximum Marks: 3)

Matching list Type (Single Option Correct) 1Q. [3 M (–1)]

23. Match List I of the nuclear processes with List II containing parent nucleus and one of the end products of each process and then select the correct answer using the codes given below the lists :

	List I		List II
(P)	Alpha decay	(1)	${}^{15}_{8}O \rightarrow {}^{15}_{7}N + ...$
(Q)	β^+ decay	(2)	${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + ...$
(R)	Fission	(3)	${}^{185}_{83}Bi \rightarrow {}^{184}_{82}Pb + ...$
(S)	Proton emission	(4)	${}^{239}_{94}Pu \rightarrow {}^{140}_{57}La + ...$

(A) P ⟶ 4;Q ⟶ 2;R ⟶ 1;S ⟶ 3

(B) P ⟶ 1;Q ⟶ 3;R ⟶ 2;S ⟶ 4

(C) P ⟶ 2;Q ⟶ 1;R ⟶ 4;S ⟶ 3

(D) P ⟶ 4;Q ⟶ 3;R ⟶ 2;S ⟶ 1

SECTION-II (ii) : (Maximum Marks: 12)

Numerical Answer type Questions (Upto Second Decimal Place) 3 Q. [4 M (0)]

1. A beam of light has three wavelengths 4144 Å, 4972 Å & 6216 Å with a total intensity of 3.6×10^{-3} W.m^{-2} equally distributed amongst the three wavelengths. The beam falls normally on an area 1.0 cm^2 of a clean metallic surface of work function 2.3 eV. Assume that there is no loss of light by reflection and that each energetically capable photon ejects one electron. The number of photoelectrons liberated in two seconds are_____$x \times 10^{12}$.

2. Assume that the de-Broglie wave associated with an electron can form a standing wave between the atoms arranged in a one dimensional array with nodes at each of the atomic sites. It is found that one such standing wave is formed if the distance 'd' between the atoms of the array is 2 Å. A similar standing wave is again formed if 'd' is increased to 2.5 Å but not for any intermediate value of d. Find the energy of the electrons in electron volts :

3. A neutron of kinetic energy 65 eV collides inelastically with a singly ionized helium atom at rest . It is scattered at an angle of 90º with respect of its original direction. (Given : Mass of He atom = 4 × (mass of neutron), ionization energy of H atom =13.6 eV). If the allowed values of the energy of the neutron is x and that of the atom after collision is y then fill (x + y) in OMR.

SECTION-III : (Maximum Marks: 32)

Numerical Grid Type (Ranging from 0 to 9) 8 Q. [4 M (0)]

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1. An electron is confined to a tube of length L. Its potential energy in first half of the tube is zero, while the potential energy in other half is 12 eV. If the total energy of electron is 16 eV, find the ratio of the longest to the shortest de Broglie wavelength of an electron in the tube.

2. A potential difference of V volts is applied on two parallel electrodes separated by a distance of 4.0×10^{-2} m. The electrons of very low energy are injected into the region between the electrodes which contains argon at low pressure. The average distance the electrons travel between collisions with argon atoms is 8×10^{-5} m. The ionization energy of argon atom is 16 eV. Estimate the minimum value of V(in kV) such that the electrons will cause ionization in argon atoms by collision -

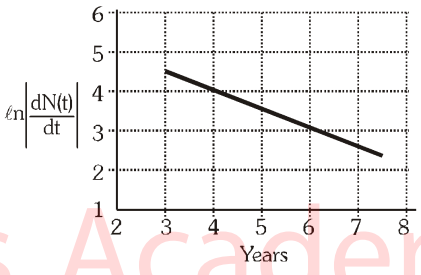
3. When 0.50 Å X-rays strike a material, the photoelectrons from the K shell are observed to move in a circle of radius 23 mm in a magnetic field of 2×10^{-2} T. The binding energy of K-shell electrons is $a^2 \times 100$ eV then write the value of a in your OMR sheet ?
(Take : mass of electron = 9.2×10^{-31} kg & hc = 12400 eV–Å)

4. Consider a universe in which the π-meson orbits around the nucleus instead of electron. Assuming a Bohr model for a π-meson of mass m_π and of the same charge as the electron is in a circular orbit of radius r about the nucleus with an orbital angular momentum $\frac{h}{2\pi}$. If the radius of a nucleus of atomic number Z is given by R = 1.6×10^{-15} $Z^{1/3}$ m. The total number of elements in this universe that can exist is given as 'N'. Fill $\left[\frac{N-1}{12}\right]$ in OMR sheet. [Given $\frac{\varepsilon_0 h^2}{\pi m_e e^2} = 0.53$ Å; $\frac{m_\pi}{m_e} = 265$; neglect any shielding effect for the havier atoms and assume non relativistic physics to be applicable and take $5^{1/4} \approx 1.5$]

5. A silver sphere of radius 1 cm and work function 4.7 eV is suspended from an insulating thread in free-space. It is under continuous illumination of 200 nm wavelength light. As photoelectrons are emitted, the sphere gets charged and acquires a potential. The maximum number of photoelectrons emitted from the sphere is $A \times 10^Z$ (where $1 < A < 10$). The value of 'Z' is

6. A small quantity of solution containing ^{24}Na radionuclide (half life 15 hours) of activity 1.0 microcurie is injected into the blood of a person. A sample of the blood of volume 1 cm^3 taken after 5 hours shows an activity of 296 distintegration per minute. Determine the total volume of blood in the body of the person. Assume that the radioactive solution mixes uniformly in the blood of the person.
(1 Curie = 3.7×10^{10} disintegrations per second)

7. To determine the half life of a radioactive element, a student plots a graph of $\ell n\left|\frac{dN(t)}{dt}\right|$ versus t. Here $\frac{dN(t)}{dt}$ is the rate of radioactive decay at time t. If the number of radioactive nuclei of this element decreases by a factor of p after 4.16 years, the value of p is



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8. Geiger counter reading of a radioactive sample is initially 6800 counts per min. The same sample gives a reading of 425 counts per min 10 hrs later. The sample's half life of α hr. Fill 2α in OMR sheet.

SECTION-IV : (Maximum Marks: 8)

Matrix Match Type (4 × 5) (One or more options correct) 1 Q. [8 M; (for each entry +2(0)]

1. Column-I shows possible inferences on events given in column-II.

	Column-I		Column-II
(A)	Not possible	(P)	Conversion of a free proton to a neutron.
(B)	Possible	(Q)	A neutron with kinetic energy 15 eV is incident on a stationary hydrogen atom in ground state on a stationary hydrogen atom in ground state such that excitation of H atom occurs.
(C)	Release of energy	(R)	Conversion of a neutron to proton outside nucleus.
(D)	A chargeless particle carrying momentum is emitted.	(S)	A nuclei emits electromagnetic radiation without emitting α or β particles.
		(T)	A thermal particle causing fission of nucleus.

PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	D	D	B	A	B	A	C	B
	Q.	11									
	A.	C									
SECTION-I (ii)	Q.	12	13	14							
	A.	A,B,D	B,C	B,D							
SECTION-I (iii)	Q.	15	16	17	18	19	20	21	22		
	A.	B	B	B	A	A	A	D	C		
SECTION-I (v)	Q.	23									
	A.	C									
SECTION-II (ii)	Q.	1	2	3							
	A.	1.1	151.00	16.640 to 24.20							
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	2	8	8	3	7	6	8	5		
SECTION-IV	Q.	1									
	A.	A->PQ,B->RST,C->RST,D->RST									

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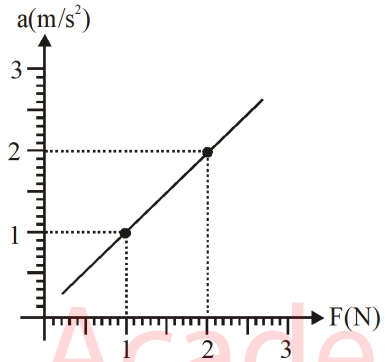
SECTION-I (i) (Maximum Marks:51)

Single Correct Answer Type 17 Q. [3 M (–1)]

1. A travelling microscope has a main scale graduated in $\frac{1}{2}$mm. 50 vernier divisions coincide with 49 main scale divisions. What will be the reading if the main scale reads 7.45 cm and the 29^{th} division of the vernier coincides with the main scale division ?

(A) 7.459 cm (B) 7.453 cm (C) 7.479 cm (D) 7.48 cm

2. In an experiment to determine the inertial mass of an object using Newton's second law, following graph is obtained between net force on the object and the acceleration produced in it. The mass of the object within error limits is :-



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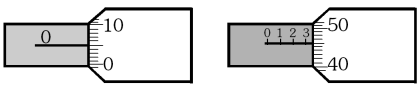
(A) 1.0 kg (B) 1 kg (C) (1.0 ± 0.1) kg (D) (1.0 ± 0.15)kg

3. The circular scale of a micrometer has 200 divisions and pitch of 2mm. Find the measured value of thickness of a thin sheet.

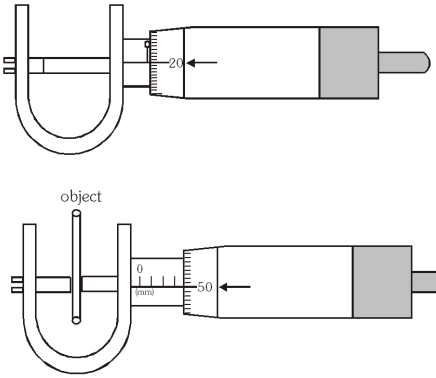


(A) 3.41 mm (B) 6.41 mm (C) 3.46 mm (D) 3.51 mm

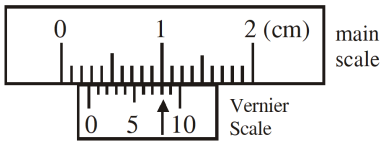
4. A screw gauge has a screw having 2 threads in 1 mm. The circular scale has 50 divisions. Find the diameter of wire, if the main scale shows 6^{th} division and the vernier reads 46.

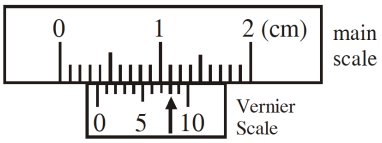
(A) 6.46 mm (B) 3.46 mm (C) 6.54 mm (D) 3.04 mm

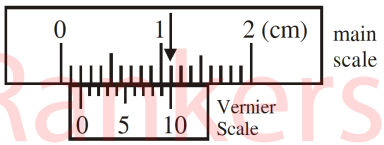
5. When a piece of wire is held diametrically in a screw gauge [pitch = 1mm, number of division on circular scale = 100]. The readings obtained are as shown :

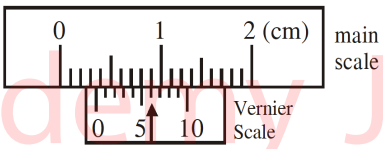


Now if we measure the same with help of vernier callipers [1 MSD = 1mm, 10 divisions of vernier coinciding with 9 divisions of main scale] having a negative zero error of 0.5 mm, then find which of the following figures correctly represents the reading.

(A) 

(B) 

(C) 

(D) 

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6. A student obtained following observations in an experiment of meter bridge to find the unknown resistance of the circuit :

S.No.	R	ℓ	$100-\ell$	$S=\left(\frac{100-\ell}{\ell}\right)R$
1	20Ω	43	57	26.51 Ω
2	30Ω	51	49	28.82 Ω
3	40Ω	59	41	27.80 Ω
4	60Ω	70	30	25.71 Ω

The most accurate value of unknown resistance is :-

(A) 26.51 Ω (B) 28.82 Ω (C) 27.80 Ω (D) 25.71 Ω

7. Relative density of the material of a body is found by weighing it first in air and then in water. The weight in air is (5.00 ± 0.05) newton and weight in water is (4.00 ± 0.05) newton. Then the relative density of material should be:-

(A) 5.01 ± 11% (B) 5.0 ± 1% (C) 5.0 ± 6% (D) 1.25 ± 5%

8. The viscosity of a fluid μ, can be determined by measuring the terminal velocity V_T of a sphere when it descends in the fluid. The fluid has a density ρ_f while the sphere has a density ρ_s and a diameter of d. The viscosity can then be calculated by the formula

$$\mu = \frac{5(\rho_s - \rho_f)}{9V_T}d^2$$

The values measured are

$V_T = (1.60 \pm 0.04)\ ms^{-1}$

$\rho_s = (2700 \pm 20)\ kg\ m^{-3}$

$\rho_f = (900 \pm 10)\ kg\ m^{-3}$

$d = (20.0 \pm 0.4)\ mm$

What is the percentage uncertainty in the value of μ?

(A) 6.2% (B) 7.1% (C) 8.2% (D) 8.4%

9. In a screw gauge, there are 100 division in circular scale and each main scale division is of 1 mm. When there is no gap between the jaws, 97^{th} division coincides with the main scale zero and zero of main scale is not visible. While measuring the diameter of a ball, the circular scale is between 3 mm mark and 4 mm mark such that the 76^{th} division of circular scale coincides with the reference line. Select the correct alternative :

(A) The least count of the micrometer is 0.01 cm

(B) The zero error is −0.04 mm

(C) The diameter of the ball is 3.79 mm

(D) There is positive zero error in the screw guage

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10. Three students make measurements of the length of a 1.50 meter rod. Each student reports an uncertainty estimate representing an independent random error applicable to the measurement.

Alice : A single measurement using a 2.0 meter long tape measure, to within ±2 mm.

Bob : Two measurements using a wooden meter stick, each to within ±2 mm, which he adds together.

Christina : Two measurements using a machinist's meter ruler, each to within ±1 mm, which she adds together.

The students' teacher prefers measurements that are likely to have less error. Which is the correct order of preference?

(A) Christina's is preferable to Alice's, which is preferable to Bob's

(B) Alice's is preferable to Christina's, which is preferable to Bob's

(C) Alice's and Christina's are equally preferable; both are preferable to Bob's

(D) Christina's is preferable to both Alice's and to Bob's, which are equally preferable

11. In an experiment to study standing waves, you use a string whose mass per length is $\mu = (1.0 \pm 0.1) \times 10^{-4}$ kg/m. You look at the fundamental mode, whose frequency f is related to the length L and tension T of the string by the following equation $L = \frac{1}{2f}\sqrt{\frac{T}{\mu}}$. You make a plot with L on the y-axis and $\sqrt{T}$ on the x-axis, and find that the best fitting line is $y = (8.0 \pm 0.3) \times 10^{-3}\, x + (0.2 \pm 0.04)$ in SI units. What is the value of the frequency of the wave (including the error)? Express your result in SI unit (Hz).

(A) 6250 ± 859.8 Hz (B) 1250 ± 32.3 Hz

(C) 6250 ± 546.9 Hz (D) 4875 ± 287 Hz

12. You measure two quantities as A = 1.0 m ± 0.2 m, B = 2.0 m ± 0.2 m. We should report correct value for $\sqrt{AB}$ as:

(A) 1.4 m ± 0.4 m (B) 1.41m ± 0.15 m (C) 1.4m ± 0.3 m (D) 1.4m ± 0.2 m

13. In a meter bridge experiment the resistance of resistance box is 16Ω, which is inserted in right gap.The null point is obtained at 36cm from the left end. The least count of meter scale is 1mm. What is the value of unknown resistance? (Error = L.C. or L.C./2)

(A) $9 \pm \frac{5}{128}\Omega$ (B) $9 \pm \frac{5}{256}\Omega$ (C) $9 \pm \frac{5}{512}\Omega$ (D) $\frac{128}{9} \pm \frac{1}{2560}\Omega$

14. In a specially designed vernier calipers, 10 divisions of the vernier scale are equal to 4 divisions of the main scale. In the following figure, the zero error and a measurement are shown. In the zero error, the 5^{th} and the 10^{th} divisions of the vernier coincides with 1 mm and 3 mm mark on the main scale respectively and in the measurement, the 2^{nd} and the 7^{th} divisions of the vernier scale coincide with the 23 mm and 25 mm mark on the main scale respectively as shown in the figure. Report the length measured with due regards to the error.

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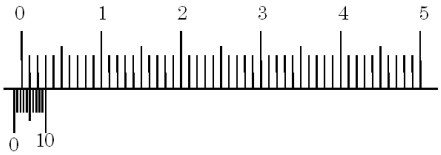
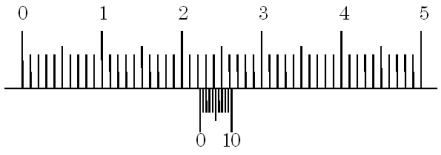
Zero Error



Reading of Measurement



(A) (21.2 ± 0.2) mm (B) (23.2 ± 0.2) mm (C) (23.8 ± 0.4) mm (D) (23.4 ± 0.6) mm

15. A student performs an experiment to determine the Young's modulus of a wire, exactly 2m long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8 mm with an uncertainty of ±0.04 mm at a load of exactly 1.0 kg. The student also measures the diameter of the wire to be 0.4 mm with an uncertainty of ±0.01mm. Take g = 10 m/s^2 (exact). The Young's modulus obtained from the reading is

(A) $(2.0 \pm 0.3) \times 10^{11}$ N/m^2 (B) $(2.0 \pm 0.2) \times 10^{11}$ N/m^2

(C) $(2.0 \pm 0.1) \times 10^{11}$ N/m^2 (D) $(2.0 \pm 0.3) \times 10^{11}$ N/m^2

16. Rank the vernier callipers/screw gauge in the order of their increasing precision :

(1) Screw gauge having main scale division 1mm and no. of circular division = 200.

(2) Screw gauge having main scale division 0.5 mm and no. of circular division = 400.

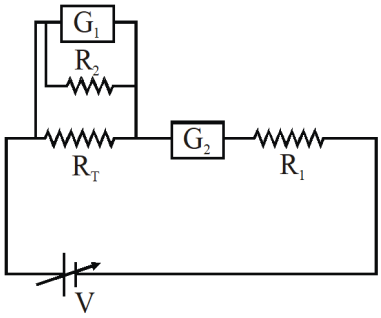
(3) Vernier callipers having main scale division = 1mm such that 9 MSD = 10 VSD.

(4) Vernier callipers having main scale division = 1mm such that 4 MSD = 5 VSD.

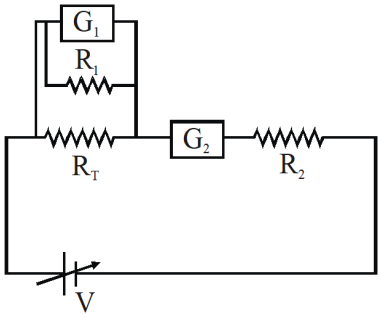
(A) $4 < 2 < 3 < 1$ (B) $2 < 4 < 1 < 3$ (C) $4 < 3 < 2 < 1$ (D) $4 < 3 < 1 < 2$

17. To verify Ohm's law, a student is provided with a test resistor R_T, a high resistance R_1, a small resistance R_2, two identical galvanometers G_1 and G_2, and a variable voltage source V. The correct circuit to carry out the experiment is :–

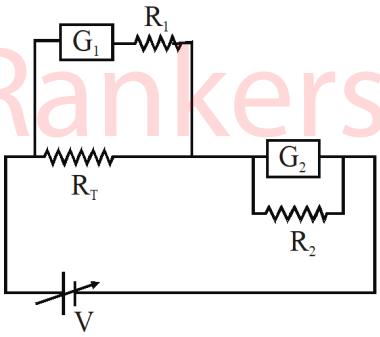
(A)



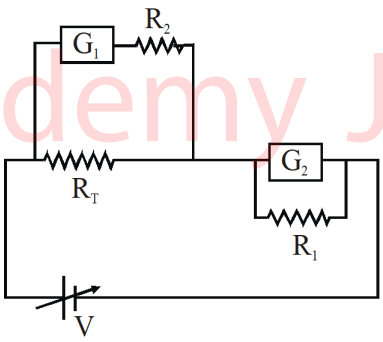
(B)



(C)



(D)



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SECTION-I (ii) : (Maximum Marks: 8)

Multiple Correct Answer Type 3 Q. [4 M (–2)]

Paragraph for Questions 18

If the measurement errors in all the independent quantities are known, then it is possible to determine the error in any dependent quantity. This is done by the use of series expansion and truncating the expansion at the first power of the error. For example, consider the relation $z = x/y$. If the errors in x, y and z are Δx, Δy and Δz, respectively, then

$$z \pm \Delta z = \frac{x \pm \Delta x}{y \pm \Delta y} = \frac{x}{y}\left(1 \pm \frac{\Delta x}{x}\right)\left(1 \pm \frac{\Delta y}{y}\right)^{-1}.$$

The series expansion for $\left(1 \pm \frac{\Delta y}{y}\right)^{-1}$, to first power in $\Delta y/y$, is $1 \mp (\Delta y/y)$. The relative errors in independent variables are always added. So the error in z will be

$$\Delta z = z\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right).$$

The above derivation makes the assumption that $\frac{\Delta x}{x} << 1$, $\frac{\Delta y}{y} << 1$. Therefore, the higher powers of these quantities are neglected.

18. Consider the ratio $r = \frac{(1-a)}{(1+a)}$ to be determined by measuring a dimensionless quantity a. If the error in the measurement of a is $\Delta a (\Delta a/a << 1)$, then what is the error Δr in determining r ?

(A) $\frac{\Delta a}{(1+a)^2}$ (B) $\frac{2\Delta a}{(1+a)^2}$ (C) $\frac{2\Delta a}{(1-a^2)}$ (D) $\frac{2a\Delta a}{(1-a^2)}$

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Paragraph for Questions 19

If the measurement errors in all the independent quantities are known, then it is possible to determine the error in any dependent quantity. This is done by the use of series expansion and truncating the expansion at the first power of the error. For example, consider the relation $z = x/y$. If the errors in x, y and z are Δx, Δy and Δz, respectively, then

$$z \pm \Delta z = \frac{x \pm \Delta x}{y \pm \Delta y} = \frac{x}{y}\left(1 \pm \frac{\Delta x}{x}\right)\left(1 \pm \frac{\Delta y}{y}\right)^{-1}.$$

The series expansion for $\left(1 \pm \frac{\Delta y}{y}\right)^{-1}$, to first power in $\Delta y/y$, is $1 \mp (\Delta y/y)$. The relative errors in independent variables are always added. So the error in z will be

$$\Delta z = z\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right).$$

The above derivation makes the assumption that $\frac{\Delta x}{x} << 1$, $\frac{\Delta y}{y} << 1$. Therefore, the higher powers of these quantities are neglected.

19. In an experiment the initial number of radioactive nuclei is 3000. It is found that 1000 ± 40 nuclei decayed in the first 1.0 s. For $|x| << 1$, In $(1 + x) = x$ up to first power in x. The error $\Delta\lambda$, in the determination of the decay constant λ, in s^{-1}, is :-

(A) 0.04 (B) 0.03 (C) 0.02 (D) 0.01

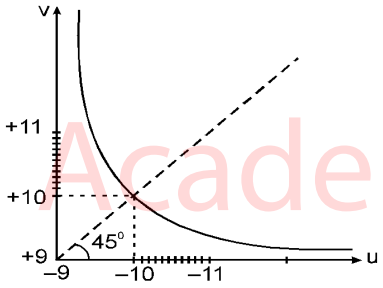
SECTION-I (iii) : (Maximum Marks: 15)

Linked Comprehension Type (Single Option Correct) 1 Para × 3 Q. [3 M (–1)]

20. A teacher asks 4 students to find the speed of sound using resonance column apparatus. The set of apparatus used by each of student was different. The frequency of tuning fork used was known precisely. He wants the error in speed of sound to be within 1%. Which of the student(s) did the experiment correctly. Their results are tabulated below.

Student	Frequency	1st resonance	2nd resonance
(A)	476 Hz	17.0 cm	50.0 cm
(B)	340 Hz	25.2 cm	76.0 cm
(C)	544 Hz	15.6 cm	46.0 cm
(D)	510 Hz	16.0 cm	48.2 cm

21. The graph between object distance u and image distance v for a lens is given below. The value of u and v are in centimeter. Then mark the correct statement(s).



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(A) The focal length of the lens is 5cm.

(B) The focal length of the lens is 0.5cm.

(C) The error in focal length is 0.05 cm

(D) The fractional error in focal length is 0.1

22. A flat circular annular disc has a mass M, outer radius a and inner radius b. The measured values of M, a and b are, M = (0.300 kg ± 0.003 kg), a = (4 cm ± 0.01 cm) & b = (2 cm ± 0.01 cm) :-

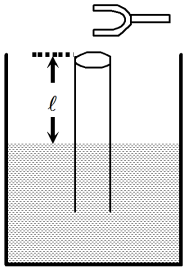
(A) Percentage error in the calculation of a and b are same.

(B) Percentage error in the calculation of M is 1%

(C) Percentage error in the calculation of moment of inertia of the disc about an axis passing through centre and perpendicular to plane of disc is 1.6 %.

(D) Percentage error in the calculation of moment of inertia of the disc about an axis passing through centre and perpendicular to plane of disc is 3.2 %.

23. A tuning fork of known frequency is held at the open end of a long tube, which is dipped into water as shown in the figure. The tuning fork of frequency 165 Hz resonates in first and second harmonic when air columns have lengths $\ell_1 = 50 \pm 0.5$ cm and $\ell_2 = 150 \pm 0.1$ cm. Then



(A) Speed of sound in air is (330 ± 1.98) m/s

(B) Speed of sound in air is (330 ± 1.2) m/s

(C) Percentage maximum error in speed of sound is 0.6%

(D) Percentage maximum error in speed of sound is 0.4%

24. Figure shows an experimental setup to find the value of an unknown resistance R_X using a meter bridge. AB is uniform meter bridge wire of length L = 100 cm. When sliding jockey is at J (AJ = x), the galvanometer shows zero deflection. Length AJ is measured by a meter scale having leastcount of 1mm. In an experiment known resistance is taken as 20 Ω length was found to be (20.0 ± 0.1)cm

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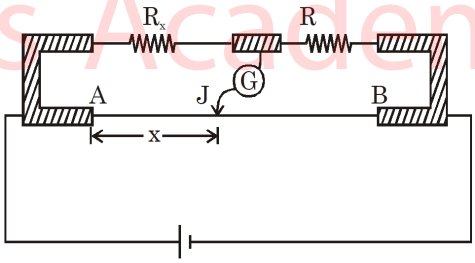

(A) Value of $R_x = (5.00 \pm 0.03)\Omega$

(B) Value of $R_x = (5.00 \pm 0.02)\Omega$

(C) To minimize fractional error the known resistance should be very close to R_X

(D) When fractional error is minimized $x = \frac{L}{2}$

SECTION-III : (Maximum Marks: 20)

Numerical Grid Type (Ranging from 0 to 9) 5 Q. [4 M (0)]

1. Volume of a gas undergoing certain process varis with temperature as shown $V = V_0\left(\frac{T_0}{T}\right)^{1/2} e^{\left[\frac{2}{3}\frac{(T-T_0)}{T_0}\right]}$ where T_0 is initial temperature, V_0 is initial volume. If apparatus rise for measuring temperature can measure temperature with an accuracy of 0.2 k, find the percentage error in calculation of volume at temp $T = \frac{3T_0}{4}$ (V_0 and T_0 are accurately known)

2. The force exerted on a length L of a wire carrying a current I by a magnetic field perpendicular to the wire is measured in an experiment and a graph is prepared between modulus F of the force and segment length L. The plot is a straight line, with slope s = (10 ± 1) × 10^{-5} A·T. The current in the wire is I = (15 ± 1) mA. The modulus of the magnetic induction in SI units, including the error is found to be (x ± y) × 10^{-3} T. Find value of x/y.

3. An open organ pipe containing air resonates in fundamental mode due to a tuning fork. The measured values of length ℓ (in cm) of the pipe and radius r (in cm) of the pipe are ℓ = 94 ± 0.1, r = 5 ± 0.05. The velocity of the sound in air is accurately known. The maximum percentage error in the measurement of the frequency of that tuning fork by this experiment is given by α^2%. Find the value of 10α.

4. A hemispherical bowl of radius R = 0.1m is rotating about its own axis (which is vertical) with an angular velocity ω. A particle of mass 10^{-2} kg on the frictionless inner surface of the bowl is also rotating with the same ω. The particle is at a height h from the bottom of the bowl. It is desired to measure 'g' using this set up, by measuring h accurately. Assuming that R and ω are known precisely and that the least-count in the measurement of h is 10^{-4} m, the numerical value of minimum possible error Δg in the measured value of g is given by $\alpha \times 10^{-2}$ m/s^2. Fill the value of α in OMR sheet.

5. The diameter of a wire of length 100 cm is measured with the help of a screw gauge. The main scale reading is 1 mm and circular scale is reading is 25. Pitch of the screw gauge is 1 mm and the total number of divisions on the circular scale is 100. This wire is used in an experiment of determination of Young's modulus of a wire by Searle's method. The following data are available: elongation in the wire l = 0.125 cm under the tension of 50 N, least count for measuring normal length of wire is 0.01 cm and for elongation in the wire is 0.001 cm. The maximum error in the calculating value of Young's modulus (Y), assuming that the force is measured very accurately, is $\frac{8n}{10}$%, where n is very nearest interger. Find the value of n.

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ANSWER KEY

GR_ERROR

PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	B	A	B	A	C	C	C
	Q.	11	12	13	14	15	16	17			
	A.	C	D	B	B	B	D	C			
SECTION-I (ii)	Q.	18	19								
	A.	B	C								
SECTION-I (iii)	Q.	20	21	22	23	24					
	A.	B ,D	A,C	B,C	A,C	A,C,D					
SECTION-III	Q.	1	2	3	4	5					
	A.	0	7	4	1	3					

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